

Split-Zero: escaping fibres and separated-zeta positivity

Proofs of 22–23 September 2026

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This edition contains the complete fourteen derivations in the order listed below. Its finite algebra, analytic explicit formulas and exact comparison maps retain their own hypotheses. The source and reading guide follows the proofs.

The escaping inverse fibre, its infinitesimal boundary, and its trace form

This derivation retains the original polynomial map, its target coordinates, and its heat constants. It computes an actual dual-number fibre at every double collision, the larger boundary at a triple collision, the exact maps under which infinitesimals vanish, and the signatures of the associated finite trace forms. It does not identify these finite trace forms with the classical Weil distribution. No off-critical zero of the Riemann zeta function is asserted.

Sources actually read

The complete original LaTeX files `tex/satellites/26_incompressible_fibre_heat.tex` and `tex/satellites/23_source_mechanism_transfer.tex` in the local Zeta-Function-Foundation repository were read for this calculation. The received source formulas are `eq:ifh-original`, `eq:ifh-det`, `eq:ifh-chart`, `eq:ifh-target-chart`, `eq:ifh-target-flow`, `eq:ifh-triple`, `eq:retained-node-map`, and `eq:retained-arithmetic-extension`. Their original-map provenance is Tao's displayed map and explanation and Speyer's marked-factor discussion, as cited in those programme sources. The full-support definition used below is reproduced in FSR1–FSR7 of full-support reconstruction proof; its first 70 lines were read. The results below supply their own algebraic proofs.

The exact read coverage was lines 1–480 of the first file, SHA256 72058517553FF3A2C35699844FA95BBA9EC695E774E0E3D4CF33CF7B84DA51DD, and lines 1–666 of the second file, SHA256 D8A65C46B78A5ACE521A164AABC99F246908058E36B60C35BCE938D51C020898. These are local programme source versions, not author-source archives.

The forward heat time called tau in the source is denoted h here, solely to keep it distinct from the global absorber τ_L . The displacement from a specified collision is $d = h - h_c$. All original spatial and target coordinates are retained.

1. Original map and projective inverse chart

Over $\mathbb{C}$, write

$$\begin{aligned} F_1 &= (1 + xy)^3 w + y^2(1 + xy)(4 + 3xy), \\ F_2 &= y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2 y - x^3 w. \end{aligned} \tag{EFI1}$$

The target is (a, b, c) . Define

$$P(r) = cr^3 - 2r^2 + br - 2a, \quad \alpha = \frac{P'(r)}{2}, \quad H(r, \alpha, c) = \left(\frac{1}{\alpha}, r - \alpha, 5\alpha^2 - 3r\alpha - c\alpha^3 \right). \tag{EFI2}$$

Direct substitution before imposing $P = 0$ gives

$$F \circ H = (r^2 + r\alpha - cr^3, 4r + 2\alpha - 3cr^2, c). \tag{EFI3}$$

Consequently $F_2 = b$ is exactly $2\alpha = P'(r)$, and then $2(F_1 - a) = P(r)$. Conversely, from a finite point with $x \neq 0$, recover $\alpha = 1/x$ and $r = y + 1/x$; solving $F_3 = c$ recovers every coordinate of EFI2. This proves that simple roots of P are in bijection with all inverse points with $x \neq 0$. No multiple root supplies a finite point in this chart. On the excluded hyperplane,

$$F(0, y, w) = (w + 4y^2, y, 0). \tag{EFI4}$$

Thus a separate finite point $(0, b, a - 4b^2)$ exists exactly when $c = 0$.

Put projective coordinates $[X : Y : W : T]$ on the compactification of the source, with affine coordinates $(x, y, w) = (X/T, Y/T, W/T)$. Multiplying the homogeneous representative of EFI2 by α gives the everywhere defined map

$$\overline{H}(r, \alpha, c) = [1 : \alpha(r - \alpha) : 5\alpha^3 - 3r\alpha^2 - c\alpha^4 : \alpha]. \tag{EFI5}$$

It agrees with the actual inverse point when $\alpha \neq 0$. Every point with $\alpha = 0$ maps to

$$p_\infty = [1 : 0 : 0 : 0]. \tag{EFI6}$$

This equality of projective points does not identify their infinitesimal neighbourhoods. Those neighbourhoods are computed below.

2. The original two escaping trajectories give exact dual numbers

On the original target line $b = c = 0$, the entire root cover is

$$a = -r^2, \quad \alpha = -2r, \quad (x, y, w) = \left(-\frac{1}{2r}, 3r, 26r^2\right) \quad (r \neq 0). \quad (\text{EFI7})$$

These equalities follow by substituting $c = b = 0$ in EFI2. In particular, the projective graph is

$$(r, a) \mapsto ([1 : -6r^2 : -52r^3 : -2r], a = -r^2). \quad (\text{EFI8})$$

In the projective chart $X = 1$, its coordinate $T/X = -2r$ recovers r . Hence this map identifies the graph closure with the affine r -line; its completed local ring at the escaping point is $\mathbb{C}[[r]]$. The base parameter map is exactly

$$\mathbb{C}[[a]] \longrightarrow \mathbb{C}[[r]], \quad a \mapsto -r^2. \quad (\text{EFI9})$$

The scheme fibre over $a = 0$ is therefore

$$\mathbb{C}[[r]]/(a) = \mathbb{C}[r]/(r^2) \xrightarrow[r \mapsto \epsilon]{r \mapsto \epsilon} D := \mathbb{C}[\epsilon]/(\epsilon^2). \quad (\text{EFI10})$$

The inverse is $\epsilon \mapsto r$. The class r is nonzero because the two residue classes $1, r$ are linearly independent. Thus dual numbers arise from the special fibre of the actual inverse map; they have not been introduced by a resemblance of notation.

The additional finite point of EFI4 is $(0, 0, a)$ and survives at $a = 0$. It is a different component of the compactified inverse fibre. The length-two point EFI10 is supported at infinity, while this extra point is supported at the finite origin. Keeping both preserves total fibre length three on this line.

The original forward heat orbit has

$$a = -\frac{1}{4} + 2h, \quad r^2 = \frac{1}{4} - 2h, \quad d = h - \frac{1}{8}, \quad r^2 = -2d. \quad (\text{EFI11})$$

For $h < 1/8$, taking $r = \mp\sqrt{1 - 8h}/2$ in EFI7 recovers exactly

$$\gamma_{\pm}(h) = \left(\pm(1 - 8h)^{-1/2}, \mp\frac{3}{2}(1 - 8h)^{1/2}, \frac{13}{2}(1 - 8h)\right). \quad (\text{EFI12})$$

Thus the dual-number boundary is attached to those same escaping trajectories with the same time constant.

The exact cancellation and the retained tangent

For a holomorphic germ $f(r) = \sum_{n \geq 0} f_n r^n$ on the graph, the sheet involution is $\sigma(r) = -r$. The trace is

$$\text{Tr}(f) = f(r) + f(-r) = 2 \sum_{n \geq 0} f_{2n} r^{2n}. \quad (\text{EFI13})$$

This is a map to $\mathbb{C}[[a]]$ under EFI9. Its kernel is exactly $r\mathbb{C}[[a]]$, and the trace divided by two has the even-section inclusion as a right inverse. This proves exactly what the sheet trace forgets.

The antisymmetric functional satisfies

$$\frac{f(r) - f(-r)}{2r} \longrightarrow f_1 = f'(0) \quad \text{as } r \longrightarrow 0. \quad (\text{EFI14})$$

The equality follows directly from its power series, which equals $f_1 + f_3 r^2 + f_5 r^4 + \dots$. On a germ pulled back from $a = -r^2$, the numerator is identically zero. On the projective coordinate $T/X = -2r$, the limit is -2 . Therefore the retained limit is a nonzero tangent functional at p_{∞} , killed by projection to the target. Its algebra map is precisely EFI10.

This limit does not assign a finite value to the pole $x = -1/(2r)$. Indeed the two sheet sums and the quadratic moment are

$$x(r) + x(-r) = 0, \quad x(r) - x(-r) = -\frac{1}{r}, \quad x(r)^2 + x(-r)^2 = \frac{1}{2r^2} = -\frac{1}{2a}. \quad (\text{EFI15})$$

Thus a first-moment cancellation coexists with a divergent second moment. The latter is the explicit retained defect in any proposed assertion that all observables cancel.

3. Every double collision in the given heat family

Fix a double root r_0 at a target (a_0, b_0, c_0) . Put

$$q_0 = 3c_0r_0 - 2 \neq 0, \quad t = r - r_0, \quad a = a_0 + 2d, \quad b = b_0 + 6c_0d, \quad c = c_0. \quad (\text{EFI16})$$

The inequality $q_0 \neq 0$ says exactly that the root has multiplicity two rather than three, since $P''(r_0) = 2q_0$. Taylor expansion of this cubic is an exact identity, with no omitted remainder:

$$P = q_0t^2 + c_0t^3 + 2q_0d + 6c_0dt. \quad (\text{EFI17})$$

In the completed local ring at $(d, t) = (0, 0)$, the factor $2q_0 + 6c_0t$ is a unit. Hence EFI17 is equivalent to

$$d = -\frac{t^2(q_0 + c_0t)}{2q_0 + 6c_0t}, \quad \alpha = q_0t + \frac{3}{2}c_0t^2 + 3c_0d. \quad (\text{EFI18})$$

The completed root ring is $\mathbb{C}[[t]]$ with exactly this base map. The coefficient of t in α is $q_0 \neq 0$. A formal series with a nonzero linear coefficient has a unique inverse series: its coefficient of degree one is determined by division by q_0 , and at every later degree the new unknown enters multiplied by the same nonzero q_0 . This recursive argument proves $\mathbb{C}[[\alpha]] = \mathbb{C}[[t]]$.

Since $T/X = \alpha$ is a coordinate of the projective graph, the graph completion is this same ring. In EFI18 the coefficient multiplying t^2 is a unit, so the special fibre is

$$\mathbb{C}[[t]]/(d) = \mathbb{C}[t]/(t^2). \quad (\text{EFI19})$$

The projective tangent has

$$T/X = q_0t, \quad Y/X = r_0q_0t, \quad W/X = 0 \quad \text{modulo } t^2. \quad (\text{EFI20})$$

This retains the original constants and the actual tangent direction. For $c_0 = 0$, one has $q_0 = -2$, and EFI18 gives exactly $d = -t^2/2$, including the quadratic calculation above.

4. Triple collision: root fibre and projective graph fibre differ by exact maps

Now fix $c \neq 0$ and the triple target

$$r_* = \frac{2}{3c}, \quad a_* = \frac{4}{27c^2}, \quad b_* = \frac{4}{3c}. \quad (\text{EFI21})$$

Along its heat orbit $a = a_* + 2d, b = b_* + 6cd$, put $z = r - r_*$. The exact equations are

$$P = c(z^3 + 6dz), \quad \alpha = \frac{3c}{2}z^2 + 3cd. \quad (\text{EFI22})$$

Let $R = \mathbb{C}[d]$ and

$$B = R[z]/(z^3 + 6dz). \quad (\text{EFI23})$$

It is free of rank three over R , with ordered basis $1, z, z^2$, by division by the displayed monic cubic. Its completed local ring is $\hat{B} = \mathbb{C}[[d, z]]/(z^3 + 6dz)$. Its special root fibre is

$$B_0 = B/(d) = \mathbb{C}[z]/(z^3). \quad (\text{EFI24})$$

This is a length-three infinitesimal, rather than a length-two one. It has the surjection $B_0 \rightarrow D, z \mapsto \epsilon$, with kernel (z^2) . It also contains the subalgebra $\mathbb{C}[z^2] \cong D$, with $\epsilon \mapsto z^2$. There is no retraction of this latter inclusion: any homomorphism $B_0 \rightarrow D$ sends z to a nilpotent multiple of ϵ , whose square is zero, whereas a retraction would have to send z^2 to $\epsilon \neq 0$.

The graph closure itself

In the projective chart $X = 1$, write

$$\eta = Y/X = \alpha(r_* + z - \alpha), \quad \omega = W/X = 5\alpha^3 - 3(r_* + z)\alpha^2 - c\alpha^4. \quad (\text{EFI25})$$

The projective graph algebra is the R -subalgebra $A = R[\alpha, \eta, \omega] \subset B$. Define

$$u = z^2 = \frac{2}{3c}(\alpha - 3cd), \quad v = dz = -\frac{\eta - r_*\alpha + \alpha^2}{6c}. \quad (\text{EFI26})$$

The second equality follows because $\alpha z = (3c/2)z^3 + 3cdz = -6cdz$. Thus $u, v \in A$. Conversely $\alpha = (3c/2)u + 3cd$, $\eta = r_*\alpha - \alpha^2 - 6cv$, and

$$\omega = 5\alpha^3 - 3r_*\alpha^2 + 18c\alpha v - c\alpha^4. \quad (\text{EFI27})$$

All projective coordinates therefore belong to $R[u, v]$, proving $A = R[u, v]$ inside B .

Products in this ring satisfy exactly

$$u^2 = -6du, \quad uv = -6dv, \quad v^2 = d^2u. \quad (\text{EFI28})$$

For example $u^2 = z^4 = -6dz^2$, and the other identities follow in the same manner. Every word in u, v reduces, by these identities, to an R -linear combination of $1, u, v$. These three elements are linearly independent in B : a relation would have coefficients $A_0 + A_1z^2 + A_2dz = 0$, and the basis $1, z, z^2$ forces $A_0 = A_1 = dA_2 = 0$; the polynomial ring R has no d -torsion. Hence

$$A \cong R[u, v]/(u^2 + 6du, uv + 6dv, v^2 - d^2u), \quad A = R \oplus Ru \oplus Rv. \quad (\text{EFI29})$$

This proves that the displayed relations are complete.

For $d \neq 0$, $z = v/d$, so $A[d^{-1}] = B[d^{-1}]$, which is the original three-point inverse graph. The algebra B is finite over $R \subset A$, hence is finite over A . The image of its map to the affine projective-coordinate chart is closed, and its coordinate algebra is exactly the subalgebra A . Equivalently, any polynomial vanishing on the graph with $d \neq 0$ is zero in $A[d^{-1}]$; since A is R -free, it was already zero in A . These two facts prove that EFI29 is the scheme-theoretic graph closure.

Its completion is obtained by replacing R with $\mathbb{C}[[d]]$ in EFI29. Its special fibre is

$$A_0 = A/(d) = \mathbb{C}[u, v]/(u^2, uv, v^2). \quad (\text{EFI30})$$

It has two independent square-zero directions. The total graph is flat of rank three over the d -line, as the proved basis shows. This is the exact scheme in which all three escaping branches converge at p_∞ .

The inclusion $A \hookrightarrow B$ induces

$$j_0 : A_0 \longrightarrow B_0, \quad u \longmapsto z^2, \quad v \longmapsto 0. \quad (\text{EFI31})$$

Its kernel is $\mathbb{C}v$, its image is $\mathbb{C} \oplus \mathbb{C}z^2 \cong D$, and its cokernel as a vector space is $\mathbb{C}z$. In particular, the vanishing class v is nonzero in the projective special fibre; its vanishing names this specific morphism, not the absence of an infinitesimal.

The failure of injectivity after specializing has a complete exact sequence. From the two R -bases,

$$0 \longrightarrow A \longrightarrow B \longrightarrow R/(d) \longrightarrow 0, \quad (\text{EFI32})$$

where the last map takes the coefficient of z modulo d . Tensoring the elementary resolution $0 \rightarrow R \xrightarrow{d} R \rightarrow R/(d) \rightarrow 0$ proves

$$0 \longrightarrow \mathbb{C}v \longrightarrow A_0 \xrightarrow{j_0} B_0 \longrightarrow \mathbb{C}z \longrightarrow 0. \quad (\text{EFI33})$$

The first map can be checked without invoking derived notation: the omitted generator $z \in B/A$ is killed by d , and $dz = v \in A$ is its connecting class. This is the precise retained defect of specialization.

Each original branch and its poles

The central root has $z = 0$, $\alpha = 3cd$. Each outer root has $z^2 = -6d$, $\alpha = -6cd$. After the common parameter substitution $d = -q^2/6$, the three roots are $z = 0, q, -q$, and

$$\alpha_{\text{central}} = -\frac{c}{2}q^2, \quad \alpha_{\text{outer}} = cq^2. \quad (\text{EFI34})$$

All projective coordinates in EFI25 converge to EFI6, with T/X of order two in q . The outer branch distinction first enters αz in order three. The exact sum and sum of squares of the original x -coordinates are

$$\sum x_i = \frac{1}{3cd} + 2 \left(-\frac{1}{6cd} \right) = 0, \quad \sum x_i^2 = \frac{1}{6c^2d^2}. \quad (\text{EFI35})$$

The second expression is a nonzero pole, so cancellation of the sum does not cancel the family of observables.

5. Exact positivity of the finite trace forms

These forms are attached to the root and graph algebras just proved. They are not identified here with the Weil form. Their value is that they calculate exactly what trace-vanishing of an infinitesimal can and cannot imply.

For real a , let $B_a = \mathbb{C}[r]/(r^2 + a)$, with the conjugation that conjugates coefficients and fixes r . Define

$$\langle f, g \rangle_a = \text{Tr}_{B_a/\mathbb{C}}(M_{\bar{f}g}), \quad (\text{EFI36})$$

where M_b means multiplication by b . In basis $1, r$, $\text{Tr}(1) = 2$, $\text{Tr}(r) = 0$, and $\text{Tr}(r^2) = -2a$. Thus the Gram matrix is

$$G_2(a) = \begin{pmatrix} 2 & 0 \\ 0 & -2a \end{pmatrix}. \quad (\text{EFI37})$$

It is positive definite for $a < 0$, has inertia $(1, 1, 0)$ for $a > 0$, and has radical $\mathbb{C}r$ at $a = 0$. Here inertia lists positive, negative, and zero dimensions. In particular, a nonzero dual-number direction can be trace-null at the collision and become negative on one side of that collision.

For real d , use $B_d = \mathbb{C}[z]/(z^3 + 6dz)$ with coefficient conjugation fixing z . Multiplication in basis $1, z, z^2$ gives

$$\text{Tr}(1) = 3, \quad \text{Tr}(z) = 0, \quad \text{Tr}(z^2) = -12d, \quad \text{Tr}(z^3) = 0, \quad \text{Tr}(z^4) = 72d^2. \quad (\text{EFI38})$$

For example the matrix of M_z has columns $(0, 1, 0)^T$, $(0, 0, 1)^T$, $(0, -6d, 0)^T$, and squaring it proves the second moment; the last two follow from $z^3 = -6dz$. The Gram matrix is

$$G_3(d) = \begin{pmatrix} 3 & 0 & -12d \\ 0 & -12d & 0 \\ -12d & 0 & 72d^2 \end{pmatrix}, \quad \det G_3(d) = -864d^3. \quad (\text{EFI39})$$

For $d \neq 0$, the $1, z^2$ block has positive first entry and determinant $72d^2 > 0$, so completing the square makes both of its directions positive. The remaining entry is $-12d$. Hence the inertia is $(3, 0, 0)$ when $d < 0$, $(2, 1, 0)$ when $d > 0$, and $(1, 0, 2)$ at $d = 0$. At the latter point the radical is the entire nilpotent ideal $(z) \subset B_0$.

For the graph algebra A_d , in the retained basis $1, u, v$, the same computation using EFI28 gives

$$G_{\text{graph}}(d) = \begin{pmatrix} 3 & -12d & 0 \\ -12d & 72d^2 & 0 \\ 0 & 0 & -12d^3 \end{pmatrix}, \quad \det G_{\text{graph}}(d) = -864d^5. \quad (\text{EFI40})$$

To verify that this is the trace intrinsic to A_d , rather than an unproved transport of trace, multiply $1, u, v$ using EFI28. The diagonal entries of M_u are $0, -6d, -6d$, and those of M_v are all zero. Thus $\text{Tr}(u) = -12d$ and $\text{Tr}(v) = 0$, from which every entry follows. The inertias are exactly those of EFI39, while at $d = 0$ the radical is the two-dimensional ideal (u, v) .

The trace forms before and after the graph inclusion are related by the exact basis map $(1, u, v) \mapsto (1, z^2, dz)$. Thus the change of determinant is the factor d^2 . The quotient of length one in EFI32 is recorded by two additional orders of discriminant vanishing, not by cancellation of the sign on $d > 0$.

The full original cubic has discriminant $c^4 \det G_3(d)$, namely $-864c^4d^3$. The factor c^4 is retained because the displayed polynomial is $c(z^3 + 6dz)$, and each of the three root differences is unchanged while a cubic discriminant has leading-coefficient power four. This agrees with its exact coefficient heat discriminant.

6. The zero-prime globalization and the infinitesimal fibre

Let L be a nontrivial bounded distributive lattice and R a nonzero commutative unital ring. The programme semiring is

$$G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}, \quad (r, \lambda) + (s, \mu) = (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu). \quad (\text{EFI41})$$

Its global additive identity is $\tau_L = (0, 0_L)$, whereas its supported zero is $e_L = (0, 1_L) \neq \tau_L$. Write $Z_L = \{(0, \lambda)\}$. Multiplying by e_L proves $e_L G_L(R) = Z_L$.

The amplitude projection

$$\rho_R : G_L(R) \longrightarrow R, \quad (r, \lambda) \longmapsto r \quad (\text{EFI42})$$

preserves addition, multiplication, and both the global zero and unit. The ideal Z_L is prime exactly when R is a domain. Indeed membership of xy in Z_L is exactly $\rho_R(x)\rho_R(y) = 0$, and every ring element occurs as a supported amplitude. This proves both implications, including the need for the domain hypothesis. For $R = \mathbb{Z}$, the supported-zero prime is therefore the established prime of the original programme; this is not a new claim.

A ring homomorphism $f : R \rightarrow R'$ induces the semiring map

$$G_L(f) : G_L(R) \longrightarrow G_L(R'), \quad (r, \lambda) \longmapsto (f(r), \lambda). \quad (\text{EFI43})$$

It is defined even when a nonzero r maps to zero: its top support remains top, so the image is e_L , not τ_L . The operation identities follow by applying f to amplitudes and leaving the lattice operations unchanged.

In particular EFI10 induces

$$G_L(\mathbb{C}[[r]]) \longrightarrow G_L(D), \quad \hat{r} \longmapsto \hat{\epsilon} =: \eta, \quad \eta^2 = e_L \neq \tau_L. \quad (\text{EFI44})$$

Here $\hat{r} = (r, 1_L)$. The image infinitesimal η is not itself e_L , because its amplitude ϵ is nonzero. Consequently Z_L is not prime in $G_L(D)$: $\eta^2 \in Z_L$ but $\eta \notin Z_L$. The same calculation holds for the two nilpotent directions of $G_L(A_0)$ and the nonzero nilpotents of $G_L(B_0)$. The change is forced by the actual amplitude ring; it cannot be avoided by relabelling e_L as τ_L .

The prime map remains completely explicit. For a ring prime $\mathfrak{p} \subset R$, set

$$Q_{\mathfrak{p}} = Z_L \cup \{(r, 1_L) : r \in \mathfrak{p}\}. \quad (\text{EFI45})$$

The product test through ρ_R proves this is prime. In the dual-number ring every prime contains ϵ , because $\epsilon^2 = 0$; the quotient by (ϵ) is the field $\mathbb{C}$. Thus its unique ring prime is (ϵ) , and the arithmetic prime at the collision is

$$Q_{(\epsilon)} \subset G_L(D), \quad G_L(f)^{-1}(Q_{(\epsilon)}) = Q_{(r)} \subset G_L(\mathbb{C}[[r]]). \quad (\text{EFI46})$$

The contraction follows from $f^{-1}((\epsilon)) = (r)$. The pure support primes Z_a , for lattice prime ideals $\mathfrak{a} \subset L$, contract to the identical support primes: a top-supported amplitude always has top support under EFI43, and a proper lattice ideal does not contain 1_L . This proves the full prime-contraction description for this map.

In the scalar support case $L = \{0, 1\}$, the domain family has

$$(\tau) \subsetneq (e) \subsetneq Q_{(r)}. \quad (\text{EFI47})$$

The infinitesimal fibre instead has the two prime ideals $(\tau) \subsetneq Q_{(\epsilon)}$. Its missing intermediate prime has not been silently deleted: (e) remains an ideal, and EFI44 is its explicit failure of primality. The spectrum map contracts these two primes to (τ) and $Q_{(r)}$, respectively.

Why the pole still needs the compactified boundary

Let $R = \mathbb{C}[[r]]$ and $K = \mathbb{C}((r))$. The inclusion $G_L(R) \hookrightarrow G_L(K)$ is injective because amplitudes and supports are individually preserved. The finite inverse expression $x = -1/(2r)$ belongs to the latter amplitude ring, not the former. If an element $X \in G_L(R)$ extended it while satisfying $-2rX = \hat{1}$, applying EFI42 would give $-2r\rho_R(X) = 1$ in R , impossible by its constant term. At the special fibre, $\eta X = \hat{1}$ is also impossible, since amplitudes would assert that the nilpotent ϵ is invertible. This proves the precise nonextension; EFI8–EFI10 supply the compactified object and its nonzero infinitesimal in its place.

7. Support-valued retention of the calculated trace signatures

Assume L is finite. Define the contracted meet algebra

$$C_L = \mathbb{C}[L, \wedge]/([0_L]). \quad (\text{EFI48})$$

For every $a \neq 0_L$, evaluation $\chi_a([\lambda]) = 1$ when $a \leq \lambda$, and zero otherwise, is multiplicative. In any linear extension of the finite order on $L \setminus \{0_L\}$, their incidence matrix is triangular with ones on the diagonal. Hence the joint evaluation is an algebra isomorphism $C_L \cong \mathbb{C}^{L \setminus \{0_L\}}$. Let E_a denote its coordinate idempotents. Thus $E_a E_b = \delta_{ab} E_a$ and $\sum_a E_a = 1$.

For any one of the finite algebras V above, extend it to $C_L \otimes V$ and give the coefficient algebra the conjugation fixing each E_a . For $x = \sum_a E_a \otimes x_a$ and $y = \sum_a E_a \otimes y_a$, define the support-valued trace form

$$\mathcal{T}_L(x, y) = \sum_a E_a \operatorname{Tr}_V(M_{\overline{x_a y_a}}). \quad (\text{EFI49})$$

This is exactly the trace of the multiplication endomorphism over C_L , because the product decomposition makes its matrix blockwise the multiplication matrix over V . Each χ_a sends EFI49 to the already calculated finite trace form on V , while $v \mapsto E_a \otimes v$ provides its supported section.

For positive real weights $w_a > 0$, applying $\ell(\sum t_a E_a) = \sum w_a t_a$ gives a Hermitian form which is a direct sum of positively weighted copies of the original form. For example, for $a > 0$ in the quadratic family and any nonzero support idempotent E_j ,

$$\ell\mathcal{T}_L(E_j \otimes r, E_j \otimes r) = -2w_j a < 0. \quad (\text{EFI50})$$

For $d > 0$, the triple root direction $E_j \otimes z$ has value $-12w_j d < 0$, and the graph direction $E_j \otimes v$ has value $-12w_j d^3 < 0$. At the collision these nonzero infinitesimal directions are radical directions in every supported component. The support extension therefore retains both the vanishing at the collision and the computed sign on either side. It does not itself prove that the corrected Weil form equals this trace extension.

8. Exact result available for the arithmetic continuation

The actual inverse map supplies a finite cover whose double special fibre is dual numbers, and whose triple projective special fibre has two independent square-zero directions. The sheet trace, graph-to-root specialization, and target projection each have different computed kernels: EFI13, EFI31–EFI33, and EFI14. Their kernels are retained objects, with explicit connecting maps. The original pole observable has the nonzero second moments EFI15 and EFI35. The trace signatures are EFI37, EFI39, EFI40, and their support-valued forms are EFI49.

The exact arithmetic interface already present in the read source is the finite scalar extension $s \mapsto -r^2/2$ of its specified arithmetic quotient, with multiplication matrix $\begin{pmatrix} 0 & -2S \\ I & 0 \end{pmatrix}$. Its square is $-2 \operatorname{diag}(S, S)$. These identities transport the two-sheet geometry while preserving the original spectral coordinate. They do not identify an arbitrary value of the collision parameter with a zero of that arithmetic operator. A corrected Weil calculation must carry its actual test-function map and trace through this interface; no positivity conclusion about that different form has been substituted for this calculation.

9. Exact bridge to the previously observed deformation

The first 135 lines of the programme's OBSERVED_COTANGENT_FROBENIUS_DERIVATION.md were additionally read. Its original coefficients in OCF1–OCF3 are $a_{\text{obs}} > 0$, $\varepsilon_{\text{obs}} > 0$, $r_{\text{obs}} \in \mathbb{C}$,

$$\lambda(s) = a_{\text{obs}} s + \varepsilon_{\text{obs}} r_{\text{obs}} = a_{\text{obs}}(s - s_*), \quad s_* = -\frac{\varepsilon_{\text{obs}} r_{\text{obs}}}{a_{\text{obs}}}, \quad A_{\text{obs}} = \mathbb{C}[s, z_{\text{obs}}]/(z_{\text{obs}}^2 - \lambda(s)z_{\text{obs}}). \quad (\text{EFI51})$$

The subscripts distinguish the observed coefficients from the original map's target a and inverse-root coordinate r . They do not alter any coefficient.

Retain the escaping cover from EFI7 as $C_{\text{esc}} = \mathbb{C}[a, r]/(r^2 + a)$ over $\mathbb{C}[a]$. Define the base map

$$\mathbb{C}[a] \longrightarrow \mathbb{C}[s], \quad a \longmapsto -\frac{\lambda(s)^2}{4}. \quad (\text{EFI52})$$

Its pullback is

$$C_{\text{esc}} \otimes_{\mathbb{C}[a]} \mathbb{C}[s] = \mathbb{C}[s, r]/\left(r^2 - \frac{\lambda(s)^2}{4}\right). \quad (\text{EFI53})$$

There is the exact $\mathbb{C}[s]$ -algebra isomorphism

$$\begin{aligned} \Psi : A_{\text{obs}} &\longrightarrow C_{\text{esc}} \otimes_{\mathbb{C}[a]} \mathbb{C}[s], & z_{\text{obs}} &\longmapsto r + \frac{\lambda}{2}, \\ \Psi^{-1} : C_{\text{esc}} \otimes_{\mathbb{C}[a]} \mathbb{C}[s] &\longrightarrow A_{\text{obs}}, & r &\longmapsto z_{\text{obs}} - \frac{\lambda}{2}. \end{aligned} \quad (\text{EFI54})$$

For proof,

$$\left(z_{\text{obs}} - \frac{\lambda}{2}\right)^2 - \frac{\lambda^2}{4} = z_{\text{obs}}^2 - \lambda z_{\text{obs}}. \quad (\text{EFI55})$$

This verifies both quotient relations, and both compositions fix their generators. Thus the previously observed deformation is the specified base change of the actual escaping cover. This is an isomorphism of these full families, rather than an identification only of their special fibres. The base map is quadratic and is part of the result.

Under EFI54 the sheet involution $r \mapsto -r$ becomes exactly

$$z_{\text{obs}} \mapsto \lambda - z_{\text{obs}}. \quad (\text{EFI56})$$

The two observed factors $z_{\text{obs}} = 0, \lambda$ correspond to $r = -\lambda/2, +\lambda/2$. Consequently the original source coordinates along these sheets, for $\lambda \neq 0$, are respectively

$$\left(\frac{1}{\lambda}, -\frac{3}{2}\lambda, \frac{13}{2}\lambda^2\right), \quad \left(-\frac{1}{\lambda}, \frac{3}{2}\lambda, \frac{13}{2}\lambda^2\right). \quad (\text{EFI57})$$

Substitution into EFI1 gives $(a, b, c) = (-\lambda^2/4, 0, 0)$, as already proved by EFI3. The additional finite point is $(0, 0, -\lambda^2/4)$. Hence the isomorphism carries the actual inverse branches and their remaining finite point, not just the root equation.

The infinitesimal deformation that vanishes to first order

Put $\delta = s - s_*$ and $\kappa = a_{\text{obs}} > 0$, so that $\lambda = \kappa\delta$. Over the first-order base $T_1 = \mathbb{C}[\delta]/(\delta^2)$, EFI54 gives

$$A_{\text{obs}} \otimes_{\mathbb{C}[s]} T_1 \xrightarrow[\sim]{r=z_{\text{obs}}-\kappa\delta/2} T_1[r]/(r^2) = T_1 \otimes_{\mathbb{C}} \mathbb{C}[r]/(r^2). \quad (\text{EFI58})$$

The relation follows because the full centered equation is

$$r^2 = \frac{\kappa^2\delta^2}{4}. \quad (\text{EFI59})$$

Thus the first-order deformation is a product. This is the exact sense in which the deformation class vanishes at first order. The class r itself is nonzero, since $1, r$ are a free T_1 -basis. The map and its inverse in EFI58 reduce to the identity on the central dual-number fibre. Its first-order triviality has therefore not been obtained by removing that fibre.

At second order let $T_2 = \mathbb{C}[\delta]/(\delta^3)$. Then the family is not isomorphic over T_2 to $T_2[\epsilon]/(\epsilon^2)$. To prove this assertion completely, every element of the centered family has a unique form $A + Br$, with $A, B \in T_2$. If it is the image of ϵ under such an isomorphism, its reduction modulo δ must generate the nilpotent ideal of $\mathbb{C}[r]/(r^2)$. Therefore B has nonzero constant term and is a unit. The equation $(A + Br)^2 = 0$ gives, in the free basis $1, r$,

$$2AB = 0, \quad A^2 + B^2 \frac{\kappa^2\delta^2}{4} = 0. \quad (\text{EFI60})$$

Since $2B$ is a unit, the first equation forces $A = 0$. The second then asserts that the nonzero class δ^2 in T_2 is zero after multiplication by the unit $B^2\kappa^2/4$, a contradiction. Thus the first-order triviality has an explicitly nontrivial second-order continuation.

This supplies a precise distinction between a vanishing tangent deformation and a vanishing family. The original escape parameter is $a = -\kappa^2\delta^2/4$: its first derivative is zero at the collision, and its second derivative there is exactly $-\kappa^2/2$.

Trace form and discriminant transported through the isomorphism

Along the real displacement line $s = s_* + \delta$, $\delta \in \mathbb{R}$, the original coefficient $\lambda = \kappa\delta$ is real, even when s_* is not real. Coefficient conjugation fixes this real parameter and fixes z_{obs} . The multiplication trace in basis $1, z_{\text{obs}}$ is

$$G_{\text{obs}}(\lambda) = \begin{pmatrix} 2 & \lambda \\ \lambda & \lambda^2 \end{pmatrix}. \quad (\text{EFI61})$$

Indeed $\text{Tr}(1) = 2$, $\text{Tr}(z_{\text{obs}}) = \lambda$, and $\text{Tr}(z_{\text{obs}}^2) = \lambda^2$, by multiplication in the free basis. For the centered basis $1, r$, the basis-change matrix from its coordinates to the original basis is

$$Q = \begin{pmatrix} 1 & -\lambda/2 \\ 0 & 1 \end{pmatrix}, \quad Q^* G_{\text{obs}} Q = \begin{pmatrix} 2 & 0 \\ 0 & \lambda^2/2 \end{pmatrix}. \quad (\text{EFI62})$$

Direct multiplication verifies this equality. It is precisely the pullback of EFI37 through $a = -\lambda^2/4$: $-2a = \lambda^2/2$. In particular, the observed real-displacement family is positive definite away from the collision and positive semidefinite at the collision, with radical $\mathbb{C}r$ there. Its real base image lies on $a \leq 0$; this is why it does not enter the negative-sign side $a > 0$ of EFI37.

The discriminants satisfy, with every coefficient retained,

$$\text{disc}(r^2 + a) = -4a, \quad (-4a)|_{a=-\lambda^2/4} = \lambda^2 = a_{\text{obs}}^2(s - s_*)^2 = \det G_{\text{obs}}. \quad (\text{EFI63})$$

The original quadratic in EFI2 is $-2(r^2 + a)$, whose polynomial discriminant is $-16a$; its pullback is $4\lambda^2$. This extra factor four belongs to the polynomial's retained leading coefficient. It is not the trace discriminant in the monic algebra basis $1, r$.

The map EFI54 also induces the isomorphism of programme semirings $G_L(\Psi)$ by EFI43, preserving every support label, τ_L , and e_L individually. At $\delta = 0$ it recovers $\hat{r}^2 = e_L \neq \tau_L$. Thus the actual bridge carries the infinitesimal, its zero-prime contraction, and its trace form; no identification of τ_L with supported zero is used.

10. Differential bridge and the primes of the full families

Retain every coordinate of EFI51–EFI55, with $\kappa = a_{\text{obs}}$. Differentiating those exact polynomial maps gives

$$dr = dz_{\text{obs}} - \frac{\kappa}{2} ds, \quad da = -\frac{\kappa\lambda}{2} ds, \quad 2r dr + da = (2z_{\text{obs}} - \lambda) dz_{\text{obs}} - \kappa z_{\text{obs}} ds. \quad (\text{EFI64})$$

For proof substitute $2r = 2z_{\text{obs}} - \lambda$ in the first two equalities: the coefficient of ds is $-\kappa(2z_{\text{obs}} - \lambda)/2 - \kappa\lambda/2 = -\kappa z_{\text{obs}}$. Thus EFI64 is exactly the differential of $r^2 + a = z_{\text{obs}}^2 - \lambda z_{\text{obs}}$. It induces the map between the differential-module presentations

$$\frac{C_{\text{esc}} da \oplus C_{\text{esc}} dr}{C_{\text{esc}}(da + 2r dr)} \longrightarrow \frac{A_{\text{obs}} ds \oplus A_{\text{obs}} dz_{\text{obs}}}{A_{\text{obs}}((2z_{\text{obs}} - \lambda)dz_{\text{obs}} - \kappa z_{\text{obs}} ds)} \quad (\text{EFI65})$$

with scalars extended by EFI52–EFI54. These are the universal modules of differentials: a derivation on a polynomial ring is determined by its values on the two generators, and descends to the quotient exactly when it annihilates the differential of the displayed relation. This proves the presentations and the induced map without assuming smoothness at a collision.

The tangent-deformation comparison can also be proved directly. Over $\mathbb{C}[\eta]/(\eta^2)$, replacing r by $r + \eta h(r)$ changes the first-order coefficient of a relation $r^2 + \eta g(r)$ by $2rh(r)$; multiplying its defining relation by the unit $1 + \eta k(r)$ changes it by $k(r)r^2$. The retained class of that coefficient is therefore

$$[g] \in \mathbb{C}[r]/(r^2, 2r) = \mathbb{C}. \quad (\text{EFI66})$$

For the original escaping unfolding $r^2 + a$, the displacement $a = \eta$ gives $[g] = [1] \neq 0$. For the observed unfolding $z_{\text{obs}}^2 - \kappa\delta z_{\text{obs}}$, the displacement $\delta = \eta$ gives $[g] = [-\kappa z_{\text{obs}}] = 0$. The explicit centering in EFI58 proves actual first-order triviality, not merely the vanishing of an invariant. Equivalently, $da/d\delta = -\kappa^2\delta/2$ vanishes at $\delta = 0$. EFI60 proves exactly what survives at second order.

Primality of supported zero must be checked on the total family as well as on each fibre. The original escape algebra $C_{\text{esc}} \cong \mathbb{C}[r]$ is a domain, so $Z_L = (e_L)$ is prime in $G_L(C_{\text{esc}})$. In the observed total family, $z_{\text{obs}}(z_{\text{obs}} - \lambda) = 0$ with both factors nonzero. Their nonvanishing follows from the free basis $1, z_{\text{obs}}$ and the nonzero polynomial $\lambda = \kappa(s - s_*)$. Hence (e_L) is not prime in $G_L(A_{\text{obs}})$.

There is a precise splitting map between the prime sets. The two ideals $\mathfrak{p}_0 = (z_{\text{obs}})$ and $\mathfrak{p}_1 = (z_{\text{obs}} - \lambda)$ are prime, since their quotients are each $\mathbb{C}[s]$. Their intersection is zero: an element $A(s) + B(s)z_{\text{obs}}$ vanishes on both branches only when $A = 0$ and $B\lambda = 0$, which force $A = B = 0$. They are the only minimal primes, since any prime must contain one factor of $z_{\text{obs}}(z_{\text{obs}} - \lambda)$. Under the composite map $\iota : C_{\text{esc}} \rightarrow A_{\text{obs}}$, $r \mapsto z_{\text{obs}} - \lambda/2$, obtained from base change and Ψ^{-1} in EFI54, r restricts to $-\kappa(s - s_*)/2$ and $+\kappa(s - s_*)/2$ on these branches. Both substitutions are injective maps $\mathbb{C}[r] \rightarrow \mathbb{C}[s]$. Consequently

$$Q_{\mathfrak{p}_0} \cap Q_{\mathfrak{p}_1} = Z_L, \quad G_L(\iota)^{-1}(Q_{\mathfrak{p}_i}) = Z_L \quad (i = 0, 1), \quad (\text{EFI67})$$

with Z_L on the right denoting the zero-amplitude ideal in $G_L(C_{\text{esc}})$. Thus the original supported-zero prime has two arithmetic prime points over it, whose intersection is the retained nonprime zero-amplitude ideal. At the collision both lie below $Q_{(z_{\text{obs}}, \lambda)}$, whose contraction is $Q_{(r)}$. The last contraction follows by evaluating r at zero and retaining the constant coefficient of any polynomial in r .

The triple total families have the same domain issue already before specialization. In B and A respectively,

$$z(z^2 + 6d) = 0, \quad u(u + 6d) = 0, \quad (\text{EFI68})$$

with both factors nonzero. For B this follows from its free basis $1, z, z^2$; for A it follows from its free basis $1, u, v$. Thus their supported-zero ideals are not prime either. Their individual arithmetic primes and their contractions are always the inverse images of the ring primes under EFI42, as proved in EFI45. This statement retains the established prime (e) over domain amplitudes such as $\mathbb{Z}$ and $\mathbb{C}[r]$, and also retains the exact zero-divisor products created by the actual base change and collision.

Exact algebra checks

The companion script `check_escaping_fibre_infinitesimal.py` passes 23 exact polynomial, rational, matrix, and discriminant identities. Its recorded output is `ESCAPING_FIBRE_EXACT_CHECKS.json`. These checks include the original determinant, original inverse composition, both observed branches in the original map, graph-algebra coordinates and products, the two trace matrices, their determinants, and EFI54–EFI63. The ring-theoretic completeness, exact-sequence, and nontrivial second-order arguments are proved in the text rather than inferred from finite sampling.

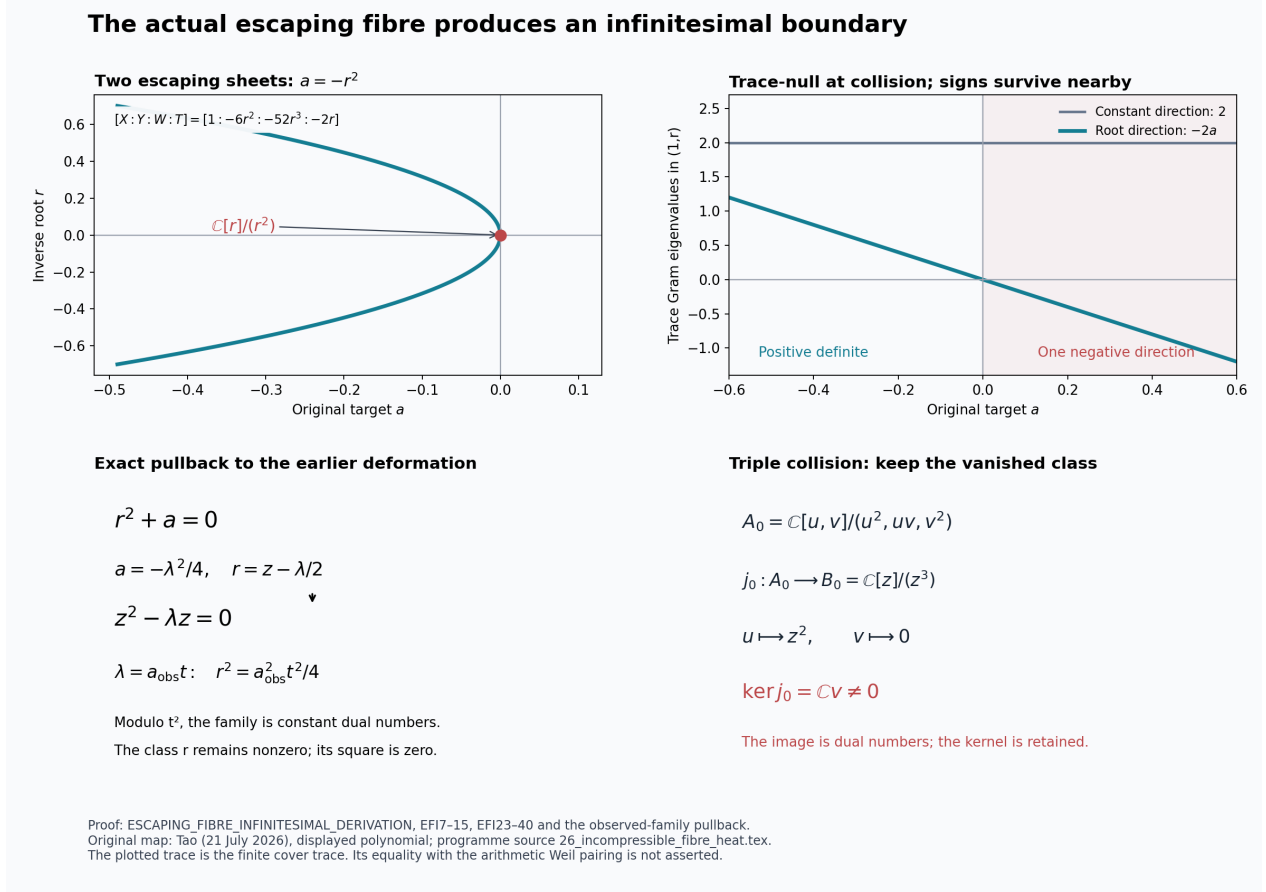


Figure 1: The exact cover $a=-r^2$ and finite trace eigenvalues; the observed-family pullback retains the collision generator. The triple graph map retains its kernel. Proof: EFI7–EFI15, EFI23–EFI40, EFI51–EFI68.

Infinitesimal vanishing, retained currents, and support-valued positivity

22 September 2026. This note computes the several different vanishing statements at the observed collision, the complete signs that survive, and the exact class of corrections capable of changing them. The calculations retain the original observation, metric, Hermitian summand, amplitude multiplier, and full support algebra. No off-line zero of the Riemann zeta function is asserted to exist.

The incoming public programme proofs are *The observed infinitesimal collision and the full arithmetic operator*, OC1–OC17; *Observed support at the shifted nilpotent collision*, OSP1–OSP42; and *Cotangent and Frobenius calculations for the observed collision*, OCF1–OCF23 and OIF1–OIF36. The additional complete local dependencies are full-support reconstruction proof, FSR1–FSR36; finite Weil packet proof, WP1–WP57; and analytic Weil packet proof, WA1–WA30. Their complete sources accompany this derivation; no public link is asserted for an unpublished dependency. The cited proofs retain their original human citations. This is a further finite derivation and an application of the already proved analytic identity WA24, not a replacement for its analytic proof.

1. Original operators and the collision

Keep the original source data and metric from OC1–OC3:

$$\begin{aligned} M &= C + R, \quad C = C^\dagger, \quad R = \epsilon f e^\dagger, \quad e^\dagger e = f^\dagger f = 1, \quad e^\dagger f = 0, \quad \epsilon > 0, \\ J &= G_N^{-1} \Lambda^* Q_B^{1/2}, \quad Q_B = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad J^\dagger J = I_B, \quad C_B = J^\dagger C J = C_B^*. \end{aligned} \quad (\text{ISP1})$$

Here the original arithmetic quotient, cutoff, and observation remain those of OC1: $q = (k+1)^2$, $k \geq 17$, $k \equiv 1 \pmod{4}$, $q-1 \leq N \leq 2q$, and the original surjection $\Lambda : E \rightarrow B$. The star is the adjoint in the Euclidean coordinates supplied by the isometry J ; it does not change the original metric.

Write

$$u = J^\dagger e, \quad v = J^\dagger f, \quad a = u^* u, \quad r = u^* v, \quad d = v^* v, \quad \Delta = ad - |r|^2 > 0.$$

The measured area statement in OC3 supplies this strict inequality. Define

$$g = u/\sqrt{a}, \quad h = \frac{v - (r/a)u}{\sqrt{\Delta/a}}, \quad b = \epsilon\sqrt{\Delta} > 0, \quad P = gg^* + hh^*. \quad (\text{ISP2})$$

The numerator defining h has inner product zero with u , because $u^* v - (r/a)u^* u = 0$. Its squared norm is $d - |r|^2/a = \Delta/a$. Thus g, h are an orthonormal pair and P is their orthogonal projector.

The exact observed family is

$$F_B(s) = (\epsilon v + su)u^*, \quad M_B(s) = C_B + F_B(s), \quad \lambda(s) = as + \epsilon r. \quad (\text{ISP3})$$

Direct multiplication gives $F_B(s)^2 = \lambda(s)F_B(s)$. On the original orthogonal plane $S = \text{im } P$, and on its orthogonal complement H , respectively, it is

$$F_B(s)|_S = \begin{pmatrix} \lambda(s) & 0 \\ b & 0 \end{pmatrix}, \quad F_B(s)|_H = 0. \quad (\text{ISP4})$$

At the collision parameter

$$s_* = -\epsilon r/a, \quad N = F_B(s_*) = \epsilon(v - (r/a)u)u^* = bhg^*, \quad N^2 = 0, \quad Ng = bh \neq 0. \quad (\text{ISP5})$$

The map $\mathbb{C}[\eta]/(\eta^2) \rightarrow \text{End}(B)$, $1 \mapsto I_B$, $\eta \mapsto N$, is injective. Indeed a relation $cI_B + \ell N = 0$ applied to h gives $c = 0$, and then applied to g gives $\ell b = 0$, hence $\ell = 0$. Its image is exactly the generated algebra. The supported version sends its unit to P and has the same proof on S .

2. Exactly what vanishes

For the isolated nilpotent operator, for every scalar t ,

$$\text{tr}(N^j) = 0 \quad (j \geq 1), \quad \det(I_B - tN) = 1, \quad (I_B - tN)^{-1} = I_B + tN. \quad (\text{ISP6})$$

In the basis g, h , N is strictly triangular, and is zero on H . This proves the trace and determinant assertions; multiplication proves the inverse. A marked matrix element retains the exact term

$$x^*((I_B - tN)^{-1} - I_B)y = tb(x^*h)(g^*y). \quad (\text{ISP7})$$

For $x = h, y = g$, this term is tb , which is nonzero when $t \neq 0$. Vanishing of the isolated trace series therefore has an explicit kernel and a nonzero receiver.

The family algebra itself is

$$\mathcal{A} = \mathbb{C}[t, z]/(z(z - at)), \quad t = s - s_*.$$

The invertible change $\xi = z - at/2$, with inverse $z = \xi + at/2$, gives

$$z(z - at) = \xi^2 - a^2t^2/4. \quad (\text{ISP8})$$

Over $\mathbb{C}[t]/(t^2)$, this is an isomorphism with the constant dual-number family, and it reduces to the identity on the fibre $t = 0$. Thus its first-order deformation class vanishes. Over $\mathbb{C}[t]/(t^3)$, the trace discriminant in the basis $1, z$ equals $a^2t^2 \neq 0$; the constant dual-number family's trace discriminant is zero. To compute it, multiplication by z has trace at , and multiplication by $z^2 = atz$ has trace a^2t^2 . The trace matrix is therefore $\begin{pmatrix} 2 & at \\ at & a^2t^2 \end{pmatrix}$, with determinant a^2t^2 . An invertible basis change multiplies its determinant by a unit square. Hence it cannot turn this nonzero discriminant into zero. The first nontrivial order is exactly two.

The original operator representation and its metric retain additional information. Work now over $\mathbb{C}[t]/(t^2)$, and put

$$K = \frac{a}{2b}gh^*, \quad T = I_B + tK, \quad T^{-1} = I_B - tK.$$

Since $[K, N] = (a/2)(gg^* - hh^*)$, direct multiplication proves

$$F_B(s_* + t) = TNT^{-1} + \frac{at}{2}P,$$

$$T^{-1}M_B(s_* + t)T = C_B + N + t\left([C_B, K] + \frac{a}{2}P\right). \quad (\text{ISP9})$$

Here C_B remains the original operator on all of B ; it need not preserve S . These formulas follow by expanding products and using $t^2 = 0$. If the formal parameter is given the real involution $t^* = t$, then

$$T^*T = I_B + t(K + K^*). \quad (\text{ISP10})$$

The coefficient $K + K^*$ is nonzero, because $(K + K^*)h = (a/(2b))g$. Thus this coordinate trivialization is not unitary for the original metric. Equations ISP9–ISP10 retain the exact operator and metric changes attached to the vanishing deformation class.

There is also a choice-dependent Frobenius vanishing. Over a torsionfree integral collision ring $D_p = K_p[\eta]/\eta^2$ with the coefficient Frobenius specified in OIF33, every lift is

$$\Phi_c(\eta) = pc\eta, \quad c \in K_p. \quad (\text{ISP11})$$

To verify this, write an image as $A + B\eta$. Its square is zero, so the coefficient domain forces $A = 0$. Its reduction must equal $\eta^p = 0$, forcing $B \in pK_p$; the converse is immediate. The corrected lift has $c = 0$, so kills η , whereas the earlier multiplicative lift has $c = 1$ and sends it to $p\eta \neq 0$. The same underlying nilpotent algebra therefore does not determine this Frobenius vanishing without the specified lift.

3. The full arithmetic determinant does not vanish infinitesimally

Keep C_B in the characteristic polynomial, and let $n = \dim B$. Rank-one determinant expansion gives

$$\det(\zeta I_B - C_B - N) = \det(\zeta I_B - C_B) - b g^* \text{adj}(\zeta I_B - C_B) h. \quad (\text{ISP12})$$

For a proof, expand the determinant column by column. Every term containing at least two columns from bhg^* vanishes because those columns are proportional to h . The terms containing exactly one such column sum to the displayed adjugate contraction, with the minus sign from the subtracted perturbation. This proof is a polynomial identity and does not require an inverse.

In particular, the isolated identity $\det(I - tN) = 1$ does not remove the second term of ISP12. On the plane, the concrete Hermitian choice $C_B|_S = \begin{pmatrix} 0 & c \\ c & 0 \end{pmatrix}$, $c \in \mathbb{R} \setminus \{0\}$, gives

$$\det(\zeta I_S - C_B|_S - N|_S) = \zeta^2 - c(c + b), \quad \det(\zeta I_S - C_B|_S) = \zeta^2 - c^2. \quad (\text{ISP13})$$

This example disproves an inference from nilpotence alone; it is not substituted for the native C_B .

For the actual family, define

$$p_*(\zeta) = \det(\zeta I_B - M_B(s_*)), \quad q_*(\zeta) = u^* \operatorname{adj}(\zeta I_B - M_B(s_*))u.$$

The same polynomial expansion proves, exactly for every t ,

$$p_{s_*+t}(\zeta) = p_*(\zeta) - tq_*(\zeta). \quad (\text{ISP14})$$

Moreover q_* has degree $n-1$ and leading coefficient $u^*u = a > 0$: the adjugate of $\zeta I_B - M_B(s_*)$ has leading coefficient $\zeta^{n-1}I_B$. Consequently the full characteristic polynomial has a nonzero first derivative in the original parameter, even though the abstract algebra deformation class in ISP8 vanishes. Equivalently, $\operatorname{tr} M_B(s_* + t) = \operatorname{tr} M_B(s_*) + at$. These statements concern the actual original family.

4. The signed current survives with both signs

Use exactly the programme convention

$$W_B(s) = i(M_B(s) - M_B(s)^*).$$

The retained Hermitian term cancels in this particular difference because $C_B = C_B^*$. It remains in ISP12–ISP14. Direct conjugate transposition of ISP4 gives

$$W_B(s)|_S = \begin{pmatrix} -2\operatorname{Im} \lambda(s) & -ib \\ ib & 0 \end{pmatrix}, \quad W_B(s)|_H = 0. \quad (\text{ISP15})$$

At the infinitesimal collision,

$$\begin{aligned} W_* &:= W_B(s_*) = i(N - N^*), & W_*^2 &= b^2 P, \\ W_* \frac{g + ih}{\sqrt{2}} &= b \frac{g + ih}{\sqrt{2}}, & W_* \frac{g - ih}{\sqrt{2}} &= -b \frac{g - ih}{\sqrt{2}}. \end{aligned} \quad (\text{ISP16})$$

These follow by multiplying the two-by-two matrix, or by using $Ng = bh$, $Nh = 0$, $N^*h = bg$, $N^*g = 0$. Its inertia on B is $(1, 1, n-2)$. For $y = c_g g + c_h h + y_H$,

$$y^* W_* y = 2b \operatorname{Im}(\overline{c_g} c_h). \quad (\text{ISP17})$$

Thus the two unit eigenvectors in ISP16 have values $+b$ and $-b$, not zero.

The current correction from the original parameter $s = 0$ is exactly

$$W_* - W_B(0) = 2\epsilon \operatorname{Im}(r) g g^*. \quad (\text{ISP18})$$

For every complex parameter s , the determinant of its active matrix in ISP15 is $-b^2 < 0$. Consequently that rank-one displacement cannot make the current positive semidefinite. Along the real displacement $s = s_* + t$, $t \in \mathbb{R}$, the full current is exactly constant, $W_B(s_* + t) = W_*$, because $\lambda(s_* + t) = at$ is real.

5. All Hermitian corrections on the current plane

Write an arbitrary Hermitian correction on S as

$$D = \begin{pmatrix} \alpha & c \\ \bar{c} & \delta \end{pmatrix}, \quad \alpha, \delta \in \mathbb{R}, \quad c \in \mathbb{C}.$$

Then the complete set of corrections giving a nonnegative form is

$$\boxed{W_* + D \succeq 0 \iff \alpha \geq 0, \delta \geq 0, \alpha\delta \geq |c - ib|^2.} \quad (\text{ISP19})$$

For necessity evaluate on g, h , obtaining the diagonal inequalities, and use the nonnegative determinant. For sufficiency, when $\alpha > 0$, complete the square:

$$\alpha \left| x + \frac{c - ib}{\alpha} y \right|^2 + \left(\delta - \frac{|c - ib|^2}{\alpha} \right) |y|^2 \geq 0.$$

When $\alpha = 0$, the determinant inequality forces $c - ib = 0$, and the form is $\delta|y|^2$. This proves every boundary case as well as the positive-definite case.

In particular, an orthogonal-support correction tP , $t \in \mathbb{R}$, gives positivity exactly for $t \geq b$. A correction supported only on gg^* , or only on hh^* , never suffices. The equality $e^2 = e$ by itself supplies no scalar coefficient t ; ISP19 states the exact correction needed by the operator it would act on.

The analogous complete classification for a larger support space is as follows. Let X, Z be finite-dimensional Hermitian spaces and let

$$\mathcal{Q} = \begin{pmatrix} H & A \\ A^* & D \end{pmatrix} \quad \text{on } X \oplus Z, \quad H = H^*, \quad D = D^*.$$

Let D^+ be the operator equal to the reciprocal of each nonzero eigenvalue of D and zero on its kernel. Then

$$\boxed{\mathcal{Q} \succeq 0 \iff D \succeq 0, \quad A(\ker D) = 0, \quad H - AD^+A^* \succeq 0.} \quad (\text{ISP20})$$

To prove necessity, first restrict to Z . For $z_0 \in \ker D$, the value on (x, tz_0) is $x^*Hx + 2\operatorname{Re}(tx^*Az_0)$. Its nonnegativity for every complex t forces $Az_0 = 0$. Thus A^*x belongs to $(\ker D)^\perp = \operatorname{im} D$. For all x, z , exact multiplication now gives

$$\langle (x, z), \mathcal{Q}(x, z) \rangle = x^*(H - AD^+A^*)x + (z + D^+A^*x)^*D(z + D^+A^*x). \quad (\text{ISP21})$$

Set $z = -D^+A^*x$ for the last necessary inequality. Conversely ISP21 proves positivity from the three displayed conditions. This proof constructs the minimizing support coordinate and includes singular D .

For a fixed Hermitian correction R to the old block, replace H in ISP20 by $H + R$; the result classifies every such corrected extension. Adding independent support coordinates and mixed blocks while leaving H unchanged cannot remove an old negative value: it is still the value on $(x, 0)$. Indeed minimizing over the added positive coordinates subtracts AD^+A^* , exactly as ISP21 records.

6. Full support and every positive scalar trace

Retain the finite bounded distributive lattice L , the actual semiring $G_L(R)$, and its supported zero e_L , with $(e_L) = Z_L$ prime when R is a domain. This is prior programme mathematics, not a new result here. The contracted multiplicative algebra is the proved product

$$\Gamma_L = \mathbb{C}[(G_L(R), \times)]/([\tau_L]) \cong C_L \times D_R, \quad C_L = \mathbb{C}[(L, \wedge)]/([0_L]). \quad (\text{ISP22})$$

In this algebra $[e_L]$ is the identity of the C_L factor. It is a multiplicative idempotent, not the square-zero operator N of ISP5. The latter comes from the separate, explicitly specified deformation/observation receiver. The exact role of the former in the following coefficient extension is multiplication by the support identity.

FSR11–FSR14 give

$$C_L = \bigoplus_{a \in L \setminus \{0\}} \mathbb{C}E_a, \quad E_a E_b = \delta_{ab} E_a, \quad E_a^* = E_a, \quad \sum_a E_a = 1_{C_L}. \quad (\text{ISP23})$$

For completeness, the characters are $\eta_a([\lambda]) = 1_{a \leq \lambda}$; their incidence matrix is triangular with diagonal one, hence invertible. Its inverse gives $E_a = \sum_{\lambda \leq a} \mu(\lambda, a)[\lambda]$. Applying every character proves the stated multiplication and unit identities.

Every positive complex-linear scalar functional on C_L is uniquely

$$\ell\left(\sum_a c_a E_a\right) = \sum_a t_a c_a, \quad t_a \geq 0. \quad (\text{ISP24})$$

Indeed $E_a = E_a^* E_a$ forces $\ell(E_a) \geq 0$; conversely the displayed coefficients give $\ell(v^*v) = \sum_a t_a |v_a|^2 \geq 0$. It is faithful on positive elements precisely when every $t_a > 0$.

Extend the actual current coefficientwise to $B_L = C_L \otimes B$, retaining $1 \otimes C_B$ in its full arithmetic operator. Its support-valued form is

$$\mathcal{W}_L(x, y) = \sum_a E_a x_a^* W_* y_a, \quad x = \sum_a E_a \otimes x_a. \quad (\text{ISP25})$$

For every a with $t_a > 0$, the exact vector

$$x_a^- = E_a \otimes (g - ih)/\sqrt{2} \quad \text{has} \quad \mathcal{W}_L(x_a^-, x_a^-) = -bE_a, \quad \ell\mathcal{W}_L(x_a^-, x_a^-) = -bt_a < 0. \quad (\text{ISP26})$$

Thus every nonzero positive functional retains a negative direction for this form. If exactly r of the weights are positive and $d_L = |L| - 1$, its scalar inertia is

$$(r, r, (n-2)r + n(d_L - r)). \quad (\text{ISP27})$$

This follows by using the two eigenvectors in ISP16 and a basis of H in every sector; the sectors of weight zero contribute their entire n -dimensional space to the radical.

The same proof applied to the full finite Weil pairing FSR28, whose scalar inertia is $(2, 2, 4m - 4)$, gives

$$\text{inertia}(\ell\mathbb{B}_L) = (2r, 2r, (4m-4)r + 4m(d_L - r)). \quad (\text{ISP28})$$

In particular, the regular trace has all weights one. Boolean branch traces select only join-irreducible sectors and put zero weights on the remaining mixed sectors; zero observation there does not prove zero full form. For $L = \mathbb{B}^2$, the mixed projector $w = [1] - [a] - [b]$ obeys $w^2 = w \neq 0$, both branch traces kill it, and FSR34–FSR35 give the exact value $-2mw$ on the specified mixed negative packet.

7. Exact maps from the collision current to a quartet packet

This section constructs a finite comparison without identifying it with the native period observation. Let $\rho = 1/2 + \delta + i\gamma$, $0 < \delta < 1/2$, $\gamma > 2$, and use the four distinct points

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (\rho, 1 - \bar{\rho}, \bar{\rho}, 1 - \rho).$$

For $m \geq 1$, let $d(s) = \prod_i (s - \alpha_i)$, $h(s) = d(s)^m$, $E_h = \mathbb{C}[s]/(h)$, and let e_i be the full primary idempotents. Let $U = M_{j_h(v)}$ be the original amplitude unit, with its full Taylor coefficients and nonzero values $v(\alpha_i)$. Let $a_h = \text{ev } U : E_h \rightarrow \mathbb{C}^4$. The finite pairing is

$$B_h(x, y) = m(\overline{(a_h x)_2}(a_h y)_1 + \overline{(a_h x)_1}(a_h y)_2 + \overline{(a_h x)_4}(a_h y)_3 + \overline{(a_h x)_3}(a_h y)_4). \quad (\text{ISP29})$$

These algebraic data can be defined for arbitrary such points and an arbitrary specified unit; they do not assert a zeta zero.

Define the typed injective linear map

$$\mathcal{I} : S \longrightarrow E_h, \quad c_g g + c_h h \longmapsto \sqrt{b/m} U^{-1} (c_g e_1 - i c_h e_2). \quad (\text{ISP30})$$

Applying a_h gives $\sqrt{b/m}(c_g, -i c_h, 0, 0)$, so injectivity follows. Substitution in ISP29 gives the exact isometry

$$B_h(\mathcal{I}x, \mathcal{I}y) = x^* W_* y \quad (x, y \in S). \quad (\text{ISP31})$$

Explicitly the right side is $-ib\overline{x_g}y_h + ib\overline{x_h}y_g$, and the left side has the same two coefficients. This is an isometry of Hermitian forms, whose signs are indefinite; it is not asserted to preserve the original positive norm.

There is also a nilpotent operator on the complete packet, including all its jets:

$$\mathcal{N}_h(x) = -ib(a_h x)_1 U^{-1} e_2. \quad (\text{ISP32})$$

Since $(a_h U^{-1} e_2)_1 = 0$, it has square zero. It is nonzero on $U^{-1} e_1$, and direct substitution proves

$$\mathcal{N}_h \mathcal{I} = \mathcal{I} N|_S. \quad (\text{ISP33})$$

This comparison preserves the constant b , its complex phase, and the original U . It sends the radical $J_h = (d)/(d^m)$ to zero by the formula for a_h ; every radical jet remains in the domain and in U . Extending ISP30–ISP33 by $C_L \otimes -$ gives the identical maps in every support sector. The construction is explicit and depends on the chosen quartet; equality with the original period receiver has not been asserted or assumed.

8. Interpolation removes endpoint terms without changing any packet jet

Since none of the four points is 0 or 1, $h(0)h(1) \neq 0$. For a polynomial q , set

$$\boxed{\mathcal{R}_h q(s) = q(s) - h(s) \left((1-s) \frac{q(0)}{h(0)} + s \frac{q(1)}{h(1)} \right)}. \quad (\text{ISP34})$$

This is a linear polynomial map. Substitution at the two endpoints gives

$$\mathcal{R}_h q(0) = \mathcal{R}_h q(1) = 0, \quad \mathcal{R}_h q \equiv q \pmod{h}. \quad (\text{ISP35})$$

The latter congruence preserves the derivatives of every order $0, \dots, m-1$ at every α_i , because h has order m there. Conversely divisibility by h is exactly the kernel of the complete quartet jet map, by successive division by the four distinct linear factors. Therefore every original packet class, including every nilpotent jet and the entire action of U , is retained by this map.

Let $s_h : E_h \rightarrow \mathbb{C}[s]$ select the unique representative of degree below $4m$, as supplied by monic polynomial division. The composite

$$\tilde{s}_h = \mathcal{R}_h s_h : E_h \longrightarrow s(s-1)\mathbb{C}[s] \quad (\text{ISP36})$$

is a linear section of polynomial reduction modulo h , with degree at most $4m+1$. Its image has dimension $4m$, and every class has exactly the stated lift. In particular this endpoint condition does not constrain any quartet jet.

The same statement holds for arbitrary specified finite endpoint orders. For $K \geq 1$, put $b_K(s) = s^K(s-1)^K$. It is a unit in E_h , because each $b_K(\alpha_i) \neq 0$. An explicit inverse is obtained by the finite Taylor reciprocal in each factor $\mathbb{C}[t_i]/(t_i^m)$, followed by the primary idempotent inverse of the jet map. Consequently

$$\tilde{s}_{h,K}(x) = b_K s_h([b_K]^{-1}x) \quad (\text{ISP37})$$

is a linear section of reduction modulo h whose values vanish to order at least K at both 0 and 1. Multiplication followed by reduction proves the section identity exactly. This constructs the connecting map for every finite endpoint jet constraint.

9. Consequence for the explicit formula and zero-prime corrections

The following application concerns the programme's counterfactual zeta packet used in WP1–WP4 and WA1–WA2: its four specified points have exact zero order m for $g = 2\xi$, so $v = g/h$ is entire and has nonzero values there. This is the established test of an off-line packet; it is not an existence claim.

For two polynomials q_1, q_2 , use exactly

$$q^\#(s) = \overline{q(1-\bar{s})}, \quad A_{q_1, q_2}(s) = q_1^\#(s)q_2(s)v(s)^2. \quad (\text{ISP38})$$

Let $\tilde{q}_i = \mathcal{R}_h q_i$. Equations ISP35 and WP21 prove that the original full primary amplitude jets of $\tilde{q}_i v$ and $q_i v$ agree. Furthermore

$$A_{\tilde{q}_1, \tilde{q}_2}(0) = A_{\tilde{q}_1, \tilde{q}_2}(1) = 0. \quad (\text{ISP39})$$

Indeed the factor $\tilde{q}_2$ vanishes at both endpoints. This conclusion also follows by evaluating the reflected first factor; no positivity inference is used.

Take the full-jet negative class

$$x_- = U^{-1}(e_1 - e_2), \quad \tilde{q}_- = \tilde{s}_h(x_-). \quad (\text{ISP40})$$

Its four amplitude values are exactly $(1, -1, 0, 0)$, hence

$$B_h([\tilde{q}_-], [\tilde{q}_-]) = -2m. \quad (\text{ISP41})$$

Its Fourier-coordinate polynomial is $\tilde{P}_-(z) = \tilde{q}_-(1/2 + iz)$; define $F_-(z) = \tilde{P}_-(z)v(1/2 + iz)$,

$$K_-(t) = |F_-(t)|^2, \quad k_-(u) = \frac{1}{2\pi} \int_{\mathbb{R}} K_-(t) e^{itu} dt.$$

WA9–WA16 apply to every polynomial, so the increased degree in ISP36 changes their constants but not the proved admissibility, exponential strip decay, or absolute convergence. Substitution into the full analytic identity WA24, using ISP39, gives

$$\boxed{-2m = \frac{1}{2\pi} \int_{\mathbb{R}} |F_-(t)|^2 \left(\operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi \right) dt - \sum_{n \geq 2} \frac{\Lambda_{\text{ar}}(n)}{\sqrt{n}} (k_-(\log n) + k_-(-\log n)).} \quad (\text{ISP42})$$

This is the exact endpoint-free arithmetic expression for that packet, with every sign and transform convention retained. It does not prove that this packet exists for zeta, and does not prove nonnegativity of the arithmetic expression.

More generally let $\mathcal{E}(q)$ be the vector of any fixed finite list of derivatives at 0, 1, and let $D = D^*$ be any matrix on that vector space. Every correction

$$\mathcal{D}(q_1, q_2) = \mathcal{E}(q_1)^* D \mathcal{E}(q_2) \quad (\text{ISP43})$$

vanishes on the image of ISP37 for sufficiently large K . Yet that image surjects onto every packet jet and contains the negative class ISP40. The same conclusion holds if $\mathcal{E}$ takes the endpoint jets of qv : multiplication by the entire function v preserves the required vanishing orders. Thus no correction factoring solely through finitely many endpoint evaluations or endpoint derivatives makes this packet form positive. This is a classification of that class of corrections, not an assumption that all effects of the zero prime belong to it.

In full support, choose any sector E_a and lift the polynomial as $E_a \tilde{q}_-$. Its exact support-valued form is $-2mE_a$, while all the specified endpoint jets are zero. A positive scalar trace with weight $t_a > 0$ reads $-2mt_a < 0$, by ISP24. Mixed sectors obey the same calculation, including sectors omitted by every separate Boolean branch.

The proved zero-prime structure therefore changes the spectrum and enlarges the retained coefficient and observation spaces, as FSR1–FSR22 specify. A corrected analytic formula must supply its actual distribution or operator on these spaces. The results above already settle three concrete possibilities: nilpotence alone does not kill the observed current; coefficientwise full-support extension does not change its surviving signs; and a finite endpoint correction cannot erase the explicit negative quartet test. The spaces and maps exposed by these statements are ISP20–ISP21, ISP23–ISP28, and ISP34–ISP43. They retain the objects on which a further arithmetic correction must act.

10. The endpoint pairing also has a computable sign

For this quartet polynomial,

$$h(0) = h(1) = (|\rho|^2 |1 - \rho|^2)^m > 0, \quad v(0) = v(1) = c_0 := 1/h(0) > 0.$$

The first equality follows by grouping the conjugate factors and using the reflection permutation of the four roots. The second uses the exact convention $g(0) = g(1) = 1$ of WA1. Hence for endpoint vectors $e(q) = (q(0), q(1))$, the pole pairing itself is

$$\begin{aligned} A_{q_1, q_2}(0) + A_{q_1, q_2}(1) &= c_0^2 (\overline{q_1(1)} q_2(0) + \overline{q_1(0)} q_2(1)) \\ &= e(q_1)^* c_0^2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} e(q_2). \end{aligned} \quad (\text{ISP44})$$

Thus the pole contribution by itself has one positive and one negative endpoint direction. Evaluation onto $\mathbb{C}^2$ is surjective by the polynomial $(1-s)c + sd$, so both directions occur. This computes the actual contribution already present in WA24; it does not assign an unproved coefficient to the additional prime (e_L). The section ISP36 kills both of these directions while preserving every packet jet.

11. Exact finite verification

The reproducible script `check_infinisimal_support_positivity.py` passed 38 exact symbolic checks, recorded in `INFINITESIMAL_SUPPORT_POSITIVITY_CHECKS.json`. The checks include the nilpotent/current identities, both signed eigenvectors, the first-order conjugation with a Hermitian matrix coupling the current plane to its complement, the full characteristic polynomial, the current-to-packet isometry and intertwiner, every retained jet in a multiplicity-two quartet example, endpoint vanishing through order two in the higher-order section, the singular-block square-completion identity, and the eight exact module/metric identities of ISP46–ISP53. The calculations use exact rational and symbolic coefficients. The finite sample points are not asserted to be zeta zeros. These checks support the displayed algebraic calculations; they do not replace the general proofs above or assert a new analytic positivity theorem.

12. The positive escaping trace and the original observed metric

The escaping-family isomorphism EFI52–EFI55 uses the exact base map $a_{\text{esc}} = -\lambda^2/4$ and coordinate $r_{\text{esc}} = z_{\text{obs}} - \lambda/2$. Along the real displacement $s - s_*$, the observed coefficient $\lambda = a(s - s_*)$ is real. Its algebra

multiplication trace in the basis 1, z_{obs} is

$$G_{\text{tr}}(\lambda) = \begin{pmatrix} 2 & \lambda \\ \lambda & \lambda^2 \end{pmatrix}. \quad (\text{ISP45})$$

Indeed $z_{\text{obs}}^2 = \lambda z_{\text{obs}}$, multiplication by z_{obs} has trace λ , and multiplication by its square has trace λ^2 , while the identity has trace two. The centered coordinate $r_{\text{esc}} = z_{\text{obs}} - \lambda/2$ gives the congruent matrix $\text{diag}(2, \lambda^2/2)$. Thus this form is positive definite for real $\lambda \neq 0$, and has one-dimensional radical at zero. The following calculation gives its exact relationship to the original metric and current.

Write $z = z_{\text{obs}}$ for the retained observed generator. Retain $S = \text{span}\{g, h\}$ with its original orthonormal frame, and $b = \epsilon\sqrt{\Delta} > 0$. The supported receiver is

$$\rho_\lambda : \mathbb{C}[z]/(z^2 - \lambda z) \longrightarrow \text{End}(S), \quad 1 \mapsto I_S, \quad z \mapsto F_\lambda = \begin{pmatrix} \lambda & 0 \\ b & 0 \end{pmatrix}. \quad (\text{ISP46})$$

It is a homomorphism because $F_\lambda^2 = \lambda F_\lambda$. It is injective: a linear relation $cI_S + dF_\lambda = 0$ applied to h gives $c = 0$, and applied to g then gives $db = 0$. The cyclic module map from its regular representation to S is

$$T_\lambda : \mathbb{C}[z]/(z^2 - \lambda z) \longrightarrow S, \quad 1 \mapsto g, \quad z \mapsto \lambda g + bh, \quad [T_\lambda]_{(1,z),(g,h)} = \begin{pmatrix} 1 & \lambda \\ 0 & b \end{pmatrix}. \quad (\text{ISP47})$$

Its determinant is $b \neq 0$. Direct multiplication proves $F_\lambda T_\lambda = T_\lambda M_z$, where $M_z = \begin{pmatrix} 0 & 0 \\ 1 & \lambda \end{pmatrix}$. Thus regular multiplication and the supported receiver have the same ordinary traces.

The algebra involution used in ISP45 fixes z when λ is real. Its image is not fixed by the original matrix adjoint:

$$\rho_\lambda(z^*) - \rho_\lambda(z)^* = F_\lambda - F_\lambda^* = -iW_*, \quad W_* = \begin{pmatrix} 0 & -ib \\ ib & 0 \end{pmatrix}. \quad (\text{ISP48})$$

The equality follows by conjugate transposition. It is nonzero for every real λ , including zero. Thus the exact map relating the forms is a faithful algebra receiver with this specified adjoint defect.

There are three retained Hermitian forms on the same two-dimensional coordinate space. The algebra multiplication trace gives ISP45. Pulling back the original vector metric through ISP47 gives

$$G_{\text{vec}} = T_\lambda^* T_\lambda = \begin{pmatrix} 1 & \lambda \\ \lambda & \lambda^2 + b^2 \end{pmatrix}. \quad (\text{ISP49})$$

Pulling back the matrix Hilbert–Schmidt metric through ISP46 gives

$$G_{\text{HS}} = \begin{pmatrix} \text{Tr}(I_S) & \text{Tr}(F_\lambda) \\ \text{Tr}(F_\lambda^*) & \text{Tr}(F_\lambda^* F_\lambda) \end{pmatrix} = \begin{pmatrix} 2 & \lambda \\ \lambda & \lambda^2 + b^2 \end{pmatrix}. \quad (\text{ISP50})$$

Every entry follows by multiplying the displayed matrix F_λ . In particular

$$G_{\text{HS}} - G_{\text{tr}} = \begin{pmatrix} 0 & 0 \\ 0 & b^2 \end{pmatrix}, \quad \|\rho_0(z)\|_{\text{HS}}^2 = b^2 > 0. \quad (\text{ISP51})$$

The trace-null generator at collision has a nonzero, exactly measured operator norm. The supported unit in ISP46 has rank two. Using instead the full unital receiver $1 \mapsto I_B$ changes the upper-left entry of ISP50 to $\dim B$; it does not change its other three entries.

The trace pairing makes multiplication by the real generator self-adjoint, as matrix multiplication verifies:

$$G_{\text{tr}} M_z = M_z^* G_{\text{tr}}.$$

For the positive matrix metric the exact defect is instead

$$G_{\text{HS}} M_z - M_z^* G_{\text{HS}} = \begin{pmatrix} 0 & -b^2 \\ b^2 & 0 \end{pmatrix}. \quad (\text{ISP52})$$

This is obtained by subtracting the two products, with no limiting argument. Thus the extra positive term in ISP51 has a specified effect on multiplication adjoints; it cannot be added while asserting unchanged adjoint compatibility.

Finally, transporting the trace form itself to the original vector plane gives

$$Q_\lambda = (T_\lambda^{-1})^* G_{\text{tr}} T_\lambda^{-1} = \begin{pmatrix} 2 & -\lambda/b \\ -\lambda/b & \lambda^2/b^2 \end{pmatrix}, \quad \det Q_\lambda = \lambda^2/b^2. \quad (\text{ISP53})$$

The inverse $T_\lambda^{-1} = \begin{pmatrix} 1 & -\lambda/b \\ 0 & 1/b \end{pmatrix}$ proves the formula by multiplication. This positive metric for $\lambda \neq 0$ makes F_λ self-adjoint; at the collision it becomes degenerate and kills h . There cannot be a positive definite metric making the nonzero $N = bhg^*$ self-adjoint: that would imply $\langle Nx, Nx \rangle = \langle x, N^2 x \rangle = 0$ for every x , forcing $N = 0$. The explicitly constructed degenerate metric ISP53 is the retained limiting object, while the original positive metric and the current W_* remain unchanged.

The Hermitian summand C_B of the full arithmetic operator remains in ISP12–ISP14. The present algebra comparison concerns F_λ , and makes no commutation assertion about C_B or identification of the trace metric with the classical Weil form.

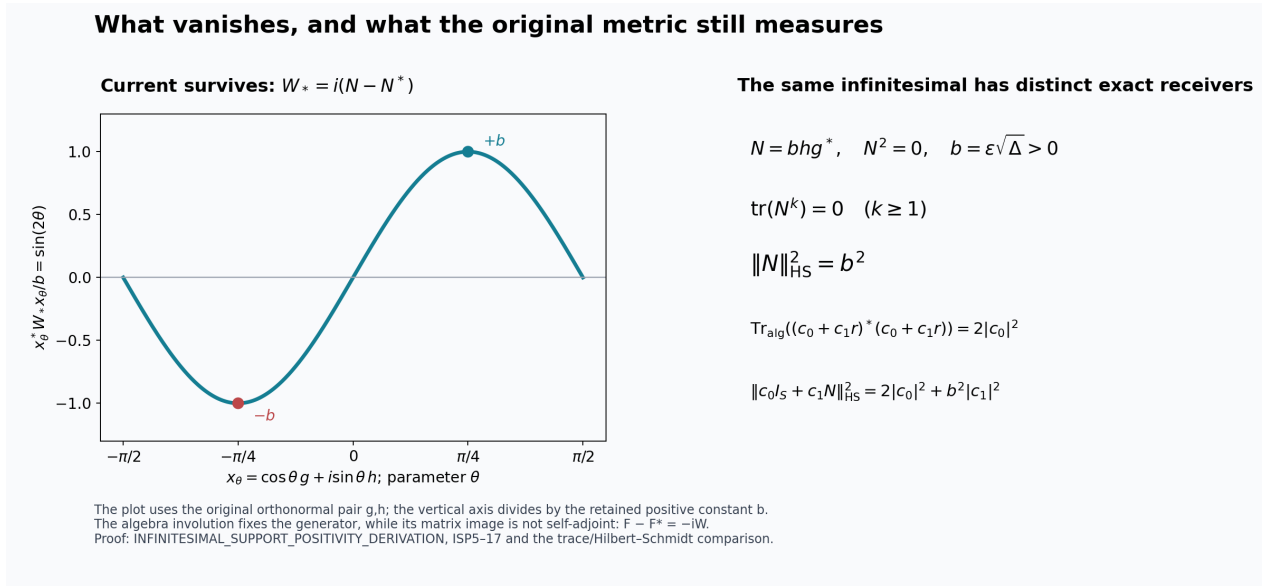


Figure 2: The same nonzero infinitesimal has zero algebraic trace and nonzero matrix norm; its current has both signs in the original metric. Proof: ISP5–ISP6, ISP18–ISP20, ISP45–ISP53.

The logarithmic arithmetic distribution of the separated-support zeta sheet

This calculation uses the actual interpolation in the programme's *Secondary Note on Split-Zero Globalization*, canonical source PUBUNIT-B1A260A5E731EA568C353AA4, equations under `def:hurwitz-specialization`, `thm:universal-taylor-deformation`, `thm:arithmetic-kernels-riemann`, and `cor:first-log-jet-explicit`. It also uses *Split-Zero Support Repair*, some `considerations.tex`, sections “Conservative ideal and zeta side” and “Shifted zeta deformation”. These are prior programme results. The supported-zero prime is also a prior result of Paper 1, attributed by the user to Gemini Pro 2.5. The calculation below derives the signed arithmetic distribution that replaces the prime-power distribution for this particular separated sheet. It does not assign an arbitrary finite norm to the supported-zero prime.

NL1. The original scalar maps and the actual analytic family

Let $R = \mathbb{Z}$, and let $G(R) = R \sqcup \{\tau\}$. Addition and multiplication on supported elements are their ring operations; $\tau + x = x$ and $\tau x = \tau$. Write $e = 0_R$, so $e \neq \tau$, $e^2 = e$, and $er = e$ for every supported r . The two maps

$$p : G(R) \rightarrow R, \quad p(r) = r, \quad p(\tau) = 0, \quad \chi : G(R) \rightarrow \mathbb{B}, \quad \chi(r) = 1, \quad \chi(\tau) = 0 \quad (\text{NL1})$$

preserve both operations and units, directly by these laws. The prime ideal $(e) = \{\tau, e\}$ is $p^{-1}(0)$; its complement consists of nonzero integers and is multiplicatively closed. This proves its primality without identifying its generator with the semiring zero.

The separated residue map $G(\mathbb{Z}) \rightarrow G(\mathbb{Z}/n\mathbb{Z})$, obtained by reducing the supported amplitude and keeping τ , has target cardinality $n + 1$. This holds also for $n = 1$: the zero ring has a supported zero and the distinct τ . Composing with amplitude gives $\mathbb{Z}/n\mathbb{Z}$, of cardinality n . Thus the two defined sums are

$$F_0(s) = \sum_{n \geq 1} n^{-s} = \zeta(s), \quad F_1(s) = \sum_{n \geq 1} (n + 1)^{-s} = \zeta(s) - 1. \quad (\text{NL2})$$

Their interpolation is

$$F_t(s) = \sum_{n \geq 1} (n + t)^{-s} = \zeta(s, 1 + t), \quad t \geq 0, \quad \Re s > 1. \quad (\text{NL3})$$

Here t is a real shift parameter, not the semiring element τ . The equality with Hurwitz zeta is the defining series with its index translated; compare the original TeX formulas at DLMF 25.11.1 and 25.11.3.

The prime (e) cannot be inserted into the old ideal Euler product with a positive finite multiplicative norm extending the integer ideal norm. Indeed $(e)I_{(n)} = (e)$; a multiplicative norm h would satisfy $h((e)) = nh((e))$, so $h((e)) = 0$ for $n = 2$. This equality specifies the obstruction. The separated residue sum (NL3) is a different, already defined analytic receiver of support. Its logarithmic distribution is now calculated in full.

NL2. A convergent expansion with every multiplicative interaction retained

Put

$$b = 1 + t, \quad r_n = \frac{n + t}{1 + t} \quad (n \geq 2), \quad H_t(s) = \sum_{n \geq 2} r_n^{-s}. \quad (\text{NL4})$$

Every $r_n > 1$, and

$$F_t(s) = b^{-s}(1 + H_t(s)). \quad (\text{NL5})$$

There exists $\sigma_0 > 1$ with $H_t(\sigma_0) < 1$. To prove this, fix any $\sigma_1 > 1$. The terms $r_n^{-\sigma}$ tend to zero as $\sigma \rightarrow \infty$, and for $\sigma \geq \sigma_1$ they are dominated by the summable sequence $r_n^{-\sigma_1}$. Dominated convergence gives $H_t(\sigma) \rightarrow 0$.

On every closed half-plane $\Re s \geq \sigma_0$ with $H_t(\sigma_0) < 1$, the series

$$\log F_t(s) = -s \log b + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \sum_{n_1, \dots, n_k \geq 2} (r_{n_1} \cdots r_{n_k})^{-s} \quad (\text{NL6})$$

uses the holomorphic logarithm of $1 + H_t$ defined by its power series. Its absolute sum is at most $\sum_{k \geq 1} H_t(\sigma_0)^k/k < \infty$. Differentiation is locally uniformly justified: for a slightly smaller abscissa still exceeding 1 and with $H_t < 1$,

$$J_t(\sigma) = \sum_{n \geq 2} (\log r_n) r_n^{-\sigma} < \infty,$$

and the absolute sum of the differentiated length terms of length k , after the factor $1/k$, is $J_t(\sigma)H_t(\sigma)^{k-1}$. Summing this geometric bound proves

$$-\frac{F'_t(s)}{F_t(s)} = \log b + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \sum_{n_1, \dots, n_k \geq 2} \log(r_{n_1} \cdots r_{n_k}) (r_{n_1} \cdots r_{n_k})^{-s}. \quad (\text{NL7})$$

No factor indexed by an ordinary prime has been assumed.

Define the signed measure on $[0, \infty)$

$$\nu_t = (\log b)\delta_0 + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \sum_{n_1, \dots, n_k \geq 2} \ell(n_1, \dots, n_k) \delta_{\ell(n_1, \dots, n_k)}, \quad \ell(n_1, \dots, n_k) = \sum_i \log r_{n_i}. \quad (\text{NL8})$$

It is locally finite: if $\ell \leq M$, then $k \leq M/\log r_2$, and every $r_{n_i} \leq e^M$, leaving finitely many tuples. Equal lengths are added with their exact signed multiplicities. The same bound proving (NL7) gives $\int e^{-\sigma \ell} |d\nu_t| < \infty$ in the stated right half-plane. Its Laplace transform is exactly $-F'_t/F_t$.

NL3. An explicit negative arithmetic atom at the separated endpoint

At $t = 1$, $b = 2$, $r_n = (n+1)/2$, and

$$H_1(4) = \sum_{m \geq 3} (2/m)^4 \leq \frac{16}{81} + \int_3^\infty 16x^{-4} dx = \frac{32}{81} < 1. \quad (\text{NL9})$$

Thus (NL7) holds, in particular, for $\Re s \geq 4$, with the convergence estimate just proved. There is a positive atom

$$\nu_1(\{\log(3/2)\}) = \log(3/2). \quad (\text{NL10})$$

Only the one-term tuple $n_1 = 2$ has that length: a tuple of at least two terms has product at least $(3/2)^2$.

There is also the exact negative atom

$$\boxed{\nu_1(\{\log(9/4)\}) = -\log(3/2)}. \quad (\text{NL11})$$

For a one-term tuple, $(n+1)/2 = 9/4$ would give $n = 7/2$, impossible. A tuple of at least three terms has product at least $27/8 > 9/4$. For two terms both factors are at least $3/2$, so equality forces both to be $3/2$, that is, the single ordered tuple $(2, 2)$. Its coefficient in (NL8) is $-\frac{1}{2} \log(9/4) = -\log(3/2)$. This proves (NL11) with no truncation.

Consequently the logarithmic arithmetic distribution for the actual separated sheet is signed. The original von Mangoldt distribution, $\sum_{n \geq 2} \Lambda(n) \delta_{\log n}$, cannot be reused for this sheet. The negative atom alone is not a negative value of the full Weil form: the gamma, divisor, reflection-defect, and boundary terms of a complete contour identity must also be retained. It is an exact arithmetic coefficient that such an identity must contain.

NL4. The first infinitesimal is nonzero and contains mixed prime data

For $|t| < 1$, the binomial series gives, locally uniformly for $\Re s > 1$,

$$F_t(s) = \zeta(s) - t s \zeta(s+1) + O(t^2). \quad (\text{NL12})$$

For completeness, fix a compact set of s and a radius $r < 1$. The Taylor remainder in $(1+t/n)^{-s}$ is bounded by a constant times $|t|^2/n^2$, uniformly on that compact set; after multiplication by n^{-s} it is summable. This proves the asserted termwise expansion and its holomorphic derivatives.

The Euler product of ζ , used only at $t = 0$, gives

$$Q(s) = \frac{\zeta(s+1)}{\zeta(s)} = \sum_{n \geq 1} \alpha(n) n^{-s}, \quad \alpha(1) = 1, \quad \alpha(n) = (-1)^{\omega(n)} \frac{\prod_{p|n} (p-1)}{n}. \quad (\text{NL13})$$

Indeed the local series is

$$\frac{1-p^{-s}}{1-p^{-s-1}} = 1 - \sum_{j \geq 1} \frac{p-1}{p^j} p^{-js}.$$

Its absolute product converges for $\Re s > 1$; multiplying the series proves (NL13). Thus

$$\partial_t \log F_t(s)|_{t=0} = -sQ(s), \quad \partial_t \left(-\frac{F'_t}{F_t} \right) \Big|_{t=0} = Q(s) + sQ'(s) = \sum_{n \geq 1} \alpha(n)(1 - s \log n)n^{-s}. \quad (\text{NL14})$$

The derivative series is absolutely convergent on compact subsets of $\Re s > 1$, since $|\alpha(n)| \leq 1$ and $\sum (1 + \log n)n^{-\sigma} < \infty$. For real $s > 1$, $sQ(s) > 0$, so the first deformation of $\log F_t$ does not vanish. This infinitesimal is not the first-order-trivial node deformation from the observed collision.

Equation (NL14) is equivalently the Laplace transform of the locally finite distribution

$$\nu_0 = \sum_{n \geq 1} \alpha(n) (\delta_{\log n} - (\log n) \delta'_{\log n}). \quad (\text{NL15})$$

Here $\langle \delta'_\ell, f \rangle = -f'(\ell)$, so its Laplace transform is $se^{-s\ell}$. This fixes the derivative sign in (NL15).

The dot also denotes the actual distributional derivative of (NL8), not only a matching transform. On a fixed compact length interval and for $0 \leq t \leq 1$, the inequalities $r_2(t) \geq 3/2$ and $r_n(t) \geq (n+1)/2$ give a uniform finite bound on the tuples which can enter. Differentiating their smooth moving atoms therefore defines the derivative on every compactly supported smooth test. For weighted Laplace tests, $|\partial_t \log r_n| \leq 1$. A tuple of length k has length derivative bounded by k ; differentiating its coefficient and its exponential introduces at most one further factor k , together with its original length. On a sufficiently far right half-plane, uniformly in a neighborhood of zero, the sums are bounded by geometric series of the forms $\sum k B^k$ and $\sum k D B^{k-1}$, with $B < 1$ and $D < \infty$ as above. Thus Laplace transformation commutes with differentiation. At zero all tuple lengths are logs of positive integers. The resulting distribution has the form $\sum_n (a_n \delta_{\log n} + b_n \delta'_{\log n})$. These coefficients are uniquely determined by its Laplace transform: multiply the difference of two such transforms by the exponential of its least remaining length and let the real argument tend to infinity; the leading polynomial of degree at most one must be zero, and induction removes every length. The exponentially weighted convergence just proved bounds the remaining tail in each step. Comparing with (NL14) proves exactly $a_n = \alpha(n)$, $b_n = -(\log n)\alpha(n)$, as in (NL15).

For distinct primes p, q ,

$$\alpha(pq) = \frac{(p-1)(q-1)}{pq} \neq 0. \quad (\text{NL16})$$

These are supported at products of distinct primes, whereas the original von Mangoldt coefficients vanish there. The deformation therefore has explicitly nonzero mixed-prime terms already at first order; they cannot be removed by keeping only a modified weight at each prime power.

NL5. Full finite support and exact specialization

For a finite bounded distributive support lattice L , the actual set is

$$G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\}. \quad (\text{NL17})$$

Let $c = |L| - 1$. Its separated residue over $\mathbb{Z}/n\mathbb{Z}$ has $n + c$ elements: n supported top elements and $|L| - 1$ lower support elements. Hence its scalar cardinality receiver is

$$F_c(s) = \sum_{n \geq 1} (n + c)^{-s} = \zeta(s) - \sum_{n=1}^c n^{-s}. \quad (\text{NL18})$$

This proves exactly how (NL3) receives the full support count. It does not identify different support labels with one another in G_L . The cardinality map forgets that distinction; the full meet-algebra coefficients and their primitive idempotents remain separate in the programme's full-support reconstruction. Formula (NL8) applies with $t = c$ to the count receiver. All logarithmic product tuples, including their collisions in the length variable, are retained.

NL6. Consequence for the positivity calculation

There are three exact facts to carry forward. First, the supported-zero prime remains a nontrivial prime of $G(\mathbb{Z})$; no positive finite multiplicative ideal norm extending the old norm can give it an ordinary new Euler factor. Second, the programme already supplies a different analytic receiver, the separated quotient-size interpolation (NL3), whose logarithmic derivative is (NL7), with the explicit signed atom (NL11). Third,

its infinitesimal variation is (NL14), not zero. A corrected explicit formula for this receiver must use these arithmetic terms and its own completed divisor. The exact escaping-fibre trace and observed-operator current are calculated in the accompanying derivations; equality with a Weil form requires the actual map, not an identification of their names.

The two primitive-defect source files in folders 3 and 4 are byte-identical (SHA256 2f3289f6a04105d3f15fe5f7c36dde286515e61f465caf1582f9808e19f3d2f7). Their negative augmentation retains the kernel of a scalar cancellation. That same instruction is followed here: the full signed length distribution (NL8) and mixed-prime distribution (NL15) are retained rather than discarded when an ordinary Euler presentation fails.

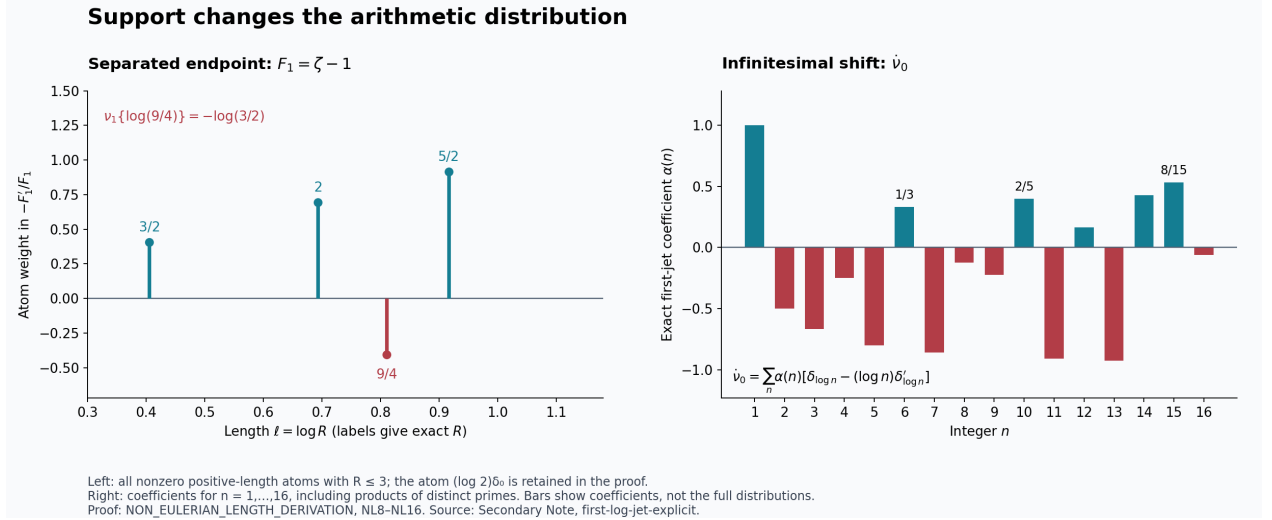


Figure 3: All nonzero positive-length atoms through ratio 3, and the exact first arithmetic coefficients; the zero-length mass is stated separately. Proof: NL7–NL16.

The actual separated zeta sheet, its explicit formula, and its positivity defect

This calculation uses the quotient-size interpolation in the supplied original LaTeX. It retains the supported arithmetic zero e , the unsupported element τ , the distinct residue quotients, the completed-function endpoint residues, every gamma pole crossed by the contour, and the functional-equation defect. The final counterexample concerns this particular shifted zeta sheet. It is not a counterexample to the classical Riemann hypothesis.

Source coverage for this derivation: `globalization nte/2/split_zero_secondary_note.tex`, lines 1111–1555, read in full; `globalization nte/4/split_zero_defect_preprint (1).tex`, lines 500–854, read in full; the standing definitions, quotient sheets, and shifted-deformation sections of `globalization nte/5/some_considerations.tex` were read in the returned source text. The whole-file read of the latter was truncated in an unrelated middle section and is not recorded as complete. The relevant named results are `thm:universal-taylor-deformation`, `cor:jets`, `thm:arithmetic-kernels-riemann`, `cor:first-log-jet-explicit`, `prop:hurwitz-standard-package`, and the defect preprint’s “Endpoint residues” and “Invariant/anti-invariant decomposition.” The secondary note lists The Clankers; `some_considerations.tex` lists “The Clankers / prepared in continuation”; the defect preprint has a blank author field. The source-reading ledger maintained by the root task records canonical source IDs. This file supplies the additional proofs below; it does not claim those older formulas as new.

SW1. The typed quotient maps and the interpolation they actually give

Let $S = G(\mathbb{Z}) = \mathbb{Z} \sqcup \{\tau\}$, with $e = 0_{\mathbb{Z}} \neq \tau$. For $n \geq 1$, the two successive surjections are

$$S \xrightarrow{\pi_n} G(\mathbb{Z}/n\mathbb{Z}) \xrightarrow{p_n} \mathbb{Z}/n\mathbb{Z}, \quad \pi_n(\tau) = \tau_n, \quad \pi_n(a) = a \bmod n, \quad p_n(\tau_n) = 0. \quad (\text{SW1})$$

The first target has $n + 1$ elements, and the second has n . Their zero fibres in the source are respectively $\{\tau\}$ and $\{\tau\} \cup n\mathbb{Z}$. These statements follow directly from the definitions, including at $n = 1$, when the first quotient has two elements.

The two Dirichlet sums, and the supplied interpolation between their numerical weights, are

$$F_t(s) = \sum_{n \geq 1} (n+t)^{-s} = \zeta(s, 1+t), \quad F_0 = \zeta, \quad F_1 = \zeta - 1, \quad 0 \leq t \leq 1. \quad (\text{SW2})$$

Initially these equalities hold for $\Re s > 1$. The identity at $t = 1$ is a change of index in an absolutely convergent series. The intermediate values are numerical interpolation of the cardinality weights; they are not asserted to count a nonintegral number of elements in a quotient.

In particular, SW2 is not the separate prime-weight Euler product $\prod_p (1 - (p+1)^{-s})^{-1}$. Its distinction from that product is proved with the actual quotient maps in TN1–TN9 of prime-norm comparison proof. For example the weights satisfy $(m+1)(n+1) - (mn+1) = m+n$; retaining the labels gives the multiplicative vector $(1, n)$, while summing its entries gives the nonmultiplicative number $n+1$.

SW2. Analytic continuation and estimates used below

Write $a = 1 + t \in [1, 2]$. For any integer $M \geq 1$, Euler summation with $2M$ derivatives gives

$$F_t(s) = \frac{a^{1-s}}{s-1} + \frac{a^{-s}}{2} + \sum_{j=1}^M \frac{B_{2j}}{(2j)!} (s)_{2j-1} a^{1-s-2j} - \frac{(s)_{2M}}{(2M)!} \int_0^\infty \tilde{B}_{2M}(x) (x+a)^{-s-2M} dx, \quad \Re s > 1 - 2M. \quad (\text{SW3})$$

Here $\tilde{B}_k(x) = B_k(\{x\})$, and $(s)_j = s(s+1) \cdots (s+j-1)$. To check the formula, apply integration by parts on every interval $[n, n+1]$, using $B'_k = kB_{k-1}$, $B_k(1) = B_k(0)$ for $k \neq 1$, and $B_1(1) - B_1(0) = 1$. Summing the first boundary jumps gives the values $(n+a)^{-s}$. The remaining boundaries at the lower end are the displayed half-value and even Bernoulli terms; the upper boundaries tend to zero for $\Re s > 1$. The derivative of order j of $(x+a)^{-s}$ is $(-1)^j (s)_j (x+a)^{-s-j}$, giving the sign of the remainder. This proves SW3 first for $\Re s > 1$. Since $\tilde{B}_{2M}$ is bounded, its integral is absolutely and locally uniformly convergent for $\Re s > 1 - 2M$, so it proves the asserted continuation there.

Letting M increase proves that F_t is meromorphic on $\mathbb{C}$ with its sole pole at 1, simple of residue 1. On each fixed vertical strip, away from that pole, SW3 bounds $F_t(s)$ by a fixed polynomial in $1 + |\Im s|$, uniformly for $t \in [0, 1]$. Indeed choose $2M$ larger than one minus the left edge; the integral of $(x + a)^{-\Re s - 2M}$ is bounded uniformly on that strip, and $(s)_{2M}$ is a polynomial. Derivatives in s satisfy the same kind of bound by Cauchy's formula in a slightly larger strip. These estimates, rather than a numerical approximation to a zero sum, justify every infinite vertical integral below.

SW3. The first infinitesimal is explicit and does not vanish

Termwise differentiation, dominated uniformly on compact subsets of $\Re s > 1$, gives

$$\partial_t F_t(s) = -s F_t(s+1), \quad \partial_t^j F_t(s) = (-1)^j (s)_j F_t(s+j). \quad (\text{SW4})$$

Analytic continuation gives these meromorphic identities everywhere. In the dual-number ring $\mathbb{C}[\epsilon]/(\epsilon^2)$, the precise specialization is

$$F_\epsilon(s) = \zeta(s) - \epsilon s \zeta(s+1), \quad \log F_\epsilon = \log \zeta - \epsilon Q, \quad Q(s) = s \frac{\zeta(s+1)}{\zeta(s)}. \quad (\text{SW5})$$

The logarithm equality is local on any open set where $\zeta \neq 0, \infty$; its differentiated form is a global meromorphic identity. In particular

$$\partial_t \frac{F'_t}{F_t} \Big|_{t=0} = -Q'(s), \quad \partial_t \left(-\frac{F'_t}{F_t} \right) \Big|_{t=0} = Q'(s). \quad (\text{SW6})$$

The nilpotence $\epsilon^2 = 0$ does not make the coefficient of ϵ zero: at $s = 2$, that coefficient in F_ϵ is $-2\zeta(3) \neq 0$.

For $\Re s > 1$, multiplying the absolutely convergent series of $1/\zeta(s)$ and $\zeta(s+1)$ yields

$$Q(s) = s \sum_{n \geq 1} \alpha(n) n^{-s}, \quad \alpha(1) = 1, \quad \alpha(n) = (-1)^{\omega(n)} \frac{\prod_{p|n} (p-1)}{n}. \quad (\text{SW7})$$

The local factor is $(1 - p^{-s})/(1 - p^{-s-1}) = 1 - \sum_{k \geq 1} (p-1) p^{-k} p^{-ks}$, which proves the displayed coefficients. Their series is absolutely convergent on $\Re s > 1$, also after one logarithmic differentiation. Thus

$$Q'(s) = \sum_{n \geq 1} \alpha(n) (1 - s \log n) n^{-s}. \quad (\text{SW8})$$

This first jet includes all squarefree-prime support combinations in $\alpha(n)$; it is not a single new prime-power term.

SW4. Endpoint residues, logarithmic orders, and the full reflection defect

Use exactly the supplied completion

$$\Lambda_t(s) = \pi^{-s/2} \Gamma(s/2) F_t(s), \quad \ell_t(s) = \Lambda'_t(s) / \Lambda_t(s). \quad (\text{SW9})$$

SW3 at zero, or the first Bernoulli polynomial, gives $F_t(0) = -1/2 - t$. Hence

$$\text{Res}_0 \Lambda_t = -1 - 2t, \quad \text{Res}_1 \Lambda_t = 1, \quad \text{PP}_{0,1} \Lambda_t = \frac{1+t}{s(s-1)} - t \frac{2s-1}{s(s-1)}. \quad (\text{SW10})$$

Both poles are simple for $0 \leq t \leq 1$. Consequently

$$\text{Res}_0 \ell_t = \text{Res}_1 \ell_t = -1. \quad (\text{SW11})$$

For proof, a nonzero principal coefficient gives $\Lambda_t = (s-b)^{-1} u_t(s)$, with $u_t(b) \neq 0$; its logarithmic derivative is $-1/(s-b) + u'_t/u_t$. Thus the weight of the endpoint in a logarithmic explicit formula is its order, not the coefficient $-1 - 2t$ of the original completed function.

At every negative even integer $-2j$, $j \geq 1$, the precise order is

$$\text{ord}_{-2j} \Lambda_t = \text{ord}_{-2j} F_t - 1. \quad (\text{SW12})$$

This formula also retains exceptional parameter values at which a Hurwitz zero cancels a gamma pole. At $t = 1$, $F_1(-2j) = -1$, so all these completed poles are present and simple. In particular multiplying only by $s(s-1)$ does not make Λ_1 entire.

Define the additive and logarithmic reflection defects by

$$\Delta_t(s) = \Lambda_t(s) - \Lambda_t(1-s), \quad d_t(s) = \ell_t(s) + \ell_t(1-s) = \frac{d}{ds} \log \frac{\Lambda_t(s)}{\Lambda_t(1-s)}. \quad (\text{SW13})$$

These are explicit meromorphic objects. Their signs follow by the chain rule. At $t = 0$, the classical functional equation gives both defects zero. SW5–SW6 give the complete infinitesimal logarithmic defect

$$\dot{d}_0(s) = -Q'(s) - Q'(1-s). \quad (\text{SW14})$$

Its endpoint residues vanish because Q is regular at both endpoints: $Q(0) = -2$, from $s\zeta(s+1) \rightarrow 1$, and $Q(1) = 0$. This vanishing is only a statement about these two residues. It does not set SW14 or the moving zero divisor to zero.

SW5. Exact contour identity before invoking any positivity

For entire functions f, g , set

$$f^\#(s) = \overline{f(1-\bar{s})}, \quad A(s) = f^\#(s)g(s), \quad K(u) = A(1/2 + iu), \quad k(v) = \frac{1}{2\pi} \int_{\mathbb{R}} K(u) e^{iuv} du. \quad (\text{SW15})$$

In the formulas below the entire tests have Gaussian decay, up to a polynomial factor, in each fixed vertical strip. We construct such tests explicitly in SW9. The more general finite-rectangle residue identity requires no decay hypothesis.

Fix $c > 1$ and let $a_c = 1 - c$. First take a finite rectangle with these two real edges and imaginary edges $\pm T$, avoiding nonremovable poles on its boundary. The residue theorem gives

$$\frac{1}{2\pi i} \oint \ell_t(s) A(s) ds = \sum_{F_t(\rho)=0 \text{ inside}} m_\rho A(\rho) - A(1) - \sum_{j \geq 0: -2j \text{ inside}} A(-2j). \quad (\text{SW16})$$

Zeros and gamma poles are written separately, so exact cancellations in SW12 occur by addition of their integer orders. All quantities in SW16 are finite.

If $\ell_t A$, $\ell_t(s)A(1-s)$, and $\ell_t(1-s)A(1-s)$ have only finitely many nonremovable poles in the strip and their horizontal integrals tend to zero, then taking $T \rightarrow \infty$ gives

$$\begin{aligned} Z_{t,c}(A) - A(1) - \sum_{j \geq 0: -2j > 1-c} A(-2j) &= \frac{1}{2\pi i} \int_{\Re s=c} \ell_t(s)(A(s) + A(1-s)) ds \\ &\quad - \frac{1}{2\pi i} \int_{\Re s=c} d_t(s) A(1-s) ds. \end{aligned} \quad (\text{SW17})$$

Here $Z_{t,c}(A)$ is the finite sum of the zero terms not annihilated by A , counted with multiplicity. The actual tests in SW9 satisfy every property in this sentence by a displayed cancellation identity, so SW17 is used below without a missing analytic assumption. For the sign, the left upward integral becomes the right upward integral of $\ell_t(1-s)A(1-s)$; the positive contour subtracts it. Substituting $\ell_t(1-s) = d_t(s) - \ell_t(s)$ proves SW17.

SW6. The non-Eulerian arithmetic term and the completed formula

Put $a = 1 + t$, $r_n = (n+t)/a$ for $n \geq 2$, and choose c so large that $\sum_{n \geq 2} r_n^{-c} < 1$. Write $r_{\mathbf{n}} = \prod_{i=1}^k r_{n_i}$. Absolute convergence of the logarithm series gives

$$-\frac{F'_t}{F_t}(s) = \log a + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \sum_{n_1, \dots, n_k \geq 2} \log r_{\mathbf{n}} r_{\mathbf{n}}^{-s}. \quad (\text{SW18})$$

To verify differentiation and absolute convergence, take a slightly smaller real part on which the sum is still less than 1. Its positive margin bounds the differentiated geometric series. Explicitly, if $B(c) = \sum r_n^{-c} < 1$ and $D(c) = \sum (\log r_n) r_n^{-c} < \infty$, the sum of the absolute differentiated terms is $\sum_{k \geq 1} D(c) B(c)^{k-1} = D(c)/(1-B(c))$. The general-length derivation and its signed atoms are developed independently in non-Eulerian length proof.

Define the absolutely convergent functional

$$\begin{aligned} \mathcal{L}_t(k) &= 2 \log(a) k(0) \\ &+ \sum_{k_0 \geq 1} \frac{(-1)^{k_0+1}}{k_0} \sum_{n_1, \dots, n_{k_0} \geq 2} \frac{\log r_{\mathbf{n}}}{\sqrt{r_{\mathbf{n}}}} [k(\log r_{\mathbf{n}}) + k(-\log r_{\mathbf{n}})]. \end{aligned} \quad (\text{SW19})$$

Gaussian strip decay implies $|k(v)| \leq C_R e^{-R|v|}$ for every fixed $R > 0$: shift the vertical contour defining k by R in the appropriate direction. Choose $R = c - 1/2$, or a slightly larger value, and the absolute majorant just proved for SW18 proves convergence of SW19. All exchanges of its sum and integral are therefore justified.

Let

$$w_\infty(u) = \Re \psi(1/4 + iu/2) - \log \pi, \quad \mathcal{D}_{t,c}(A) = \frac{1}{2\pi i} \int_{\Re s=c} d_t(s) A(1-s) ds. \quad (\text{SW20})$$

Then SW17 is exactly

$$\begin{aligned} Z_{t,c}(A) &= A(1) + \sum_{j \geq 0: -2j > 1-c} A(-2j) \\ &+ \frac{1}{2\pi} \int_{\mathbb{R}} K(u) w_\infty(u) du - \mathcal{L}_t(k) - \mathcal{D}_{t,c}(A). \end{aligned} \quad (\text{SW21})$$

For completeness, $\ell_t = -\frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2) + F'_t/F_t$. The gamma part can be moved from c to $1/2$ without crossing a gamma pole, and reflection of the integration variable gives w_∞ . The digamma series on $\Re(s/2) > 0$ bounds it polynomially, so Gaussian decay kills the horizontal edges. A term r^{-s} in SW18 integrates to $r^{-1/2} [k(-\log r) + k(\log r)]$, by shifting its entire integrand to the critical line. The constant gives $2 \log(a) k(0)$. Finally the minus sign of $\mathcal{D}$ is the sign derived in SW17. This proves every sign and endpoint contribution in SW21.

At $t = 0$, $d_0 = 0$, and SW18 reduces to the usual logarithmic prime-power series. For tests annihilating the classical trivial zeros, $A(-2j) = 0$ for $j \geq 1$. SW21 then has exactly the two endpoints, archimedean term, and prime term in WA24. At $t = 1$, neither the negative-even terms nor $\mathcal{D}_{1,c}$ can be dropped merely because the original endpoint principal part has been computed.

SW7. Local variation of the original zero packet

Let ρ be an actual zero of ζ , of multiplicity m , and take a small circle containing no other zero and neither endpoint. Its continuation as a zero cluster of F_t is counted by

$$T_t(A) = \frac{1}{2\pi i} \oint \frac{F'_t(s)}{F_t(s)} A(s) ds. \quad (\text{SW22})$$

The compact boundary is zero-free at $t = 0$, so uniform continuity makes it zero-free for all sufficiently small t . The argument principle proves that this is the sum of A over the moving cluster, counted with multiplicities. Differentiating on that compact contour and integrating an exact derivative gives

$$\dot{T}_0(A) = -\frac{1}{2\pi i} \oint Q'(s) A(s) ds = \text{Res}_{s=\rho} (Q(s) A'(s)). \quad (\text{SW23})$$

The integration-by-parts equality follows from the zero contour integral of $(QA)'$. It does not require the zero to be simple. If it is simple, the result is

$$\dot{\rho}(0) = \frac{\rho \zeta'(\rho + 1)}{\zeta'(\rho)}, \quad \dot{T}_0(A) = \dot{\rho}(0) A'(\rho). \quad (\text{SW24})$$

For a nontrivial zero, the numerator is nonzero because $\rho \neq 0$ and $\Re(\rho + 1) > 1$. Thus a simple nontrivial zero has a nonzero first displacement in this interpolation. For a multiple zero, SW23 retains the full residue of order up to m , rather than assuming a differentiable labeling of its individual branches.

The arithmetic first variation on the right-hand side is explicit as well. With the conventions of SW15, the integral of $-Q'(s)(A(s) + A(1-s))$ on a right line is

$$\sum_{n \geq 1} \frac{\alpha(n)}{\sqrt{n}} \left[\left(\frac{\log n}{2} - 1 \right) (k(-\log n) + k(\log n)) + \log n (k'(-\log n) - k'(\log n)) \right]. \quad (\text{SW25})$$

To verify it, differentiation of the Fourier integral gives $\frac{1}{2\pi} \int (1/2 + iu) K(u) e^{iuv} du = \frac{1}{2} k(v) + k'(v)$. Reflection replaces the second derivative term by its negative. Insert SW7 and differentiate termwise. Absolute convergence follows from the same exponential contour bounds for k, k' . The reflection correction SW14 remains alongside SW25; omitting it is not a deformation of the same contour identity.

SW8. The exact symmetric lift and the missing Hermitian property

Let

$$G_t(s) = (s-1)F_t(s), \quad H_t(s) = G_t(s)G_t(1-s) = -s(s-1)F_t(s)F_t(1-s). \quad (\text{SW26})$$

Both G_t and H_t are entire by SW3. They have real conjugation symmetry, and $H_t(1-s) = H_t(s)$. The actual map is the multiplicative norm for the involution on entire functions,

$$N : \mathcal{O}(\mathbb{C}) \longrightarrow \mathcal{O}(\mathbb{C})^{s \mapsto 1-s}, \quad N(G)(s) = G(s)G(1-s). \quad (\text{SW27})$$

It satisfies $N(G_1G_2) = N(G_1)N(G_2)$, and its divisor is the original divisor plus its reflected divisor, with multiplicities. It is not asserted to be additive or to preserve the zero count without doubling. On the critical line $H_t(s) = |G_t(s)|^2 \geq 0$.

The zero pairing of H_t , on finite packets closed under $\rho \mapsto 1-\bar{\rho}$, is Hermitian. Proof: applying complex conjugation and this permutation to $\sum m_\rho f(1-\bar{\rho})g(\rho)$ exchanges f and g . By contrast, a divisor lacking that permutation symmetry need not define a Hermitian form. The next calculation proves the failure, and a negative direction in the symmetric lift, for the specific endpoint $t=1$.

SW9. A fully explicit negative test for the separated endpoint

The exact special values are

$$F_1(-20) = -1, \quad F_1(-19) = \frac{174611}{6600} - 1 = \frac{168011}{6600} > 0. \quad (\text{SW28})$$

They follow from $\zeta(-n) = -B_{n+1}/(n+1)$, $B_{21} = 0$, and $B_{20} = -174611/330$, also obtainable from SW3. Continuity therefore gives at least one real zero in $(-20, -19)$. Define α to be the least such real zero. This is well-defined: the endpoint values exclude neighborhoods of the two endpoints, and a nonzero holomorphic function has only finitely many zeros in the remaining compact interval. Write $m \geq 1$ for its exact multiplicity, and put $\beta = 1 - \alpha \in (20, 21)$.

Since $F_1(\sigma) = \sum_{n \geq 2} n^{-\sigma} > 0$ for real $\sigma > 1$, β is not a zero of F_1 . The divisor of H_1 , however, has both α and β , each with multiplicity m . Define

$$h(s) = (s-\alpha)^m(s-\beta)^m, \quad W(s) = e^{(s-1/2)^2} \frac{H_1(s)}{h(s)}. \quad (\text{SW29})$$

The quotient is entire, $W(\alpha), W(\beta) \neq 0$, and $W^\# = W$. Indeed reflection exchanges the two factors of h with total sign $(-1)^{2m} = 1$, and all the coefficients are real. SW3 proves that W and every fixed derivative are bounded by a polynomial times $e^{-(\Im s)^2}$ on each fixed vertical strip; away from a bounded region the denominator is a polynomial bounded below by a positive constant times $|\Im s|^{2m}$, and inside that region removability gives boundedness.

For arbitrary prescribed $z_\alpha, z_\beta \in \mathbb{C}$, take the degree-at-most-one polynomial

$$q(s) = \frac{z_\alpha}{W(\alpha)} \frac{s-\beta}{\alpha-\beta} + \frac{z_\beta}{W(\beta)} \frac{s-\alpha}{\beta-\alpha}, \quad f = qW, \quad A = f^\# f. \quad (\text{SW30})$$

Then $f(\alpha) = z_\alpha$ and $f(\beta) = z_\beta$. At every other zero of H_1 , W vanishes at least to the full multiplicity. Therefore A annihilates all of those zero terms. Take the strip $-21 < \Re s < 22$. Its native and symmetric zero functionals are exactly

$$Z_{1,22}(A) = m\bar{z}_\beta z_\alpha, \quad Z_{H_1}(A) = m(\bar{z}_\beta z_\alpha + \bar{z}_\alpha z_\beta). \quad (\text{SW31})$$

No other zero contributes, including a multiple zero or a zero with large imaginary part, because of the preceding vanishing orders.

For $(z_\alpha, z_\beta) = (1, -1)$, this gives

$$\boxed{Z_{1,22}(f^\# f) = -m < 0, \quad Z_{H_1}(f^\# f) = -2m < 0.} \quad (\text{SW32})$$

For $(z_\alpha, z_\beta) = (1, i)$, it gives

$$\boxed{Z_{1,22}(f^\# f) = -im,} \quad (\text{SW33})$$

which is not real. Thus the original shifted divisor pairing is not Hermitian, and the exact symmetric lift has a genuine negative direction. Nonnegativity of H_1 as a function on the critical line, proved in SW8, does not imply nonnegativity of its zero pairing.

Here are the promised contour checks for the actual tests. Since

$$\frac{H'_1}{H_1}A = q^\# q e^{2(s-1/2)^2} \frac{H'_1 H_1}{h^2}, \quad (\text{SW34})$$

the only possible nonremovable poles are at α, β . Every other zero is canceled as an identity of meromorphic functions. SW3 and its differentiated estimate bound the numerator polynomially on each strip, so the horizontal edges tend to zero with Gaussian speed. Likewise, using $H_1 = G_1 G_1(1-s)$, each occurrence of F'_1/F_1 or its reflected counterpart multiplied by A cancels a factor of F_1 in H_1^2 . The same assertion holds for $A(1-s)$, since its denominator is the same reflection-invariant h^2 . The remaining digamma factors have finitely many poles in this strip and at most polynomial growth on high horizontal edges, by the digamma recurrence and its convergent right-half-plane series. Consequently SW17–SW21 apply with no unproved horizontal zero-avoidance estimate.

At $t = 1, c = 22$, the absolute logarithm expansion required in SW18 holds. For instance

$$\sum_{n \geq 2} \left(\frac{2}{n+1} \right)^{22} \leq \left(\frac{2}{3} \right)^{22} + \int_3^\infty \left(\frac{2}{x} \right)^{22} dx = \left(\frac{2}{3} \right)^{22} \left(1 + \frac{3}{21} \right) < 1. \quad (\text{SW35})$$

Thus the corrected explicit formula evaluated on the negative test is the exact equality

$$-m = A(1) + \sum_{j=0}^{10} A(-2j) + \frac{1}{2\pi} \int_{\mathbb{R}} |f(1/2 + iu)|^2 w_\infty(u) du - \mathcal{L}_1(k) - \mathcal{D}_{1,22}(A). \quad (\text{SW36})$$

Every term is defined in SW15 and SW19–SW20, and every series and integral converges by the estimates above. This is a concrete positivity analysis of the actual non-Eulerian separated sheet, including its retained completion and reflection terms. It does not insert an arbitrary Euler factor for the supported-zero prime.

The tested divisor is the full shifted divisor in the strip $-21 < \Re s < 22$, and its selected real zero lies outside the classical critical strip. Thus SW32 does not establish nonpositivity for a separately defined pairing restricted to $0 < \Re s < 1$. Nor does it rule out a different boundary subtraction or a selected positivity cone. Such an operation must specify its map and retain the component it removes. The exact maps for this detected component are given next.

SW10. The retained two-point object and its positive and negative projections

Keep the actual function W of SW29 and define the vector space

$$V = W\mathbb{C}[s], \quad E : V \longrightarrow \mathbb{C}^2, \quad E(f) = (f(\alpha), f(\beta)). \quad (\text{SW37})$$

Let $j : \mathbb{C}^2 \rightarrow V$ be the interpolation formula SW30. Direct evaluation proves $Ej = \text{id}$. Since $W(\alpha), W(\beta) \neq 0$, a polynomial multiple qW lies in $\ker E$ precisely when q is divisible by both distinct linear factors. Hence

$$K := \ker E = W(s - \alpha)(s - \beta)\mathbb{C}[s], \quad V = j(\mathbb{C}^2) \oplus K. \quad (\text{SW38})$$

The decomposition is explicit: $f = jE(f) + (f - jE(f))$, and its second term lies in K . An element of the intersection has evaluation both equal to its coefficient vector and zero, proving directness.

For $z = E(f)$ and $w = E(g)$, the native and reflected pairings factor through these maps as

$$B_{\text{nat}}(f, g) = m \overline{z_\beta} w_\alpha, \quad B_{\text{sym}}(f, g) = m(\overline{z_\beta} w_\alpha + \overline{z_\alpha} w_\beta). \quad (\text{SW39})$$

The same cancellation used in SW31 proves this for arbitrary $f, g \in V$. Both forms vanish when either input lies in K . The descended matrix of B_{sym} is $m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which is invertible, so its full radical is exactly K . Also $B_{\text{sym}} = B_{\text{nat}} + B_{\text{nat}}^*$, where $B^*(f, g) = \overline{B(g, f)}$; this is the exact passage to the Hermitian part, including its factor of two.

Define

$$\Pi_+ = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \Pi_- = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad P_\pm = j\Pi_\pm E, \quad P_K = 1 - jE. \quad (\text{SW40})$$

Multiplication gives $\Pi_\pm^2 = \Pi_\pm$, $\Pi_+\Pi_- = 0$, and $\Pi_+ + \Pi_- = 1$. Using $Ej = 1$ proves that P_+, P_-, P_K are mutually annihilating idempotents summing to 1_V . Their images are respectively $j\mathbb{C}(1, 1)$, $j\mathbb{C}(1, -1)$, and K . These three summands are orthogonal for B_{sym} . To verify all signs and the orthogonality, write

$$c_+(f) = \frac{f(\alpha) + f(\beta)}{2}, \quad c_-(f) = \frac{f(\alpha) - f(\beta)}{2}.$$

Substitution in SW39 gives

$$B_{\text{sym}}(f, g) = 2\overline{mc_+(f)}c_+(g) - 2\overline{mc_-(f)}c_-(g). \quad (\text{SW41})$$

Thus the nonnegative cone is exactly $\{|c_-| \leq |c_+|\}$, with the radical directions retained, and $j\mathbb{C}(1, 1) \oplus K$ is a maximal nonnegative linear subspace. Indeed any vector outside it has $c_- \neq 0$; subtracting its positive component, already in this subspace, gives a negative vector by SW41, so it cannot be adjoined while preserving nonnegativity.

A particular boundary subtraction can now be defined, rather than left implicit:

$$B_+(f, g) := B_{\text{sym}}(f, g) + 2\overline{mc_-(f)}c_-(g) = B_{\text{sym}}(P_+f, P_+g) = 2\overline{mc_+(f)}c_+(g). \quad (\text{SW42})$$

It is positive semidefinite, and its removed negative direction is exactly the recorded image of P_- . Equations SW37–SW42 identify the defect object, its quotient, section, radical, orthogonal summands, and one exact corrected form. They do not claim that this particular subtraction is the arithmetic or prismatic choice required elsewhere in the programme; they provide the complete finite object such a comparison would have to map.

SW11. Consequences for the programme’s vanishing-infinitesimal question

The exact distinctions now have proved maps. The quotient-size construction SW1 maps to the interpolation SW2; its first-order base change is SW5; its completed divisor is measured by SW16; its failure of reflection is the meromorphic function SW13; and its reflected Hermitian lift is the norm map SW27. The first-order endpoint residue variation of the logarithmic derivative is zero, while the first-order function, logarithmic derivative, and moving zero cluster are respectively SW5, SW6, and SW23 and need not vanish. SW24 proves nonvanishing of the displacement at every simple nontrivial classical zero.

The previously derived prime-weight formula TN21 concerns a different function, a holomorphic-unit multiple of classical ζ in $\Re s > 0$. Its unchanged zero divisor and corrected archimedean term cannot be transferred to $F_1 = \zeta - 1$. The actual map from this interpolation to a symmetric divisor is SW27, which adds the reflected divisor and exposes SW32. A prismatic or deformation comparison must carry these explicit objects and the reflection defect through its maps; nilpotence of a parameter or vanishing of an endpoint residue does not erase them.

Reconstruction over the full support lattice

This calculation starts from the programme's actual support-amplitude semiring. It proves the precise comparison with the earlier scalar calculations, identifies information lost by all separate Boolean branch maps after linearization, and constructs a support-valued extension of the finite Weil packet form. It does not assert that a zeta zero off the critical line exists, or that positivity for the classical Weil distribution has been proved.

The source definition and the existing spectrum theorem are in *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, and Frobenius-Perfect Quantization*, The Clankers, June 2026, Version 11. The file read for this derivation is `split_support_geometry_arithmetic_curve_v11910.tex`, SHA256 8443cc0401b18d373939d992f0a2f5384ee2fa1130f059753d5f3972eec5052d, Definition `def:lattice-split`, Theorems `thm:lattice-normal-form`, `thm:birkhoff-support-coordinates`, `thm:lattice-prime-classification`, `thm:spectral-ordinal-sum`, and `thm:support-base-change`, source lines1180–1871. Those results are prior programme mathematics. The different file `split_support_geometry_arithmetic_curve_v11.tex`, SHA256 5fa6a55aa9d1a2e57599e206087d3d2466b25dd99fa1a90d5f86367aafab1b4d, has the explicit branch-spectrum theorem at lines1925–1985. The editions are not treated as byte-identical.

1. The base and its split comparison maps

Let R be a nonzero commutative unital ring and L a nontrivial bounded distributive lattice. Write its bounds as 0_L and 1_L . Set

$$S_L = G_L(R) = \{z_\lambda = (0, \lambda) : \lambda \in L\} \cup \{\widehat{r} = (r, 1_L) : r \in R\}, \quad z_{1_L} = \widehat{0} = e_L, \quad z_{0_L} = \tau_L. \quad (\text{FSR1})$$

The overlap in this union consists of the one element e_L . The operations are

$$(r, \lambda) + (s, \mu) = (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu). \quad (\text{FSR2})$$

In particular, $\widehat{r} + z_\lambda = \widehat{r}$ and $\widehat{r}z_\lambda = z_\lambda$, including $r=0$. Closure follows because every nonzero amplitude has top support. The semiring laws follow from those of R and the distributive lattice law. Its zero is τ_L and its unit is $\widehat{1}$.

Put $S = G_L(R)$, with $B = \{0, 1\}$. The endpoint inclusion and each Boolean branch give maps

$$\iota : S \longrightarrow S_L, \quad \tau \mapsto \tau_L, \quad r \mapsto \widehat{r}, \quad \beta_\epsilon : S_L \longrightarrow S, \quad z_\lambda \mapsto z_{\epsilon(\lambda)}, \quad \widehat{r} \mapsto r, \quad \beta_\epsilon \iota = \text{id}_S. \quad (\text{FSR3})$$

Here $\epsilon : L \rightarrow B$ preserves the bounds, joins, and meets. Both maps preserve every operation in FSR2: on the zero fibre this is the lattice-map identity; on two supported amplitudes it is the identity of R ; on a mixed pair it follows from the two identities after FSR2. Thus ι is injective and every β_ϵ is surjective. The kernel congruence of β_ϵ identifies exactly the zero-fibre elements with the same ϵ value, and no two distinct nonzero amplitudes. Its inverse image of the actual scalar zero is

$$\beta_\epsilon^{-1}(\tau) = \{z_\lambda : \epsilon(\lambda) = 0\}. \quad (\text{FSR4})$$

Its inverse image of the supported scalar zero is the complementary set of zero-fibre elements. This is the exact distinction between a zero ideal and a congruence in this comparison.

The amplitude projection $p_L : S_L \rightarrow R$ has zero fibre $Z_L = \{z_\lambda\}$. Every homomorphism to a ring kills Z_L : the equality $e_L + z_\lambda = e_L$ implies this by additive cancellation in the target. The restriction to the supported copy then gives a unique ring homomorphism $R \rightarrow A$. This proves the universal property of p_L , without identifying e_L with τ_L in S_L .

For a domain R , the supported zero is a prime element in the divisibility sense:

$$e_L S_L = Z_L, \quad e_L \mid xy \iff p_L(x)p_L(y) = 0 \iff e_L \mid x \text{ or } e_L \mid y. \quad (\text{FSR5})$$

The first equality follows from $e_L z_\lambda = z_\lambda$ and $e_L \widehat{r} = e_L$. The next equivalence follows from the first equality and the multiplication law. The last uses precisely the absence of zero divisors in R . The ideal Z_L is proper because $\widehat{1}$ is outside it. The element e_L is not a unit, since every product with it has amplitude zero. It is reducible, since $e_L = e_L e_L$ with two nonunits. Its primality and its multiplicative idempotence are simultaneous facts.

For completeness the entire semiring-prime comparison can be obtained directly. If an ideal contains a supported amplitude, multiplication by e_L puts e_L in it, hence puts all of Z_L in it. Its supported

amplitudes form a ring ideal J . An ideal containing no supported amplitude is Z_a for a proper lattice ideal a : joins follow from addition and downward closure follows from $z_\mu z_\lambda = z_\mu$ when $\mu \leq \lambda$. Thus all ideals are Z_a and

$$Q_J = Z_L \cup \{\hat{r} : r \in J\}. \quad (\text{FSR6})$$

Testing FSR2 shows that Z_a is prime exactly when a is a prime lattice ideal, and Q_J is prime exactly when J is a ring prime. Every support prime is strictly below every arithmetic prime. For $R=Z$ this includes

$$Z_a \subsetneq Z_L = (e_L) = Q_{(0)} \subsetneq Q_{(p)}. \quad (\text{FSR7})$$

The basic opens are $D(z_\lambda) = \{Z_a : \lambda \notin a\}$ and $D(\widehat{r}) = \text{Spec_lat}(L) \cup D_R(r)$. These formulas prove the topological ordinal sum, including its topology. Contraction along β_ϵ sends the scalar prime $\{\tau\}$ to $Z_{(\epsilon^{-1})}$ and Q_q to Q_q . Contraction along ι sends every Z_a to $\{\tau\}$ and Q_q to Q_q . Consequently the scalar spectrum is a retract with an explicitly chosen section; it is not the entire support spectrum.

2. Joint reconstruction before linearization

Now assume L finite and let $J(L)$ be its nonzero join-irreducible elements. For $j \in J(L)$ put $\theta_j(\lambda) = 1$ if $j \leq \lambda$ and 0 otherwise. A join-irreducible is join-prime: if $j \leq \lambda \vee \mu$, distributivity gives $j = (j \wedge \lambda) \vee (j \wedge \mu)$, so one term equals j . Therefore θ_j is a bounded lattice homomorphism. Every nonzero element is the join of the join-irreducibles below it, by induction in the finite partially ordered set. Hence these characters jointly distinguish elements of L .

Conversely, a Boolean branch has a prime filter whose least element j is the meet of its elements. If $j = x \vee y$ with neither x nor y equal to j , the primality of the filter contradicts the minimality of j . Thus j is join-irreducible, and the branch is θ_j . This proves the entire branch list.

The product of all branch maps is the injective semiring map

$$B : S_L \longrightarrow \prod_{j \in J(L)} S, \quad B(x) = (\beta_j(x))_j. \quad (\text{FSR8})$$

Its image is exactly the following set. All coordinate amplitudes have a common value r . If $r \neq 0$, all coordinates are the same supported element r . If $r = 0$, each coordinate is τ or e , and the set of coordinates equal to e is an order ideal in $J(L)$. Indeed it is $\{j : j \leq \lambda\}$ for the source z_λ . Conversely, an order ideal A gives $\lambda = \text{join } A$; join-primality shows that $\{j : j \leq \lambda\} = A$. This proves both surjectivity onto the displayed image and injectivity.

For $L = B^d$ the poset $J(L)$ is an antichain of d atoms. Every subset is an order ideal, so FSR8 becomes an isomorphism

$$G_{B^d}(R) \cong \underbrace{G(R) \times_R \cdots \times_R G(R)}_{d \text{ factors}}, \quad (\text{FSR9})$$

where the maps to R are amplitude projections. This is different from the product of independent amplitude coordinates: the common-amplitude condition is part of the isomorphism, not an omitted restriction.

For a general finite distributive L , FSR8 is the subsemiring of that fibre product whose zero-amplitude masks satisfy the stated order-ideal relations. The additive and multiplicative laws remain FSR2. All earlier scalar identities pull back along each β_j . Conversely, if two S_L -valued expressions have equal images under every β_j , FSR8 proves equality in S_L . This last conclusion concerns semiring-valued expressions; it does not yet concern their free additive envelopes.

3. The exact extra kernel created by linearization

For a commutative coefficient ring k define the contracted meet algebra

$$C_L = k[(L, \wedge)] / ([0_L]). \quad (\text{FSR10})$$

It is free as a k -module on $u_\lambda = [\lambda]$ for $\lambda \neq 0_L$, with $u_0 = 0$, unit u_1 , and $u_\lambda u_\mu = u_{(\lambda \wedge \mu)}$. It retains multiplication from support and introduces its own additive linear combinations. In particular $u_{(\lambda \vee \mu)}$ is not defined to be $u_\lambda + u_\mu$.

For every nonzero $a \in L$, whether join-irreducible or not, define

$$\eta_a(u_\lambda) = \mathbf{1}_{\{a \leq \lambda\}}. \quad (\text{FSR11})$$

This is a unital k -algebra homomorphism: $a \leq \lambda \wedge \mu$ holds exactly when both inequalities hold. With any linear extension of the order, its matrix has entries $1_{a \leq \lambda}$, is triangular with diagonal 1, and is invertible over the integers. Consequently

$$\eta : C_L \xrightarrow{\sim} \prod_{a \in L \setminus \{0\}} k. \quad (\text{FSR12})$$

No division in k is used.

An explicit inverse uses the Möbius function of the finite order. Define $\mu(a,a)=1$ and, for $a < b$, $\mu(a,b) = -\sum_{a \leq c < b} \mu(a,c)$. The upper triangular incidence matrix $Z_{ab} = 1_{a \leq b}$ has inverse with these entries: the recurrence proves $MZ=I$, and finite triangular matrices then also satisfy $ZM=I$. Define

$$E_a = \sum_{\lambda \leq a} \mu(\lambda, a) u_\lambda. \quad (\text{FSR13})$$

Then $\eta_b(E_a) = \sum_{b \leq \lambda \leq a} \mu(\lambda, a) = \delta_{ab}$. The term $\lambda=0$ contributes zero. Thus

$$E_a E_b = \delta_{ab} E_a, \quad \sum_{a \neq 0} E_a = u_1, \quad u_\lambda = \sum_{0 < a \leq \lambda} E_a. \quad (\text{FSR14})$$

These identities follow by applying the isomorphism FSR12, so hold over every coefficient ring.

The linearized Boolean branch θ_j is precisely η_j . Write

$$K_L = \bigcap_{j \in J(L)} \ker \eta_j = \bigoplus_{a \in L \setminus (\{0\} \cup J(L))} k E_a. \quad (\text{FSR15})$$

This proves the full kernel, not only its dimension. Over a field its dimension is $|L| - 1 - |J(L)|$. It is a direct product of copies of k , with multiplicatively idempotent coordinate projectors. It is zero exactly when every nonzero lattice element is join-irreducible, equivalently when L is a chain. To verify the last equivalence, a chain has that property. If L is not a chain, choose incomparable a, b ; then $a \vee b$ is nonzero and join-reducible.

For $L = B^2 = \{0, a, b, 1\}$, with $a \wedge b = 0$ and $a \vee b = 1$,

$$E_a = u_a, \quad E_b = u_b, \quad w = E_1 = u_1 - u_a - u_b, \quad w^2 = w \neq 0, \quad \eta_a(w) = \eta_b(w) = 0. \quad (\text{FSR16})$$

Indeed the three terms are distinct basis vectors, proving nonzero, and direct expansion gives $(u_1 - u_a - u_b)^2 = u_1 + u_a + u_b - 2u_a - 2u_b = u_1 - u_a - u_b$. Thus the semiring branch maps in FSR8 distinguish every support element, but their separately linearized maps omit w . This is a proved failure of linearized reconstruction from separate branches. Its retained object is K_L with inclusion, coordinate projectors, and quotient given in FSR15.

The nonzero a that are not join-irreducible give genuine mixed observables. For a with join-irreducible decomposition $a = \text{join}_{j \leq a} j$,

$$\eta_a(u_\lambda) = \prod_{j \leq a} \theta_j(\lambda). \quad (\text{FSR17})$$

Both sides are 1 exactly when every $j \leq a$ lies below λ , equivalently when $a \leq \lambda$. Thus a joint product of branch tests restores this missing coordinate. Separate branch expectations do not determine expectations of these products. FSR12 gives the complete recovery map using all these joint tests.

4. Contracted multiplicative coefficients and the scalar retract

Let $\Gamma_L = k[(S_L, \text{multiplication})]/([\tau_L])$. Put $E = [e_L]$ and $F = 1 - E$. Multiplication by E sends $[z_\lambda]$ to itself and $[\widehat{r}]$ to $[e_L]$. Thus $E \Gamma_L$ is C_L , embedded by $u_\lambda \mapsto [z_\lambda]$. The complementary factor $F \Gamma_L$ has basis

$$v_r = [\widehat{r}] - [e_L], \quad r \in R \setminus \{0\}, \quad (\text{FSR18})$$

and $v_0 = 0$. Their multiplication is $v_r v_s = v_{rs}$, with zero when $rs = 0$. Indeed expansion gives $[\widehat{rs}] - [e_L] - [e_L] + [e_L]$. The two sets $\{[z_\lambda] : \lambda \neq 0\}$ and $\{v_r : r \neq 0\}$ form a basis: each original nonzero supported basis element is $v_r + [e_L]$, and comparison of the distinct $[\widehat{r}]$ coefficients proves linear independence. Hence

$$\Gamma_L \cong C_L \times D_R, \quad D_R = k[(R, \times)]/([0_R]). \quad (\text{FSR19})$$

The formulas prove this for rings with zero divisors as well as domains. For $R=\mathbb{Z}$ the second factor is

$$D_{\mathbb{Z}} \cong k[t, X_p : p \text{ rational prime}]/(t^2 - 1), \quad v_{-1} \mapsto t, \quad v_p \mapsto X_p. \quad (\text{FSR20})$$

Unique prime factorization of every nonzero integer proves a bijection of the displayed monomial bases and compatibility with multiplication.

For scalar support the same construction is $\text{Gamma_B}=k \times D_R$. Under FSR19 each branch and the scalar inclusion become

$$\Gamma(\beta_j) : (c, d) \mapsto (\eta_j(c), d), \quad \Gamma(\iota) : (a, d) \mapsto (a u_1, d). \quad (\text{FSR21})$$

These formulas follow on z_lambda and $\widehat{r}$, which generate the algebras. The composite is the identity. The intersection of all branch kernels is exactly $K_L \times 0$. Thus the earlier arithmetic prime-label algebra D_R is preserved as an actual direct factor, and the missing support coefficient object is explicitly identified. This is stronger than saying that earlier work might survive.

The evaluation map $\text{Gamma_L} \rightarrow R$ when $k=\mathbb{Z}$ sends $[\widehat{r}]$ to r and every $[z_lambda]$ to 0. For $R=\mathbb{Z}$, FSR19–20 identify its kernel as

$$C_L \times (t + 1, X_p - p \text{ for every prime } p) \subset C_L \times D_{\mathbb{Z}}. \quad (\text{FSR22})$$

The quotient of D_Z by the displayed ideal is $\mathbb{Z}$: every polynomial reduces to its integer evaluation, and that evaluation is the inverse of the resulting inclusion of $\mathbb{Z}$. This proves equality of the kernel ideal. In particular all of C_L , including K_L , is killed by arithmetic addition. It remains present before that quotient. The inclusion and quotient in FSR19 and FSR22 give its exact relation to the arithmetic factor.

5. Extension of complexes, operators, and all primary jets

Take $k=\mathbb{C}$ for this section. Let $(V^\bullet, d)$ be any specified complex of complex vector spaces, with its original differentials. Its full support extension is

$$V_L^\bullet = C_L \otimes_{\mathbb{C}} V^\bullet = \bigoplus_{a \neq 0} E_a \otimes V^\bullet, \quad d_L(E_a \otimes v) = E_a \otimes dv. \quad (\text{FSR23})$$

This is a defined construction, not an identification with a prismatic complex. Decomposition FSR14 proves directly

$$\ker d_L = \bigoplus_a E_a \otimes \ker d, \quad \text{im } d_L = \bigoplus_a E_a \otimes \text{im } d, \quad H^q(V_L) = \bigoplus_a E_a \otimes H^q(V). \quad (\text{FSR24})$$

The first two equalities follow by equating the coefficients of the independent E_a . The third is their quotient. Thus a nonzero cohomology class of a specified old complex remains nonzero in every individual support sector. For every linear chain map T , the map $1 \otimes T$ preserves all sectors. A homotopy $dH + Hd = T - T'$ extends to $1 \otimes H$ and satisfies the identical formula. The maps $\eta_j \otimes \text{id}$ are chain maps, with joint kernel $K_L \otimes V^\bullet$. The scalar inclusion $v \mapsto u_1 \otimes v$ is split by every $\eta_j \otimes \text{id}$. These formulas specify exactly which part of this statement is functorial and which prismatic comparison still requires identifying the actual input complex.

For the finite packet take the four distinct points

$$\alpha_1 = \rho, \quad \alpha_2 = 1 - \bar{\rho}, \quad \alpha_3 = \bar{\rho}, \quad \alpha_4 = 1 - \rho,$$

where $\rho = 1/2 + \delta + i\gamma$, $0 < \delta < 1/2$ and $\gamma > 2$. Put $d(s) = \prod_{i=1}^4 (s - \alpha_i)$, $h = d^m$ with $m \geq 1$, and $A_h = \mathbb{C}[s]/(h)$. These parameters define a finite algebra for every such ρ ; they do not assert that ρ is a zeta zero. Pairwise coprimality gives the isomorphism to the four rings $\mathbb{C}[t_i]/(t_i^m)$ by Taylor coefficients. To see its inverse explicitly, take $h_i = h/(s - \alpha_i)^m$ and multiply h_i by the degree- $(m-1)$ Taylor polynomial of $1/h_i$ at α_i . The resulting class e_i is 1 modulo $(s - \alpha_i)^m$ and 0 modulo each other primary power. Then

$$\sum_{i=1}^4 \sum_{b=0}^{m-1} c_{i,b} e_i (s - \alpha_i)^b \quad (\text{FSR25})$$

is the inverse image of the four Taylor polynomials. Injectivity follows because a polynomial vanishing to order m at all four points is divisible by h . This proves the complete primary decomposition and retains every jet.

Evaluation $\text{ev} : A_h \rightarrow \mathbb{C}^4$ is surjective with kernel $J_h = (d)/(d^m)$, of dimension $4m - 4$. Surjectivity follows from the e_i . The kernel statement follows by divisibility by the four distinct linear factors. Choose any

specified unit $u \in A_h$ and write $U = M_u$. Its four constant Taylor coefficients are nonzero; conversely that condition makes it a unit by the finite inverse of each truncated Taylor series. Then $a_h = \text{ev } U$ is onto with kernel J_h . The old programme choice $u = j_h(g/h)$, wherever that entire quotient defines a unit, is retained exactly by this construction; it is not substituted by 1.

Extend A_h, J_h, U and a_h coefficientwise as in FSR23. If $u_{i,b}$ denotes its Taylor coefficient, the full operator remains

$$U_L((x_{a,i,b})_{a,i,b})_{a,i,b} = \sum_{c=0}^b u_{i,c} x_{a,i,b-c}. \quad (\text{FSR26})$$

Thus no amplitude derivative, multiplicity, or support sector is omitted. Multiplication by s still acts on each basis element by α_i times that element plus the next jet, with the last next jet zero. The kernel of the extended a_h is $C_L \otimes J_h$, while the joint kernel of all branch maps on the entire algebra is $K_L \otimes A_h$. These are distinct explicitly defined subspaces.

6. The complete support-valued finite Weil pairing

Let σ exchange 1 with 2 and 3 with 4. On $\mathbb{C}^4$ set

$$b(c, c') = m \sum_{i=1}^4 \overline{c_{\sigma i}} c'_i, \quad B_h(x, y) = b(a_h x, a_h y). \quad (\text{FSR27})$$

Complex conjugation and reindexing by σ prove Hermitian symmetry. The matrix of b is the sum of two blocks $m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which is nonsingular and has one positive and one negative direction per block. Since a_h is onto with kernel J_h , the radical of B_h is J_h and its inertia is $(2, 2, 4m - 4)$. A full diagonal basis is

$$\frac{U^{-1}(e_1 + e_2)}{\sqrt{2m}}, \quad \frac{U^{-1}(e_1 - e_2)}{\sqrt{2m}}, \quad \frac{U^{-1}(e_3 + e_4)}{\sqrt{2m}}, \quad \frac{U^{-1}(e_3 - e_4)}{\sqrt{2m}},$$

together with $U^{-1}e_i(s - \alpha_i)^b$ for $1 \leq i \leq 4$ and $1 \leq b < m$. Direct evaluation gives squared values $+1, -1$ and 0 as stated. This proves the finite result independently of any analytic zeta assertion.

Give C_L the involution conjugating coefficients and fixing each E_a . For $x = \sum_a E_a \otimes x_a$ and $y = \sum_a E_a \otimes y_a$ define

$$\mathbb{B}_L(x, y) = \sum_{a \neq 0} E_a B_h(x_a, y_a). \quad (\text{FSR28})$$

The definition is unambiguous by FSR14. It is C_L -Hermitian and sesquilinear by the scalar identities and orthogonality of the E_a . Its radical is exactly $C_L \otimes J_h$: if its values against all y vanish, choosing y supported on one sector shows that the corresponding x_a lies in the scalar radical. For every branch j ,

$$\eta_j(\mathbb{B}_L(x, y)) = B_h((\eta_j \otimes \text{id})x, (\eta_j \otimes \text{id})y). \quad (\text{FSR29})$$

This is the proved comparison with the earlier scalar packet form.

The intrinsic positive cone of the finite product algebra C_L is

$$C_L^+ = \left\{ \sum_a c_a E_a : c_a \in \mathbb{R}_{\geq 0} \right\}. \quad (\text{FSR30})$$

It equals $\{v^*v : v \in C_L\}$, since each coordinate is an absolute square, and conversely each nonnegative coordinate has its nonnegative real square root. The regular-representation trace is $\text{tr}_{\text{reg}}(\sum_a c_a E_a) = \sum_a c_a$: multiplication by E_a is the projection onto its one-dimensional coordinate. Consequently $\text{tr}_{\text{reg}} \mathbb{B}_L$ has inertia

$$(2(|L| - 1), 2(|L| - 1), (4m - 4)(|L| - 1)). \quad (\text{FSR31})$$

This follows from the direct sum of the explicit diagonal bases after FSR27. No claim identifies this finite trace with the adelic trace without the corresponding analytic map.

Retaining only the sum of Boolean-branch traces means using $\sum_{j \in J(L)} \eta_j$. The radical of that form is

$$\left(\bigoplus_{j \in J(L)} E_j \otimes J_h \right) \oplus (K_L \otimes A_h), \quad (\text{FSR32})$$

and its inertia is

$$(2|J(L)|, 2|J(L)|, (4m-4)|J(L)| + 4m(|L| - 1 - |J(L)|)). \quad (\text{FSR33})$$

Each observed sector has exactly the scalar radical and each unobserved sector has its entire $4m$ -dimensional space in the radical; this proves both formulas.

For B^2 let w be FSR16 and choose the fully specified element

$$x = w \otimes U^{-1}(e_1 - e_2). \quad (\text{FSR34})$$

Both branch images of x vanish, but

$$\mathbb{B}_{B^2}(x, x) = -2m w \neq 0. \quad (\text{FSR35})$$

The equality follows from FSR27 and $w^2 = w$. Thus zero results from all separate branch observations do not make the full support-valued form zero or positive. Replacing the minus by a plus in FSR34 gives $+2mw$, so the omitted space carries both signs in this finite packet model. This is a concrete comparison result about the defined form. It is not a counterexample to RH: the arbitrary ρ in FSR25 has not been asserted to be a zeta zero.

Finally the scalar inclusion $x \mapsto u_1 \otimes x$ obeys

$$\mathbb{B}_L(u_1 \otimes x, u_1 \otimes y) = u_1 B_h(x, y). \quad (\text{FSR36})$$

It therefore preserves the old calculation as a diagonal subspace. Full support adds exact sectors and additional observations; it does not by itself repair a negative scalar value. This specifies the part of the old Weil construction that survives and the additional calculation required for an actual full-support adelic realization.

The localized Weil packet: values, jets, and the observation multiplier

This paper computes the finite Hermitian form specified by the zero side of the Weil pairing on the programme's entire functions. It keeps the polynomial coordinate change, all primary jets, and the actual multiplier $U = M_{j_h(v_h)}$. The form has a four-dimensional nondegenerate quotient and a fully described nilpotent radical. Every jet remains present in the source algebra and in the multiplier, although the zero-side pairing evaluates only the zeroth jets.

The definitions of the quartet, E_h , the Taylor remainder j_h , the unit U , and the primary basis are those of the programme's OPG1 and OPG10–OPG11 in *Full observation proofs*. They are restated and proved to the extent needed here. The starting suggestion is Proposition 3 in the supplied *Split-Zero programme: orientation and new angles*, 22 September 2026. The formula there changes from the Fourier coordinate to the s -coordinate without changing the polynomial's name. Equations (WP5)–(WP11) below give the exact coordinate maps.

No analytic explicit-formula identity, Fourier inversion assertion, or positivity theorem is assumed in the finite calculation. Section 10 specifies the precise zero-side identity proved here and its analytic scope.

1. The complete-order quartet and the two polynomial coordinates

Retain the programme's hypotheses

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad m \in \mathbb{Z}_{\geq 1}. \quad (\text{WP1})$$

Order its four distinct points as

$$\alpha_1 = \rho, \quad \alpha_2 = 1 - \bar{\rho}, \quad \alpha_3 = \bar{\rho}, \quad \alpha_4 = 1 - \rho, \quad \mathcal{R} = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}. \quad (\text{WP2})$$

The real parts distinguish points with the same imaginary part because $\delta > 0$, and the imaginary parts distinguish the upper and lower pairs because $\gamma > 0$. Let $g = 2\xi$ have a zero of exact order m at each $\alpha \in \mathcal{R}$. Put

$$d(s) = \prod_{\alpha \in \mathcal{R}} (s - \alpha), \quad h(s) = d(s)^m, \quad n = 4m, \quad v_h(s) = \frac{g(s)}{h(s)}. \quad (\text{WP3})$$

Here s is the original zeta coordinate. At each α , write $g(s) = (s - \alpha)^m b_\alpha(s)$, where b_α is holomorphic and $b_\alpha(\alpha) \neq 0$. If $h_\alpha(s) = h(s)/(s - \alpha)^m$, then $h_\alpha(\alpha) \neq 0$, and

$$v_h(s) = \frac{b_\alpha(s)}{h_\alpha(s)} \quad \text{near } \alpha, \quad v_h(\alpha) = \frac{g^{(m)}(\alpha)}{m! h_\alpha(\alpha)} \neq 0. \quad (\text{WP4})$$

Thus all the apparent singularities of g/h are removable and v_h is entire. At every zero $\rho' \notin \mathcal{R}$ of g , it vanishes because $h(\rho') \neq 0$.

Define the Fourier coordinate and its inverse by

$$\theta(s) = \frac{s - 1/2}{i}, \quad \iota(z) = \frac{1}{2} + iz, \quad \theta\iota = \text{id}_{\mathbb{C}}, \quad \iota\theta = \text{id}_{\mathbb{C}}. \quad (\text{WP5})$$

For $P \in \mathbb{C}[z]$, the associated s -polynomial is

$$q_P(s) = P(\theta(s)), \quad F_P(z) = P(z)v_h(\iota(z)). \quad (\text{WP6})$$

In particular $F_P(\theta(s)) = q_P(s)v_h(s)$. The two polynomials P and q_P have the same degree, but their coefficients and variables differ by (WP5). A factor $P(1 - \bar{\alpha})$ is therefore not the value obtained from F_P unless P has first been replaced by q_P .

Put $\beta_\alpha = \theta(\alpha)$. In the order (WP2),

$$(\beta_{\alpha_1}, \beta_{\alpha_2}, \beta_{\alpha_3}, \beta_{\alpha_4}) = (\gamma - i\delta, \gamma + i\delta, -\gamma - i\delta, -\gamma + i\delta). \quad (\text{WP7})$$

Let $\sigma\alpha = 1 - \bar{\alpha}$. This involution interchanges α_1, α_2 and α_3, α_4 . Direct substitution gives

$$\overline{\theta(\alpha)} = \theta(\sigma\alpha), \quad \iota(\overline{\beta_\alpha}) = \sigma\alpha. \quad (\text{WP8})$$

Define

$$H(z) = \prod_{\alpha \in \mathcal{R}} (z - \beta_\alpha)^m = i^{-n} h(\iota(z)), \quad E_z = \mathbb{C}[z]/(H), \quad E_h = \mathbb{C}[s]/(h). \quad (\text{WP9})$$

Here $n = 4m$, so $i^{-n} = 1$; the scalar is displayed to record how the defining polynomials transform. The maps

$$\mathcal{C} : E_z \longrightarrow E_h, \quad [P] \longmapsto [P \circ \theta], \quad \mathcal{C}^{-1} : E_h \longrightarrow E_z, \quad [q] \longmapsto [q \circ \iota] \quad (\text{WP10})$$

are inverse algebra isomorphisms. Indeed substitution sends H to $i^{-n}h$ and h to $i^n H$, so the maps descend to the quotients. Their two composites fix every polynomial by (WP5). They also retain the multiplication operators exactly:

$$\mathcal{C} M_z = M_{\theta(s)} \mathcal{C}, \quad \mathcal{C} M_{\iota(z)} = M_s \mathcal{C}. \quad (\text{WP11})$$

2. All the jets and the entire-function remainder

For each $\alpha \in \mathcal{R}$, introduce an independent nilpotent coordinate t_α and the algebra

$$A_\alpha = \mathbb{C}[t_\alpha]/(t_\alpha^m). \quad (\text{WP12})$$

There is an algebra isomorphism

$$\mathcal{J} : E_h \longrightarrow \prod_{\alpha \in \mathcal{R}} A_\alpha, \quad [q] \longmapsto \left(\sum_{a=0}^{m-1} \frac{q^{(a)}(\alpha)}{a!} t_\alpha^a \right)_{\alpha \in \mathcal{R}}. \quad (\text{WP13})$$

Every derivative in this formula is part of the specified map.

Here is a direct proof including its inverse. Define

$$T_\alpha(s) = \sum_{a=0}^{m-1} \frac{(1/h_\alpha)^{(a)}(\alpha)}{a!} (s - \alpha)^a, \quad e_\alpha = [h_\alpha T_\alpha] \in E_h. \quad (\text{WP14})$$

Modulo $(s - \alpha)^m$, the product $h_\alpha T_\alpha$ equals 1, by multiplication of the Taylor expansion and its truncated inverse. Modulo $(s - \eta)^m$ for $\eta \neq \alpha$, it equals 0, because h_α is divisible by that factor. Thus $\mathcal{J}(e_\alpha)$ is 1 in the α -factor and 0 elsewhere. The inverse of (WP13) is

$$\mathcal{J}^{-1} \left(\left(\sum_{a=0}^{m-1} c_{\alpha,a} t_\alpha^a \right)_\alpha \right) = \sum_{\alpha \in \mathcal{R}} \sum_{a=0}^{m-1} c_{\alpha,a} e_\alpha (s - \alpha)^a. \quad (\text{WP15})$$

This proves surjectivity. For injectivity, a polynomial in the kernel has a zero of order at least m at each of the four distinct points. Successive division by their linear factors shows it is divisible by their product h . The kernel is therefore zero in E_h . Taylor multiplication modulo t_α^m proves multiplicativity. Equations (WP13)–(WP15) also prove

$$e_\alpha e_\eta = 0 \ (\alpha \neq \eta), \quad e_\alpha^2 = e_\alpha, \quad \sum_{\alpha} e_\alpha = 1, \quad (\text{WP16})$$

and give the basis $e_\alpha (s - \alpha)^a$, $0 \leq a < m$, of E_h .

For an entire function f , define its finite Taylor remainder by

$$j_h(f) = \mathcal{J}^{-1} \left(\left(\sum_{a=0}^{m-1} \frac{f^{(a)}(\alpha)}{a!} t_\alpha^a \right)_\alpha \right) \in E_h. \quad (\text{WP17})$$

This is an algebra homomorphism: addition and scalar multiplication follow coefficient by coefficient, and the Leibniz rule gives multiplication of the truncated Taylor series. It agrees with the quotient map on polynomials. The class $j_h(f)$ has a unique representative of degree less than n , by division by the monic polynomial h .

In the programme notation $j_h v_h$, the symbol j_h denotes this map applied to v_h . It is not a second scalar function multiplied by v_h . Put

$$u_h = j_h(v_h), \quad U = M_{u_h} : E_h \longrightarrow E_h. \quad (\text{WP18})$$

In the α -factor of (WP13), set

$$v_{\alpha,a} = \frac{v_h^{(a)}(\alpha)}{a!}, \quad u_\alpha(t_\alpha) = \sum_{a=0}^{m-1} v_{\alpha,a} t_\alpha^a. \quad (\text{WP19})$$

All constants $v_{\alpha,0}$ are nonzero by (WP4). Write $u_\alpha = v_{\alpha,0}(1 + b_\alpha)$, with $b_\alpha \in (t_\alpha)$. Since $b_\alpha^m = 0$, its inverse is

$$u_\alpha^{-1} = v_{\alpha,0}^{-1} \sum_{r=0}^{m-1} (-b_\alpha)^r. \quad (\text{WP20})$$

Multiplication of this finite series by u_α gives $1 - (-b_\alpha)^m = 1$. Hence u_h is a unit and U an automorphism. In full jet coordinates its formula is

$$(Ux)_{\alpha,a} = \sum_{b=0}^a v_{\alpha,b} x_{\alpha,a-b} \quad (0 \leq a < m). \quad (\text{WP21})$$

It therefore keeps every derivative of the amplitude through order $m-1$. The determinant in the primary basis is $\prod_{\alpha \in \mathcal{R}} v_h(\alpha)^m \neq 0$, because each block in (WP21) is triangular with that constant diagonal.

3. The zeroth-jet quotient and its exact multiplier map

Let $V_{\mathcal{R}} = \mathbb{C}^{\mathcal{R}}$, with coordinatewise multiplication, and define

$$e : E_h \longrightarrow V_{\mathcal{R}}, \quad e([q]) = (q(\alpha))_{\alpha \in \mathcal{R}}. \quad (\text{WP22})$$

This is a surjective algebra homomorphism: it is the constant-term map in each factor of (WP13), and (WP15) lifts any tuple by $\sum c_\alpha e_\alpha$. Its kernel is

$$J = \ker e = (d)/(d^m) \subset E_h, \quad \dim_{\mathbb{C}} J = 4m - 4. \quad (\text{WP23})$$

Indeed a polynomial has value zero at every α precisely when it is divisible by d . In the primary coordinates, J is $\prod_{\alpha} (t_\alpha)$; it has basis $e_\alpha(s - \alpha)^a$, $1 \leq a < m$, and the asserted dimension follows. Every element of J has its m -th power zero. Conversely, a nilpotent element must have value zero in each complex coordinate, since a complex number with a vanishing positive power is zero. Thus J is exactly the nilradical of E_h .

Write

$$V_h : V_{\mathcal{R}} \longrightarrow V_{\mathcal{R}}, \quad (c_\alpha)_\alpha \longmapsto (v_h(\alpha)c_\alpha)_\alpha, \quad a_h = eU : E_h \longrightarrow V_{\mathcal{R}}. \quad (\text{WP24})$$

Then

$$eU = V_h e, \quad \ker a_h = J, \quad a_h \text{ is surjective.} \quad (\text{WP25})$$

The identity follows either from evaluation of a product or from the $a=0$ case of (WP21). The map V_h is invertible because its four diagonal entries are nonzero, so the kernel and surjectivity assertions follow from (WP22)–(WP23). Thus the exact comparison with the programme's multiplier is the commuting diagram of typed maps

$$\begin{array}{ccc} E_h & \xrightarrow{U} & E_h \\ e \downarrow & & \downarrow e \\ V_{\mathcal{R}} & \xrightarrow{V_h} & V_{\mathcal{R}} \end{array} \quad (\text{WP26})$$

The left and right vertical maps retain the same kernel J ; the horizontal map on the upper row retains all the additional coefficients (WP21).

The entire-function family is a further, distinct vector space

$$\mathcal{F}_h = \{F_P : P \in \mathbb{C}[z]\}. \quad (\text{WP26a})$$

The map $P \mapsto F_P$ is a linear isomorphism onto this family. Surjectivity is its definition. For injectivity, v_h is nonzero at a packet root, hence nonzero on a neighborhood of that root. If F_P vanishes identically, P vanishes on the corresponding open set in the z -coordinate. A nonzero polynomial has only finitely many roots, so $P = 0$.

The full-jet map from this entire-function family is

$$\begin{aligned} \mathcal{T}_h : \mathcal{F}_h &\longrightarrow E_h, & \mathcal{T}_h(F) &= j_h(F \circ \theta), \\ \mathcal{T}_h(F_P) &= U\mathcal{C}[P], & (\mathcal{J}\mathcal{T}_h(F))_{\alpha,a} &= i^{-a} \frac{F^{(a)}(\beta_\alpha)}{a!}. \end{aligned} \quad (\text{WP26b})$$

The first identity on the second line follows from (WP6), (WP17), and (WP18). The second follows by differentiating the affine change of variable θ exactly a times; its derivative is $1/i$. Thus the factor i^{-a} is retained in every jet coordinate.

The map $\mathcal{T}_h$ is onto and its kernel is

$$\ker \mathcal{T}_h = \{F_{HQ} : Q \in \mathbb{C}[z]\}. \quad (\text{WP26c})$$

Indeed U and $\mathcal{C}$ are isomorphisms, so $\mathcal{T}_h(F_P) = 0$ exactly when P is divisible by H . There is a separate surjection

$$e\mathcal{T}_h : \mathcal{F}_h \longrightarrow V_{\mathcal{R}}, \quad F \longmapsto (F(\beta_\alpha))_\alpha, \quad \ker(e\mathcal{T}_h) = \{F_{D_z Q} : Q \in \mathbb{C}[z]\}, \quad D_z(z) = \prod_\alpha (z - \beta_\alpha). \quad (\text{WP26d})$$

The coordinate formula follows from (WP26b) at $a = 0$. It is onto by (WP25). Since $v_h(\alpha) \neq 0$, its value at F_P is zero exactly when P vanishes at all the distinct β_α , which is equivalent to divisibility by D_z . Thus the entire functions themselves are not identified with their finite jets or with their values: (WP26b)–(WP26d) give the actual surjections and both exact kernels.

4. The finite Hermitian form and its radical

Use the convention that a Hermitian form is conjugate-linear in its first argument and linear in its second. On $V_{\mathcal{R}}$, define

$$b_{\mathcal{R}}(c, c') = m \sum_{\alpha \in \mathcal{R}} \overline{c_{\sigma\alpha}} c'_\alpha. \quad (\text{WP27})$$

It is Hermitian: the conjugate of $b_{\mathcal{R}}(c', c)$, after replacing the summation index α by $\sigma\alpha$, is exactly (WP27). It is nondegenerate. If it vanishes against every c' , selecting c' with a single nonzero coordinate at α gives $m\overline{c_{\sigma\alpha}} = 0$, hence every coordinate of c is zero.

Define the packet form on the full algebra by the pullback

$$B_h(x, y) = b_{\mathcal{R}}(a_h x, a_h y) = m \sum_{\alpha \in \mathcal{R}} \overline{v_h(\sigma\alpha)x(\sigma\alpha)} v_h(\alpha)y(\alpha), \quad x, y \in E_h. \quad (\text{WP28})$$

Evaluation is independent of polynomial representatives by (WP22), so the formula is well defined. It is Hermitian because (WP27) is. Let $\text{rad } B_h$ mean the set of x with $B_h(x, y) = 0$ for every $y \in E_h$.

Theorem WP1 (exact radical and quotient). There is an exact sequence

$$0 \longrightarrow J \longrightarrow E_h \xrightarrow{a_h} V_{\mathcal{R}} \longrightarrow 0, \quad \text{rad } B_h = J. \quad (\text{WP29})$$

The induced map $\bar{a}_h : E_h/J \rightarrow V_{\mathcal{R}}$ is an algebra-multiplier comparison and a linear isometry from the descended Hermitian form to $b_{\mathcal{R}}$. More precisely, $e : E_h/J \rightarrow V_{\mathcal{R}}$ is an algebra isomorphism, and $\bar{a}_h = V_h \bar{e}$ is a linear isomorphism; $\bar{a}_h$ need not preserve the unit or multiplication.

Proof. Exactness is (WP25). If $x \in J$, then $a_h x = 0$, so (WP28) vanishes for every y . Conversely, if (WP28) vanishes for every y , surjectivity of a_h implies that $b_{\mathcal{R}}(a_h x, c') = 0$ for every c' . Nondegeneracy of (WP27) gives $a_h x = 0$, hence $x \in J$. The quotient map is an isomorphism by exactness. Formula (WP28) proves the isometry identity. The distinction between $\bar{e}$ and $\bar{a}_h$ follows from (WP24): multiplying all coordinates by a fixed nonconstant unit is a linear automorphism, and is not in general an algebra homomorphism. $\square$

This theorem does not remove the nilpotent directions from the original packet. It identifies the precise quotient seen by this particular Hermitian form, and identifies every omitted direction as an element of the specified ideal J .

5. Complete inertia and explicit negative representatives

In the order (WP2), write $c = (a, b, c_3, c_4)$. Then

$$b_{\mathcal{R}}(c, c') = m(\bar{b}a' + \bar{a}b' + \bar{c}_4 c'_3 + \bar{c}_3 c'_4). \quad (\text{WP30})$$

The matrix is the direct sum of two blocks $m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. In each pair the two vectors $(1, 1)/\sqrt{2m}$ and $(1, -1)/\sqrt{2m}$ have squared values $+1$ and -1 , and their mutual pairing is zero. They form a basis of that two-dimensional pair. Consequently

$$\text{inertia}(b_{\mathcal{R}}) = (2, 2, 0), \quad \text{inertia}(B_h) = (2, 2, 4m - 4), \quad (\text{WP31})$$

where inertia lists positive, negative, and radical dimensions in that order. This is inertia under congruence of Hermitian forms, rather than an assertion that an arbitrary coefficient-frame matrix of B_h has eigenvalues $\pm m$.

For a complete diagonal basis of the original packet, use

$$x_1^\pm = \frac{1}{\sqrt{2m}} U^{-1}(e_{\alpha_1} \pm e_{\alpha_2}), \quad x_2^\pm = \frac{1}{\sqrt{2m}} U^{-1}(e_{\alpha_3} \pm e_{\alpha_4}), \quad (\text{WP32})$$

together with

$$x_{\alpha,a} = U^{-1}(e_\alpha(s - \alpha)^a), \quad \alpha \in \mathcal{R}, \quad 1 \leq a < m. \quad (\text{WP33})$$

These form a basis because the original primary vectors do, the two changes of basis at order zero are invertible, and U is invertible. The last $4m - 4$ vectors lie in J . Formula (WP28) gives $B_h(x_j^+, x_j^+) = 1$, $B_h(x_j^-, x_j^-) = -1$, all pairings between distinct displayed positive or negative vectors zero, and all pairings with (WP33) zero. This proves (WP31) on the entire vector space, including all primary multiplicities.

These representatives are explicit finite Taylor expressions. For example

$$U^{-1}e_\alpha = e_\alpha \sum_{a=0}^{m-1} \frac{(1/v_h)^{(a)}(\alpha)}{a!} (s - \alpha)^a. \quad (\text{WP34})$$

The reciprocal is taken only in a neighborhood of α , where it is holomorphic by (WP4). Taylor multiplication proves (WP34), as also follows from (WP20). Formula (WP34) prescribes the full jet of a representative whose amplitude has constant primary jet 1 at one root and zero at the others.

There are also degree-three representatives, without imposing any condition on their higher amplitude jets. Define the Lagrange polynomials

$$\ell_\alpha(s) = \prod_{\eta \in \mathcal{R} \setminus \{\alpha\}} \frac{s - \eta}{\alpha - \eta}. \quad (\text{WP35})$$

Their denominators are nonzero, and $\ell_\alpha(\eta) = \delta_{\alpha\eta}$. Put

$$q_-(s) = \frac{\ell_{\alpha_1}(s)}{v_h(\alpha_1)} - \frac{\ell_{\alpha_2}(s)}{v_h(\alpha_2)}, \quad P_-(z) = q_-(1/2 + iz). \quad (\text{WP36})$$

Then $a_h[q_-] = (1, -1, 0, 0)$, so

$$B_h([q_-], [q_-]) = -2m. \quad (\text{WP37})$$

Dividing either polynomial in (WP36) by $\sqrt{2m}$ gives value -1 ; dividing by $\sqrt{m}$ gives value -2 . The second negative direction is obtained with α_3, α_4 . This formula includes the exact Fourier-coordinate representative P_- .

For $m > 1$, the class $[q_-]$ in (WP36) and the full-jet representative $U^{-1}(e_{\alpha_1} - e_{\alpha_2})$ need not agree. Their difference lies in J , because their images under a_h agree and $\ker a_h = J$. Thus the two explicit choices have the same pairings with every packet class, with their difference retained as a specified nilpotent class.

6. The surviving jet filtration and multiplication operators

The radical has the finite filtration

$$E_h = J^0 \supset J^1 \supset \dots \supset J^{m-1} \supset J^m = 0, \quad J^r = (d^r)/(d^m). \quad (\text{WP38})$$

In primary coordinates, $J^r = \prod_\alpha (t_\alpha^r)$. To verify this from d , write $d(s) = (s - \alpha)c_\alpha(s)$, with $c_\alpha(\alpha) \neq 0$. The factor c_α is a unit in A_α , so the ideal generated by d^r in that primary factor is exactly (t_α^r) . The isomorphism (WP13) then proves (WP38) for every r .

For $0 \leq r < m$, the exact layer map is

$$\lambda_r : J^r/J^{r+1} \longrightarrow V_{\mathcal{R}}, \quad [x] \longmapsto \left(\frac{x^{(r)}(\alpha)}{r!} \right)_\alpha. \quad (\text{WP39})$$

For $r = 0$, this means the map e on E_h/J . A polynomial representative of an element of J^r has all derivatives of orders less than r zero at each root. Changing it by a multiple of h changes none of the derivatives in (WP39), since $r < m$. Changing its class by J^{r+1} also leaves those coefficients unchanged. Formula (WP13)

shows that the kernel is exactly J^{r+1} , and that the elements $e_\alpha(s - \alpha)^r$ map to the coordinate basis. Hence (WP39) is an isomorphism and every layer has dimension 4.

All these layers are preserved by U , its inverse, and $A_h = M_s$. Their induced operators are

$$\lambda_r \operatorname{gr}_r(U) = V_h \lambda_r, \quad \lambda_r \operatorname{gr}_r(A_h) = \operatorname{diag}(\alpha) \lambda_r. \quad (\text{WP40})$$

For the first equality, use (WP21): the coefficient of order r in a product with an element whose lower coefficients are zero is $v_{\alpha,0} x_{\alpha,r}$. For the second, multiply its Taylor series by $\alpha + t_\alpha$; the extra term only enters the next layer. Before passage to a layer, the complete formulas remain (WP21) and

$$A_h(e_\alpha(s - \alpha)^a) = \alpha e_\alpha(s - \alpha)^a + e_\alpha(s - \alpha)^{a+1}, \quad (\text{WP41})$$

where the last term is zero for $a = m - 1$. Thus the original Jordan chains and the amplitude derivatives survive in the source and its filtered maps. The form (WP28) reads only the layer $r = 0$; this follows from its evaluated formula and does not assert that higher derivatives are unavailable to the observation map.

7. The exact adjoint relation for every polynomial multiplier

For $q \in \mathbb{C}[s]$, define

$$q^\#(s) = \overline{q(1 - \bar{s})}. \quad (\text{WP42})$$

This is a polynomial: if $q(s) = \sum_k c_k s^k$, then $q^\#(s) = \sum_k \bar{c}_k (1 - s)^k$. It is an antilinear, multiplicative involution. Since σ permutes $\mathcal{R}$,

$$h^\#(s) = \prod_{\alpha \in \mathcal{R}} (1 - s - \bar{\alpha})^m = (-1)^{4m} \prod_{\alpha \in \mathcal{R}} (s - \sigma\alpha)^m = h(s). \quad (\text{WP43})$$

It therefore descends to an antilinear algebra involution on E_h . Its action on all the jets is

$$(x^\#)_{\alpha,a} = (-1)^a \overline{x_{\sigma\alpha,a}}. \quad (\text{WP44})$$

Indeed, substitute $s = \alpha + t$ into (WP42), expand q at $\sigma\alpha$, and conjugate its expansion at the increment $-\bar{t}$. This gives the displayed coefficient, including its sign.

For the original completed zeta function, its functional identities are $g(1 - s) = g(s)$ and $g(\bar{s}) = \overline{g(s)}$. They imply $g^\# = g$. The quartet polynomial has exactly the same two identities: conjugation permutes its four factors, and replacing s by $1 - s$ gives the sign $(-1)^{4m} = 1$ and the permutation $\alpha \mapsto 1 - \alpha$. Therefore

$$v_h^\# = v_h, \quad v_h(\sigma\alpha) = \overline{v_h(\alpha)}, \quad (j_h v_h)^\# = j_h v_h. \quad (\text{WP44a})$$

The first identity follows by division off the finite zero set of h and then by continuity at the removable points (WP4). The second is its evaluation, and the third follows from the signed jet formula (WP44). No scalar factor accompanies these identities for the monic degree- $4m$ polynomial in (WP3).

Theorem WP2 (multiplier adjoints). For every $q, x, y \in E_h$,

$$B_h(M_q x, y) = B_h(x, M_{q^\#} y). \quad (\text{WP45})$$

In particular, if $Y_h = (A_h - \frac{1}{2}I)/i = M_{\theta(s)}$, then

$$B_h(A_h x, y) = B_h(x, (I - A_h)y), \quad B_h(Y_h x, y) = B_h(x, Y_h y). \quad (\text{WP46})$$

Proof. At each root α , $q^\#(\alpha) = \overline{q(\sigma\alpha)}$. Inserting a factor $q(\sigma\alpha)$ in the conjugated first entry of (WP28) therefore has exactly the same effect as inserting $q^\#(\alpha)$ in its second entry. This proves (WP45) term by term. For the coordinate polynomial, $s^\# = 1 - s$. For $\theta(s) = -i(s - 1/2)$, antilinearity and this equality give $\theta^\#(s) = i((1 - s) - 1/2) = -i(s - 1/2) = \theta(s)$. These substitutions prove both assertions in (WP46). $\square$

The operator Y_h is consequently self-adjoint with respect to the specified Hermitian form. Its induced eigenvalues on E_h/J are exactly (WP7), since (WP40) gives the diagonal multiplier $(\alpha - 1/2)/i$. The nonreal values $\gamma \pm i\delta$, $-\gamma \pm i\delta$ are paired by the two hyperbolic blocks (WP30). There is no claim that self-adjointness for this indefinite form makes those values real. Equations (WP40), (WP44), and (WP46) give the exact relation between the full primary operator and its value quotient.

In particular the original amplitude unit is itself self-adjoint for this form, by (WP44a) and (WP45). For polynomial inputs, the entire function

$$\mathcal{A}_{P,Q}(s) = q_P^\#(s)q_Q(s)v_h(s)^2 \quad (\text{WP46a})$$

has the packet values

$$\mathcal{A}_{P,Q}(\alpha) = \overline{q_P(\sigma\alpha)v_h(\sigma\alpha)}q_Q(\alpha)v_h(\alpha). \quad (\text{WP46b})$$

This follows from $q_P^\#(\alpha) = \overline{q_P(\sigma\alpha)}$ and (WP44a). On the complement of the zeros of g , the exact logarithmic-derivative identity is

$$\frac{\xi'(s)}{\xi(s)}\mathcal{A}_{P,Q}(s) = q_P^\#(s)q_Q(s)\frac{g'(s)g(s)}{h(s)^2}. \quad (\text{WP46c})$$

Here $g = 2\xi$, so $\xi'/\xi = g'/g$; multiplication by $(g/h)^2$ proves the formula with no additional factor of two. At a zero of g outside the packet, the right side is holomorphic because h does not vanish there. At $\alpha \in \mathcal{R}$, writing $g = (s - \alpha)^m b_\alpha$ and $h = (s - \alpha)^m h_\alpha$ shows that g'/h^2 has the Laurent term $mv_h(\alpha)^2/(s - \alpha)$. Thus its residue after multiplication by $q_P^\#q_Q$ is exactly $m\mathcal{A}_{P,Q}(\alpha)$. These are local meromorphic identities; no contour estimate or prime-side identity is used to prove them.

8. The original period observation and what each map reads

The map $a_h = eU$ applies the same full-jet unit as the original observation and then takes values. To make the relation to the period stage explicit, let $\Pi : E_h \rightarrow \mathbb{C}^n$ be the programme's OPG4 linear map, defined on the unique degree-less-than- n representative by its prescribed convergent period integrals. Retain its actual contours, exponential weight, and basis; none is changed here. Put

$$\mathbf{b}_{\alpha,a} = \Pi(e_\alpha(s - \alpha)^a). \quad (\text{WP47})$$

For every $x \in E_h$, linearity and (WP21) give the identity of vectors in $\mathbb{C}^n$

$$\Pi Ux = \sum_{\alpha \in \mathcal{R}} \sum_{a=0}^{m-1} \mathbf{b}_{\alpha,a} \sum_{b=0}^a v_{\alpha,b} x_{\alpha,a-b}. \quad (\text{WP48})$$

This proof uses a finite sum of the original primary classes; it does not assign a period integral to an arbitrary entire function. The value map, with the same coefficients before projection, is

$$a_h x = (v_{\alpha,0} x_{\alpha,0})_\alpha. \quad (\text{WP49})$$

Thus (WP48) and (WP49) are two exact maps out of the same source after the same multiplication. In particular, for $x = j_h(f)$, the ring-homomorphism property (WP17) gives

$$Uj_h(f) = j_h(v_h f), \quad \Pi Uj_h(f) = \Pi j_h(v_h f), \quad a_h j_h(f) = (v_h(\alpha)f(\alpha))_\alpha. \quad (\text{WP50})$$

The middle formula evaluates Π on a finite Taylor remainder, exactly as in OPG11. It is not the integral of the unreduced entire function unless a separate equality proves that assertion.

There is also a precise test for whether any proposed observation $T : E_h \rightarrow W$, to a complex vector space W , factors through the value map a_h . The factorization exists exactly when $T(J) = 0$; when it exists its only possible formula is

$$\overline{T}(c) = T \left(U^{-1} \sum_{\alpha \in \mathcal{R}} c_\alpha e_\alpha \right). \quad (\text{WP51})$$

To prove this, the expression in parentheses has a_h -image c by (WP15) and (WP24). If $T(J) = 0$, any two lifts with that image differ by (WP29) in J , so their T -images agree, and (WP51) gives a well-defined linear factorization. Surjectivity of a_h forces uniqueness. Conversely a factorization through a_h kills its kernel J . This computes the exact space of factorizing maps as the injective image

$$\text{Hom}_{\mathbb{C}}(V_{\mathcal{R}}, W) \xrightarrow{a_h^*} \text{Hom}_{\mathbb{C}}(E_h, W), \quad \overline{T} \mapsto \overline{T}a_h, \quad \text{im}(a_h^*) = \{T : T(J) = 0\}. \quad (\text{WP52})$$

It does not presume that the period map, or the later cyclic tensor observation, lies in that subspace.

9. Why the off-line and on-line cases have different packet algebras

The hypotheses (WP1) guarantee four distinct roots. Substituting $\delta = 0$ into a formula which assumes four distinct roots merges α_1 with α_2 and α_3 with α_4 . If one retains the product of four factors literally, their orders double. Exact zero order m would then no longer justify dividing g by that new polynomial. The on-line packet must instead be defined from its distinct roots and their actual orders.

For completeness, the general finite value calculation is as follows. Let $\mathcal{A}$ be a finite set of distinct complex points stable under $\sigma\alpha = 1 - \bar{\alpha}$. Assign positive integer orders m_α , with $m_{\sigma\alpha} = m_\alpha$, and suppose an entire g has these exact orders there. Set

$$h_{\mathcal{A}}(s) = \prod_{\alpha \in \mathcal{A}} (s - \alpha)^{m_\alpha}, \quad v_{\mathcal{A}} = g/h_{\mathcal{A}}. \quad (\text{WP53})$$

With the corresponding jet algebra, remainder multiplier, and amplitude-value map, define

$$B_{\mathcal{A}}(x, y) = \sum_{\alpha \in \mathcal{A}} m_\alpha \overline{v_{\mathcal{A}}(\sigma\alpha)} x(\sigma\alpha) v_{\mathcal{A}}(\alpha) y(\alpha). \quad (\text{WP54})$$

Let f be the number of fixed points of σ in $\mathcal{A}$, and let p be the number of its two-element orbits. Then

$$\text{inertia}(B_{\mathcal{A}}) = \left(f + p, p, \sum_{\alpha \in \mathcal{A}} (m_\alpha - 1) \right). \quad (\text{WP55})$$

To prove this directly, use the local rings $\mathbb{C}[t_\alpha]/t_\alpha^{m_\alpha}$. The interpolation proof (WP14)–(WP15) applies with the individual order in each factor, because the other factors are still units at that root. Evaluation has kernel $\prod_\alpha (t_\alpha)$ of the stated dimension, and amplitude multiplication is invertible because its constants are nonzero. On its value quotient, a fixed point contributes the positive form $m_\alpha |c_\alpha|^2$. A two-element orbit contributes the block $m_\alpha \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, with one positive and one negative direction. Distinct orbits are orthogonal since their coordinates never occur together in (WP54). This proves the radical and all three entries of (WP55).

In particular, a packet supported entirely on the critical line has $p = 0$, and (WP54) is positive semidefinite. Its radical still contains all higher jet directions. A distinct off-line quartet has $f = 0, p = 2$, giving (WP31). Both assertions concern the respective actual packet algebras and the exact orders in (WP53).

10. The exact zero-side localization and analytic scope

Index the distinct nontrivial zeros of ξ by ρ' , and let $m_{\rho'}$ be their multiplicities. For the entire functions (WP6), form the discrete zero pairing

$$\mathcal{W}_0(F_P, F_Q) = \sum_{\rho'} m_{\rho'} \overline{F_P(\theta(\rho'))} F_Q(\theta(\rho')). \quad (\text{WP56})$$

This particular sum has only four potentially nonzero summands. Indeed, when $\rho' \notin \mathcal{R}$, (WP4) gives $F_Q(\theta(\rho')) = q_Q(\rho') v_h(\rho') = 0$. Thus no issue of summation order remains in (WP56). At the quartet, the multiplicities equal m , and (WP8) gives the exact identity

$$\begin{aligned} \mathcal{W}_0(F_P, F_Q) &= m \sum_{\alpha \in \mathcal{R}} \overline{q_P(\sigma\alpha)} v_h(\sigma\alpha) q_Q(\alpha) v_h(\alpha) \\ &= B_h(\mathcal{C}[P], \mathcal{C}[Q]). \end{aligned} \quad (\text{WP57})$$

The bracketed arguments on the second line are classes in E_z , mapped into E_h by (WP10). If a polynomial is changed by a multiple of H , its amplitude has all its first m jets zero at each Fourier root. Therefore (WP57) descends to E_z exactly as claimed. In fact the form already factors through the smaller value quotient $\mathbb{C}[z]/\prod_\alpha (z - \beta_\alpha)$; the explicit kernel of the map from E_z to that quotient is $\mathcal{C}^{-1}J$.

For $m > 1$, the multiplicity in (WP56) multiplies a value. It does not differentiate F_P or F_Q . This is why the correct inertia on the full $4m$ -dimensional packet is (WP31), and why the higher jet directions appear as its precisely computed radical. Their maps remain (WP21), (WP38)–(WP44), and (WP48).

The proof above does not identify (WP56) with pole, Gamma, and prime terms for this test family. Such an identity involves its analytic test class and convergence statements; none follows from the finite quotient alone. The finite algebra supplies a complete target for that calculation: the Hermitian form (WP28), with the explicit negative representatives (WP32) or (WP36), the complete radical (WP23), and the exact relation to the native amplitude multiplier (WP26). No assertion here establishes positivity of the prime-side expression or a conclusion about the existence of an off-line zero of ξ .

The explicit formula on the original entire-function packet

This note proves the analytic identity needed to connect the finite packet to the arithmetic side of Weil’s explicit formula. It uses the programme’s actual multiplier, retains both polynomial coordinates, and proves convergence of every term. The result is an identity, not a proof of positivity or of the Riemann hypothesis.

The classical source for the formula and the Fourier/Mellin conventions is Alain Connes and Caterina Consani, *Weil positivity and Trace formula, the archimedean place*, arXiv:2006.13771v1, original author source `weil-compo.tex`, Appendix A, “Fourier versus Mellin transforms” (`appenmellinapp`), and Appendix B, “Explicit formula” (`appendix2`), source lines 2011–2070. In particular `bombieriexplicit`, `bombieriexplicit1`, and `burnolexPLICIT1` fix the pole terms, prime terms, and archimedean sign. That source states its formula for compactly supported functions. The proof below establishes the required extension directly for the present functions; it does not infer admissibility from a numerical residual.

WA1. Objects, involutions, and the exact integrand

Let

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{WA1})$$

Use the entire continuation of this classical function, with $g(1-s) = g(s)$, $\overline{g(\bar{s})} = g(s)$, and $g(0) = g(1) = 1$. Retain the programme’s counterfactual quartet

$$\rho = \tfrac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \tfrac{1}{2}, \quad \gamma > 2, \quad \mathcal{R} = (\rho, 1 - \bar{\rho}, \bar{\rho}, 1 - \rho).$$

Each of these four distinct points is a zero of exact order $m \geq 1$ of g . This is the original counterfactual input, not an assertion that such a zero has been found. Put

$$h(s) = \prod_{\alpha \in \mathcal{R}} (s - \alpha)^m, \quad v(s) = g(s)/h(s), \quad \theta(s) = (s - \tfrac{1}{2})/i, \quad \iota(z) = \tfrac{1}{2} + iz. \quad (\text{WA2})$$

All apparent singularities of v are removable. At each packet point,

$$v(\alpha) = \frac{g^{(m)}(\alpha)}{m! \prod_{\eta \in \mathcal{R} \setminus \{\alpha\}} (\alpha - \eta)^m} \neq 0.$$

For polynomials $P, Q \in \mathbb{C}[z]$, define

$$q_P = P \circ \theta, \quad q_Q = Q \circ \theta, \quad F_P(z) = P(z)v(\iota(z)), \quad F_Q(z) = Q(z)v(\iota(z)).$$

For an entire function a in the s -coordinate, write $a^\#(s) = \overline{a(1 - \bar{s})}$. Conjugation followed by coefficient conjugation makes this entire, and the operation respects products. Reflection and conjugation permute the four factors of h ; the reflection contributes the sign $(-1)^{4m} = 1$. Thus $h^\# = h$ and $v^\# = v$. The exact pairing integrand is

$$A(s) = q_P^\#(s)q_Q(s)v(s)^2. \quad (\text{WA3})$$

It is entire. On the critical line,

$$K(t) := A(\tfrac{1}{2} + it) = \overline{F_P(t)}F_Q(t), \quad t \in \mathbb{R}. \quad (\text{WA4})$$

At a packet point it is

$$A(\alpha) = \overline{q_P(1 - \bar{\alpha})v(1 - \bar{\alpha})}q_Q(\alpha)v(\alpha). \quad (\text{WA5})$$

These identities follow by substitution in the displayed involution. They do not replace q_P by P .

WA2. Uniform decay in every fixed vertical strip

Here is a decay proof that needs no zero-free horizontal line or estimate of ζ'/ζ inside the critical strip.

For $\Re s > 1$, integration of the absolutely convergent series for ζ gives

$$\zeta(s) = s \int_1^\infty [x]x^{-s-1} dx = \frac{s}{s-1} - s \int_1^\infty \{x\}x^{-s-1} dx. \quad (\text{WA6})$$

For the first equality, write $[x] = \sum_{n \geq 1} 1_{[n, \infty)}(x)$, exchange the absolutely convergent sum and integral, and integrate each term. The last integral is holomorphic for $\Re s > 0$, by uniform domination on compact subsets there. It therefore gives the meromorphic continuation to this half-plane and the bound

$$|\zeta(s)| \leq |s/(s-1)| + |s|/(\Re s). \quad (\text{WA7})$$

In particular it is bounded by a constant times $1 + |\Im s|$ when the real part ranges in a fixed compact interval contained in $(0, \infty)$ and $|\Im s| \geq 2$.

For $z = \sigma + it$, $\sigma > 0$, rotate the Euler integral for $\Gamma(z)$ through the angle $\vartheta \operatorname{sgn}(t)$, where $0 < \vartheta < \pi/2$. The integral on the small circular arc tends to zero because $\sigma > 0$; the large arc tends to zero because the real part of its ray is positive. Cauchy's theorem gives

$$\Gamma(z) = e^{i\vartheta \operatorname{sgn}(t)z} \int_0^\infty e^{-re^{i\vartheta \operatorname{sgn}(t)}} r^{z-1} dr, \quad |\Gamma(\sigma + it)| \leq e^{-\vartheta|t|} \frac{\Gamma(\sigma)}{(\cos \vartheta)^\sigma}. \quad (\text{WA8})$$

The bound is uniform for σ in any positive compact interval. Apply it with $\vartheta = \pi/3$ to $\Gamma(s/2)$. Equations WA1 and WA7 show, uniformly in every fixed interval $1/2 \leq \Re s \leq B$,

$$|g(s)| \leq C_B(1 + |\Im s|)^3 e^{-\pi|\Im s|/6} \quad (|\Im s| \geq 2). \quad (\text{WA9})$$

Reflection $g(s) = g(1 - s)$ gives the same type of estimate on every fixed vertical strip, including negative real parts. The constant may depend on that strip. Applying Cauchy's derivative formula on circles of radius $1/4$, inside a slightly larger strip, gives the same estimate with an adjusted constant for every fixed derivative of g . The exponential changes on such a circle by at most a constant factor.

For $|\Im s|$ larger than twice the largest modulus of a packet point plus the strip width, every factor of h has modulus bounded below by a fixed positive multiple of $1 + |\Im s|$. Division by this polynomial and multiplication by the finite polynomials $q_P^\#, q_Q$ therefore give

$$|A^{(j)}(s)| \leq C_{B,P,Q,j}(1 + |\Im s|)^{D_j} e^{-\pi|\Im s|/3} \quad (\text{WA10})$$

on each fixed vertical strip, for each fixed derivative order j . On its bounded remaining part, entireness supplies a finite maximum. The same argument gives exponential strip decay of F_P, F_Q and every fixed derivative, with exponent $\pi/6$. These are estimates on the actual functions, not assertions of uniform constants as the packet or the polynomials vary.

WA3. The inverse transform, convolution, and prime-tail convergence

Define

$$f_P(u) = \frac{1}{2\pi} \int_{\mathbb{R}} F_P(t) e^{itu} dt, \quad f_Q(u) = \frac{1}{2\pi} \int_{\mathbb{R}} F_Q(t) e^{itu} dt, \quad k(u) = \frac{1}{2\pi} \int_{\mathbb{R}} K(t) e^{itu} dt. \quad (\text{WA11})$$

Exponential decay and its derivative estimates show that these inverse transforms are Schwartz functions. More explicitly, differentiate under the integral for derivatives in u , and integrate by parts any number of times for powers of u . All boundary terms vanish by WA10, and all resulting integrals converge absolutely. Fourier inversion on Schwartz functions gives $\widehat{f}_P = F_P$, $\widehat{f}_Q = F_Q$ with the convention $\widehat{f}(t) = \int f(u) e^{-iut} du$.

For any real $R > 0$, shift the t -contour in the last integral of WA11 to $\Im t = R$ for $u \geq 0$, and to $\Im t = -R$ for $u < 0$. The vertical connecting pieces tend to zero by WA10; multiplication by e^{itu} is bounded on each connecting segment for fixed u . Thus

$$|k(u)| \leq C_R e^{-R|u|}, \quad C_R = \frac{1}{2\pi} \max_{\epsilon=\pm 1} \int_{\mathbb{R}} |A(\tfrac{1}{2} - \epsilon R + it)| dt < \infty. \quad (\text{WA12})$$

The same proof applies to f_P, f_Q , and to their derivatives after inserting powers of the integration variable. Fubini's theorem on Schwartz functions gives the exact convolution

$$k = \widetilde{f}_P * f_Q, \quad \widetilde{f}_P(u) = \overline{f_P(-u)}. \quad (\text{WA13})$$

Its Fourier transform is $\overline{F_P(t)} F_Q(t) = K(t)$, which proves the equality by Fourier inversion.

Let $\Lambda_{\text{ar}}(n) = \log \ell$ when $n = \ell^r$ for a prime ℓ and $r \geq 1$, and zero otherwise. This is the arithmetic von Mangoldt function, not the observation map. Since $0 \leq \Lambda_{\text{ar}}(n) \leq \log n$, taking any $R > 1/2$ in WA12 proves absolute convergence of

$$\sum_{n \geq 2} \frac{\Lambda_{\text{ar}}(n)}{\sqrt{n}} (k(\log n) + k(-\log n)). \quad (\text{WA14})$$

There is also the explicit bound for every integer $N \geq 2$:

$$\sum_{n > N} \frac{\Lambda_{\text{ar}}(n)}{\sqrt{n}} (|k(\log n)| + |k(-\log n)|) \leq 2C_R \sum_{n > N} \frac{\log n}{n^{R+1/2}}. \quad (\text{WA15})$$

For example, with $R = 2$, the right side is at most

$$2C_2N^{-3/2} \left(\frac{2}{3} \log N + \frac{4}{9} \right). \quad (\text{WA16})$$

Indeed $(\log x)x^{-5/2}$ is decreasing for $x \geq 2$, so its sum beyond N is bounded by its integral from N to infinity; integration by parts gives the displayed expression. A numerical error certificate would additionally require a certified numerical bound for C_2 ; the existence proof alone is not such a certificate.

WA4. Cancellation of every unwanted zero

Set

$$L(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s) = \frac{g(s)}{s(s-1)}, \quad \ell(s) = L'(s)/L(s). \quad (\text{WA17})$$

This meromorphic completed function has simple poles at $0, 1$, is invariant under $s \mapsto 1-s$, and otherwise has exactly the nontrivial zeta zeros with their multiplicities. These assertions also follow from WA1 and the stated entire continuation of g . As a meromorphic identity,

$$\ell(s)A(s) = q_P^\#(s)q_Q(s) \frac{g'(s)g(s)}{h(s)^2} - \frac{A(s)}{s} - \frac{A(s)}{s-1}. \quad (\text{WA18})$$

Away from the roots of g , this is the logarithmic derivative identity and cancellation of one factor of g . Both sides then have the same meromorphic continuation.

At a zero outside $\mathcal{R}$, the first term in WA18 is holomorphic: its denominator does not vanish there. At $\alpha \in \mathcal{R}$, the logarithmic derivative of g has residue m ; multiplication by the entire function A gives residue $mA(\alpha)$, also when that residue is zero. At $0, 1$, the residues are respectively $-A(0), -A(1)$. These are the only possible poles. This proves exact cancellation at every other zero, regardless of its location or multiplicity.

Choose a rectangle with vertical sides $\Re s = 3/2$ and $\Re s = -1/2$, and horizontal sides $\Im s = \pm T$, for $T > \gamma + 1$. There are no poles on its boundary after the removable singularities in WA18 have been filled. Equations WA9–WA10 bound its horizontal integrals by a polynomial in T times $e^{-\pi T/3}$. They therefore tend to zero for all sufficiently large T , without selecting a sequence avoiding other zeta zeros. The residue theorem and $\ell(1-s) = -\ell(s)$ give

$$m \sum_{\alpha \in \mathcal{R}} A(\alpha) - A(0) - A(1) = \frac{1}{2\pi i} \int_{\Re s = 3/2} \ell(s) (A(s) + A(1-s)) ds. \quad (\text{WA19})$$

The integral is oriented upwards. To verify the sign, the positively oriented left edge is downward. Writing it as minus an upward integral and substituting $w = 1-s$ changes $\ell(1-w)$ to $-\ell(w)$; its total contribution is therefore the plus $A(1-s)$ term in WA19.

WA5. The arithmetic integral and its exact signs

On $\Re s = 3/2$, the absolutely convergent Euler product gives

$$\ell(s) = -\frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2) - \sum_{n \geq 2} \Lambda_{\text{ar}}(n) n^{-s}, \quad \psi = \Gamma'/\Gamma. \quad (\text{WA20})$$

For completeness, absolute convergence follows from $\sum_{n \geq 2} (\log n) n^{-3/2} < \infty$. Expanding each Euler factor as a geometric series gives ζ ; expanding its logarithm and differentiating on any strictly smaller closed half-plane gives the last series with the displayed coefficients. Uniform convergence justifies that differentiation and shows the product is nonzero there.

The gamma factor in WA20 is holomorphic between real parts $1/2$ and $3/2$. Its growth is at most polynomial, which is sufficient here. One direct bound uses

$$\psi(z) = -\gamma_E + \sum_{j=0}^{\infty} \left(\frac{1}{j+1} - \frac{1}{j+z} \right), \quad \Re z > 0. \quad (\text{WA21})$$

This expansion follows by logarithmic differentiation of Euler's gamma product, uniformly on compact subsets of this half-plane. For $\Re z \geq a > 0$, its summand has modulus at most $|z-1|/((j+1)(j+a))$, so the series gives $|\psi(z)| \leq C_a(1+|z|)$. Gamma conjugation gives $\psi(\bar{z}) = \overline{\psi(z)}$. Shift the gamma integral in WA19 to

$\Re s = 1/2$; the horizontal pieces vanish by this bound and WA10. Then substitute $s = 1/2 + it$, and in the term with $A(1 - s)$ replace t by $-t$. The result is

$$\frac{1}{2\pi} \int_{\mathbb{R}} K(t) \left(\Re \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi \right) dt. \quad (\text{WA22})$$

All these integrals are absolutely convergent.

For the Dirichlet series part, WA10 and its absolute convergence on $\Re s = 3/2$ permit termwise integration by absolute domination. For each fixed n , shift the entire integrand $n^{-s}(A(s) + A(1 - s))$ to the critical line. WA11 then gives

$$\frac{1}{2\pi i} \int_{\Re s=3/2} n^{-s} (A(s) + A(1 - s)) ds = \frac{k(-\log n) + k(\log n)}{\sqrt{n}}. \quad (\text{WA23})$$

The first sign comes from $n^{-it} = e^{-it \log n}$; reflection changes it for the second term. Thus the prime contribution is exactly minus WA14. Its convergence can be justified either before the shift by the absolutely convergent Dirichlet series or afterwards by WA12–WA15.

Combining WA19, WA22 and WA23 proves the complete identity

$$\boxed{\begin{aligned} m \sum_{\alpha \in \mathcal{R}} \bar{q}_P(1 - \bar{\alpha}) v(1 - \bar{\alpha}) q_Q(\alpha) v(\alpha) &= A(0) + A(1) \\ &+ \frac{1}{2\pi} \int_{\mathbb{R}} \overline{F_P(t)} F_Q(t) \left(\Re \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi \right) dt \\ &- \sum_{n \geq 2} \frac{\Lambda_{\text{ar}}(n)}{\sqrt{n}} (k(\log n) + k(-\log n)). \end{aligned}} \quad (\text{WA24})$$

It holds for every pair of polynomials and every packet satisfying the retained original hypotheses. No assumption of Weil positivity enters its proof.

WA6. Exact comparison with the published multiplicative convention

Define a function on $\mathbb{R}_{>0}$ by

$$b_A(x) = x^{-1/2} k(-\log x). \quad (\text{WA25})$$

It has decay sufficient for all the following integrals by WA12. Substituting $x = e^u$ in its Mellin transform gives

$$\int_0^\infty b_A(x) x^{s-1} dx = \int_{\mathbb{R}} k(-u) e^{(s-1/2)u} du = A(s). \quad (\text{WA26})$$

On the critical line this is Fourier inversion, and on the whole plane it follows by the same absolutely convergent transform and analytic continuation. Its involution satisfies

$$b_A^\sharp(x) = x^{-1} b_A(x^{-1}) = x^{-1/2} k(\log x). \quad (\text{WA27})$$

Consequently the two pole integrals in `bombieriexplicit` are precisely $A(1), A(0)$; for the second integral use the substitution $y = x^{-1}$. At a prime power $x = \ell^r$, its two finite-place terms are exactly $\ell^{-r/2}(k(-r \log \ell) + k(r \log \ell))$. The opposite of the archimedean distribution in `burnolexplicit1` is the integral in WA24. Equations WA25–WA27 prove the complete change of conventions, including its inverse $k(u) = e^{-u/2} b_A(e^{-u})$. The function is not asserted compactly supported; WA19–WA24 provide the required analytic extension.

WA7. Consequence for the programme's finite form, with the radical retained

Let $E_h = \mathbb{C}[s]/(h)$, let $U = M_{j_h(v)}$ be the literal finite Taylor-remainder multiplier, and let $e : E_h \rightarrow \mathbb{C}^{\mathcal{R}}$ evaluate the zeroth jets. The finite proofs WP13–WP25 give its complete jet formula and establish

$$a_h = eU, \quad a_h([q]) = (q(\alpha) v(\alpha))_\alpha, \quad \ker a_h = (d)/(d^m), \quad d = \prod_{\alpha \in \mathcal{R}} (s - \alpha). \quad (\text{WA28})$$

The elementary proof of the kernel is that all $v(\alpha)$ are nonzero and a polynomial vanishes at the four distinct points exactly when divisible by d . Surjectivity follows from interpolation at those points. The actual multiplier retains the higher jets, as specified in WP21; WA28 takes their stated quotient.

In the displayed order of $\mathcal{R}$, the left side of WA24 is the pullback under a_h of

$$b(c, d) = m(\bar{c}_2 d_1 + \bar{c}_1 d_2 + \bar{c}_4 d_3 + \bar{c}_3 d_4). \quad (\text{WA29})$$

Each pair has matrix $m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, with eigenvectors $(1, 1)$, $(1, -1)$ and eigenvalues $m, -m$. Hence this form has inertia $(2, 2)$, and its pullback has inertia $(2, 2, 4m - 4)$ and radical exactly WA28. The degree-three polynomial

$$q_-(s) = \frac{\prod_{\eta \neq \alpha_1} (s - \eta)}{v(\alpha_1) \prod_{\eta \neq \alpha_1} (\alpha_1 - \eta)} - \frac{\prod_{\eta \neq \alpha_2} (s - \eta)}{v(\alpha_2) \prod_{\eta \neq \alpha_2} (\alpha_2 - \eta)} \quad (\text{WA30})$$

has amplitude values $(1, -1, 0, 0)$. Taking $P_- = q_- \circ \iota$, and $P = Q = P_-$ in WA24, proves that its full arithmetic right side is exactly $-2m$. This is the arithmetic identity associated with the assumed off-line packet. It is not a constructed off-line zero and does not establish that the arithmetic right side is nonnegative for zeta.

The analytic identity closes the admissibility question for these noncompact test functions. The remaining RH question is an actual sign statement for the right side of WA24 on this programme family. Neither positive source norms nor the nonzero divided-power classes alone prove that sign. Their comparison must supply a proved map and an inequality before any RH conclusion can be drawn.

Tau residue norms and the exact change in the Weil formula

The original semiring is $S = G(\mathbb{Z}) = \{\tau\} \sqcup \mathbb{Z}$, with the original integer operations, $\tau + x = x$, and $\tau x = \tau$. Write $e = 0_{\mathbb{Z}}$, retaining $e \neq \tau$. This calculation starts from its residue and support maps. It does not replace a chosen residue norm by a different one without recording the map.

The originating programme's `globalization_note.tex`, `ideal/congruence/zeta` sections, proves the original arithmetic ideal norm and its zeta function. Its prime spectrum is two-dimensional. The companion `TAU_PRIME_SPECTRUM_DERIVATION.md` checks that classification, its localizations and the coefficient maps in full. Here the calculation continues from the different residue constructions to their exact prime distributions and Weil forms.

TN1. Three numerical constructions, with their quotient maps

For $n \geq 1$, reduction of supported integers defines the support-preserving semiring map

$$\pi_n : S \longrightarrow G(\mathbb{Z}/n\mathbb{Z}), \quad \tau \longmapsto \tau_n, \quad a \longmapsto (a \bmod n)^{\bullet}. \quad (\text{TN1})$$

It is onto and preserves both operations by their definitions. Its equivalence classes keep τ separate from every integer. Its global-zero fibre is only $\{\tau\}$. Arithmetic projection gives a second surjection

$$G(\mathbb{Z}/n\mathbb{Z}) \longrightarrow \mathbb{Z}/n\mathbb{Z}, \quad \tau_n \longmapsto 0, \quad b^{\bullet} \longmapsto b. \quad (\text{TN2})$$

The composite has zero fibre $I_n = \{\tau\} \cup n\mathbb{Z}$. The two target cardinalities are $n + 1$ and n , respectively. At $n = 1$ the first target is the two-element Boolean semiring and the second is the one-element ring; this case is retained.

The usual ideal product is $I_m I_n = I_{mn}$. Every finite sum of products of multiples of m, n is a multiple of mn , and every multiple of mn is itself such a product; the adjoined identity τ causes no further integer values. Hence the second cardinality is multiplicative, whereas the first has the exact defect

$$(m + 1)(n + 1) - (mn + 1) = m + n. \quad (\text{TN3})$$

Consequently the two actual ideal sums on I_n , for $\Re s > 1$, are

$$Z_{\text{ar}}(s) = \sum_{n \geq 1} n^{-s} = \zeta(s), \quad Z_{\text{card}}(s) = \sum_{n \geq 1} (n + 1)^{-s} = \zeta(s) - 1. \quad (\text{TN4})$$

The second equality is an index change in an absolutely convergent series. It does not have the prime Euler product obtained by replacing every p by $p + 1$.

There is also a precisely defined multiplicative prime weight

$$N_{\text{pr}}(n) = \prod_{p|n} (p + 1)^{v_p(n)}, \quad N_{\text{pr}}(1) = 1. \quad (\text{TN5})$$

This agrees with the first residue cardinality at prime integers, but not at general integers or at 1. Unique factorization gives its separate Euler product

$$E_{\text{pr}}(s) = \sum_{n \geq 1} N_{\text{pr}}(n)^{-s} = \prod_p \frac{1}{1 - (p + 1)^{-s}}, \quad \Re s > 1. \quad (\text{TN6})$$

Absolute convergence follows from $N_{\text{pr}}(n) \geq n$; expanding finite products and then passing to the absolutely convergent limit proves the identity. Thus all three functions in TN4 and TN6 are defined, and none is substituted for another.

TN2. The exact mixed-support reason for TN3

For commutative rings R, T , let $\chi_R : G(R) \rightarrow \mathbb{B}$ send τ to 0 and every supported element to 1; similarly for T . There is a semiring isomorphism

$$G(R \times T) \xrightarrow{\sim} G(R) \times_{\mathbb{B}} G(T), \quad \tau \longmapsto (\tau_R, \tau_T), \quad (r, t)^{\bullet} \longmapsto (r^{\bullet}, t^{\bullet}). \quad (\text{TN7})$$

The fibre product contains exactly the displayed two types, because equality of support means either both entries are absent or both are supported. The map is a bijection. Addition and multiplication on two

supported entries are the ring product operations; entries involving the absent pair obey the external identity and absorber laws. These checks prove preservation of both operations, both identities, and the inverse laws. They also prove the pullback universal property: a pair of semiring maps with equal support takes values in exactly this subset, and therefore factors uniquely through it.

If m, n are coprime, choose integers u, v with $um + vn = 1$. The usual reduction map $\mathbb{Z}/mn \rightarrow \mathbb{Z}/m \times \mathbb{Z}/n$ is injective because divisibility by both coprime integers implies divisibility by their product; it is surjective because $bvn + cum$ has residues b, c . Applying TN7 gives

$$G(\mathbb{Z}/mn) \cong G(\mathbb{Z}/m) \times_{\mathbb{B}} G(\mathbb{Z}/n). \quad (\text{TN8})$$

Its embedding into the full product omits precisely

$$(\mathbb{Z}/m)^{\bullet} \times \{\tau_n\}, \quad \{\tau_m\} \times (\mathbb{Z}/n)^{\bullet}. \quad (\text{TN9})$$

These disjoint mixed-support subsets have sizes m, n . TN3 is therefore the count of explicitly identified subsets. No supported cancellation is renamed as absence. In support-separated counts the vector is $(1, n)$; fibre-product multiplication is componentwise, $(1, m)(1, n) = (1, mn)$. Forgetting the labels by summing coordinates gives $n + 1$, and TN3 measures exactly the failure of that sum map to be multiplicative.

TN3. Exact analytic comparison for the prime weight

On $\Re s > 0$, use the branch

$$\log(1 - x^{-s}) = - \sum_{j \geq 1} \frac{x^{-js}}{j}, \quad x \geq 2,$$

and define

$$L_{\tau}(s) = \sum_p (\log(1 - p^{-s}) - \log(1 - (p+1)^{-s})), \quad R_{\tau}(s) = e^{L_{\tau}(s)}. \quad (\text{TN10})$$

This definition has no branch ambiguity because $|x^{-s}| < 1$. To prove convergence, differentiate the summand with respect to the real variable x :

$$\log(1 - p^{-s}) - \log(1 - (p+1)^{-s}) = -s \int_p^{p+1} \frac{x^{-s-1}}{1 - x^{-s}} dx. \quad (\text{TN11})$$

Fix $a > 0$, and put $d_a = 1 - 2^{-a} > 0$. On $\Re s \geq a$, the absolute value of the integrand without s is at most x^{-a-1}/d_a . The prime intervals are disjoint subsets of $[2, \infty)$, so

$$|L_{\tau}(s)| \leq \frac{|s|2^{-a}}{ad_a}. \quad (\text{TN12})$$

The same domination proves local uniform convergence of the series on the open half-plane. It therefore defines a holomorphic function, and its exponential is holomorphic and nowhere zero. On $\Re s > 1$, the absolutely convergent Euler products give

$$E_{\text{pr}}(s) = \zeta(s)R_{\tau}(s). \quad (\text{TN13})$$

This supplies a meromorphic continuation of the particular function TN6 to $\Re s > 0$. Its zero and pole divisor there equals that of ζ , with multiplicities, because R_{τ} is a holomorphic unit. This is a statement about TN6 with TN5, not about TN4's function $\zeta - 1$, and not about an isomorphism of spectra or trace spaces.

The retained logarithmic-derivative correction is

$$r_{\tau}(s) = \frac{R'_{\tau}(s)}{R_{\tau}(s)} = \sum_p \left(\frac{\log p}{p^s - 1} - \frac{\log(p+1)}{(p+1)^s - 1} \right). \quad (\text{TN14})$$

One may differentiate locally uniformly in TN10 by holomorphy and Cauchy's formula on slightly larger compact sets. A direct formula, including a vertical bound, is

$$r_{\tau}(s) = \sum_p \int_p^{p+1} x^{-s-1} \left(\frac{s \log x}{(1 - x^{-s})^2} - \frac{1}{1 - x^{-s}} \right) dx, \quad (\text{TN15})$$

$$|r_{\tau}(s)| \leq \frac{2^{-a}}{ad_a} + \frac{|s|2^{-a} \left(\frac{\log 2}{a} + \frac{1}{a^2} \right)}{d_a^2}, \quad \Re s \geq a. \quad (\text{TN16})$$

For TN15, differentiate $(\log x)/(x^s - 1)$ in x and integrate its negative derivative from p to $p+1$. The estimates use $\int_2^{\infty} x^{-a-1} dx = 2^{-a}/a$ and $\int_2^{\infty} (\log x)x^{-a-1} dx = 2^{-a}(\log 2/a + 1/a^2)$. They prove absolute local uniform convergence, and show that r_{τ} has at most linear vertical growth in every fixed strip inside this half-plane. All quantities in TN10–TN16 are retained; equality of a zero divisor does not remove this nonzero correction.

TN4. The corresponding prime distributions

Use the programme's actual entire test functions from WEIL_PACKET_ANALYTIC_DERIVATION.md:

$$A(s) = q_P^\#(s)q_Q(s)v(s)^2, \quad K(t) = A(\tfrac{1}{2} + it) = \overline{F_P(t)}F_Q(t), \quad k(u) = \frac{1}{2\pi} \int_{\mathbb{R}} K(t)e^{itu}dt. \quad (\text{TN17})$$

Its proof WA2–WA3 establishes exponential decay of A in every fixed vertical strip and $|k(u)| \leq C_R e^{-R|u|}$ for every fixed $R > 0$. Define the two explicit prime functionals

$$\begin{aligned} \mathcal{P}(k) &= \sum_p \sum_{j \geq 1} (\log p) p^{-j/2} (k(j \log p) + k(-j \log p)), \\ \mathcal{P}_\tau(k) &= \sum_p \sum_{j \geq 1} \log(p+1) (p+1)^{-j/2} (k(j \log(p+1)) + k(-j \log(p+1))). \end{aligned} \quad (\text{TN18})$$

Both sums converge absolutely: for $R = 2$, each is dominated by a constant times $\sum_p (\log(p+1))(p+1)^{-5/2}/(1 - (p+1)^{-5/2})$, or the analogous expression with p ; each is bounded by a convergent sum over all integers.

The exact difference is

$$\boxed{\mathcal{P}(k) - \mathcal{P}_\tau(k) = \frac{1}{2\pi} \int_{\mathbb{R}} K(t) 2\Re r_\tau(\tfrac{1}{2} + it) dt.} \quad (\text{TN19})$$

To prove it, start with

$$\frac{1}{2\pi i} \int_{\Re s = 3/2} r_\tau(s) (A(s) + A(1-s)) ds.$$

On this line, expand both terms in TN14 as absolutely convergent geometric series and integrate term by term. Fourier inversion and an entire contour shift give exactly the difference in TN18, with the factors $p^{-j/2}$ and $(p+1)^{-j/2}$. Alternatively shift the integral itself to $\Re s = 1/2$. No poles are crossed because r_τ is holomorphic for positive real part; its vertical bound TN16 and the strip decay of A make the horizontal integrals tend to zero. Reflect $t \mapsto -t$ in the second summand. Since TN10 has real coefficients under conjugation, $r_\tau(\bar{s}) = r_\tau(s)$. This gives TN19, including its sign and factor 2.

TN5. Weil's condition with the full tau correction

Let

$$w_\infty(t) = \Re \psi(\tfrac{1}{4} + \tfrac{it}{2}) - \log \pi, \quad w_{\infty, \tau}(t) = w_\infty(t) - 2\Re r_\tau(\tfrac{1}{2} + it). \quad (\text{TN20})$$

The complete analytic identity WA24 and TN19 give

$$\boxed{B_h([q_P], [q_Q]) = A(0) + A(1) + \frac{1}{2\pi} \int_{\mathbb{R}} \overline{F_P(t)} F_Q(t) w_{\infty, \tau}(t) dt - \mathcal{P}_\tau(k).} \quad (\text{TN21})$$

Here B_h is the literal full-jet packet form with its exact radical, as constructed in WP28–WP29. To check the sign directly, TN19 says $\mathcal{P} = \mathcal{P}_\tau + D$, where D is the integral of $2\Re r_\tau$. Substitution into the original term $-\mathcal{P}$ subtracts this integral, giving TN20. Thus altering the prime norms alone while keeping the previous infinite-place term does not produce the same form; TN20 is its required exact correction.

For $P = Q$, the right side of TN21 is real. Its nonnegativity is exactly the original packet's Weil positivity condition, since TN21 is an equality of forms. This calculation does not prove that condition. It supplies a full new expression of the condition through the separately defined prime-residue weight, preserving the support cardinality defect and the correcting distribution. In particular the degree-three negative representative WP36 of an assumed off-line packet still gives the exact value $-2m$ in TN21.

TN6. What this does and does not identify

There are exact maps from the original tau semiring to two different residue targets, an exact support fibre product explaining the failure of scalar count multiplicativity, and a complete analytic comparison from one specified prime-residue Euler product to its Weil functional. These statements use the original addition and keep its supported zero. They do not replace the two-dimensional semiring spectrum by the one-dimensional arithmetic spectrum, or identify either with Connes's adèle-class orbit space.

The further Connes calculation must retain the supported coordinate-zero hyperplanes, their stabilizers and transverse additive fields, and the new absence strata. Those are the objects entering Section VI,

equations (5)–(15), of Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, arXiv:math/9811068v1. The independent endpoint and adelic comparison continues in TAU_CONNES_ENDPOINT_DERIVATION.md and TAU_PRIME_SPECTRUM_DERIVATION.md. No equality of their Hilbert-space traces is assumed from the scalar function identity TN13.

The original three-point fibre, the prime-zero endpoints, and the heat sign change

This calculation answers the specific comparison between the target $(-1/4, 0, 0)$ of the original polynomial map and the pole pairing at $s = 0, 1$. It proves an isomorphism preserving the specified involution and trace, retains the third inverse point and the infinitesimal at infinity, and computes the exact change to the completed zeta function when its endpoint factor is moved. Positivity proved here concerns these finite trace forms. The complete zeta-zero form is retained in the completion calculation.

The original polynomial and three points are Tao, displayed formula (1) and the following equalities. Earlier programme proofs are the complete inverse chart, `eq:retained-cubic`, `eq:retained-inverse`, and `eq:retained-heat`, the original heat trajectories, and EFI7–EFI15, including the projective graph and dual-number fibre. The complete arguments needed for this comparison follow.

1. Original coordinates and the entire target line

Write the third source coordinate as W ; this is the identity renaming of the w in the cited source, to distinguish it from quotient generators. The original map is

$$\begin{aligned} F_1 &= (1 + xy)^3 W + y^2(1 + xy)(4 + 3xy), \\ F_2 &= y + 3x(1 + xy)^2 W + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2y - x^3W. \end{aligned} \tag{HEB1}$$

Its target coordinates remain (a, b, c) . On $b = c = 0$, the complete inverse curve has the scheme decomposition

$$F^{-1}\{(a, 0, 0) : a \in \mathbb{C}\} \simeq \text{Spec } \mathbb{C}[a] \sqcup \text{Spec } \mathbb{C}[r, r^{-1}], \tag{HEB2}$$

with maps to the target respectively $a \mapsto a$ and $a \mapsto -r^2$. The source coordinates on the two components are

$$(x, y, W) = (0, 0, a), \quad (x, y, W) = \left(-\frac{1}{2r}, 3r, 26r^2\right). \tag{HEB3}$$

Proof. Put $K = 2 - 3xy - x^2W$. Then $F_3 = xK$ and $K + x(3y + xW) = 2$. In the coordinate ring defined by $F_1 - a, F_2$, the ideals (x) and (K) are comaximal. Their intersection is their product: if $v \in (x) \cap (K)$, multiply v by a representation $1 = j_x + j_K$ with $j_x \in (x)$ and $j_K \in (K)$; each summand is in (xK) . The converse inclusion is immediate. The Chinese remainder map therefore splits the quotient by F_3 into these two components, with no lost scheme structure.

On $x = 0$, $F_2 = y$ and $F_1 = W + 4y^2$, giving the first component. On $K = 0$, x is a unit with inverse $(3y + xW)/2$. Set $\alpha = 1/x$ and $r = y + 1/x$. Solving $F_3 = 0$ gives $W = 5\alpha^2 - 3r\alpha$. Direct substitution gives

$$F_2 = 4r + 2\alpha, \quad F_1 = r^2 + r\alpha. \tag{HEB4}$$

Consequently $\alpha = -2r$, $a = -r^2$, and (HEB3) follows. Conversely (HEB3) satisfies all three target equations and recovers $r = y + 1/x$. These mutually inverse coordinate substitutions prove (HEB2). Explicit component idempotents are $K/2$ for the finite component and $(3xy + x^2W)/2$ for the escaping component. Their sum is one and their product is zero. $\square$

At $a = -1/4$, the two roots are $r = -1/2, +1/2$. The complete fibre is precisely

component coordinate	(x, y, W)	
finite	$(0, 0, -1/4)$	
$r = -1/2$	$(1, -3/2, 13/2)$	
$r = +1/2$	$(-1, 3/2, 13/2)$	(HEB5)

Thus the number $1/4$ in the root relation is exactly the negative of the original target coordinate, rather than a new choice of scale.

This target is not the only noninjective one. Every $a \neq 0$ has the separate finite point and two distinct complex points in (HEB3). For example at $a = +1/4$ these are

$$(0, 0, 1/4), \quad (i, 3i/2, -13/2), \quad (-i, -3i/2, -13/2). \tag{HEB6}$$

Direct substitution or (HEB3) proves all three images are $(1/4, 0, 0)$. The two nonreal points cannot be dropped when discussing complex injectivity.

2. Compactification and the involution

In source projective coordinates $[X : Y : Z : T]$, whose affine coordinates are $(x, y, W) = (X/T, Y/T, Z/T)$, the escaping graph extends to

$$r \longmapsto ([1 : -6r^2 : -52r^3 : -2r], a = -r^2). \quad (\text{HEB7})$$

The displayed graph is closed in $\mathbb{P}^3 \times \mathbb{A}^1$: it is the scheme defined by $Y = 6aX$, $Z = -26aT$, and $T^2 + 4aX^2 = 0$. These equations have no projective point with $X = 0$, because they would successively force $Y = T = Z = 0$. In the chart $X = 1$ its coordinate ring is $\mathbb{C}[a, T]/(T^2 + 4a)$, and the mutually inverse substitutions $r = -T/2$, $a = -r^2$ identify this ring with $\mathbb{C}[r]$. Substitution recovers exactly all four coordinates in (HEB7), proving the asserted scheme equality. The coordinate $T/X = -2r$ is its inverse coordinate, so the graph is exactly the affine r -line. Its fibre at $a = 0$ is the length-two algebra $\mathbb{C}[r]/(r^2)$ supported at $[1 : 0 : 0 : 0]$. The finite component remains $[0 : 0 : a : 1]$, disjoint from (HEB7). Hence the compactified inverse graph over the target line is finite flat of rank three, with coordinate algebra

$$\mathcal{C} = \mathbb{C}[a] \times \mathbb{C}[a, r]/(r^2 + a). \quad (\text{HEB8})$$

Freeness of the second factor in the basis $1, r$ follows by division by its monic relation. The original affine inverse curve (HEB2) is obtained by removing the point $r = 0$ from the second component, not by quotienting away its infinitesimal.

The original source involution and its target map are

$$R(x, y, W) = (-x, -y, W), \quad F \circ R = (F_1, -F_2, -F_3). \quad (\text{HEB9})$$

Every term of (HEB1) verifies this equality: xy, y^2, x^2W remain fixed and the indicated odd factors change sign. Thus R preserves our target line, fixes the finite component, and sends $r = y + 1/x$ to $-r$. For real a , compose this algebraic reflection with coefficient conjugation to obtain

$$r^\# = -r, \quad z^\# = \bar{z} \quad (z \in \mathbb{C}). \quad (\text{HEB10})$$

The relation $r^2 + a$ is stable and $\#^2 = 1$, so this is a well-defined conjugate-linear ring involution. It is not ordinary coefficient conjugation with r fixed; (HEB9) is the exact map connecting the two.

3. An isometry with the actual endpoint pairing

For a compactly supported smooth complex test f on $\mathbb{R}$, retain

$$\widehat{f}(z) = \int_{\mathbb{R}} f(v) e^{-ivz} dv, \quad M_f(s) = \widehat{f}((s - 1/2)/i). \quad (\text{HEB11})$$

Set $A = M_f(0) = \widehat{f}(i/2)$ and $B = M_f(1) = \widehat{f}(-i/2)$. The endpoint contribution is

$$W_0(f, g) = \overline{A}_f B_g + \overline{B}_f A_g. \quad (\text{HEB12})$$

Use uppercase A, B here to avoid confusion with the original target a, b, c .

At the exact target $a = -1/4$, there are mutually inverse algebra maps

$$\begin{aligned} \mathbb{C}[s]/(s(s-1)) &\longrightarrow \mathbb{C}[r]/(r^2 - 1/4), & s &\longmapsto 1/2 + r, \\ \mathbb{C}[r]/(r^2 - 1/4) &\longrightarrow \mathbb{C}[s]/(s(s-1)), & r &\longmapsto s - 1/2. \end{aligned} \quad (\text{HEB13})$$

They respect the relations by expansion and are inverse on their generators. Under (HEB13), $s^\# = 1 - s$. Evaluation at $r = -1/2, +1/2$ maps the second algebra isomorphically to $\mathbb{C}^2$, with inverse

$$(A, B) \longmapsto p_{A,B} = (A + B)/2 + (B - A)r. \quad (\text{HEB14})$$

Both evaluations and the coefficient formula prove the inverse assertion. The involution acts on endpoint values as $(A, B)^\# = (\overline{B}, \overline{A})$.

For $E_a = \mathbb{C}[r]/(r^2 + a)$, define the regular trace by the trace of its multiplication operator. In the basis $(1, r)$,

$$L_r = \begin{pmatrix} 0 & -a \\ 1 & 0 \end{pmatrix}, \quad \text{Tr}_{E_a}(c + dr) = 2c. \quad (\text{HEB15})$$

Multiplying $(\bar{c} - \bar{d}r)(e + kr)$ gives scalar coefficient $\bar{c}e + a\bar{d}k$, and hence

$$q_a(c + dr, e + kr) := \text{Tr}_{E_a}((c + dr)^\#(e + kr)) = 2\bar{c}e + 2a\bar{d}k. \quad (\text{HEB16})$$

In particular substitution of (HEB14) gives

$$q_{-1/4}(p_{A,B}, p_{A,B}) = \frac{1}{2}|A + B|^2 - \frac{1}{2}|A - B|^2 = 2\text{Re}(A\bar{B}). \quad (\text{HEB17})$$

Polarization, or the same multiplication with two inputs, proves equality with (HEB12) in both slots. These are isomorphisms of algebras with involution and isometries of the trace pairings. They do not identify the separate third component with either endpoint. On the whole algebra (HEB8) the form is

$$q_a^{\text{full}}(v, c + dr) = |v|^2 + 2|c|^2 + 2a|d|^2. \quad (\text{HEB18})$$

4. The original heat clock crosses the sign boundary

Retain the original forward heat parameter h , with its original speed and initial target:

$$a(h) = -1/4 + 2h, \quad b = c = 0, \quad r^2 = 1/4 - 2h. \quad (\text{HEB19})$$

The original cubic on this line is $P = -2r^2 - 2a$; it satisfies $\partial_h P = -4 = \partial_r^2 P$. Thus (HEB19) is the original coefficient heat flow. The exact pulled-back form in the free basis $(1, r)$ is

$$G_h = \begin{pmatrix} 2 & 0 \\ 0 & 4h - 1/2 \end{pmatrix}, \quad q_h(r, r) = 4h - 1/2. \quad (\text{HEB20})$$

It has one positive and one negative direction for $h < 1/8$; at $h = 1/8$ it has one positive direction and radical $\mathbb{C}r$; for $h > 1/8$ it is positive definite. This follows directly from the two real diagonal entries. In the full rank-three algebra (HEB18), the finite point adds a positive direction in every case.

At the collision $r \neq 0$ as an algebra element and $r^2 = 0$: division by the monic polynomial gives independent classes $1, r$. It is the trace pairing of this nonzero infinitesimal that vanishes. Under the supported embedding into $G(E_0)$ its square is the supported zero e , distinct from the global unsupported zero τ . No positivity conclusion follows from identifying either of these zeros with the other.

The two spectral coordinates obtained from (HEB13) throughout the family are

$$s_\pm(h) = 1/2 \pm \sqrt{1/4 - 2h}. \quad (\text{HEB21})$$

At $h = 0$ they are the actual endpoints $0, 1$. At $h = 1/8$ they coincide at $1/2$ with multiplicity two. At $h > 1/8$ they lie on the critical line, at $1/2 \pm i\sqrt{2h - 1/4}$. In the original source, however, the corresponding x, y have become nonreal; the original real trajectories escape to infinity at the collision. Formula (HEB7) gives the continuation as a complex algebraic graph, not a finite continuation in $\mathbb{R}^3$.

There is also an operator version with the metric retained. Put $S = I/2 + L_r$ and $D = (S - I/2)/i = -iL_r$. Direct multiplication gives

$$L_r^* G_h = -G_h L_r, \quad D^* G_h = G_h D. \quad (\text{HEB22})$$

For $h > 1/8$, G_h is positive definite, so if $Dv = \lambda v$ with $v \neq 0$, the equality of the two pairings gives $\lambda = \bar{\lambda}$. Thus S has real part $1/2$ on every eigenvalue. At the collision, D is a nonzero nilpotent; it cannot be self-adjoint for any positive definite metric, because that would give $\|Dv\|^2 = \langle v, D^2 v \rangle = 0$ for every v . Its specified semidefinite trace metric retains exactly the radical already computed. This is a complete finite-dimensional positivity and spectral calculation.

5. Moving this factor in the completed zeta function

Define

$$q_h(s) = s(s - 1) + 2h, \quad q_0 = s(s - 1), \quad \Lambda(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad 2\xi(s) = q_0(s) \Lambda(s). \quad (\text{HEB23})$$

The entire endpoint test M_f has a unique class modulo q_h : for two distinct roots use their two values, and at a double root use its value and first derivative. Existence follows by degree-one interpolation; uniqueness follows because a polynomial of degree at most one with the two required vanishing conditions is zero. The involution is the same $s^\# = 1 - s$ with conjugate coefficients. Its trace pairing is

$$P_h(f, g) = \sum_{q_h(\rho)=0}^{\text{mult}} \overline{M_f(1 - \bar{\rho})} M_g(\rho). \quad (\text{HEB24})$$

This equals (HEB16) for those interpolating classes: multiplication matrices at distinct roots are diagonalized by evaluation, and at a double root their trace is twice their scalar coefficient. Therefore $P_0 = W_0$, $P_{1/8}(f, f) = 2|M_f(1/2)|^2$, and

$$P_h(f, f) = |M_f(s_+(h))|^2 + |M_f(s_-(h))|^2 \geq 0 \quad (h > 1/8). \quad (\text{HEB25})$$

An exact realization of this moving denominator is

$$\Lambda_h(s) = \frac{2\xi(s)}{q_h(s)} = \Lambda(s) \frac{q_0(s)}{q_h(s)}. \quad (\text{HEB26})$$

As identities of meromorphic functions and signed divisors,

$$\frac{\Lambda'_h}{\Lambda_h} = \frac{\Lambda'}{\Lambda} + \frac{q'_0}{q_0} - \frac{q'_h}{q_h}, \quad \text{div } \Lambda_h = \text{div } \xi - \text{div } q_h. \quad (\text{HEB27})$$

Logarithmic differentiation proves the first identity; factoring at each point proves the second, including cancellations when a zero of q_h coincides with a zero of ξ . One must keep those cancellations; at such a parameter, the factor trace (HEB24) is not a list of uncanceled poles of Λ_h .

Here is the exact effect on Weil's formula. Write its original decomposition as

$$W(f, g) = P_0(f, g) + \mathcal{A}(f, g) - \mathcal{P}(f, g), \quad (\text{HEB28})$$

where $\mathcal{A}$ is the original archimedean term and $\mathcal{P}$ the original rational-prime term. Define the archimedean term for the modified completion by

$$\mathcal{A}_h = \mathcal{A} + P_0 - P_h. \quad (\text{HEB29})$$

Then exactly

$$W = P_h + \mathcal{A}_h - \mathcal{P}. \quad (\text{HEB30})$$

This is both an algebraic identity of forms and the residue identity supplied by the rational logarithmic correction in (HEB27). Indeed q'_j/q_j has residue equal to the root multiplicity at every root of q_j , so pairing these finite divisors with $M_f^\# M_g$ gives $P_0 - P_h$. The remaining zeta factor and its prime logarithmic derivative are unchanged. Thus the sign change of the endpoint receiver can be made exact while preserving the complete original zero form W , but it is accompanied by the equally exact correction (HEB29). Positivity of (HEB25) alone is not positivity of the complete form (HEB30).

For clarity, Λ_h in (HEB26) is a rationally modified completion of the original ζ . It is not the Hurwitz family $\zeta(s, 1+t)$ and it is not the de Bruijn–Newman heat evolution of ξ . Each has its own proved map and parameter. Replacing any of them by (HEB26) without its correction would change the calculation.

6. The integral coefficient before quadratic base change

Before choosing the heat clock, the integral algebra is

$$E_{\mathbb{Z}} = \mathbb{Z}[u, r]/(r^2 - u), \quad \sigma(r) = -r, \quad \sigma(u) = u. \quad (\text{HEB31})$$

The geometric substitutions are $u = -a$ and, over $\mathbb{Z}[1/2]$, $u = 1/4 - 2h$. The ramified coefficient family used for the reflected analytic pair is the exact base change $u \mapsto d^2$, $r \mapsto w$. The image has $u \geq 0$ when d is real. The original heat line instead has access to both signs of u . These are different maps from the same coefficient space; neither map is omitted.

For the integral Frobenius lift $\Phi_p(u) = u^p$, $\Phi_p(r) = r^p$, coefficient integers fixed, the relation is preserved and the reduction modulo p is Frobenius. The ring is torsionfree, so $\delta_p(x) = (\Phi_p(x) - x^p)/p$ is defined. Direct computation on the generator at $p = 2$ gives

$$(\sigma\Phi_2 - \Phi_2\sigma)(r) = 2u, \quad (\sigma\delta_2 - \delta_2\sigma)(r) = u, \quad q_u(r, r) = -2u. \quad (\text{HEB32})$$

Thus the divided arithmetic defect is the same coefficient whose sign controls the real reflected form, through explicit integral and complex realizations. In the collision quotient $u = 0$, r survives with $r^2 = 0$. There is no claim that a p -adic coefficient has an intrinsic real sign, or that quotienting by u chooses the positive real half-line. The complete quotient, conormal and prism maps are calculated in WRF; (HEB31)–(HEB32) give the exact common source needed here.

The accompanying `check_endpoint_heat_bridge.py` verifies the original three images, the inverse chart, source reflection, matrix isometry, and rational logarithmic correction using exact rational symbolic arithmetic. Its finite verification scope is recorded in `ENDPOINT_HEAT_BRIDGE_CHECKS.json`. All arguments above are supplied independently of that check.

The earlier first-order vanishing in the ramified observed family also has an exact clock map. Under $u = d^2$ and $u = 1/4 - 2h$,

$$h = \frac{1}{8} - \frac{d^2}{2}, \quad r^2 = d^2, \quad q(r, r) = -2d^2. \quad (\text{HEB32a})$$

Thus modulo d^2 the algebra is the constant dual-number algebra and this paired value vanishes, while its coefficient of d^2 is exactly -2 . The first derivative of the displayed heat clock in d is zero at zero and its second derivative is -1 . This is why first-order vanishing in d does not itself choose the positive side of the heat clock: real d maps to $h \leq 1/8$. The original real heat parameter and this quadratic pullback are both retained through the displayed polynomial map.

7. An admissible test through the vanishing infinitesimal and positive quarter

The sign comparison can be evaluated on an actual compactly supported smooth test. Let

$$\psi(v) = C \begin{cases} \exp(-1/(1-v^2)), & |v| < 1, \\ 0, & |v| \geq 1, \end{cases} \quad \int_{\mathbb{R}} \psi(v) dv = 1, \quad f = \psi'. \quad (\text{HEB33})$$

The constant C is the reciprocal of the positive finite integral of the displayed function. The function is even, nonnegative, smooth and compactly supported: each one-sided derivative at $|v| = 1$ is an exponential of a negative reciprocal times a polynomial in reciprocal powers, hence tends to zero. This last assertion follows from $t^{-m}e^{-1/t} \rightarrow 0$ for every nonnegative integer m as $t \downarrow 0$, proved by the exponential series bound $e^{1/t} \geq t^{-(m+1)}/(m+1)!$. Thus f is an admissible real odd test in (HEB11).

Define the entire function $B(t) = \int \psi(v)e^{-tv} dv$. Entirety and differentiation under the integral follow uniformly on every compact set of t from $|v| \leq 1$. Integration by parts, with zero boundary values, gives

$$M_f(s) = (s - 1/2)B(s - 1/2), \quad B(-t) = B(t), \quad B(0) = 1. \quad (\text{HEB34})$$

The first equality uses the exact Fourier convention (HEB11); the second follows by $v \mapsto -v$. In particular $M_f(1/2) = 0$ and $M'_f(1/2) = 1$. Hence its class at the collision is precisely the nonzero class r in $\mathbb{C}[r]/r^2$. Its value vanishes, its first derivative survives, and its reflected regular trace pairing is zero, by (HEB16).

For every real a , introduce the entire coefficient function

$$b(a) = \sum_{k=0}^{\infty} \frac{(-a)^k}{(2k)!} \int \psi(v)v^{2k} dv. \quad (\text{HEB35})$$

The moments have absolute value at most one, so the series converges uniformly on bounded subsets of a . The even series of B and the relation $r^2 = -a$ give exactly $[M_f] = b(a)r$ in E_a , including its double-root class at zero. Equivalently $b(a) = \int \psi(v) \cos(\sqrt{a}v) dv$ for $a \geq 0$ and $b(a) = \int \psi(v) \cosh(\sqrt{-a}v) dv$ for $a \leq 0$. The exponential series and uniform convergence on $|v| \leq 1$ prove both formulas, with no choice of complex square root needed.

Put $b_0 = b(-1/4) = \int \psi(v) \cosh(v/2) dv$ and $b_1 = b(1/4) = \int \psi(v) \cos(v/2) dv$. Both are strictly positive: the integrands are nonnegative, and $\cos(v/2) \geq \cos(1/2) > 0$ on the support. Formula (HEB16) consequently gives the exact values and the actual compensation

$$\begin{aligned} P_0(f, f) &= -\frac{1}{2}b_0^2 < 0, \\ P_{1/8}(f, f) &= 0, \\ P_{1/4}(f, f) &= \frac{1}{2}b_1^2 > 0, \\ (\mathcal{A}_{1/4} - \mathcal{A})(f, f) &= -\frac{1}{2}(b_0^2 + b_1^2) < 0. \end{aligned} \quad (\text{HEB36})$$

The fourth equality follows from (HEB29), and cancels exactly the gain $P_{1/4}(f, f) - P_0(f, f)$. All values belong to the same admissible test f , without replacing it by an arbitrary endpoint vector or a polynomial outside the test space. This proves both the actual sign crossing and its complete explicit-formula compensation. It does not assign a sign to the remaining total $W(f, f)$: that total is unchanged by (HEB30).

The zero-prime endpoint form through a nilpotent collision

23 September 2026. This note starts with the specified endpoint contribution $W_0(a, b) = 2 \operatorname{Re}(a\bar{b})$. It constructs the exact double-root family carrying that form, computes its sign on every real fibre, proves what a nilpotent thickening can retain, and constructs the positive correction by reflection. The calculation concerns this endpoint contribution. It does not identify the reflected endpoint contribution with the full corrected Weil distribution.

The original observed-family coordinates used in Section 8 are those of the complete escaping-to-observed comparison, EFI51–EFI63. Their operator/metric distinctions are proved in the original metric comparison, ISP45–ISP53, which was read for this calculation. Every finite algebra and form used below is defined and calculated here.

1. The endpoint form and its negative direction

Let $V = \mathbb{C}^2$, with coordinates $x = (a, b)$, coordinatewise multiplication, and unit $(1, 1)$. Define the antilinear algebra involution and linear trace

$$x^\# = (\bar{b}, \bar{a}), \quad \operatorname{tr}(a, b) = a + b. \quad (\text{PZS1})$$

Applying the involution twice fixes each coordinate; it preserves products because multiplication is coordinatewise. Its trace-Hermitian form is

$$W_0(x, y) = \operatorname{tr}(x^\# y) = \bar{b} a' + \bar{a} b', \quad y = (a', b'). \quad (\text{PZS2})$$

It is conjugate-linear in its first entry, linear in its second, and Hermitian by direct conjugation and reordering of the two summands. Its matrix in the coordinate basis is

$$J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad W_0(x, y) = x^* J y. \quad (\text{PZS3})$$

In particular

$$W_0(x, x) = 2 \operatorname{Re}(a\bar{b}) = \frac{1}{2}(|a + b|^2 - |a - b|^2). \quad (\text{PZS4})$$

Expansion of the absolute squares proves the last identity. The vectors $(1, 1)/\sqrt{2}$ and $(1, -1)/\sqrt{2}$ are orthonormal eigenvectors of J with eigenvalues $+1, -1$; hence the form has inertia $(1, 1, 0)$, in the order positive, negative, radical dimensions. This keeps the whole negative direction, rather than only its numerical value.

2. A flat algebra family with the required involution

For each real number u , define

$$A_u = \mathbb{C}[w]/(w^2 - u), \quad (c + dw)^\# = \bar{c} - \bar{d}w. \quad (\text{PZS5})$$

The involution is well-defined because $(w^2 - u)^\# = w^2 - u$ for real u . It is multiplicative, antilinear, and has square the identity. The complete family is

$$\mathcal{A} = \mathbb{C}[u, w]/(w^2 - u), \quad \mathbb{C}[u] \longrightarrow \mathcal{A}, \quad u \longmapsto w^2. \quad (\text{PZS6})$$

Monic division gives the free $\mathbb{C}[u]$ -basis $1, w$; therefore the family is flat of rank two, including at $u = 0$. Its fibre is the stated quotient A_u . The family has the coefficient-conjugating involution fixing u and sending w to $-w$; its real fibres inherit PZS5.

In basis $1, w$, multiplication by $c + dw$ has matrix

$$M_{c+dw} = \begin{pmatrix} c & ud \\ d & c \end{pmatrix}. \quad (\text{PZS7})$$

Its regular trace is $2c$. Thus for $p = c + dw$, $q = c' + d'w$,

$$\begin{aligned} q_u(p, q) &:= \operatorname{Tr}_{A_u/\mathbb{C}}(M_{p^\# q}) \\ &= 2\bar{c}c' - 2u\bar{d}d', \end{aligned} \quad [q_u]_{(1, w)} = \begin{pmatrix} 2 & 0 \\ 0 & -2u \end{pmatrix}. \quad (\text{PZS8})$$

To verify this, expand $(\bar{c} - \bar{d}w)(c' + d'w)$, whose constant coefficient is $\bar{c}c' - u\bar{d}d'$, and apply PZS7. The displayed matrix also proves Hermitian symmetry.

3. The original endpoints occur at the exact fibre $u = 1/4$

Evaluation at the two roots gives an algebra map

$$E : A_{1/4} \longrightarrow V, \quad p \longmapsto (p(-1/2), p(1/2)). \quad (\text{PZS9})$$

Its explicit inverse is

$$E^{-1}(a, b) = \frac{a+b}{2} + (b-a)w. \quad (\text{PZS10})$$

Substitution verifies both endpoint values. Conversely, for $p = c + dw$, these values are $(c - d/2, c + d/2)$, and PZS10 recovers c, d . This proves the algebra isomorphism, since evaluation is multiplicative.

One has the commuting identities

$$E(p^\#) = E(p)^\#, \quad \text{Tr}_{A_{1/4}}(M_p) = \text{tr}(E(p)), \quad q_{1/4}(p, q) = W_0(E(p), E(q)). \quad (\text{PZS11})$$

The first follows because negating the real roots interchanges them; the second follows by summing $c - d/2$ and $c + d/2$; the third follows from the first two and multiplicativity. It also follows directly by substituting PZS10 into PZS8:

$$q_{1/4}(E^{-1}x, E^{-1}x) = \frac{1}{2}|a+b|^2 - \frac{1}{2}|b-a|^2 = W_0(x, x). \quad (\text{PZS12})$$

Thus the constant, variable, involution, and endpoint ordering are all fixed by exact maps.

4. The sign changes through the actual collision

The matrix PZS8 gives the full list

$$\text{inertia}(q_u) = \begin{cases} (1, 1, 0), & u > 0, \\ (1, 0, 1), & u = 0, \\ (2, 0, 0), & u < 0. \end{cases} \quad (\text{PZS13})$$

For $u \neq 0$, the two complex roots of $w^2 - u$ are distinct, each with multiplicity one. For $u = 0$, the fibre is

$$A_0 = \mathbb{C}[w]/w^2, \quad w \neq 0, \quad \text{rad } q_0 = \mathbb{C}w. \quad (\text{PZS14})$$

Here $w \neq 0$ follows from the free basis, and the radical follows from the diagonal matrix. The collision has length two; no multiplicity is removed when its two points merge.

For $u > 0$, put $r = \sqrt{u} > 0$. Evaluation at $-r, r$ identifies A_u with two coordinate factors, and the involution interchanges their conjugates. Its trace form is therefore the same hyperbolic endpoint form as PZS2. For $u < 0$, put $r = \sqrt{-u} > 0$. Evaluation at $-ir, ir$ instead has

$$(p^\#)(\pm ir) = \overline{p(\pm ir)}, \quad q_u(p, p) = |p(-ir)|^2 + |p(ir)|^2. \quad (\text{PZS15})$$

Indeed $(p^\#)(z) = \overline{p(-\bar{z})}$, and each imaginary root satisfies $-\bar{z} = z$. The sum of the values still equals the regular trace. Evaluation is invertible because the roots are distinct. This proves directly why the involution and sign change together while the total length remains two.

There is an explicit continuing negative vector: the polynomial $p = -2w$ corresponds at $u = 1/4$ to $(a, b) = (1, -1)$, and

$$q_u(-2w, -2w) = -8u. \quad (\text{PZS16})$$

It is negative for $u > 0$, zero for $u = 0$, and positive for $u < 0$. The fixed free module $\mathbb{C}[u] \oplus \mathbb{C}[u]w$ supplies this precise continuation through the collision. A sign change is therefore possible in the real family. The nilpotent fibre itself makes this direction radical; positivity of that direction occurs after continuing to the other side.

The two signs are not connected by an involution-preserving isomorphism of the endpoint algebras. For $u > 0$, the two nontrivial primitive idempotents $(1 \pm w/\sqrt{u})/2$ are exchanged by $\#$. For $u < 0$, the two primitive idempotents associated with the imaginary roots are each fixed by $\#$, because evaluation has componentwise conjugation by PZS15. An algebra isomorphism permutes the two primitive idempotents, so cannot change an exchanged pair into a fixed pair while preserving the involution. There are nevertheless explicit algebra isomorphisms: for $u > 0, v < 0$,

$$A_u \longrightarrow A_v, \quad w \longmapsto i\sqrt{u/(-v)}w', \quad (\text{PZS17})$$

with inverse multiplication by the reciprocal coefficient. Its square is u because $(w')^2 = v$. Its precise involution defect on w is

$$\phi(w^\#) - \phi(w)^\# = -2i\sqrt{u/(-v)}w' \neq 0. \quad (\text{PZS18})$$

This proves both the relationship and the information not preserved by that isomorphism.

5. A nilpotent thickening of the original endpoints keeps their negative residue

Let R be a commutative real algebra with a nilpotent ideal I and a fixed isomorphism $R/I \cong \mathbb{R}$. Extend conjugation to $R_{\mathbb{C}} = R \otimes_{\mathbb{R}} \mathbb{C}$ by fixing R . Let $\tilde{V} = R_{\mathbb{C}}^2$ and let $\tilde{G}$ be any Hermitian matrix whose reduction modulo I is J of PZS3. This includes every matrix deformation of the original endpoint pairing over this nilpotent base.

For every lift $\tilde{x}$ of $(1, -1)$, reduction gives

$$\tilde{x}^* \tilde{G} \tilde{x} \pmod{I} = -2. \quad (\text{PZS19})$$

This follows by applying the quotient homomorphism to the entire matrix product. On the other hand every element of the Hermitian-square cone

$$\Sigma_R = \left\{ \sum_{j=1}^n z_j^* z_j : n \geq 0, z_j \in R_{\mathbb{C}} \right\} \subset R \quad (\text{PZS20})$$

has nonnegative residue, because its residue is $\sum_j |\pi(z_j)|^2$, where $\pi : R_{\mathbb{C}} \rightarrow \mathbb{C}$ is the quotient homomorphism. Therefore the value PZS19 is outside Σ_R , for every possible nilpotent correction. No deformation of the pairing reducing to the original endpoint form can make all its values Hermitian-square-positive.

This result also has a scalar receiver: compose the deformed pairing with $R \rightarrow \mathbb{R}$. The resulting scalar form is exactly W_0 and retains value -2 on every such lift. Thus the negative direction cannot be removed by adding terms that vanish under this residue map. Crossing the family from $u = 1/4$ to $u < 0$ in Section 4 changes the residue parameter; it is not a nilpotent thickening with residue $u = 1/4$.

6. Positive functionals at the collision and their exact radical

On the involutive algebra A_0 , call a complex-linear functional ℓ positive when $\ell(p^\# p)$ is a nonnegative real number for every p . Then all such functionals are exactly

$$\ell(c + dw) = tc, \quad t \in \mathbb{R}_{\geq 0}. \quad (\text{PZS21})$$

Here is a direct proof. Positivity at $p = 1$ gives $t = \ell(1) \geq 0$. For $p = 1 + dw$, the product is $p^\# p = 1 + (d - \bar{d})w$. With $d = iy$, $y \in \mathbb{R}$, positivity requires $t + 2iy\ell(w)$ to be real and nonnegative for every real y . Its slope must be real because the whole expression is real, and must be zero because otherwise it takes negative values for one sign of sufficiently large y . Hence $\ell(w) = 0$. Linearity proves the asserted form. Conversely $\ell(p^\# p) = t|c|^2 \geq 0$, proving sufficiency.

For $t > 0$, the induced form $\ell(p^\# q)$ has radical exactly $\mathbb{C}w$; for $t = 0$, its radical is all of A_0 . Thus every positive functional kills the nonzero infinitesimal direction, with the explicit augmentation map

$$\epsilon : A_0 \rightarrow \mathbb{C}, \quad c + dw \mapsto c, \quad \ker \epsilon = \mathbb{C}w, \quad \ell = t\epsilon. \quad (\text{PZS22})$$

The scalar section $c \mapsto c$ splits this map and leaves its square-zero kernel specified.

More generally, let B be a positive semidefinite Hermitian form on the regular vector space of A_0 , compatible with the specified multiplication adjoints:

$$B(M_a x, y) = B(x, M_{a^\#} y). \quad (\text{PZS23})$$

Then $B(w, w) = B(1, -w^2) = 0$. A zero-norm vector of a positive semidefinite form lies in its radical: expanding $B(x + tz, x + tz) \geq 0$ with $B(z, z) = 0$ and varying complex t forces $B(x, z) = 0$. Apply this to $z = w$. Thus compatible positivity cannot give a positive norm to the nilpotent regular generator. A positive norm on that generator requires a changed adjoint relation, whose concrete original-observation example is ISP51–ISP52.

7. An explicit positive correction on the fixed endpoint space

The linear involution $J(a, b) = (b, a)$ is also an algebra automorphism of the endpoint algebra. Define

$$W_+(x, y) = W_0(x, Jy). \quad (\text{PZS24})$$

Using $J^2 = I$ in PZS3 gives

$$W_+(x, y) = x^* y, \quad W_+(x, x) = |a|^2 + |b|^2. \quad (\text{PZS25})$$

Thus it is positive definite. Its exact difference from the original form is the positive rank-one form

$$W_+(x, y) - W_0(x, y) = \overline{(a - b)}(a' - b'), \quad W_+(x, x) - W_0(x, x) = |a - b|^2. \quad (\text{PZS26})$$

Expansion proves these formulas. The correction has kernel $\{(a, a) : a \in \mathbb{C}\}$ and takes value 4 on $(1, -1)$, changing the old value -2 to $+2$. This is an explicit way to flip the negative direction while retaining its removed contribution as a known form.

The same change can be expressed as a changed algebra involution. Put

$$x^\star = J(x^\#) = (\bar{a}, \bar{b}). \quad (\text{PZS27})$$

Then $\text{tr}(x^\star y) = W_+(x, y)$. The original involution $\#$ and the new $\star$ are related by the exact automorphism J , but are not equal. On $A_{1/4}$, the automorphism is the sheet map $w \mapsto -w$. Under PZS27 the generator satisfies $w^\star = w$; under the original involution it satisfies $w^\# = -w$. This records the data altered by the correction.

The endpoint multiplication operator corresponding to the coordinate s is

$$S = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \quad (\text{PZS28})$$

Its adjoint for W_0 is

$$S^{\dagger 0} = JS^\star J = I - S, \quad W_0(Sx, y) = W_0(x, (I - S)y), \quad (\text{PZS29})$$

whereas its adjoint for W_+ is $S^{\dagger +} = S$. These follow by the displayed matrix products. Also $JSJ = I - S$, proving the exact operator intertwining with the sheet switch.

Consequently the centered Fourier coordinate $Y = (S - I/2)/i$ is self-adjoint for W_0 , but skew-adjoint for W_+ :

$$Y^{\dagger 0} = Y, \quad Y^{\dagger +} = -Y. \quad (\text{PZS30})$$

The sign in the first assertion follows because adjoints conjugate $1/i$ and replace $S - I/2$ by its negative. The second uses the ordinary adjoint of the real diagonal matrix $S - I/2$. Thus positivity of the corrected form does not retain the old adjoint identity for this operator.

Let $\mathcal{P} = \mathbb{C}[s]$, let $\text{ev}_{0,1}(f) = (f(0), f(1))$, and define the reflection automorphism $(Tf)(s) = f(1 - s)$. Then

$$\text{ev}_{0,1} T = J \text{ev}_{0,1}, \quad T^2 = I, \quad TM_s = (I - M_s)T. \quad (\text{PZS31})$$

Each identity follows by evaluation or direct substitution. Therefore

$$W_0(\text{ev}_{0,1} f, \text{ev}_{0,1}(Tg)) = \overline{f(0)}g(0) + \overline{f(1)}g(1). \quad (\text{PZS32})$$

This proves the positive correction as an exact reflected-test map for the endpoint contribution alone. Applying T to a test in a full analytic explicit formula also changes its other terms; PZS32 does not assert their invariance or positivity.

8. The observed quadratic pullback selects a side and an involution

Retain the observed family

$$B_\lambda = \mathbb{C}[z]/(z^2 - \lambda z), \quad \lambda = a_{\text{obs}}(s - s_*), \quad a_{\text{obs}} > 0. \quad (\text{PZS33})$$

For a real displacement $s - s_*$, $\lambda \in \mathbb{R}$. The exact coordinate map is

$$w = z - \lambda/2, \quad u = \lambda^2/4, \quad w^2 - u = z^2 - \lambda z. \quad (\text{PZS34})$$

It identifies B_λ with $A_{\lambda^2/4}$. The inverse is $z = w + \lambda/2$. Under the endpoint involution $w^\# = -w$, one has

$$z^\# = \lambda - z, \quad [q_\lambda^-]_{(1,w)} = \begin{pmatrix} 2 & 0 \\ 0 & -\lambda^2/2 \end{pmatrix}, \quad [q_\lambda^-]_{(1,z)} = \begin{pmatrix} 2 & \lambda \\ \lambda & 0 \end{pmatrix}. \quad (\text{PZS35})$$

For the second matrix, trace of z is λ and $z^\# z = (\lambda - z)z = 0$. Its determinant is $-\lambda^2$. Thus along this original real observed path, the endpoint form is indefinite for every $\lambda \neq 0$, and becomes radical at $\lambda = 0$. The base map $u = \lambda^2/4$ stays on $u \geq 0$ and cannot reach the positive side $u < 0$ in PZS13.

The positive escaping trace previously calculated in EFI61–EFI62 and ISP45 instead uses the coefficient-conjugating involution $w^\star = w$, equivalently $z^\star = z$. With that involution its matrices are

$$[q_\lambda^+]_{(1,w)} = \begin{pmatrix} 2 & 0 \\ 0 & \lambda^2/2 \end{pmatrix}, \quad [q_\lambda^+]_{(1,z)} = \begin{pmatrix} 2 & \lambda \\ \lambda & \lambda^2 \end{pmatrix}. \quad (\text{PZS36})$$

This is the same algebra and the same regular trace, with the different stated involution. Their connecting automorphism is

$$\sigma(w) = -w, \quad \sigma(z) = \lambda - z, \quad \star = \sigma \circ \#, \quad q_\lambda^+(p, q) = q_\lambda^-(p, \sigma q). \quad (\text{PZS37})$$

To prove the form identity, regular trace is invariant under σ : conjugation by an algebra automorphism sends a multiplication matrix to a similar matrix. Hence $\text{Tr}(p^\# \sigma q) = \text{Tr}(\sigma(p^\#) q) = \text{Tr}(p^\star q)$. One can also verify it using

$$[\sigma]_{(1,z)} = \begin{pmatrix} 1 & \lambda \\ 0 & -1 \end{pmatrix}, \quad [q_\lambda^+] - [q_\lambda^-] = \begin{pmatrix} 0 & 0 \\ 0 & \lambda^2 \end{pmatrix}. \quad (\text{PZS38})$$

Thus the positive escaping trace has an exact relationship to the original negative endpoint form: it performs the fundamental-symmetry correction on the same fibres. This is not evidence that the original endpoint involution already had the positive trace.

At $\lambda = 0$, both trace forms have matrix $\text{diag}(2, 0)$, although their involutions still differ on the nonzero nilpotent generator. For $\lambda \neq 0$, their signs differ exactly as PZS35–PZS36 calculate. This specifies the coincidence at the collision and the data that remain distinct there.

There is also a real obstruction to reaching the positive side through this fixed observed base map, valid over any real nilpotent parameter algebra with residue field $\mathbb{R}$: the residue of $\lambda^2/4$ is nonnegative. To access $u < 0$, one must change the real base path, or change the involution as in PZS37. Both constructions are explicitly available above, and neither is identified with a nilpotent correction leaving the original endpoint form unchanged.

9. What the calculation establishes for the proposed sign flip

The free family PZS6 and its original involution provide an actual passage from a negative direction, through a nonzero trace-null dual-number direction, to a positive direction, as PZS13–PZS16 prove. The original observed quadratic base map restricts this family to its nonnegative- u side. Its previously positive trace uses the alternative involution, connected by PZS37. On the fixed endpoint space, PZS24–PZS32 construct the positive correction and identify exactly its rank-one contribution and altered multiplication adjoint. A nilpotent thickening that reduces to the original endpoint pairing retains its negative residue by PZS19; at the collision, every positive functional has the explicit radical and augmentation in PZS21–PZS23.

These are complete calculations of the proposed endpoint mechanism. They retain the nilpotent, the full multiplicity, the sign-changing continuation, and the correction maps. The analytic effect of applying the same reflection or continuation to every other term of the corrected Weil formula is a separate calculation on those actual terms; no endpoint-only identity above asserts that effect.

10. Exact verification

The reproducible script `check_prime_zero_sign_deformation.py` passed 16 exact checks, recorded in `PRIME_ZERO_SIGN_DEFORMATION_CHECKS.json`. They verify the endpoint interpolation and its inverse, the involution and trace isometries, the continuing negative vector, the rank-one fundamental-symmetry correction, both endpoint multiplication adjoints, the two observed-coordinate trace forms and their sheet-switch relation, and preservation of a negative residue under a specified matrix deformation with nilpotent parameters. These checks use symbolic arithmetic. They support the calculations but do not replace the proofs for arbitrary parameters and nilpotent bases, and do not assert positivity of the full Weil formula.

The ES quarter spectrum, its tetrahedral metric, and the original heat trace

This note starts with the actual four raw ES cells and their quarter-inversion operator. It constructs a form-preserving linear isomorphism with two positive scalar directions and the reflected trace of the original quadratic heat family. It proves the transported symmetry actions, the precise multiplication maps, and the surviving collision algebra. The ES form is not presumed to be the complete Weil form.

The original source read for this calculation is `ES-Fable-C123/es_fable_c123_preprint.tex`: the definition `eq:Hermitian-circle-coefficients`; Corollaries `cor:raw-blade-cells-quarter-three-plus-one` and the following raw-cell projector corollary, including `eq:four-cell-explicit-quarter-images`, `eq:raw-negative-cell-projector`, `eq:raw-positive-cell-projector`; Theorem `thm:Fable-D3-raw-cell-action`; Corollary `cor:Fable-three-character-tetrahedron`; Theorem `thm:Fable-forced-regular-four-simplex`; and `eq:quarter-inversion-matrix` through `eq:quarter-negative-eigenline`. These are actual passages of the source, rather than identifications inferred from a four-element count. The receiving programme proof read in full is the complete heat endpoint sign bridge proof, HEB1–HEB32. The original heat equations also have the public proof *The original heat trajectories*, `eq:ifh-target-flow` and the ensuing trajectories, while EFI7–EFI15 give their projective graph and collision algebra. All finite-dimensional claims made below have their proofs here.

1. The source operator and its complete intertwiner

Let

$$V = M_2(\mathbb{R}), \quad x = x_{11}E_{11} + x_{12}E_{12} + x_{21}E_{21} + x_{22}E_{22}, \quad (\text{ESQ1})$$

with the displayed ordering of the four basis vectors. We retain both the vector space and its original matrix multiplication. The symbol $I_2 = E_{11} + E_{22}$ denotes the matrix identity, whereas I_V denotes the identity linear operator on the four-dimensional vector space.

The quarter-inversion operator pulled back from the ES source is

$$A_0 = \text{diag}(1/4, 1/4, -1/4, 1/4). \quad (\text{ESQ2})$$

Its two invariant spaces and its negative projector are

$$U_+ = \text{span}_{\mathbb{R}}\{E_{11}, E_{12}, E_{22}\}, \quad U_- = \mathbb{R}E_{21}, \quad P_-x = x_{21}E_{21} = \left(\frac{1}{2}I_V - 2A_0\right)x. \quad (\text{ESQ3})$$

These formulas follow by evaluating the diagonal operator on all four basis vectors. In particular $P_-^2 = P_-$, its kernel is U_+ , and its image is U_- .

Here is the full source intertwiner, including its inverse. In Hermitian coefficient coordinates write

$$H(\alpha, \beta_0, \beta_1, \gamma) = \begin{pmatrix} \alpha & \beta_0 + i\beta_1 \\ \beta_0 - i\beta_1 & \gamma \end{pmatrix}. \quad (\text{ESQ4})$$

The coefficient vector of $\mathcal{T}_{\square}x$ is

$$\begin{aligned} \alpha &= x_{11} + 2x_{21} + x_{22}, & \beta_0 &= (x_{11} - x_{22})/2, \\ \beta_1 &= x_{12}, & \gamma &= (x_{11} + x_{22})/4 - x_{21}/2. \end{aligned} \quad (\text{ESQ5})$$

This is exactly the linear extension of the four source images. Its inverse is

$$x_{11} = \alpha/4 + \gamma + \beta_0, \quad x_{22} = \alpha/4 + \gamma - \beta_0, \quad x_{12} = \beta_1, \quad x_{21} = \alpha/4 - \gamma. \quad (\text{ESQ6})$$

Substitution in either direction returns all four starting coefficients, proving that it is an isomorphism of real vector spaces. The source coefficient inversion is

$$\mathcal{I}_- = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1/4 & 0 & 0 \\ 0 & 0 & 1/4 & 0 \\ 1/16 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{ESQ7})$$

Applying this matrix to the four columns of (ESQ5) multiplies the E_{21} column by $-1/4$ and each other column by $1/4$. Since the four columns form a basis, this proves the entire operator identity

$$\mathcal{T}_{\square}A_0 = \mathcal{I}_-\mathcal{T}_{\square}. \quad (\text{ESQ8})$$

The negative line is therefore the specific raw E_{21} line, not an unspecified member of a quartet. In the source channel order $(++)$, $(+-)$, $(-+)$, $(--)$, it is the $(-, +)$ channel. The source specializations $\mathcal{M}(53) = 2E_{21}$ and $\mathcal{M}(109) = E_{12}$ retain their different eigenspaces under (ESQ8).

2. The tetrahedral metric and the rank-one heat motion

Retain the source's forced metric

$$G = \hat{G}_F = \begin{pmatrix} 1/16 & 0 & 0 & -7/144 \\ 0 & 2/9 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -7/144 & 0 & 0 & 1/16 \end{pmatrix}. \quad (\text{ESQ9})$$

On U_+ , the sum and difference of E_{11}, E_{22} have respective eigenvalues $1/72, 1/9$, and E_{12} has eigenvalue $2/9$. The orthogonal fourth direction E_{21} has eigenvalue 1. These four positive numbers prove that G is positive definite. Its tetrahedral origin is proved again in Section 7.

Use the original forward heat parameter with its original constants,

$$a(h) = -\frac{1}{4} + 2h, \quad h \in \mathbb{R}. \quad (\text{ESQ10})$$

Define the actual deformation of the source operator by

$$A_h = A_0 + 2hP_- = \frac{1}{4}(I_V - P_-) + a(h)P_-. \quad (\text{ESQ11})$$

It is G -self-adjoint: the decomposition $U_+ \oplus U_-$ is G -orthogonal, and the operator is scalar on each summand. Equivalently $A_h^\top G = GA_h$, as is also immediate from the displayed matrices. Consequently

$$B_h(x, y) = x^\top GA_h y \quad (\text{ESQ12})$$

is a symmetric bilinear form. Its full matrix and quadratic expression are

$$GA_h = \begin{pmatrix} 1/64 & 0 & 0 & -7/576 \\ 0 & 1/18 & 0 & 0 \\ 0 & 0 & a(h) & 0 \\ -7/576 & 0 & 0 & 1/64 \end{pmatrix}, \quad (\text{ESQ13})$$

$$B_h(x, x) = \frac{x_{11}^2 + x_{22}^2}{64} - \frac{7}{288}x_{11}x_{22} + \frac{x_{12}^2}{18} + a(h)x_{21}^2. \quad (\text{ESQ14})$$

Matrix multiplication proves every coefficient. The unchanged positive eigenvalues of this matrix are $1/288, 1/36, 1/18$, and its remaining eigenvalue is $a(h)$. Thus it has signature $(3, 1)$ for $h < 1/8$, has three positive directions and radical $\mathbb{R}E_{21}$ for $h = 1/8$, and is positive definite for $h > 1/8$. At $h = 1/4$, $A_h = I_V/4$ and $B_h = G/4$. The exact difference formula is

$$B_{h_2}(x, y) - B_{h_1}(x, y) = 2(h_2 - h_1)x_{21}y_{21}. \quad (\text{ESQ15})$$

This follows because $GP_- = P_-$. It proves that the entire changing part has rank one and is increasing as a quadratic form when heat time increases.

For the two marked source specializations this gives the exact values

$$B_h(\mathcal{M}(53), \mathcal{M}(53)) = 4a(h) = -1 + 8h, \quad B_h(\mathcal{M}(109), \mathcal{M}(109)) = 1/18. \quad (\text{ESQ15a})$$

They follow from $\mathcal{M}(53) = 2E_{21}$, $\mathcal{M}(109) = E_{12}$, and (ESQ13), retaining the factor two on the first source vector.

The original intertwiner also transports the entire heat family, not only its initial value. Define

$$\mathcal{I}_h = \mathcal{I}_- + 2h \left(\frac{1}{2}I - 2\mathcal{I}_- \right) = hI + (1 - 4h)\mathcal{I}_-. \quad (\text{ESQ16})$$

Substitution of (ESQ3) in (ESQ8) proves $\mathcal{T}_\square A_h = \mathcal{I}_h \mathcal{T}_\square$. Transport the metric by $G_T = (\mathcal{T}_\square^{-1})^\top G \mathcal{T}_\square^{-1}$. In the coordinates of (ESQ4) its quadratic expression is

$$G_T(y, y) = \frac{(\alpha + 4\gamma)^2}{576} + \frac{2}{9}(\beta_0^2 + \beta_1^2) + (\alpha/4 - \gamma)^2. \quad (\text{ESQ17})$$

This is obtained by inserting (ESQ6) in (ESQ9). It proves the metric transport exactly; no identification with the Lorentz form $-\det H$ is being substituted for it. Likewise the transported heat form is

$$B_{T,h}(y, y) = \frac{(\alpha + 4\gamma)^2}{2304} + \frac{\beta_0^2 + \beta_1^2}{18} + a(h)(\alpha/4 - \gamma)^2. \quad (\text{ESQ18})$$

3. The exact trace receiver and the form-preserving isomorphism

For real a , let

$$E_a = \mathbb{R}[r]/(r^2 + a). \quad (\text{ESQ19})$$

Every element has a unique expression $c + dr$, by division by the monic quadratic. The real algebra involution is $(c + dr)^\# = c - dr$. It preserves the relation and squares to the identity. The regular trace is the trace of multiplication on the basis $(1, r)$. Since

$$L_r = \begin{pmatrix} 0 & -a \\ 1 & 0 \end{pmatrix}, \quad \text{Tr}_{E_a}(c + dr) = 2c, \quad (\text{ESQ20})$$

the full reflected trace pairing is

$$\frac{1}{2} \text{Tr}_{E_a}((c + dr)^\#(e + fr)) = ce + adf. \quad (\text{ESQ21})$$

Indeed the product has scalar coefficient $ce + adf$, and (ESQ20) gives twice it. At complex coefficients the involution also conjugates c, d , and the same proof gives $\bar{c}e + a\bar{d}f$.

Define the real vector space

$$W_a = \mathbb{R}^2 \oplus E_a \quad (\text{ESQ22})$$

with symmetric form

$$Q_a((z_1, z_2, p), (t_1, t_2, q)) = z_1 t_1 + z_2 t_2 + \frac{1}{2} \text{Tr}_{E_a}(p^\# q). \quad (\text{ESQ23})$$

The map proposed for the raw ES cells is exactly

$$\begin{aligned} \Theta_h : V &\longrightarrow W_{a(h)}, \\ z_1 &= \frac{x_{11} + x_{22}}{24}, \quad z_2 = \frac{x_{11} - x_{22}}{6\sqrt{2}}, \\ p &= c + dr = \frac{x_{12}}{3\sqrt{2}} + x_{21}r. \end{aligned} \quad (\text{ESQ24})$$

It is a vector-space isomorphism for every h , including the collision. Its inverse is

$$x_{11} = 12z_1 + 3\sqrt{2}z_2, \quad x_{22} = 12z_1 - 3\sqrt{2}z_2, \quad x_{12} = 3\sqrt{2}c, \quad x_{21} = d. \quad (\text{ESQ25})$$

These formulas are mutually inverse by substitution, without dividing by a . They prove in particular that no vector is dropped when $a = 0$.

The exact isometry identity is

$$B_h(x, y) = Q_{a(h)}(\Theta_h x, \Theta_h y). \quad (\text{ESQ26})$$

For the diagonal terms, (ESQ21) gives $c^2 + ad^2 = x_{12}^2/18 + ax_{21}^2$. The remaining two squares are

$$\frac{(x_{11} + x_{22})^2}{576} + \frac{(x_{11} - x_{22})^2}{72} = \frac{x_{11}^2 + x_{22}^2}{64} - \frac{7}{288} x_{11} x_{22}. \quad (\text{ESQ27})$$

Expansion proves the equality with (ESQ14). For two arguments the identical expansion with products proves (ESQ26), or it follows by real polarization. Complex-linear extension gives the corresponding Hermitian identity, with all squared terms replaced by absolute squares and first coefficients conjugated.

The scalar factor one half in (ESQ23) is essential: HEB16 uses the full regular trace. More precisely, if the whole HEB rank-three form is

$$q_a^{\text{full}}(v, p) = |v|^2 + \text{Tr}_{E_a}(p^\# p),$$

then the explicit map

$$(z_1, z_2, p) \longmapsto (z_1, (z_2, p/\sqrt{2})) \quad (\text{ESQ28})$$

is an isometry from Q_a to one additional positive scalar form plus q_a^{full} . Its inverse multiplies the last coordinate by $\sqrt{2}$. This gives the exact relation with HEB's whole finite receiver while retaining the additional positive dimension.

The raw involution $4A_0 = I_V - 2P_-$ is also intertwined:

$$\Theta_h(4A_0 x) = (z_1, z_2, p^\#) \quad \text{over real coefficients.} \quad (\text{ESQ29})$$

Over complex coefficients the matching conjugate-linear raw involution is $(I_V - 2P_-)\bar{x}$. Each formula follows because exactly x_{21} changes sign, with coefficient conjugation added in the latter case. The heat operator itself becomes the diagonal map

$$(z_1, z_2, c, d) \mapsto (z_1/4, z_2/4, c/4, a(h)d). \quad (\text{ESQ30})$$

Under the same coordinates the fixed positive metric is $4|z_1|^2 + 4|z_2|^2 + 4|c|^2 + |d|^2$. This is obtained from G by (ESQ25), and together with (ESQ30) reproduces Q_a .

The trace-factor projection and its kernel are completely explicit:

$$0 \longrightarrow \text{span}\{E_{11}, E_{22}\} \longrightarrow V \xrightarrow{\pi_h} E_{a(h)} \longrightarrow 0, \quad \pi_h(x) = x_{12}/(3\sqrt{2}) + x_{21}r. \quad (\text{ESQ31})$$

It is onto with linear section $c + dr \mapsto 3\sqrt{2}cE_{12} + dE_{21}$, and its kernel follows from the independence of $1, r$. The kernel is orthogonal to that section for B_h , and carries the two positive squares (ESQ27). All assertions hold at the collision as well.

4. The endpoint algebra and the changing spectral pair

The exact all-parameter polynomial map to the HEB endpoint family is

$$\mathbb{R}[s]/(s(s-1)+2h) \xrightarrow{\sim} E_{a(h)}, \quad s \mapsto \frac{1}{2} + r, \quad \text{inverse: } r \mapsto s - \frac{1}{2}. \quad (\text{ESQ32})$$

The two relations correspond because $(s-1/2)^2 + a(h) = s(s-1) + 2h$. Substitution proves that the displayed maps are inverse unital algebra maps. The involution corresponds to $s^\# = 1-s$ over the reals, with conjugate coefficients in the complex version. Thus this is the actual HEB polynomial receiver, with the original heat speed 2 retained.

At $h = 0$, evaluation of $p = c + dr$ at $r = -1/2, +1/2$ gives

$$A = c - d/2, \quad B = c + d/2, \quad c = (A+B)/2, \quad d = B-A. \quad (\text{ESQ33})$$

The inverse formulas prove the algebra identification with the two endpoints. For two inputs the half-trace term becomes

$$\frac{1}{2} \text{Tr}(p^\# q) = \frac{1}{2} (\bar{A}_p B_q + \bar{B}_p A_q). \quad (\text{ESQ34})$$

To check the constants directly, substitute (ESQ33) into $\bar{c}e - \bar{d}f/4$ from (ESQ21); the diagonal endpoint terms cancel and the two cross terms have coefficient one half. Therefore the ES form is the two positive squares plus **one half** the HEB endpoint pairing, through the explicit invertible coordinate map, not through an assumed equality of names.

For all heat times the complex roots are

$$s_\pm(h) = \frac{1}{2} \pm \sqrt{\frac{1}{4} - 2h}. \quad (\text{ESQ35})$$

For $h > 1/8$, writing $t = \sqrt{2h - 1/4}$ gives $s_\pm = 1/2 \pm it$. The two evaluations of p are $c \pm itd$, and their squared absolute values have sum $2|c|^2 + 2a(h)|d|^2$. Thus the half-trace is half the sum of these two squares, exactly as in HEB25. At $h = 1/8$, evaluation alone sees the scalar coefficient c , while the quotient algebra still retains the independent derivative coefficient d . The relation is $r^2 = 0$, not $r = 0$.

This algebra and trace comparison does not replace the full Weil identity. HEB28–HEB30 retain its other terms when the endpoint factor is moved. No positivity claim for those other terms, or for all zeta zeros, is implied by (ESQ26) or (ESQ35).

5. Matrix multiplication and the exact collision algebra

There is no nonzero multiplicative ring map from $M_2(\mathbb{R})$ into a commutative ring. To prove this without invoking simplicity, let f be such a map. Commutativity of the target and the products $E_{12}E_{21} = E_{11}$, $E_{21}E_{12} = E_{22}$ give $f(E_{11}) = f(E_{22}) = e$. Since $E_{11}^2 = E_{11}$, one has $e^2 = e$; since $E_{11}E_{22} = 0$, one also has $e^2 = 0$. Therefore $e = 0$, and $f(I_2) = 0$. For every x , multiplicativity gives $f(x) = f(I_2x) = 0$. This proves the assertion and, in particular, excludes a unital algebra map from the full raw matrix algebra to $\mathbb{R} \times \mathbb{R} \times E_a$. Consequently (ESQ24), (ESQ28), and (ESQ31) have their stated linear meanings; their nonzero values could not obey that multiplication law.

There is nevertheless an exact algebra map in the opposite direction for the whole heat family:

$$\rho_a : E_a \hookrightarrow M_2(\mathbb{R}), \quad \rho_a(c + dr) = \begin{pmatrix} c & -ad \\ d & c \end{pmatrix} = cI_2 + dK_a, \quad K_a = E_{21} - aE_{12}. \quad (\text{ESQ36})$$

Since $K_a^2 = -aI_2$, the map respects the relation. Products of two displayed matrices have diagonal coefficient $ce - adf$ and lower-left coefficient $cf + de$, precisely the coefficients of $(c + dr)(e + fr)$. It is unital, and its diagonal and lower-left entries recover c, d , proving injectivity. This construction is already defined over $\mathbb{Z}[a]$; it uses no square roots and no division by the parameter.

Put $J = \text{diag}(1, -1)$. Direct multiplication gives

$$JK_a J^{-1} = -K_a, \quad \rho_a(p^\#) = \overline{J\rho_a(p)} J^{-1} \quad (\text{ESQ37})$$

in the complex version, with the bar omitted over the reals. The ordinary matrix trace of $\rho_a(p)$ is $2c$, hence exactly the regular trace (ESQ20). It follows, using the proved multiplicativity, that

$$\frac{1}{2} \text{tr} \left(\overline{J\rho_a(p)} J^{-1} \rho_a(q) \right) = \frac{1}{2} \text{Tr}_{E_a}(p^\# q). \quad (\text{ESQ38})$$

Thus the multiplication and reflected trace have an exact matrix realization, even though the linear isometry of Section 3 is not an algebra map.

The algebra involution in (ESQ37) is different from the raw linear reflection $\Omega_\square = 4A_0$, which changes the sign of E_{21} alone. The latter is not multiplicative on the entire matrix algebra: $\Omega_\square(E_{12}E_{21}) = E_{11}$, whereas $\Omega_\square(E_{12})\Omega_\square(E_{21}) = -E_{11}$. Their exact defect on the algebra embedding is

$$\Omega_\square \rho_a(c + dr) - \rho_a((c + dr)^\#) = -2adE_{12} \quad \text{over real coefficients.} \quad (\text{ESQ38a})$$

Indeed the left upper-right entry is $-ad$, and the subtracted entry is $+ad$; every other entry cancels. At complex coefficients, apply coefficient conjugation before $\Omega_\square$, and the corresponding right side is $-2a\bar{d}E_{12}$. At the collision this entire defect vanishes, so the raw reflection restricts there to the actual algebra involution of the collision corner. Away from the collision the reflected trace remains exactly represented by (ESQ37)–(ESQ38), with the precise difference from the raw reflection given here.

Its relation with the original ES metric is also calculable. Substituting $x_{11} = x_{22} = c$, $x_{12} = -ad$, $x_{21} = d$ into (ESQ14), and polarizing, gives

$$B_h(\rho_a(c + dr), \rho_a(e + fr)) = \frac{ce}{144} + \left(a + \frac{a^2}{18} \right) df, \quad a = a(h). \quad (\text{ESQ39})$$

The complex formula conjugates the first coefficients. This retains the exact difference between the algebra embedding and the form-preserving linear map; it does not identify their images without calculating the coefficients.

At the collision $a = 0$, $K_0 = E_{21}$, so (ESQ36) identifies the image with the actual unital matrix subalgebra

$$\mathbb{R}[I_2, E_{21}] = \{cI_2 + dE_{21} : c, d \in \mathbb{R}\} \xrightarrow{\sim} \mathbb{R}[r]/(r^2), \quad I_2 \mapsto 1, \quad E_{21} \mapsto r. \quad (\text{ESQ40})$$

Its inverse is ρ_0 . The multiplication law on both sides is $(c, d)(e, f) = (ce, cf + de)$; this proves every part of the algebra isomorphism and proves that the nilpotent remains nonzero. Complexification gives exactly $\mathbb{C}[I_2, E_{21}] \simeq \mathbb{C}[r]/(r^2)$. On this corner the ES collision form is $\bar{c}e/144$, whereas the half-trace form under (ESQ40) is $\bar{c}e$. The difference is the explicit scalar factor $1/144$. The global linear isometry (ESQ24) instead sends $cI_2 + dE_{21}$ to $(c/12, 0, dr)$, which confirms that its unit and nilpotent coordinates are distributed differently.

6. The native tetrahedral action and its trace-coordinate transport

The actual two ES generators are

$$R = \begin{pmatrix} 1/4 & \sqrt{3}/2 & 0 & 3/4 \\ -\sqrt{3}/4 & -1/2 & 0 & \sqrt{3}/4 \\ 0 & 0 & 1 & 0 \\ 3/4 & -\sqrt{3}/2 & 0 & 1/4 \end{pmatrix}, \quad J_\square = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{ESQ41})$$

Their third columns fix E_{21} , and their other columns preserve U_+ . They consequently commute with P_- and A_h for every h . For a full verification of their group and metric properties, put $(z_1, z_2, c, d) = \Theta_h x$ in the scalar basis of its target. Substitution of (ESQ41) into (ESQ24) gives

$$R : (z_1, z_2, c, d) \mapsto \left(z_1, -\frac{1}{2}z_2 + \frac{\sqrt{3}}{2}c, -\frac{\sqrt{3}}{2}z_2 - \frac{1}{2}c, d \right), \quad (\text{ESQ42})$$

$$J_\square : (z_1, z_2, c, d) \mapsto (z_1, -z_2, c, d). \quad (\text{ESQ43})$$

The two-by-two rotation in (ESQ42) has square obtained by changing both off-diagonal signs and has cube the identity. Conjugating it by the reflection of z_2 changes exactly those two signs, giving its inverse. Thus $R^3 = I$, $J_\square^2 = I$, and $J_\square R J_\square = R^{-1}$. The six maps $I, R, R^2, J_\square, R J_\square, R^2 J_\square$ are distinct: the first three restrict to distinct rotations of the (z_2, c) -plane with determinant 1, and the last three have determinant minus one and distinct restrictions. They are therefore precisely a D_3 action.

The same displayed rotation and reflection preserve $z_2^2 + c^2$, fixing z_1, d . They preserve both the positive metric in (ESQ30) and the trace-coordinate heat form $z_1^2 + z_2^2 + c^2 + ad^2$. Pulling back by the proved isomorphism gives

$$R^\top G R = G, \quad J_\square^\top G J_\square = G, \quad B_h(Rx, Ry) = B_h(x, y), \quad B_h(J_\square x, J_\square y) = B_h(x, y). \quad (\text{ESQ44})$$

This proves the full native symmetry of the deformation, not only preservation of its unique negative line.

These are linear symmetries of the metric and operator. They do not assert automorphisms of the full matrix algebra: for example the displayed matrix gives $(RE_{12})^2 = 3I_2/4$, although $E_{12}^2 = 0$. Nor does R preserve the integral raw-cell lattice, since its E_{12} column has coefficients $\sqrt{3}/2$. There is nevertheless an exact algebra-compatible restriction at the collision. Both generators fix $I_2 = E_{11} + E_{22}$, as summing their first and fourth columns proves, and both fix E_{21} . They therefore fix $\mathbb{R}[I_2, E_{21}]$ pointwise. The collision algebra isomorphism (ESQ40) is consequently equivariant for this native D_3 with the trivial action on $\mathbb{R}[r]/(r^2)$.

It also proves an exact qualification about the trace projection. The decomposition $\mathbb{R}^2 \oplus E_a$ selected by (ESQ24) is not preserved factor by factor under the native rotation: the scalar coefficient c of E_a mixes with z_2 . In fact the vector with coordinates $(0, 1, 0, 0)$ lies in $\ker \pi_h$, whereas its image under R has trace scalar coordinate $-\sqrt{3}/2$. Therefore no linear action on E_a alone can make this particular projection π_h equivariant under R . The full four-dimensional map Θ_h is equivariant with the explicit actions (ESQ42)–(ESQ43). This supplies the exact symmetry map instead of omitting the mixing.

7. The genuine regular tetrahedron and its full vertex symmetry

The three positive source rays and their forced fourth ray are

$$\begin{aligned} r_0 &= 2E_{11} + 2E_{12} + 2E_{22}, \\ r_+ &= (2 + \sqrt{3})E_{11} - E_{12} + (2 - \sqrt{3})E_{22}, \\ r_- &= (2 - \sqrt{3})E_{11} - E_{12} + (2 + \sqrt{3})E_{22}, \\ r_\infty &= -(r_0 + r_+ + r_-) = -6(E_{11} + E_{22}). \end{aligned} \quad (\text{ESQ45})$$

The source's original $3 + 1$ tetrahedron is $\Delta_F = \text{conv}\{E_{21}, r_0, r_+, r_-\}$, as stated in `cor:Table-D3-tetrahedral-realization`, which was read directly. It is distinct from the centered regular tetrahedron $\text{conv}\{r_\infty, r_0, r_+, r_-\}$ entirely inside U_+ . In the first tetrahedron, the three edges from E_{21} have squared G -length 2, while the opposite-face edges have squared length $8/3$, by orthogonality to E_{21} and the pairings proved below. Thus every vertex isometry of this first tetrahedron fixes E_{21} and permutes the other three vertices, giving exactly the native D_3 . No exchange with an apex is a further isometry of that tetrahedron. The second tetrahedron has the full geometric S_4 calculated below.

Their positive Euclidean coordinates $2(z_1, z_2, c)$ under (ESQ24) are respectively

$$\begin{aligned} v_0 &= (1/3, 0, 2\sqrt{2}/3), \\ v_+ &= (1/3, \sqrt{2/3}, -\sqrt{2}/3), \\ v_- &= (1/3, -\sqrt{2/3}, -\sqrt{2}/3), \\ v_\infty &= (-1, 0, 0). \end{aligned} \quad (\text{ESQ46})$$

Their norms are 1, and the dot product of any two different vectors is $-1/3$, by multiplication of these coordinates. For example the two products involving $v_0, v_\pm$ equal $1/9 - 4/9 = -1/3$, while $v_+ \cdot v_- =$

$1/9 - 2/3 + 2/9 = -1/3$; products with v_∞ are immediate. Since $G|_{U_+}$ is exactly the Euclidean form on these doubled coordinates by (ESQ30), this proves

$$\langle r_a, r_b \rangle_G = \begin{cases} 1 & a = b, \\ -1/3 & a \neq b, \end{cases} \quad a, b \in \{\infty, 0, +, -\}. \quad (\text{ESQ47})$$

The first three rays form a basis of U_+ : their coordinate matrix in (E_{11}, E_{12}, E_{22}) has determinant $-24\sqrt{3}$, obtained by subtracting its second column from its third and expanding. Therefore specifying their Gram matrix uniquely specifies the bilinear form. Since (ESQ9) has just been shown to give those entries, this proves the source metric's uniqueness as well as positivity and the regular-tetrahedron assertion.

Every permutation π of these four rays extends uniquely to a G -orthogonal linear operator S_π on U_+ . To see existence, the only linear dependence of the four rays is their sum zero: rank three gives a one-dimensional relation space, and this is a nonzero relation. A permutation preserves that relation, so induces an invertible map on their three-dimensional span. It preserves all the Gram entries (ESQ47), hence the form. Extend it to V by fixing E_{21} . The maps obey $S_\pi S_\rho = S_{\pi\rho}$ on the spanning vectors and are distinct on them, giving a faithful geometric S_4 action. They preserve G and commute with every A_h , since they preserve the two eigenspaces in (ESQ3).

The resulting map on every vector also has the finite formula

$$S_\pi = P_- + \frac{3}{4} \sum_{a \in \{\infty, 0, +, -\}} r_{\pi(a)} r_a^\top G. \quad (\text{ESQ47a})$$

It fixes E_{21} because the rays are orthogonal to that vector. On r_b , (ESQ47) makes the sum equal to $(3/4)(r_{\pi(b)} - (1/3) \sum_{a \neq b} r_{\pi(a)}) = r_{\pi(b)}$, using the zero sum of the rays. These spanning-vector computations prove the stated formula and all its prescribed values.

The native generators act on these rays by

$$R : r_0 \mapsto r_+ \mapsto r_- \mapsto r_0, \quad J_\square : r_0 \mapsto r_0, \quad r_+ \leftrightarrow r_-, \quad r_\infty \mapsto r_\infty. \quad (\text{ESQ48})$$

These follow either by substitution in (ESQ41) or by (ESQ42) on (ESQ46). Thus the native D_3 is exactly the stabilizer of r_∞ in this geometric tetrahedral S_4 .

For a fully explicit additional generator, put

$$K = \begin{pmatrix} 1/3 & -8/3 & 0 & -2/3 \\ -1/6 & 1/3 & 0 & -1/6 \\ 0 & 0 & 1 & 0 \\ -2/3 & -8/3 & 0 & 1/3 \end{pmatrix}. \quad (\text{ESQ49})$$

Let $v = r_0 - r_\infty = (8, 2, 0, 8)^\top$. From (ESQ47), $v^\top G v = 8/3$, and direct multiplication gives

$$K = I_V - \frac{3}{4} v v^\top G. \quad (\text{ESQ50})$$

For any x , this subtracts $2\langle v, x \rangle_G v / \langle v, v \rangle_G$; expansion of its inner product on two vectors cancels the two cross terms with the final quadratic term, proving it is a G -isometry. The formula sends r_0 to r_∞ , r_∞ to r_0 , and fixes $r_\pm, E_{21}$, since their pairings with v vanish. Hence K is the transposition extending the native stabilizer to the full geometric S_4 . Conjugating this transposition by the native three-cycle supplies the transpositions of r_∞ with each other vertex, and these generate every permutation.

There is a useful exact orthonormal four-frame that makes this action a permutation representation. For each tetrahedral vertex put

$$w_a = \frac{1}{2} E_{21} + \frac{\sqrt{3}}{2} r_a, \quad a \in \{\infty, 0, +, -\}, \quad (\text{ESQ51})$$

and let W be the four-column matrix of these vectors in that order. By (ESQ47) and orthogonality to E_{21} , their pairings are 1 on the diagonal and $1/4 + (3/4)(-1/3) = 0$ off it. Therefore

$$W^\top G W = I_4, \quad W^{-1} E_{21} = \frac{1}{2} \mathbf{1}, \quad \mathbf{1} = (1, 1, 1, 1)^\top. \quad (\text{ESQ52})$$

The second identity follows also from $\sum_a w_a = 2E_{21}$. Since each w_a has coefficient $1/2$ on E_{21} , its orthogonal projector in these coordinates is $\mathbf{1}\mathbf{1}^\top/4$. It follows that

$$W^{-1} A_h W = \frac{1}{4} I_4 + \frac{2h - 1/2}{4} \mathbf{1}\mathbf{1}^\top, \quad (\text{ESQ53})$$

and

$$B_h(Wy, Wy) = \frac{1}{4} \sum_{j=1}^4 y_j^2 + \frac{2h - 1/2}{4} \left(\sum_{j=1}^4 y_j \right)^2. \quad (\text{ESQ54})$$

Thus the changing line is the uniform line in this genuine tetrahedral four-frame; the orthogonal sum-zero three-space retains eigenvalue $1/4$. Permuting these four frame vectors is exactly S_π , so (ESQ53) is invariant under all four-vertex permutations. These frame vectors are not the four raw cells: (ESQ51) supplies the exact change of basis between the two presentations.

8. Permuting the four raw cells transports both operator and metric

Let P be a permutation matrix of the original raw basis (ESQ1), rather than a permutation of the rays or frame vectors of Section 7. The transported data are

$$A_h^{(P)} = P A_h P^{-1}, \quad G^{(P)} = P^{-\top} G P^{-1}, \quad B_h^{(P)} = P^{-\top} (G A_h) P^{-1}. \quad (\text{ESQ55})$$

Multiplying the first two formulas gives the third, and substitution yields

$$B_h^{(P)}(Px, Py) = B_h(x, y), \quad \Theta_h^{(P)} = \Theta_h P^{-1}. \quad (\text{ESQ56})$$

Hence there is an exact transported isometry for every raw permutation, with the negative line sent to $P\mathbb{R}E_{21}$. This proves the correspondence without pretending the fixed coordinate metric is invariant under every such permutation.

Indeed the only raw permutation matrices preserving the displayed G are the identity and the swap $E_{11} \leftrightarrow E_{22}$. A metric-preserving raw permutation must preserve its diagonal entries. The values 1 and $2/9$ occur only at E_{21} and E_{12} , so those two positions must be fixed. The remaining two positions have equal diagonal entries $1/16$, giving only their identity or exchange; the off-diagonal entries in (ESQ9) are preserved by both. This proves the assertion exactly. In particular a raw permutation moving the negative cell cannot preserve this fixed metric, even though it always has the transported realization (ESQ55).

There is a stronger fixed-metric check away from the scalar time. If $h \neq 1/4$, a diagonal operator that assigns its changing eigenvalue $a(h)$ to one of the raw diagonal cells E_{11}, E_{22} while retaining G fails to be G -self-adjoint: the nonzero metric entry $G_{14} = -7/144$ connecting those cells forces their assigned eigenvalues to agree, whereas $a(h) \neq 1/4$. Formula (ESQ55) avoids that change of pairing by transporting the whole metric. At the exceptional scalar time $h = 1/4$, $A_h = I_V/4$, so every linear map commutes with that operator; preservation of the fixed G still remains a separate condition.

9. The regular four-simplex and the scalar time

The original source completes the positive tetrahedron to five vectors in the full space:

$$s_* = E_{21}, \quad s_a = \frac{\sqrt{15}}{4} r_a - \frac{1}{4} E_{21} \quad (a \in \{\infty, 0, +, -\}). \quad (\text{ESQ57})$$

Using (ESQ47), each has norm one. Distinct s_a, s_b have pairing $(15/16)(-1/3) + 1/16 = -1/4$, and $\langle s_*, s_a \rangle = -1/4$. Also

$$\sum_{\alpha \in \{*, \infty, 0, +, -\}} s_\alpha = 0, \quad \frac{1}{4} \sum_{a \neq *} s_a = -\frac{1}{4} E_{21}. \quad (\text{ESQ58})$$

These follow by summing (ESQ57) and using $\sum r_a = 0$. They prove the regular four-simplex and its exact opposite-face centroid. The source intertwiner sends that centroid to

$$\mathcal{T}_\square(-E_{21}/4) = \mathcal{I}_- \mathcal{T}_\square(E_{21}) = -\frac{1}{8}(4, 0, 0, -1). \quad (\text{ESQ59})$$

The first equality is (ESQ8) on the negative eigenvector; the final coordinate follows from (ESQ5).

Every permutation of the five simplex vectors gives a unique G -orthogonal map of V . For proof, they span V : s_* gives E_{21} , and the other vectors then give the spanning positive rays. Their sole linear relation is the sum zero, so a permutation induces an invertible map, and their common Gram entries prove isometry on every vector. This realizes S_5 as the simplex's vertex-symmetry group.

For $h \neq 1/4$, the two eigenvalues in (ESQ11) differ, so an operator commuting with A_h must preserve its one-dimensional eigenspace $\mathbb{R}E_{21}$. Among the five simplex vertices only s_* lies on that line: each s_a has a

nonzero positive component. Thus the simplex permutations commuting with A_h are precisely the stabilizer of s_* , namely the geometric S_4 in Section 7. At $h = 1/4$, the operator is scalar and every simplex permutation commutes with it and preserves $B_h = G/4$. This is the exact symmetry enlargement at that heat time; it does not relabel the originally marked source cell.

10. The nilpotent with its support retained

The collision algebra isomorphism (ESQ40) preserves its nonzero nilpotent, its unit and its supported zero. For a fixed bounded distributive support lattice L , any of the proved ring maps f between these commutative algebras induces

$$G_L(f) : (r, \lambda) \mapsto (f(r), \lambda). \quad (\text{ESQ60})$$

This is well defined because a nonzero amplitude can only have top support and a zero image with that same top support is allowed. The ring homomorphism laws preserve amplitudes, and unchanged joins and meets preserve the support operations, proving it is the corresponding support-semiring map. In particular the image of E_{21} in the commutative collision corner is the nonzero supported amplitude r ; its square is the supported zero e , not the unsupported element τ . The radical of the collision form, the nilpotent in the algebra, and absence in the support semiring therefore have exact connecting maps and retain their different definitions.

11. The actual moving source tetrahedron requires its own metric transport

The fixed raw-cell metric G and the coefficient deformation A_h above do not by themselves identify the metric altitude of the moving three-point source fibre. Here is that spatial comparison with its full map. The original source's affine map and metric were read directly in `eq:Fable-collision-to-spatial-affine-map`, `eq:Fable-collision-pullback-metric`, `eq:Fable-source-face-and-axis`, and `eq:Fable-source-signed-quarter-altitude`. In this section (x, y, z) are the three original polynomial-source coordinates; this z is the coordinate denoted W in HEB3. It is not one of the four raw-cell coefficients.

For $a < 0$, let $t = \sqrt{-a} > 0$. The exact three real points from HEB3 and their target are

$$p_0(a) = (0, 0, a), \quad p_r(a) = \left(-\frac{1}{2r}, 3r, -26a\right) \quad (r = \pm t), \quad q_a = (a, 0, 0). \quad (\text{ESQ61})$$

Their centroid, plane, and distinguished displacement are

$$c_a = (0, 0, -17a), \quad H_a = \{(x, y, z) : y - 6ax = 0\}, \quad q_a - c_a = av_F, \quad v_F = (1, 0, 17). \quad (\text{ESQ62})$$

The centroid is the sum of the three displayed points divided by three: the first two coordinate sums vanish and the third is $a - 52a = -51a$. For an escaping point, $y - 6ax = 3r + 3a/r = 3(r^2 + a)/r = 0$, and it is also zero at $p_0(a)$. The two difference vectors from $p_0(a)$ to $p_t(a), p_{-t}(a)$ are independent because their first two coordinates are negatives and their common third coordinate is $-27a \neq 0$; hence this really is the plane of the face. Finally the difference of q_a, c_a is the displayed multiple of v_F .

The original affine map at $a = -1/4$ has linear part

$$L_0 = \begin{pmatrix} 68/9 & 136/27 & -4/9 \\ 1 & 2/3 & 0 \\ 0 & -2\sqrt{3}/3 & 0 \end{pmatrix}, \quad b_0 = (17/9, 3/2, 0)^\top, \quad (\text{ESQ63})$$

with source metric $g_0 = L_0^\top L_0$. Direct multiplication gives $L_0 v_F = (0, 1, 0)$ and

$$g_0 v_F = (1, 2/3, 0)^\top. \quad (\text{ESQ64})$$

Therefore the fixed g_0 -normal covector of this axis is $x + (2/3)y$. It is proportional to the moving face covector $y - 6ax$ exactly when $a = -1/4$: comparing the y -coefficients forces the proportionality constant to be $2/3$, and then the x -coefficients require $1 = -4a$. Consequently the scalar displacement a in (ESQ62) cannot be called a signed normal altitude in the fixed original metric for arbitrary heat time.

For the entire real region $a < 0$, the exact replacement is the explicitly moving affine map

$$\mathcal{A}_a(x) = L_a x + b_0, \quad L_a = \begin{pmatrix} -17/(9a) & 17/(54a^2) & 1/(9a) \\ 1 & -1/(6a) & 0 \\ 0 & -\sqrt{3}/(3\sqrt{-a}) & 0 \end{pmatrix}. \quad (\text{ESQ65})$$

It agrees with (ESQ63) when $a = -1/4$, by substituting that value in all its entries. Its determinant is $-\sqrt{3}/(27a\sqrt{-a}) \neq 0$, by expansion along the third row, so

$$g_a = L_a^\top L_a \quad (\text{ESQ66})$$

is a positive definite real metric: $v^\top g_a v = \|L_a v\|^2 > 0$ for every nonzero v .

The four images, with their signs retained, are

$$\begin{aligned} \mathcal{A}_a(p_0(a)) &= (2, 3/2, 0), \\ \mathcal{A}_a(p_t(a)) &= (-1, 3/2, -\sqrt{3}), \\ \mathcal{A}_a(p_{-t}(a)) &= (-1, 3/2, \sqrt{3}), \\ \mathcal{A}_a(q_a) &= (0, 3/2 + a, 0). \end{aligned} \quad (\text{ESQ67})$$

For the first and fourth these follow by direct multiplication. For p_r , the first coordinate simplifies to $17/(18ar) + 17r/(18a^2) - 1 = -1$ because $r^2 + a = 0$; the second differs from $3/2$ by $-1/(2r) - r/(2a) = 0$; and the third is $-\sqrt{3}r/t$. These prove both escaping images, including the displayed ordering. Thus the three points give a fixed equilateral face in these transported coordinates.

Finally $L_a v_F = (0, 1, 0)$ for every $a < 0$, and the second row gives

$$\langle v_F, v_F \rangle_{g_a} = 1, \quad \langle v_F, v \rangle_{g_a} = x - \frac{y}{6a} \quad (v = (x, y, z)). \quad (\text{ESQ68})$$

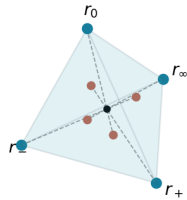
The latter covector vanishes exactly on the plane H_a . Hence v_F is its chosen unit normal in g_a , and (ESQ62) has signed normal altitude exactly a for this transported metric. The map L_a diverges as $a \rightarrow 0^-$, for example through its $17/(54a^2)$ entry. No nonsingular continuation of this specified real spatial transport at the collision is claimed. The fixed four-dimensional coefficient form and the algebraic collision receiver of Sections 2–5 remain defined there by their own displayed maps.

The calculation realizes the original ES quarter 3 + 1 operator with the original heat coefficient and a specified reflected trace. It retains the native tetrahedral symmetry, its action on trace coordinates, the exact algebra embedding and collision isomorphism, and the separate moving spatial metric required by the original three-point geometry. It proves finite forms, operators, and their maps; it makes no identification of the ES form with an unspecified global Weil pairing.

The source edition is available as the 488-page ES reader and its original TeX source and build archive. The TeX entry is `reader/erdos_straus_project_reader.tex`; the source identity and exact reading locations appear in the source guide.

The exact regular tetrahedron in the positive three-space

The four-dimensional map retains the extra direction



$$w_j = \frac{1}{2}E_{21} + \frac{\sqrt{3}}{2}r_j$$

$$\langle w_i, w_j \rangle_G = \delta_{ij}, \quad E_{21} = \frac{1}{2} \sum_j w_j$$

$$W^{-1}A_h W = \frac{1}{4}I_4 + \frac{2h-1/2}{4} \mathbf{1}\mathbf{1}^\top$$

The sum-zero space has eigenvalue $1/4$.

The uniform line has eigenvalue $-1/4 + 2h$.

Face centres: $f_j = -r_j/3$. The line $\mathbb{R}E_{21}$ is orthogonal to this space.

Figure 4: Exact tetrahedral vertices and face centres, with the four-dimensional orthonormal frame. ESQ45–ESQ54 prove the coordinates and the uniform changing direction; the orthogonal E21 direction is retained outside the displayed three-space.

The quarter heat family in the actual ES free-shell operator

23 September 2026. This note transports the quarter family through the spectral projectors of the actual raw ES operator into its free-shell carrier. It retains the original forced metric, the full raw matrix product, the two ordered sheets, all three rotations, every orbit label, the ambient simplex kernel, and both directed blades. It constructs the bounded operator on the full orbit Hilbert space, its exact quadratic algebra receiver and collision pairing, and the full Figure 10 support-corner and modular transport. Every finite orbit compression is compatible with these maps. No equality with a Weil distribution is asserted.

The original source read is `ES-Fable-C123/es_fable_c123_preprint.tex`. The precise inputs are `eq:raw-negative-cell-projector`, `eq:raw-positive-cell-projector`, `eq:pulled-back-quarter-cell-signs`, `thm:quarter-three-plus-one`, `eq:origin-resolved-support`, `eq:origin-involution`, `eq:star-kneser-D3-cover-hypotheses`, `thm:general-shell-123-linking-decomposition`, `cor:general-shell-simplex-Lorentz-pullback`, and `thm:general-shell-control-determinant-volume`. The corresponding public reading edition is *Erdős–Straus Project Reader*, pinned edition 20709bee. The mathematical reading for this derivation used the original TeX, not extracted PDF text. All finite and infinite operator claims below are proved here from the explicit source matrices and labels.

1. The original arithmetic labels and the actual shell spaces

For the ES source, let p be a prime, let a_{ar} be a positive integer, and retain

$$R = 4a_{\text{ar}} - p > 2, \quad \gcd(R, pa_{\text{ar}}) = 1, \quad N = pa_{\text{ar}}. \quad (\text{ESHL1})$$

These are `eq:star-kneser-D3-cover-hypotheses`. They imply that p is odd and $p \nmid a_{\text{ar}}$: for $p = 2$, R and pa_{ar} are both even; for odd $p \mid a_{\text{ar}}$, the same prime divides R . Define exactly the origin-resolved set

$$\mathcal{X}_{R,p} = \{(j, u) : j \in \{0, 1, 2\}, u \mid a_{\text{ar}}^2, \quad ua_{\text{ar}}^{-1} = -p^{1-j} \text{ in } (\mathbb{Z}/R\mathbb{Z})^\times\}. \quad (\text{ESHL2})$$

All inverses in this formula exist by ESHL1. Its involution is

$$\iota(j, u) = (2 - j, a_{\text{ar}}^2/u). \quad (\text{ESHL3})$$

It preserves the defining congruence because taking its multiplicative inverse turns $-p^{1-j}$ into $-p^{j-1}$. Applying it twice gives the identity. A fixed point would have $j = 1, u = a_{\text{ar}}$; the congruence would then give $1 = -1 \pmod{R}$, impossible for $R > 2$. Thus all its orbits have size two.

Let $\mathcal{O} = \mathcal{X}_{R,p}/\langle \iota \rangle$ and $q = |\mathcal{O}|$. The two divisors $d(j, u) = p^j u$ in any orbit have product N^2 . They are unequal: uniqueness of the representation $p^j u$, using $p \nmid a_{\text{ar}}$, would otherwise give a fixed point of ι . Thus each orbit has a unique low member $x_{\mathcal{O}}^-$ with divisor below N and high member $x_{\mathcal{O}}^+$ with divisor above N . This proves the original ordering used by the source.

The complete free-cover labels are (O, ϵ, m) , where $\epsilon \in \{0, 1\}$ names the low/high sheet and $m \in \mathbb{Z}/3\mathbb{Z}$ names the rotation. The original bijection and its unitary linearization are

$$\theta_{123}(m, x_{\mathcal{O}}^\epsilon) = (O, \epsilon, m), \quad \Theta_{123} : \mathbb{C}[\tilde{\mathcal{X}}_{R,p}] \longrightarrow \mathcal{H}_q := \mathbb{C}[\mathcal{O}] \otimes \mathbb{C}^2 \otimes \mathbb{C}^3, \quad e_{(m, x_{\mathcal{O}}^\epsilon)} \mapsto e_O \otimes f_\epsilon \otimes g_m. \quad (\text{ESHL4})$$

They are bijections because the three labels recover the original point. The map is unitary for the stated orthonormal label bases. Hence $\dim \mathcal{H}_q = 6q$, retaining every label.

Let $P_2 = I_3 - \mathbf{1}\mathbf{1}^*/3$ be the source triangle support projector. The original support unitary $U_2 : \text{Ran } P_2 \rightarrow \mathbb{C}^2$ satisfies $U_2^* U_2 = P_2$ and $U_2 U_2^* = I_2$, as proved in `eq:simplex-123-unitary-identities`. Write $\eta_\epsilon = U_2^* f_\epsilon$. These are an orthonormal basis of the original triangle support, with their original coordinate choice retained. The actual ambient simplex space and its support are

$$\mathcal{K}_q = \mathbb{C}[\mathcal{O}] \otimes \mathbb{C}^3 \otimes \mathbb{C}^3, \quad \mathcal{H}_q^{\text{simp}} = \mathbb{C}[\mathcal{O}] \otimes \text{Ran } P_2 \otimes \mathbb{C}^3. \quad (\text{ESHL5})$$

The isometry $V_q = I \otimes U_2^* \otimes I_3$ identifies $\mathcal{H}_q$ with $\mathcal{H}_q^{\text{simp}}$. Its range projector is $P_q^{\text{simp}} = I \otimes P_2 \otimes I_3$. The retained complementary space is

$$\mathcal{Z}_q = \mathbb{C}[\mathcal{O}] \otimes \mathbb{C}\mathbf{1} \otimes \mathbb{C}^3, \quad \mathcal{K}_q = \mathcal{H}_q^{\text{simp}} \oplus \mathcal{Z}_q, \quad \dim \mathcal{Z}_q = 3q. \quad (\text{ESHL6})$$

This ambient kernel is different from the source map Π_{123} , which sends all six labels in an orbit to one support label and has kernel dimension $5q$. Explicitly that latter kernel is the subspace whose six coefficients sum to zero on every orbit. Neither of these kernels is substituted for the other.

2. The raw quarter operator and its spectral algebra

Give the raw four-cell vector space $\mathcal{V}_{\text{raw}} = M_2(\mathbb{C})$ its Hilbert–Schmidt inner product, with orthonormal basis $E_{11}, E_{12}, E_{21}, E_{22}$. The original forced ES metric is retained by the explicit isometry ESHL43–ESHL44 below. The source’s pulled-back quarter reflection acts by

$$\mathfrak{Q}_{\square}(E_{11}) = E_{11}, \quad \mathfrak{Q}_{\square}(E_{12}) = E_{12}, \quad \mathfrak{Q}_{\square}(E_{21}) = -E_{21}, \quad \mathfrak{Q}_{\square}(E_{22}) = E_{22}. \quad (\text{ESHL7})$$

Consequently its orthogonal spectral projectors are

$$P_{-}(M) = M_{21}E_{21}, \quad P_{+} = I_{\mathcal{V}_{\text{raw}}} - P_{-}. \quad (\text{ESHL8})$$

They satisfy $P_{\pm}^2 = P_{\pm} = P_{\pm}^*$, $P_{+}P_{-} = 0$, with ranks three and one. Define the exact raw quarter operator and its heat family by

$$A_0 = \frac{1}{4}P_{+} - \frac{1}{4}P_{-}, \quad A_h = \frac{1}{4}P_{+} + a(h)P_{-}, \quad a(h) = -\frac{1}{4} + 2h. \quad (\text{ESHL9})$$

The original coefficient inversion matrix $\mathcal{I}_{-}$ is transported to this raw presentation by the source map $\mathcal{T}_{\square}$, since `eq:pulled-back-quarter-reflection` states $\mathfrak{Q}_{\square} = \mathcal{T}_{\square}^{-1}(4\mathcal{I}_{-})\mathcal{T}_{\square}$. Thus A_0 is the actual pulled-back quarter operator.

The fixed spectral algebra is

$$\mathcal{D} = \mathbb{C}[A_0] = \{cP_{+} + dP_{-} : c, d \in \mathbb{C}\} \cong \mathbb{C}^2. \quad (\text{ESHL10})$$

Indeed $P_{+} = 2A_0 + I/2$, $P_{-} = -2A_0 + I/2$, so both projectors belong to the generated algebra. Their orthogonality proves multiplication is coordinatewise and the involution conjugates c, d . The representation is faithful because both projectors have nonzero range.

The original forced metric in `eq:Fable-forced-full-metric`, also used in ESQ9, has matrix

$$G = \begin{pmatrix} 1/16 & 0 & 0 & -7/144 \\ 0 & 2/9 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -7/144 & 0 & 0 & 1/16 \end{pmatrix}. \quad (\text{ESHL43})$$

Let $v_{+} = (E_{11} + E_{22})/\sqrt{2}$, $v_{-} = (E_{11} - E_{22})/\sqrt{2}$, and let $\Pi_{+}, \Pi_{-}, \Pi_{12}, \Pi_{21}$ be the Hilbert–Schmidt projections on these two unit vectors and on E_{12}, E_{21} , respectively. Multiplying ESHL43 by these four vectors gives the respective eigenvalues $1/72, 1/9, 2/9, 1$. Hence the fully specified positive square root is

$$T = G^{1/2} = \frac{1}{6\sqrt{2}}\Pi_{+} + \frac{1}{3}\Pi_{-} + \frac{\sqrt{2}}{3}\Pi_{12} + \Pi_{21}, \quad T^*T = G, \quad TP_{\pm} = P_{\pm}T, \quad TA_h = A_hT. \quad (\text{ESHL44})$$

The four orthogonal projections sum to the identity. Squaring their positive coefficients proves $T^*T = G$, and replacing the coefficients by their reciprocals defines the inverse. Each projection preserves the decomposition into $\text{Ran } P_{+}$ and $\text{Ran } P_{-}$, proving both commutation identities. Thus $T : (\mathcal{V}_{\text{raw}}, \langle x, Gy \rangle_{\text{HS}}) \rightarrow (\mathcal{V}_{\text{raw}}, \langle x, y \rangle_{\text{HS}})$ is an isometry, and the original heat form is exactly $\langle x, GA_h y \rangle_{\text{HS}} = \langle Tx, A_h Ty \rangle_{\text{HS}}$. Conjugation by T fixes every member of the spectral algebra $\mathcal{D}$. This is a vector-metric transport, and does not change the raw matrix product: for example $T(I_2) = I_2/(6\sqrt{2})$, so T is not a unital algebra map. The separate map preserving that product is ESHL45.

The spectral algebra $\mathcal{D}$ consists of endomorphisms of the raw four-cell vector space. Its negative projector is the two-sided corner operation $P_{-}(X) = E_{22}XE_{11}$, as direct entry multiplication proves. It is not a left multiplication operator: P_{-} has rank one, whereas left multiplication by a nonzero rank-one matrix on $M_2(\mathbb{C})$ has rank two, since each of two columns independently ranges over a fixed one-dimensional image. Section 3 uses the spectral algebra as its domain; ESHL45–ESHL47 construct the full raw matrix algebra map and intertwine this corner operation exactly.

3. The faithful spectral lift into the actual ordered sheets

In $\mathcal{H}_q$, define the original low and high projectors

$$Q_{<} = I_{\mathbb{C}[\mathcal{O}]} \otimes E_{11}^{(2)} \otimes I_3, \quad Q_{>} = I_{\mathbb{C}[\mathcal{O}]} \otimes E_{22}^{(2)} \otimes I_3. \quad (\text{ESHL11})$$

They are exactly **eq:general-shell-ordered-sheet-projectors** under ESHL4. Their ranges are orthogonal, they sum to the identity on $\mathcal{H}_q$, and each has dimension $3q$. For $q \geq 1$, define

$$\boxed{\Psi_q : \mathcal{D} \longrightarrow B(\mathcal{H}_q), \quad cP_+ + dP_- \longmapsto cQ_< + dQ_>} \quad (\text{ESHL12})$$

This is a faithful unital $*$ -homomorphism. Orthogonality gives $(cQ_< + dQ_>)(c'Q_< + d'Q_>) = cc'Q_< + dd'Q_>$, matching the source product. Adjoints conjugate the same two coefficients. The source unit maps to $Q_< + Q_> = I$. If an image vanishes, applying it to any low and any high basis vector gives $c = d = 0$. This proves faithfulness and also the isometry $\|\Psi_q(cP_+ + dP_-)\| = \max(|c|, |d|)$.

The map retains the two spectral labels, but changes their representation multiplicities. Its exact trace comparison is

$$\text{Tr}_{\text{raw}}(cP_+ + dP_-) = 3c + d, \quad \text{Tr}_{\mathcal{H}_q} \Psi_q(cP_+ + dP_-) = 3q(c + d). \quad (\text{ESHL13})$$

Thus this is not a trace-preserving or Hilbert-space unitary identification of the four-cell space with the shell. Both multiplicity lists remain explicit. When $q = 0$, the shell space and all its operators are zero; the resulting map is zero and is not faithful. The nonzero-shell assertion has therefore not been extended to an empty arithmetic shell.

Define the fixed positive weight and its inverse on the support by

$$W = 16Q_< + 4Q_>, \quad W^{-1} = \frac{1}{16}Q_< + \frac{1}{4}Q_>. \quad (\text{ESHL14})$$

Both are verified by multiplication. The weight commutes with $\Psi_q(\mathcal{D})$, and its positive square root is $4Q_< + 2Q_>$. The lifted heat carrier is

$$\boxed{H_h := W\Psi_q(A_h) = 4Q_< + t(h)Q_>, \quad t(h) = -1 + 8h.} \quad (\text{ESHL15})$$

At zero it equals the actual ES carrier

$$H_0 = I_{\mathbb{C}[\mathcal{O}]} \otimes \text{diag}(4, -1) \otimes I_3 = H_{R,p}^{\text{sh}}, \quad (\text{ESHL16})$$

in the source's unitary presentation **eq:general-shell-restricted-Lorentz-carrier**. The map $A \mapsto W\Psi_q(A)$ is a specified positive linear map on self-adjoint elements, since it also equals $W^{1/2}\Psi_q(A)W^{1/2}$; it is not asserted to be an algebra homomorphism. The faithful homomorphism is Ψ_q itself.

The original simplex presentation is $\tilde{H}_h = V_q H_h V_q^*$ on $\mathcal{K}_q$, with zero action on $\mathcal{Z}_q$. This gives the exact commuting relation $\tilde{H}_h V_q = V_q H_h$, and the support-corner homomorphism $D \mapsto V_q \Psi_q(D) V_q^*$ has unit P_q^{simp} . It is unital as a map to that corner, not to the whole ambient algebra with identity $I_{\mathcal{K}_q}$.

4. Positivity, the sharp gap, and every kernel

For $x = x_< + x_>$ in the orthogonal sheet decomposition,

$$\langle x, H_h x \rangle = 4\|x_<\|^2 + t(h)\|x_>\|^2. \quad (\text{ESHL17})$$

Thus for a nonempty finite shell

$$\text{inertia}(H_h) = \begin{cases} (3q, 3q, 0), & h < 1/8, \\ (3q, 0, 3q), & h = 1/8, \\ (6q, 0, 0), & h > 1/8. \end{cases} \quad (\text{ESHL18})$$

In particular $H_h \succeq 0$ exactly for $h \geq 1/8$. For $h > 1/8$, the sharp lower and upper bounds are

$$\boxed{\min(4, -1 + 8h)I \preceq H_h \preceq \max(4, -1 + 8h)I.} \quad (\text{ESHL19})$$

Both constants are attained on the appropriate sheet vectors in ESHL17. The lower bound is independent of q . At the critical time, the kernel on the shell is exactly $\text{Ran } Q_>$.

On the full simplex ambient space, the kernel is $\mathcal{Z}_q$ for $h \neq 1/8$, and is

$$\ker \tilde{H}_{1/8} = \mathcal{Z}_q \oplus V_q \text{Ran } Q_>, \quad \dim \ker \tilde{H}_{1/8} = 6q. \quad (\text{ESHL20})$$

The original ambient kernel has not disappeared when the support form becomes positive. The sharp lower bound on the ambient space is the support inequality $\tilde{H}_h \succeq \min(4, t(h))P_q^{\text{simp}}$ for $h > 1/8$; strict positivity is a statement on $\mathcal{H}_q^{\text{simp}}$.

There are also two distinct times at which a single operator loses its ability to recover the fixed spectral labels. At $h = 1/4$, $A_h = I/4$, so $\mathbb{C}[A_h] = \mathbb{C}I$ although the fixed algebra $\mathcal{D}$ and its $P_{\pm}$ remain. At $h = 5/8$, $H_h = 4I$, so its generated algebra is scalar although $Q_{<}, Q_{>}$ remain present. These follow by equating their two coefficients. The lift uses $\mathcal{D} = \mathbb{C}[A_0]$ and the retained labelled projectors throughout; it does not reconstruct them from the possibly scalar single operator at these times.

5. The actual mixed blades remain nonzero

Retain exactly the directed blades of `eq:general-shell-low-high-blade-transport` and `eq:general-shell-high-low-blade-transport`:

$$D = I \otimes 2E_{21}^{(2)} \otimes I_3, \quad U = I \otimes 2E_{12}^{(2)} \otimes I_3 = D^*. \quad (\text{ESHL21})$$

Here D is the original low-to-high sheet map and U its adjoint; these symbols do not denote the amplitude multiplier from a different packet calculation. Matrix-unit multiplication proves

$$D^2 = U^2 = 0, \quad UD = 4Q_{<}, \quad DU = 4Q_{>}, \quad \|D\| = \|U\| = 2. \quad (\text{ESHL22})$$

Their norms follow because each takes a unit vector in its source sheet to a vector of norm two and kills the other sheet. They are nonzero for $q \geq 1$, at every heat time. Their precise source and target spaces are

$$D : \text{Ran } Q_{<} \rightarrow \text{Ran } Q_{>}, \quad U : \text{Ran } Q_{>} \rightarrow \text{Ran } Q_{<}. \quad (\text{ESHL23})$$

The carrier consequently retains the original directed expression

$$H_h = UD + t(h)Q_{>}. \quad (\text{ESHL24})$$

At $h = 0$, this is exactly `eq:general-shell-directed-Gram-support-carrier`. At $h = 1/8$, the high sheet becomes the carrier's kernel, but D still takes the positive sheet onto that kernel and U takes it back. The commutators are

$$[H_h, D] = (t(h) - 4)D, \quad [H_h, U] = (4 - t(h))U. \quad (\text{ESHL25})$$

Both follow by multiplying the two diagonal coefficients with the two matrix units. Thus the mixed maps have not been discarded by diagonalizing the form.

Their generated algebra is exactly

$$I_{\mathbb{C}[\mathcal{O}]} \otimes M_2(\mathbb{C}) \otimes I_3 : \quad E_{11} \mapsto Q_{<}, \quad E_{22} \mapsto Q_{>}, \quad E_{21} \mapsto D/2, \quad E_{12} \mapsto U/2. \quad (\text{ESHL26})$$

The four matrix-unit laws hold by ESHL22 and the sheet actions; independence follows by applying a proposed relation to one low and one high vector. This is an actual copy of the raw matrix algebra $M_2(\mathbb{C})$, with its diagonal subalgebra containing $\Psi_q(\mathcal{D})$. The faithful algebra isomorphism onto this image is R_q in ESHL45. Section 2 identifies the spectral projector P_- with a two-sided corner operation on that algebra; it does not exclude the matrix algebra identification.

There is a faithful dual-number receiver $\mathbb{C}[\eta]/\eta^2 \rightarrow B(\mathcal{H}_q)$, $1 \mapsto I$, $\eta \mapsto D$. Its multiplicativity follows from $D^2 = 0$; applying $cI + dD = 0$ first to a high vector and then to a low vector proves injectivity. These nilpotents are present in the mixed operator algebra at all times. The diagonal homomorphism Ψ_q does not itself create a nilpotent in its reduced domain $\mathbb{C}^2$.

The original rotation acts only on the third factor and commutes with D, U, H_h . The original reflection swaps the two sheet factors and reverses rotation labels, giving

$$\mathbf{S}D\mathbf{S}^{-1} = U, \quad \mathbf{S}Q_{<}\mathbf{S}^{-1} = Q_{>}, \quad \mathbf{S}H_h\mathbf{S}^{-1} = t(h)Q_{<} + 4Q_{>}. \quad (\text{ESHL27})$$

These follow directly from the source label action in `eq:general-shell-123-label-action`. The directed carrier is invariant under that reflection only when $t(h) = 4$. Its exact reflected partner, rather than an unproved invariance, is retained in ESHL27.

6. Exact finite traces, lengths, and determinant ratios

For finite $q \geq 1$, the two eigenvalues and their multiplicities give

$$\text{Tr } H_h = 3q(4+t), \quad \|H_h\|_{\text{HS}}^2 = 3q(16+t^2), \quad \det H_h = 4^{3q}t^{3q}, \quad t = -1 + 8h. \quad (\text{ESHL28})$$

The Hilbert–Schmidt formula is the trace of the square because H_h is Hermitian. The determinant is the product of all eigenvalues, including its sign. For $t \neq 0$, its absolute determinant volume and the exact ratio to the original carrier are

$$V_h^{\det} = |4t|^{3q}, \quad \frac{V_h^{\det}}{V_0^{\det}} = |t|^{3q}, \quad \det(H_h H_0^{-1}) = (-t)^{3q} = (1-8h)^{3q}. \quad (\text{ESHL29})$$

The last formula follows also from $H_h H_0^{-1} = Q_{<} - tQ_{>}$. At the critical time the determinant is zero; no logarithm of that zero is treated as finite.

The exact traceless part is

$$K_h = H_h - \frac{4+t}{2}I = \frac{4-t}{2}(Q_{<} - Q_{>}), \quad \|K_h\|_{\text{HS}}^2 = \frac{3q}{2}(4-t)^2. \quad (\text{ESHL30})$$

For the constant-generator paths $\exp(-i\sigma H_h)$, $\exp(-i\sigma K_h)$, $0 \leq \sigma \leq 1$, the Hilbert–Schmidt speed is the norm of the generator: multiplying the derivative by the inverse unitary removes the unitary factor and leaves the same Hilbert–Schmidt norm. Thus their squared lengths equal the two squared norms in ESHL28 and ESHL30. At $h = 0$ they are $51q$ and $75q/2$, exactly the ES source constants.

All per-orbit quantities have exact finite formulas, with the full factor q retained:

$$\frac{\text{Tr } H_h}{q} = 3(4+t), \quad \frac{\|H_h\|_{\text{HS}}^2}{q} = 3(16+t^2), \quad \frac{\log V_h^{\det}}{q} = 3 \log |4t| \quad (t \neq 0). \quad (\text{ESHL31})$$

For $t \neq 0$, these imply

$$\log V_h^{\det} = \frac{\log |4t|}{16+t^2} \|H_h\|_{\text{HS}}^2. \quad (\text{ESHL32})$$

For $t \neq 0, 4$, they also imply

$$\log V_h^{\det} = \frac{2 \log |4t|}{(4-t)^2} \|K_h\|_{\text{HS}}^2. \quad (\text{ESHL33})$$

At $t = -1$, these coefficients are $2 \log 2/17$ and $4 \log 2/25$, reproducing the original determinant-length identities. At $t = 4$, $K_h = 0$ while $V_h^{\det} = 16^{3q}$, so the second quotient is not defined; the original exact formulas still apply. This handles the exceptional scalar carrier explicitly.

7. The actual infinite orbit-space operator

Let $\mathcal{O}_\infty$ be any nonempty countable set of the retained labelled complement orbits, with shell labels included when orbits from different shells are combined. The empty case is the zero space already specified after ESHL13. Form the actual Hilbert completion

$$\mathcal{H}_\infty = \ell^2(\mathcal{O}_\infty) \otimes \mathbb{C}^2 \otimes \mathbb{C}^3. \quad (\text{ESHL34})$$

This definition also covers a finite orbit set. If it is infinite, the following statements concern that same labelled Hilbert space; no assertion that every arithmetic shell is occupied is being introduced. The projectors $Q_{<}, Q_{>}$, blades D, U , weight W , and Ψ_∞ are defined by the identical tensor formulas ESHL11–ESHL15 and ESHL21.

For example, if $x = (x_{O,<,m}, x_{O,>,m})$, then

$$(H_h x)_{O,<,m} = 4x_{O,<,m}, \quad (H_h x)_{O,>,m} = t(h)x_{O,>,m}. \quad (\text{ESHL35})$$

The square sum of these coordinates is bounded by $\max(16, t(h)^2) \|x\|^2$, proving that H_h is everywhere defined and bounded. Choosing a single sheet basis vector proves

$$\|H_h\| = \max(4, |t(h)|), \quad \|H_h - H_{h'}\| = 8|h - h'|. \quad (\text{ESHL36})$$

Its adjoint is itself by the real diagonal coefficients. Its quadratic form is the convergent sum of ESHL17 over all labels, so the positivity criterion and sharp uniform lower bound ESHL19 remain valid without a

dimension-dependent error. In particular, for every fixed $h > 1/8$, H_h is boundedly invertible on the entire orbit support, with

$$H_h^{-1} = \frac{1}{4}Q_{<} + \frac{1}{t(h)}Q_{>}, \quad \|H_h^{-1}\| = \frac{1}{\min(4, t(h))}. \quad (\text{ESHL37})$$

This is an explicit inverse, verified by multiplication. At $h = 1/8$, its kernel is the whole high-sheet Hilbert subspace and its range is the low-sheet subspace. The blades remain bounded with norm two and retain every identity ESHL22–ESHL27.

The infinite ambient simplex space is $\mathcal{K}_\infty = \ell^2(\mathcal{O}_\infty) \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$. Its orthogonal decomposition is the completed version of ESHL5–ESHL6, and $V_\infty = I \otimes \mathbf{U}_2^* \otimes I_3$ gives the same isometry and support projector. The zero action on the entire complementary Hilbert space is retained, including when H_h is uniformly positive on its support.

For completeness, every local matrix unit also remains an actual bounded operator. For basis labels $\alpha = (O, \epsilon, m)$, $\beta = (O', \epsilon', m')$, define

$$E_{\alpha\beta}x = e_\alpha \langle e_\beta, x \rangle. \quad (\text{ESHL38})$$

Its norm is one, its adjoint is $E_{\beta\alpha}$, and $E_{\alpha\beta}E_{\gamma\delta} = \delta_{\beta\gamma}E_{\alpha\delta}$, all by direct application to a vector. Thus the mixed maps, including maps between different orbit labels, are present in $B(\mathcal{H}_\infty)$. The particular global blades are the strong sums $D = 2 \sum_{O, m} E_{(O, >, m), (O, <, m)}$ and its adjoint, whose orthogonal source and target coordinates give norm two.

8. Exact compatibility with all finite compressions

For a finite subset $F \subset \mathcal{O}_\infty$, take all two-sheet and three-rotation labels over its orbits. Let $j_F : \mathcal{H}_F \rightarrow \mathcal{H}_\infty$ be inclusion by zero outside F , and $P_F = j_F j_F^*$. It is an isometry. The tensor formulas give

$$H_h j_F = j_F H_{h,F}, \quad j_F^* H_h j_F = H_{h,F}, \quad P_F H_h = H_h P_F, \quad \Psi_\infty(A) j_F = j_F \Psi_F(A). \quad (\text{ESHL39})$$

The same equations hold for D, U, W and the original rotation and reflection operators because none changes the orbit label. Thus these are the actual finite lifts and their exact inclusions, not new finite approximations with altered sheets or rotations.

For nested finite orbit sets F_n exhausting $\mathcal{O}_\infty$, the square-summable tail identity gives $\|(I - P_{F_n})x\| \rightarrow 0$ for every x . Therefore

$$P_{F_n} H_h P_{F_n} \longrightarrow H_h \quad \text{strongly on } \mathcal{H}_\infty. \quad (\text{ESHL40})$$

Indeed the difference on x has norm at most $\|H_h\| \|(I - P_{F_n})x\|$, since the projections commute with H_h . Positivity and its uniform bound already hold on every finite compression and on the limiting operator by ESHL17; there is no hidden loss of the lower bound in this limit. If every F_n has a nonempty complement, this convergence is not in operator norm: a unit tail vector in a sheet attaining the maximum eigenvalue magnitude has difference norm $\max(4, |t(h)|)$. The exact topology of convergence is thereby specified.

9. Infinite traces and the retained finite-volume data

When $\mathcal{O}_\infty$ is infinite, H_h is never Hilbert–Schmidt: its value on each of infinitely many low-sheet basis vectors has norm four, so the sum of their squared norms diverges. It is not trace class for the same reason, since $|H_h|$ has eigenvalue four infinitely often. For $h \geq 1/8$, its extended positive trace is $+\infty$. For $h < 1/8$, its positive and negative parts both have infinite trace, so an ordinary signed trace would be the undefined difference $\infty - \infty$. None of these facts changes its bounded operator norm or the positive lower bound at $h > 1/8$.

The ordinary finite-dimensional determinant has no direct infinite-dimensional value here. More precisely, for $h \neq 0$, the relative operator is

$$H_h H_0^{-1} = I - 8h Q_{>}. \quad (\text{ESHL41})$$

Its difference from the identity is not compact: it takes an infinite orthonormal sequence in the high sheet to an orthogonal sequence of fixed nonzero norm. Hence that difference is not trace class, and the usual trace-class Fredholm determinant is unavailable. At $h = 0$ the relative operator is the identity and its determinant is one.

The exact finite determinants and per-orbit data remain ESHL28–ESHL33 on every finite subset. Their absolute values can tend to zero, infinity, or remain one, depending on $|4t(h)|$; none of those sequence behaviours alone defines an ordinary determinant of the infinite operator. On the specified constant-fibre algebra $I_{\ell^2(\mathcal{O}_\infty)} \otimes M_6(\mathbb{C})$, there is instead the exact finite fibre trace $\tau_{\text{fibre}}(I \otimes B) = \text{Tr}_{\mathbb{C}^6} B$. It equals

$\text{Tr}_{\mathcal{H}_F}(j_F^*(I \otimes B)j_F)/|F|$ on every nonempty finite orbit set. Thus, for $t(h) \neq 0$, the explicitly defined fibre determinant is

$$\exp(\tau_{\text{fibre}}(\log |H_h|)) = |4t(h)|^3. \quad (\text{ESHL42})$$

This is a finite-fibre observable with its exact comparison map; it is not presented as an ordinary infinite determinant. The full orbit labels, all local injections, the fixed spectral representation, and the ambient kernel remain in the construction.

The result is a lift of the stated quarter heat family into the actual ES free-shell carrier, with a bounded infinite-orbit extension and a uniform positive support bound after $h = 1/8$. The full raw multiplication and quadratic receiver are constructed in ESHL45–ESHL50, the collision pairing in ESHL51–ESHL53, and the complete Figure 10 transport in ESHL54–ESHL64. These maps do not identify the ES spectral algebra or its infinite orbit trace with an arithmetic Weil pairing.

10. Exact finite verification

The reproducible script `check_es_shell_heat_lift.py` passed **217 exact symbolic checks**, recorded in `ES_SHELL_HEAT_LIFT_CHECKS.json`. They verify the raw spectral projectors and their multiplicities, the original forced metric isometry, the full raw matrix product and corner operation, the quadratic receiver and its original involution, the complete collision pairing, both blade products and nilpotence, all stated trace and determinant formulas, the original control-length constants, and the ambient kernel dimensions for one, two, and three complete orbit blocks. They also check finite inclusions retaining all six labels per orbit; the original displayed segment, triangle, and tetrahedron synthesis matrices; all nine rectangular corners and their 27 typed compositions; all 36 density ratios; the full Hilbert–Schmidt map; inter-orbit transport; and the exact ambient multiplicativity defect. The bounded infinite extension, its sharp lower bound, and strong convergence are proved in ESHL34–ESHL40 and ESHL62–ESHL63; they are not inferred from the finite checks.

11. The full raw matrix product and its exact corner operation

For every nonempty finite or countable orbit set $\mathcal{O}$, use its Hilbert space $\mathcal{H}_{\mathcal{O}} = \ell^2(\mathcal{O}) \otimes \mathbb{C}^2 \otimes \mathbb{C}^3$, where $\ell^2(\mathcal{O}) = \mathbb{C}[\mathcal{O}]$ when the set is finite. Define

$$R_{\mathcal{O}} : M_2(\mathbb{C}) \longrightarrow B(\mathcal{H}_{\mathcal{O}}), \quad R_{\mathcal{O}}(X) = I_{\ell^2(\mathcal{O})} \otimes X \otimes I_3. \quad (\text{ESHL45})$$

The tensor product laws give $R(XY) = R(X)R(Y)$, $R(I_2) = I$, and $R(X^*) = R(X)^*$. If $R(X) = 0$, fix an orbit O and rotation m ; its restriction to the span of $e_O \otimes f_0 \otimes g_m, e_O \otimes f_1 \otimes g_m$ is the matrix X , so $X = 0$. This proves faithfulness. The same restriction, together with the square-sum norm bound over all (O, m) , proves $\|R(X)\| = \|X\|$. The inverse on its image is precisely that two-dimensional restriction for any fixed (O, m) , independent of the choice. Its image is the whole blade algebra ESHL26, since it sends $E_{11}, E_{22}, E_{21}, E_{12}$ to $Q_<, Q_>, D/2, U/2$, respectively.

Entry multiplication in the original raw matrix algebra gives $P_-(X) = E_{22}XE_{11}$. Applying the multiplicative map just proved gives

$$R(P_-X) = Q_>R(X)Q_<, \quad R(A_hX) = \frac{1}{4}R(X) + \left(a(h) - \frac{1}{4}\right)Q_>R(X)Q_<. \quad (\text{ESHL46})$$

Thus the quarter operator acts on the blade algebra by the explicitly displayed linear operation on operators. This is the exact relation between the spectral endomorphism and the matrix multiplication. It is different from left multiplication by the heat carrier H_h : the former fixes the upper-right matrix cell with coefficient $1/4$, while $H_hR(E_{12}) = 4R(E_{12})$. Both actions are retained with their domains and constants. The spectral lift $\Psi(A_h)$ of ESHL12 and the operator-valued map $X \mapsto R(A_hX)$ of ESHL46 also have different domains; ESHL46 is their explicitly calculated underlying raw-cell action, not an asserted equality of those maps.

For finite q , the matrix trace on q identical orbit blocks and three identical rotation blocks gives

$$\text{Tr}(R(X)^*R(Y)) = 3q \text{tr}_2(X^*Y), \quad \text{Tr}(R(TX)^*R(TY)) = 3q \langle X, GY \rangle_{\text{HS}}. \quad (\text{ESHL47})$$

The second equality follows from the first and $T^*T = G$; TX means the four-dimensional linear metric map ESHL44 applied to the raw matrix viewed as a vector. If $\hat{A}_h$ denotes the linear operator in the right side of ESHL46, then $\hat{A}_hR(TY) = R(A_hTY) = R(TA_hY)$. Inserting this in ESHL47 gives $\text{Tr}(R(TX)^*\hat{A}_hR(TY)) = 3q \langle X, GA_hY \rangle_{\text{HS}}$. This proves the original metric and form transport alongside the product-preserving map. For an infinite orbit set, the same formulas with the finite fibre trace τ_{fibre} replace $3q$ by 3; the ordinary traces of these repeated operators need not be finite.

12. The original quadratic algebra, its involution, and the vanishing infinitesimal

For real a , define $E_a = \mathbb{C}[r]/(r^2 + a)$, with coefficient-conjugating involution $(c + dr)^\# = \bar{c} - \bar{d}r$. The basis $(1, r)$ is independent because the relation is monic of degree two. Multiplication by r on this basis has matrix $\begin{pmatrix} 0 & -a \\ 1 & 0 \end{pmatrix}$, so the regular trace is $\text{Tr}_{E_a}(c + dr) = 2c$. The original matrix receiver, proved in ESQ36 and re-established by the following multiplication, is

$$\begin{aligned} \rho_a(c + dr) &= cI_2 + d(E_{21} - aE_{12}), \\ \rho_{a,\mathcal{O}} &:= R_{\mathcal{O}}\rho_a, \quad \rho_{a,\mathcal{O}}(c + dr) = cI + \frac{d}{2}(D - aU), \quad K_{a,\mathcal{O}} := \frac{1}{2}(D - aU), \quad K_{a,\mathcal{O}}^2 = -aI. \end{aligned} \quad (\text{ESHL48})$$

For the last identity, ESHL22 gives $(D - aU)^2 = -a(DU + UD) = -4aI$. Therefore the coefficient product is $(c, d)(e, f) = (ce - adf, cf + de)$, exactly the multiplication in E_a . The unit maps to I . Applying the displayed matrix on one fixed orbit and rotation recovers c from either diagonal entry and d from the lower-left entry; hence the map is faithful even when $a = 0$. The map is an isomorphism onto the explicit commutative subalgebra $\mathbb{C}[I, K_{a,\mathcal{O}}]$, with inverse recovering those two entries. Its constants require no division by a , so the isomorphism includes the collision. This construction is initially defined over $\mathbb{Z}[a]$, before specializing the parameter or extending coefficients.

Let $\mathcal{C}$ be coefficient conjugation in the original orbit-sheet-rotation basis and $J_s = Q_{<} - Q_{>}$. The symbol J_s is the sheet-sign operator and is unrelated to the adjunction map on a Hilbert–Schmidt space in Section 13. Both J_s and $\mathcal{C}$ square to the identity and commute. The matrix-unit actions give $J_s D J_s = -D$, $J_s U J_s = -U$, and coefficient conjugation fixes the matrices D, U . Consequently

$$\rho_{a,\mathcal{O}}(p^\#) = J_s \mathcal{C} \rho_{a,\mathcal{O}}(p) \mathcal{C} J_s. \quad (\text{ESHL49})$$

On the full operator algebra the right side defines a conjugate-linear multiplicative order-two map: the two adjacent factors $\mathcal{C} J_s J_s \mathcal{C}$ cancel in a product. It is not the Hilbert adjoint, which reverses product order. On the commutative image in ESHL48 it realizes exactly the source involution $\#$; commutativity makes both product-order descriptions agree there. These facts prove the type of the involution without changing it to the positive Hilbert adjoint.

In a finite shell its trace and reflected pairing retain their full multiplicity:

$$\begin{aligned} \text{Tr}_{\mathcal{H}_q} \rho_{a,q}(p) &= 3q \text{Tr}_{E_a}(p), \\ \text{Tr}_{\mathcal{H}_q}(\rho_{a,q}(p^\#) \rho_{a,q}(q')) &= 6q(\bar{c}e + a\bar{d}f), \quad p = c + dr, \quad q' = e + fr. \end{aligned} \quad (\text{ESHL50})$$

Indeed each orbit and rotation contains the same two-dimensional regular matrix, and $p^\# q'$ has scalar coefficient $\bar{c}e + a\bar{d}f$. The finite fibre trace on an infinite orbit space gives the same two equations with $q = 1$; this does not turn the repeated operator into a trace-class operator. In the basis $(1, r)$, the full finite reflected Gram matrix is $6q \text{diag}(1, a)$. Its inertia is $(1, 1, 0)$ for $a < 0$, $(1, 0, 1)$ for $a = 0$, and $(2, 0, 0)$ for $a > 0$, because the two real diagonal entries have exactly those signs. These are the two algebra-coordinate multiplicities, alongside the actual shell-vector multiplicities ESHL18.

Set $a = a(h) = -1/4 + 2h$; then $t(h) = 4a(h)$, so the algebra collision and the carrier collision occur at the same original heat time $h = 1/8$. The two maps giving them are the quadratic algebra map ESHL48 and the weighted spectral map $H_h = W\Psi(A_h)$, respectively. They are related by their actual common blades and the formulas already proved; they have not been declared to be the same map. At the collision put

$$N := \rho_{0,\mathcal{O}}(r) = D/2, \quad N \neq 0, \quad N^2 = 0, \quad N^* N = Q_{<}, \quad N N^* = Q_{>}, \quad H_{1/8} N = 0, \quad N H_{1/8} = 4N. \quad (\text{ESHL51})$$

The two products with N^* follow from ESHL22. In particular N is nonzero and maps the low sheet isometrically onto the high sheet. Multiplication by $H_{1/8} = 4Q_{<}$ on the left kills its high-sheet image; multiplication on the right multiplies its low-sheet source by four. This proves both asymmetric mixed products and proves that the vanishing does not erase the infinitesimal.

For every pair of vectors v, w in the complete shell Hilbert space, finite or infinite, the complete carrier pairing is

$$\begin{aligned} \langle Nv, H_h Nw \rangle &= t(h) \langle Q_{<} v, Q_{<} w \rangle, \\ \langle Nv, H_{1/8} Nw \rangle &= 0, \quad \langle x, H_{1/8} Nv \rangle = \langle Nv, H_{1/8} x \rangle = 0 \quad \text{for every } x. \end{aligned} \quad (\text{ESHL52})$$

To prove the first identity, $H_h N = t(h)N$ by its high-sheet image, and $N^* N = Q_{<}$; hence $N^* H_h N = t(h)Q_{<}$. The remaining identities follow either by setting $t = 0$ or by using the self-adjointness of the carrier and $H_{1/8} N = 0$. After the collision, $\langle Nv, H_h Nv \rangle > 0$ holds exactly when $Q_{<} v \neq 0$. Before the collision it is

strictly negative for the same nonzero source. At the collision the original form vanishes on that whole image, whereas its Hilbert norm is still $\|Nv\|^2 = \|Q_{<}v\|^2$. These formulas specify the vanishing observable and the surviving norm in the same coordinates.

There is also an exact regular-trace vanishing. In a finite shell,

$$\mathrm{Tr} N^j = 0 \ (j \geq 1), \quad \det(I - zN) = 1, \quad \mathrm{Tr}(N^*N) = \mathrm{Tr}(NN^*) = 3q. \quad (\text{ESHL53})$$

For $j = 1$ the matrix diagonal is zero; for $j \geq 2$ the operator is zero. On each two-dimensional block $I - zN$ is triangular with both diagonal entries one, proving the determinant, including every rotation and orbit. The last equalities count the ranks of the two projections. In an infinite orbit set, the fibre trace gives the corresponding values zero and three, while N is not trace class and $I - zN$ has no trace-class Fredholm determinant for $z \neq 0$: its difference from the identity sends an infinite orthonormal low-sheet sequence to an orthogonal sequence of norm $|z|$. Thus neither the surviving mixed product nor the infinite trace domain is hidden by the zero regular trace. The receiver $\eta \mapsto D$ in Section 5 is the same collision algebra after the explicit scalar algebra isomorphism $\eta \mapsto 2r$, whose inverse sends $r \mapsto \eta/2$.

13. The full Figure 10 corner map and its modular transport

This section reads and proves the source result `thm:simplex-123-modular-intertwiner` displayed in Figure `fig:simplex-123-modular-linking`. Its nine-dimensional source is the direct sum $\mathbb{C}^2 \oplus \mathbb{C}^3 \oplus \mathbb{C}^4$, rather than the tensor ambient space $\mathbb{C}^3 \otimes \mathbb{C}^3$ of ESHL5. Their exact support transport will be given below, so neither ambient presentation is silently substituted for the other.

For $n = 1, 2, 3$, retain the original regular-simplex vectors $v_0, \dots, v_n$ in the n -dimensional space V_n : their inner products are one on the diagonal and $-1/n$ off the diagonal. For $n = 1$ the two vectors are exactly $1, -1$. Let $S_n : \mathbb{C}^{n+1} \rightarrow V_n$ send e_i to v_i , and define

$$P_n = I_{n+1} - \frac{1}{n+1} \mathbf{1}\mathbf{1}^*, \quad U_n = \sqrt{\frac{n}{n+1}} S_n P_n, \quad P = P_1 \oplus P_2 \oplus P_3, \quad U = U_1 \oplus U_2 \oplus U_3. \quad (\text{ESHL54})$$

The Gram entries give $S_n^* S_n = (n+1)P_n/n$. Thus $U_n^* U_n = P_n$. Its rank is n , equal to $\dim V_n$; its restriction to $\mathrm{Ran} P_n$ is therefore an onto isometry, proving $U_n U_n^* = I_{V_n}$. It follows that $U^* U = P$, $U U^* = I_{\mathcal{V}}$, where $\mathcal{V} = V_1 \oplus V_2 \oplus V_3$ has dimension six and P has rank six. The full ambient kernel is the direct sum of the three constant lines $\mathbb{C}\mathbf{1}_{n+1}$.

The full map and inverse are

$$\Phi : PM_9(\mathbb{C})P \longrightarrow B(\mathcal{V}), \quad \Phi(X) = UXU^*, \quad \Phi^{-1}(Y) = U^* Y U. \quad (\text{ESHL55})$$

The products of the two maps in either order reduce to $PXP = X$ and $IYI = Y$. For supported X, Y , $\Phi(X)\Phi(Y) = UX(U^*U)YU^* = UXPYU^* = \Phi(XY)$. Taking adjoints gives $\Phi(X)^* = \Phi(X^*)$, and $\Phi(P) = I_{\mathcal{V}}$. This proves a unital $*$ -isomorphism from the support corner with its own unit, including surjectivity and injectivity.

For $a, b, c \in \{1, 2, 3\}$, the rectangular maps and their full typed products are

$$\begin{aligned} \Phi_{ab} : P_a M_{a+1, b+1}(\mathbb{C}) P_b &\longrightarrow B(V_b, V_a), & \Phi_{ab}(X) &= U_a X U_b^*, \\ \Phi_{ab}^{-1}(Y) &= U_a^* Y U_b, & \Phi_{ab}(X) \Phi_{bc}(Y) &= \Phi_{ac}(XY), \\ \Phi_{ab}(X)^* &= \Phi_{ba}(X^*), & \dim_{\mathbb{C}} B(V_b, V_a) &= ab. \end{aligned} \quad (\text{ESHL56})$$

The inverse and adjoint follow from the same support identities; in the product the middle factor is $U_b^* U_b = P_b$, and $XP_b Y = XY$. For orthonormal bases $e_{n,k}$ of the original spaces V_n , the ab operators $|e_{a,i}\rangle\langle e_{b,j}|$ form a basis of that rectangle; their inverse images under the stated inverse form a basis of the source rectangle. The nine dimensions are therefore the full array $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$, summing to 36. Every composable rectangular path, including all six orders visiting the three sectors, is carried to its corresponding product by repeated application of the proved product law.

Retain any original positive density b_n invertible on $\mathrm{Ran} P_n$, and write $b = b_1 \oplus b_2 \oplus b_3$, $d_n = U_n b_n U_n^*$, $d = \oplus d_n$. The symbol b here denotes this density, not the quadratic parameter a of Section 12. The corner inverse is $b^{-1, P}$, with product unit P ; its image is d^{-1} , since $\Phi(b)\Phi(b^{-1, P}) = I$. On the two Hilbert–Schmidt spaces define

$$\Delta_b(X) = bXb^{-1, P}, \quad J_P(X) = X^*, \quad \Delta_d(Y) = dYd^{-1}, \quad J_{\mathcal{V}}(Y) = Y^*. \quad (\text{ESHL57})$$

These J 's are conjugate-linear maps on operator Hilbert spaces, rather than the linear sheet-sign matrix J_s in ESHL49. The map Φ is Hilbert–Schmidt unitary: for supported X, Y , cyclically moving the rectangular factors in a finite trace gives $\text{Tr}_{\mathcal{V}}(\Phi(X)^*\Phi(Y)) = \text{Tr}_{\mathbb{C}^9}(X^*PY) = \text{Tr}_{\mathbb{C}^9}(X^*Y)$. This trace identity can also be obtained by summing the equal matrix entries in the orthonormal support bases $\mathbf{U}^*e_{n,k}$.

Choose an orthonormal eigenbasis of the positive matrix d , with eigenvalues $\lambda_i > 0$. On the matrix unit E_{ij} , Δ_d acts by λ_i/λ_j , a positive real number. The corresponding inverse images under Φ give the same eigenvalues for Δ_b , by the multiplicativity already proved. Therefore Δ_b, Δ_d are positive, invertible, self-adjoint operators on their Hilbert–Schmidt spaces, and

$$\begin{aligned} \Phi\Delta_b &= \Delta_d\Phi, & \Phi J_P &= J_{\mathcal{V}}\Phi, & \Phi(-\log \Delta_b) &= (-\log \Delta_d)\Phi, \\ \Phi_{ab}(b_a X b_b^{-1, P_b}) &= d_a \Phi_{ab}(X) d_b^{-1}. \end{aligned} \tag{ESHL58}$$

The first and fourth identities are the explicit product law with the inverse. The second is the adjoint identity. For the third, define the logarithm on each positive eigenvalue; the two sides send each corresponding matrix unit to the same multiple $-\log(\lambda_i/\lambda_j)$, proving it on a basis. This also proves the equality of all imaginary powers $\Phi\Delta_b^{it} = \Delta_d^{it}\Phi$. The vectors $b^{1/2}$ and $d^{1/2}$ correspond by the same eigenbasis argument, and the antilinear maps $X \mapsto b^{-1/2, P} X^* b^{1/2}$, $Y \mapsto d^{-1/2} Y^* d^{1/2}$ equal $J\Delta^{1/2}$ and are intertwined. These are the full finite Tomita maps, as substitution on $Xb^{1/2}$ gives $X^*b^{1/2}$, and likewise on the target.

The source's specific density is retained without replacing its six eigenvalues:

$$d_1 = (1), \quad d_2 = \text{diag}(2, 3), \quad d_3 = \text{diag}(5, 7, 11), \quad b_n = \mathbf{U}_n^* d_n \mathbf{U}_n. \tag{ESHL59}$$

The modular eigenvalue on $|e_{a,i}\rangle\langle e_{b,j}|$ is exactly $\lambda_{a,i}/\lambda_{b,j}$. In a closed path $a \rightarrow b \rightarrow c \rightarrow a$ with matching intermediate basis labels, the product of these ratios is one, because its three numerators and denominators cancel pairwise; the corresponding logarithmic weights sum to zero. These statements concern the actual nine rectangles and their compositions. For the separate source density $b_n = P_n/n$, each diagonal sector has $\Delta = I$, but the (a, b) rectangle has eigenvalue b/a , obtained from $(1/a)/(1/b)$. Thus vanishing of the three diagonal modular generators does not discard the inter-sector modular data.

To transport Figure 10 into the ordered shell, specify a basis choice rather than identifying its sectors by dimension alone. Use the original orthonormal coordinate bases of V_n , and define the following unitary $C : \mathcal{V} \rightarrow \mathbb{C}^2 \otimes \mathbb{C}^3$:

source vector	$e_{1,1}$	$e_{2,1}$	$e_{2,2}$	$e_{3,1}$	$e_{3,2}$	$e_{3,3}$
image under C	$f_0 \otimes g_0$	$f_0 \otimes g_1$	$f_0 \otimes g_2$	$f_1 \otimes g_0$	$f_1 \otimes g_1$	$f_1 \otimes g_2$

(ESHL60)

The table is a bijection between orthonormal bases, so its linear extension is unitary and its inverse reverses the table. These are coordinate basis vectors, not selected simplex vertices: the latter have nonzero off-diagonal Gram entries and could not be sent to an orthonormal basis by a unitary. It is a chosen coordinate transport. In particular the one- and two-dimensional sectors together occupy the low sheet and the three-dimensional sector occupies the high sheet under this choice. This fact is specified by the table; no arithmetic rule identifying sector number with sheet or rotation has been assumed.

Let $L = CU : \mathbb{C}^9 \rightarrow \mathbb{C}^2 \otimes \mathbb{C}^3$. Then $L^*L = P$, $LL^* = I_6$, and the exact shell-corner map is

$$\Lambda(X) = C\Phi(X)C^* = LXL^*, \quad \Lambda^{-1}(Y) = L^*YL. \tag{ESHL61}$$

The proof of ESHL55 applies verbatim using the two displayed identities, and every rectangle in ESHL56 has image $C_a B(V_b, V_a) C_b^*$, where C_n is the restriction of the table to V_n . Its source and target projections are exactly $C_n C_n^*$. The transported density is CdC^* , and all identities ESHL58 follow by this unitary conjugation. To recover any previously specified shell operator Y , including $Q_<, Q_>, D, U, H_h, K_{a,\mathcal{O}}$ on one orbit, its unique supported Figure 10 source is L^*YL ; hence the relation is onto and retains every mixed operator.

This also gives a precise bridge between the two nine-dimensional ambient presentations. On one orbit, let $V_{\Delta} = \mathbf{U}_2^* \otimes I_3$ be the actual simplex support isometry from ESHL5. The map $Z = V_{\Delta}L$ obeys $Z^*Z = P$, $ZZ^* = P_2 \otimes I_3$, because both L and V_{Δ} have the initial and final projections already proved. Thus $X \mapsto ZXZ^*$ is an isomorphism of their two support corners, with inverse $Y \mapsto Z^*YZ$. It sends the Figure 10 kernel to zero and has range orthogonal to the tensor-ambient kernel; both three-dimensional kernels are retained as the specified complements. It is a partial isometry of ambient spaces and a unitary on the six-dimensional supports, not an unlabelled identification of the ambient coordinates.

The exact effect of using the whole nine-dimensional matrix algebra can also be retained. Extend the same formula as the linear map $\Lambda^{\text{amb}}(X) = LXL^*$ for every $X \in M_9(\mathbb{C})$. For arbitrary X, Y , direct subtraction

gives

$$\Lambda^{\text{amb}}(XY) - \Lambda^{\text{amb}}(X)\Lambda^{\text{amb}}(Y) = LX(I - P)YL^*. \quad (\text{ESHL64})$$

Indeed the second term is $LXPYL^*$ because $L^*L = P$. Thus ESHL64 records exactly the part of a composition passing through the retained ambient kernel. It vanishes on the support corner, recovering the proved algebra isomorphism, but that corner hypothesis has not been omitted for arbitrary ambient matrices.

For an explicit nonzero instance, choose any unit shell vector e , put $u = L^*e$, and take the unit vector $k = (1, 1, 0, \dots, 0)/\sqrt{2}$ in the segment's constant-line kernel. Then $Lu = e$, $Lk = 0$, and u, k are orthogonal unit vectors. The ambient operators $X = uk^*$, $Y = ku^*$ each have zero image under Λ^{amb} , but $XY = uu^*$ has image $ee^* \neq 0$. This computes a composition through the retained kernel, with its nonzero support return, rather than deleting it.

Finally tensor these exact maps over the retained orbit set. Write

$$\begin{aligned} \mathcal{A}_{\mathcal{O}} &= \ell^2(\mathcal{O}) \otimes \mathbb{C}^9, & P_{\mathcal{O}}^{123} &= I \otimes P, & L_{\mathcal{O}} &= I \otimes L, \\ \Lambda_{\mathcal{O}} : P_{\mathcal{O}}^{123} B(\mathcal{A}_{\mathcal{O}}) P_{\mathcal{O}}^{123} &\longrightarrow B(\mathcal{H}_{\mathcal{O}}), & \Lambda_{\mathcal{O}}(X) &= L_{\mathcal{O}} X L_{\mathcal{O}}^*, & \Lambda_{\mathcal{O}}^{-1}(Y) &= L_{\mathcal{O}}^* Y L_{\mathcal{O}}. \end{aligned} \quad (\text{ESHL62})$$

The operators are bounded, with $L_{\mathcal{O}}^* L_{\mathcal{O}} = P_{\mathcal{O}}^{123}$, $L_{\mathcal{O}} L_{\mathcal{O}}^* = I$, by the square-sum coordinate formula. The multiplication, inverse, and adjoint proofs of ESHL55 now use bounded-operator composition with these same identities and prove a unital $*$ -isomorphism of the full corners. It includes inter-orbit operators: $|O\rangle\langle O'| \otimes X_{ab}$ maps to $|O\rangle\langle O'| \otimes C_a \Phi_{ab}(X_{ab}) C_b^*$. Their products multiply the orbit matrix units and the original rectangles simultaneously. Thus this map retains more than the constant-fibre algebra. Finite complete-orbit inclusions commute with $L_{\mathcal{O}}$ and its adjoint by their coordinate formulas, so their compressions commute with $\Lambda_{\mathcal{O}}$ as well.

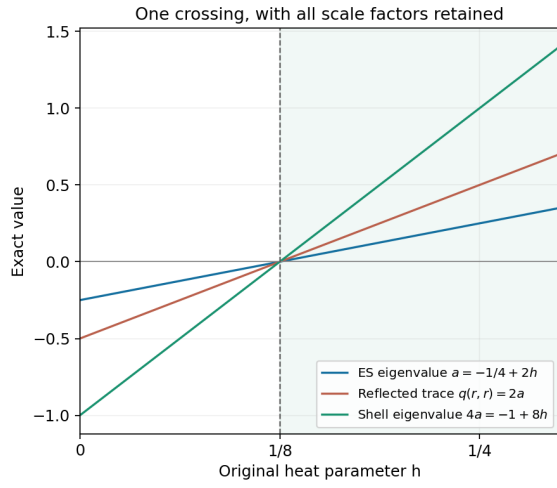
The same map restricts to a unitary between the Hilbert–Schmidt spaces of these support corners. Indeed in the orthonormal support basis $e_{\mathcal{O}} \otimes L^*(f_{\epsilon} \otimes g_m)$, the matrix entries of X equal the entries of $\Lambda_{\mathcal{O}}(X)$ in the original shell basis. Equality of their square sums proves both the Hilbert–Schmidt isometry and its surjectivity. Define the bounded positive support-invertible densities $B_{\mathcal{O}} = I \otimes b$ and $D_{\mathcal{O}} = I \otimes C d C^*$. They correspond under $\Lambda_{\mathcal{O}}$. Their inverses have the same tensor form. Consequently the conjugation operators and adjunction on the complete Hilbert–Schmidt spaces satisfy

$$\begin{aligned} \Lambda_{\mathcal{O}}(B_{\mathcal{O}} X B_{\mathcal{O}}^{-1, P}) &= D_{\mathcal{O}} \Lambda_{\mathcal{O}}(X) D_{\mathcal{O}}^{-1}, \\ \Lambda_{\mathcal{O}}(X^*) &= \Lambda_{\mathcal{O}}(X)^*, \\ \Lambda_{\mathcal{O}}(-\log \Delta_{B_{\mathcal{O}}}) &= (-\log \Delta_{D_{\mathcal{O}}}) \Lambda_{\mathcal{O}}. \end{aligned} \quad (\text{ESHL63})$$

For a complete proof of the operator domains, choose the six finite-fibre density eigenvectors. The complete Hilbert–Schmidt basis is indexed by pairs $(O, i), (O', j)$. Conjugation acts by λ_i/λ_j , independent of the two orbit labels. These ratios lie between $\lambda_{\min}/\lambda_{\max} > 0$ and $\lambda_{\max}/\lambda_{\min} < \infty$. Thus Δ , its inverse, its square root, and its logarithm are all bounded everywhere on the Hilbert–Schmidt space. The logarithm identity follows first on each such basis vector by its identical ratio, then on all square-summable series by boundedness. Adjunction reverses the two labels and conjugates their coefficients, proving its antiunitarity and the displayed intertwining. For the original density ESHL59 the ratio bounds are $1/11, 11$ and the logarithmic norm is $\log 11$, attained on the $1, 11$ matrix unit.

When the orbit set is infinite, $D_{\mathcal{O}}$ is not a trace-class density and $D_{\mathcal{O}}^{1/2}$ is not a Hilbert–Schmidt vector: each of infinitely many orbits contributes the positive sum of its six eigenvalues. ESHL63 is an exact statement about the bounded density-conjugation operators on the complete Hilbert–Schmidt space. It does not introduce a nonexistent finite-trace state vector. On each finite orbit compression the density square root is a Hilbert–Schmidt vector and the finite Tomita computation after ESHL58 applies with the same maps. This states the finite and infinite domains while retaining their exact connecting inclusions.

The source edition is available as the 488-page ES reader and its original TeX source and build archive. The TeX entry is `reader/erdos_straus_project_reader.tex`; the source identity and exact reading locations appear in the source guide.



The same lift on finitely or infinitely many orbits

$$\mathcal{H} = \mathcal{L}^2(\mathcal{O}) \otimes \mathbb{C}^2 \otimes \mathbb{C}^3$$

$$\Psi(P_+) = Q_-, \quad \Psi(P_-) = Q_+$$

$$H_h = 4Q_- + (-1 + 8h)Q_+$$

$$\langle v, H_h v \rangle \geq \min(4, -1 + 8h) \|v\|^2$$

for every $v \in \mathcal{H}$ and $h > 1/8$.

At $h = 1/8$: the negative-sector form vanishes;
 the two square-zero blades and their mixed products remain.

Figure 5: The three exact changing coefficients and their common crossing. The quarter coefficient is a , its reflected trace is $2a$, and the shell coefficient is $4a$. ESHL12–ESHL25 and ESHL34–ESHL42 prove the finite and infinite maps, sharp bound, surviving blades and trace scope.

The actual Weil correction has the ES three-plus-one form

This calculation identifies the finite correction in the full explicit formula with the original ES quarter form. It uses actual compactly supported smooth tests, proves a section onto all four evaluation coordinates, and retains the kernel. It does not identify this finite correction with the entire Weil pairing. The needed endpoint conventions and completion identity are proved in the complete heat endpoint sign bridge proof, HEB11–HEB17 and HEB23–HEB36; the exact original ES metric and its isometry are proved in the complete es quarter heat trace derivation proof, ESQ9–ESQ27. Every comparison needed here is calculated below.

1. Four evaluation coordinates of an admissible test

Let $\mathcal{T} = C_c^\infty(\mathbb{R}; \mathbb{C})$ and retain

$$M_f(s) = \int_{\mathbb{R}} f(v) e^{-(s-1/2)v} dv, \quad (s_1, s_2, s_3, s_4) = (0, 1, 1/2 + i/2, 1/2 - i/2). \quad (\text{WEC1})$$

Set $t_j = s_j - 1/2$, $k_j(v) = e^{-t_j v}$, and define

$$L : \mathcal{T} \longrightarrow \mathbb{C}^4, \quad Lf = (A, B, C, D) = (M_f(s_1), M_f(s_2), M_f(s_3), M_f(s_4)). \quad (\text{WEC2})$$

The integral defines an entire function of s , since its derivatives converge uniformly on each compact set and the support in v is compact.

Here is a concrete linear section for every four-tuple. Choose the normalized positive even smooth bump ψ on $(-1, 1)$ in HEB33 and put

$$G_{ij} = \int_{\mathbb{R}} \psi(v) k_i(v) \overline{k_j(v)} dv, \quad S(y)(v) = \psi(v) \sum_{j=1}^4 (G^{-1}y)_j \overline{k_j(v)}. \quad (\text{WEC3})$$

The matrix G is positive definite. Indeed, for $z \in \mathbb{C}^4$, $z^* G z = \int \psi(v) |\sum_i \bar{z}_i k_i(v)|^2 dv$. If this equals zero, the continuous sum vanishes on $(-1, 1)$, where $\psi > 0$. Its derivatives of orders 0, 1, 2, 3 at zero give $\sum_i \bar{z}_i (-t_i)^m = 0$. The four numbers $-t_i$ are distinct, so this Vandermonde matrix is invertible: its determinant is $\prod_{i < j} ((-t_j) - (-t_i)) \neq 0$, obtained by the usual polynomial alternating-factor expansion (both sides have the same degree and leading coefficient). Thus $z = 0$. This proves positive definiteness and existence of the displayed inverse.

The section $S(y)$ is smooth and compactly supported because it is a finite sum of smooth exponentials times ψ . Direct integration gives

$$LS(y) = GG^{-1}y = y. \quad (\text{WEC4})$$

Consequently L is onto, and $\mathcal{T} = \ker L \oplus S(\mathbb{C}^4)$ as complex vector spaces, with projection SL . This proves that no hidden condition on admissible tests prevents the four coordinates used below.

2. The exact endpoint gain and its full-form compensation

At heat time $h = 1/4$, the moving endpoint polynomial is $s(s-1) + 1/2$, whose roots are s_3, s_4 . The two forms on actual tests are therefore

$$\begin{aligned} P_0(f, g) &= \overline{A_f} B_g + \overline{B_f} A_g, \\ P_{1/4}(f, g) &= \overline{C_f} C_g + \overline{D_f} D_g. \end{aligned} \quad (\text{WEC5})$$

This follows from the endpoint involution $s^\# = 1 - s$ together with coefficient conjugation: it exchanges 0, 1 and fixes each critical-line point in the reflected evaluation. Define the finite gain form

$$\mathcal{C} = P_{1/4} - P_0. \quad (\text{WEC6})$$

In the actual four values its quadratic expression is

$$\mathcal{C}(f, f) = |C|^2 + |D|^2 + \frac{1}{2}|A - B|^2 - \frac{1}{2}|A + B|^2. \quad (\text{WEC7})$$

Expansion of the last two squares gives $-\overline{A}B - \overline{B}A$, proving the formula. In coordinates $(C, D, (A-B)/\sqrt{2}, (A+B)/\sqrt{2})$, its matrix is $\text{diag}(1, 1, 1, -1)$. Hence its quotient has signature $(3, 1)$. More precisely its radical on $\mathcal{T}$ is exactly $\ker L$: the inclusion follows from (WEC5), while if $Lf \neq 0$, nondegeneracy of this displayed matrix supplies a vector y with nonzero pairing against Lf , and the admissible test $S(y)$ detects it by (WEC4).

Let $\mathcal{A}$ and $\mathcal{P}$ denote the original archimedean and rational-prime terms. The full Weil form remains

$$W = P_0 + \mathcal{A} - \mathcal{P} = P_{1/4} + (\mathcal{A} - \mathcal{C}) - \mathcal{P}. \quad (\text{WEC8})$$

This is the exact completion correction, not a discarded residual: logarithmic differentiation of $\Lambda_{1/4} = 2\xi/[s(s-1) + 1/2]$ adds $q'_0/q_0 - q'_{1/4}/q_{1/4}$, and the residues at the four simple roots yield $P_0 - P_{1/4} = -\mathcal{C}$. Thus the correction subtracted from the archimedean term is the same nondegenerate $(3, 1)$ quotient just computed; its negative has signature $(1, 3)$.

3. An explicit isometry to the original ES quarter form

Retain the original raw ordering $(E_{11}, E_{12}, E_{21}, E_{22})$ and the forced ES metric and initial quarter operator

$$G_{\text{ES}} = \begin{pmatrix} 1/16 & 0 & 0 & -7/144 \\ 0 & 2/9 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -7/144 & 0 & 0 & 1/16 \end{pmatrix}, \quad A_0 = \text{diag}(1/4, 1/4, -1/4, 1/4). \quad (\text{WEC9})$$

They are the original source matrices used in ESQ9 and ESQ2. Define the linear map $J : \mathbb{C}^4 \rightarrow M_2(\mathbb{C})$ by

$$\begin{aligned} x_{11} &= 12C + 3\sqrt{2}D, & x_{22} &= 12C - 3\sqrt{2}D, \\ x_{12} &= 3(A - B), & x_{21} &= \sqrt{2}(A + B). \end{aligned} \quad (\text{WEC10})$$

It is invertible, with

$$C = \frac{x_{11} + x_{22}}{24}, \quad D = \frac{x_{11} - x_{22}}{6\sqrt{2}}, \quad A = \frac{1}{2} \left(\frac{x_{21}}{\sqrt{2}} + \frac{x_{12}}{3} \right), \quad B = \frac{1}{2} \left(\frac{x_{21}}{\sqrt{2}} - \frac{x_{12}}{3} \right). \quad (\text{WEC11})$$

Substitution in both orders gives each of the four starting coordinates. From the displayed matrix multiplication,

$$x^* G_{\text{ES}} A_0 x = \left| \frac{x_{11} + x_{22}}{24} \right|^2 + \left| \frac{x_{11} - x_{22}}{6\sqrt{2}} \right|^2 + \frac{|x_{12}|^2}{18} - \frac{|x_{21}|^2}{4}. \quad (\text{WEC12})$$

For example the first two squares expand to $(|x_{11}|^2 + |x_{22}|^2)/64 - (7/288) \text{Re}(\overline{x_{11}}x_{22})$, exactly the two diagonal entries and mixed term of $G_{\text{ES}}A_0$. Substituting (WEC10) in (WEC12) gives (WEC7). Polarization, or the same expansion in two slots, therefore proves the isometry

$$\boxed{\mathcal{C}(f, g) = (JLf)^* G_{\text{ES}} A_0 (JLg)}. \quad (\text{WEC13})$$

The map JL is onto, has kernel $\ker L$, and has explicit section SJ^{-1} . This proves an isomorphism of the nondegenerate Hermitian quotient $\mathcal{T}/\ker L$ with the actual complexified ES quarter form, including its coordinates, constants and sign. It is a linear form isomorphism, not an asserted ring isomorphism of a test-function multiplication with raw matrix multiplication.

The form identity links the tetrahedral description to an actual term of the Weil formula: it is the finite endpoint-change correction. It does not identify the remaining $\mathcal{A} - \mathcal{P}$, or the full form W , with this four-dimensional quotient. Equation (WEC8) records exactly where that quotient enters.

4. Actual positive and negative gain tests

The section (WEC3) supplies explicit admissible tests for both signs. For

$$f_- = S(1, 1, 0, 0), \quad f_+ = S(1, -1, 0, 0), \quad (\text{WEC14})$$

one has $P_{1/4}(f_{\pm}, f_{\pm}) = 0$, while $P_0(f_-, f_-) = 2$ and $P_0(f_+, f_+) = -2$. Thus $\mathcal{C}(f_-, f_-) = -2$ and $\mathcal{C}(f_+, f_+) = 2$. These are tests of the endpoint gain; they do not give negative values of the full classical Weil form. The bump derivative in HEB33–HEB36 supplies a single test whose endpoint term passes from negative, through a nonzero nilpotent with zero pairing, to positive, with the exact opposite change in the archimedean term.

5. The actual test involution and its exact ES intertwiner

The involution on tests is $f^\#(v) = \overline{f(-v)}$. Changing variables in (WEC1) gives $M_{f^\#}(s) = \overline{M_f(1 - \bar{s})}$, so its action on the four coordinates is

$$(A, B, C, D)^\# = (\bar{B}, \bar{A}, \bar{C}, \bar{D}). \quad (\text{WEC15})$$

Under the map J in (WEC10), the transported action on raw coordinates is $T_0 x = \text{diag}(1, -1, 1, 1)\bar{x}$: only $A-B$ changes sign. The original ES anti-linear quarter involution is instead $T_{\text{ES}} x = (4A_0)\bar{x} = \text{diag}(1, 1, -1, 1)\bar{x}$. Their exact comparison is

$$T_{\text{ES}} = C_0 T_0, \quad C_0 = \text{diag}(1, -1, -1, 1). \quad (\text{WEC16})$$

On raw matrices C_0 is conjugation by $\text{diag}(1, -1)$, as direct multiplication shows. Both involutions and their relation are thereby retained.

There is also an isometry that intertwines the two anti-linear actions. Put

$$U = \text{diag}(1, i, i, 1), \quad \hat{J} = UJ. \quad (\text{WEC17})$$

The only off-diagonal entries of $G_{\text{ES}} A_0$ connect the first and fourth positions, on which U is the identity. On the other two positions multiplication by i preserves the absolute squares. Hence $U^*(G_{\text{ES}} A_0)U = G_{\text{ES}} A_0$. For the anti-linear comparison the matrix is $U \text{diag}(1, -1, 1, 1) \bar{U}^{-1} = \text{diag}(1, 1, -1, 1)$, by multiplication of its four diagonal entries. Consequently

$$\mathcal{C}(f, g) = (\hat{J}L f)^* G_{\text{ES}} A_0 (\hat{J}L g), \quad \hat{J}L(f^\#) = T_{\text{ES}} \hat{J}L(f). \quad (\text{WEC18})$$

The inverse and section are $J^{-1}U^{-1}$ and $SJ^{-1}U^{-1}$. Explicitly $\hat{J}$ changes just two entries of (WEC10) to $x_{12} = 3i(A - B)$, $x_{21} = i\sqrt{2}(A + B)$. Thus both the Hermitian form and the specified involution are preserved by a proved map. The map U is not a raw-matrix algebra automorphism: it fixes E_{11} , while $U(E_{12})U(E_{21}) = -E_{11}$. No multiplication compatibility beyond its stated linear and involution properties is assumed.

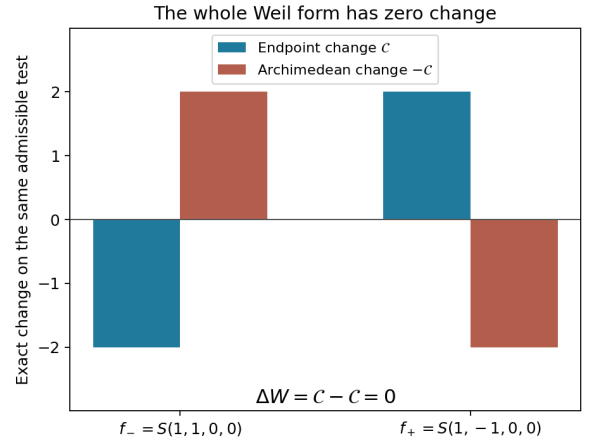
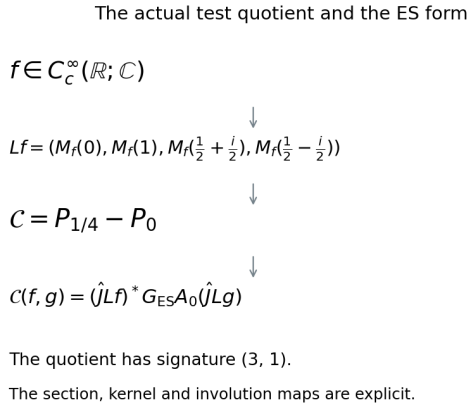


Figure 6: The actual four-value correction, its exact ES isometry, and its cancellation in the full Weil form. WEC1–WEC18 give all maps, the section and the two admissible tests.

The signed eight-state completion and its exact heat receivers

This note compares the original seven-point affine chart with its eight-state signed completion, keeping both original involutions. It calculates the complete trace forms, identifies the missing infinity state, and constructs a target path whose marked-root quotient is exactly the HEB heat family. The whole eight-state form is calculated on that path, including its length-four collision algebra.

The original sources actually read are *The ES–Fable inverse correspondence in the original weighted conductor*, FC36–FC47, and *The finite signed-root completion with all original fibre algebras*, FS1–FS27 and FS30–FS31. The latter pinned TeX has the same bytes as the local source used here. Its reciprocal-coordinate construction, chart transitions, trace and local algebra proofs are retained. The receiving HEB equations are HEB15–HEB20 and HEB31–HEB32 of the complete heat endpoint sign bridge proof, read in full during the quarter comparison. The original heat clock also has the public proof *The original heat trajectories*, eq:ifh-target-flow. All new comparisons and their hypotheses are proved below.

1. The original coordinates and the two completion charts

Write the first source coordinate called a in FS1 as α , solely to distinguish it from the HEB coefficient a_H . This is the identity renaming of that coordinate. The other source coordinates remain (y, z, w) . The original factors are

$$\begin{aligned} b &= i + \alpha y, & c &= -i + 2\alpha y + \alpha^2 z, \\ d &= -iy - \alpha(iz + 2y^2) - \alpha^2 yz, & e &= 2z - 7iy^2 + \alpha w, \\ f &= iw + 3iy^3 - 4yz + \alpha(6iy^2z + wy + 4y^4) + 2\alpha^2 y^3 z. \end{aligned} \quad (\text{ESH1})$$

The target is $\mathbf{U} = (A, B, C, D)$, where

$$P = (\alpha c, \alpha e + bd, \alpha f + be, bf), \quad H_{\mathbf{U}}(U, V) = AU^4 + U^3V + BU^2V^2 + CUV^3 + DV^4. \quad (\text{ESH2})$$

The retained factor identities are $\alpha d + bc = 1$ and $\alpha^3 f - \alpha^2 be + \alpha b^2 d - b^3 c = 1$. They follow by substituting (ESH1), and are the exact identities proved in FS1 and FC44–FC44b. Thus $(\alpha U + bV)(cU^3 + dU^2V + eUV^2 + fV^3) = H_{\mathbf{U}}$, with resultant one. The cubic coefficient is fixed at one; it is not another free parameter.

Use R for the finite marked root called r in FS2, and T for its reciprocal factor called t . The finite completed chart and the infinity chart are

$$F_{\mathbf{U}}(R) = AR^4 + R^3 + BR^2 + CR + D, \quad F_{\mathbf{U}}(R) = 0, \quad T^2 = F'_{\mathbf{U}}(R), \quad (\text{ESH3})$$

$$k_{\mathbf{U}}(v) = A + v + Bv^2 + Cv^3 + Dv^4, \quad k_{\mathbf{U}}(v) = 0, \quad \Theta^2 = -k'_{\mathbf{U}}(v). \quad (\text{ESH4})$$

On their overlap the maps are

$$v = R^{-1}, \quad \Theta = -T/R, \quad R = v^{-1}, \quad T = -\Theta/v. \quad (\text{ESH5})$$

They are inverse by substitution. Differentiating $k(v) = v^4 F(1/v)$ at a root gives $k'(v) = -v^2 F'(1/v)$, which verifies the squared-coordinate relation and its sign. On the original finite factor chart,

$$R = -b/\alpha = -y - i/\alpha, \quad T = 1/\alpha, \quad \alpha^2 F'_{\mathbf{U}}(R) = 1. \quad (\text{ESH6})$$

The finite completion allows $T = 0$; it does not declare $1/\alpha$ to be an invertible coordinate at that boundary.

For reference, the inverse factor formulas on $T \neq 0$ are

$$\begin{aligned} \alpha &= T^{-1}, & b &= -R/T, & c &= AT, \\ d &= (1 + AR)T, & e &= (B + R + AR^2)T, & f &= (C + BR + R^2 + AR^3)T. \end{aligned} \quad (\text{ESH7})$$

They give $\alpha d + bc = 1$ directly. Their four product coefficients are A, B, C , and $-R(C + BR + R^2 + AR^3) = D$, the last by $F(R) = 0$. Their resultant is $T^{-2} F'(R) = 1$. Thus these formulas prove the factor reconstruction with its original reciprocal coordinate.

At $v = 0$, the infinity equations give

$$A = 0, \quad \Theta^2 = -1, \quad \Theta = -i \text{ is included, } \Theta = i \text{ is omitted.} \quad (\text{ESH8})$$

Indeed the infinity factor chart has $b = \Theta^{-1}$; the original source at $\alpha = 0$ has $b = i$ by (ESH1), so its value is $\Theta = -i$. The other sign has $b = -i$ and does not belong to that original chart. This is the precise meaning of the missing eighth state.

The missing infinity point is unramified. The relative derivative matrix of (ESH4) with respect to (v, Θ) has determinant $2\Theta k'(v)$; at $v = 0$ its value is $2\Theta \neq 0$, since $k'(0) = 1$. This matrix is invertible there, proving the vanishing of the relative differential module and the absence of ramification in that chart. On the finite chart the same determinant is $2TF'(R) = 2T^3$, so its ramification support is $T = 0$. The infinity points never belong to that support. These calculations reproduce the relevant part of FS16–FS17 and prove that the omitted infinity sign is not the nilpotent collision.

2. The entire completed fibre at the original seven-point target

At

$$\mathbf{U}_* = (0, 0, -1, 0), \quad H_*(U, V) = UV(U - V)(U + V), \quad F_*(R) = R^3 - R, \quad (\text{ESH9})$$

the projective roots are $\infty, 0, 1, -1$, all simple. At the finite roots the derivative values are

$$F'_*(0) = -1, \quad F'_*(1) = F'_*(-1) = 2. \quad (\text{ESH10})$$

The completed complex fibre algebra is therefore exactly

$$\mathcal{A}_* = \frac{\mathbb{C}[T_0]}{(T_0^2 + 1)} \times \frac{\mathbb{C}[T_+]}{(T_+^2 - 2)} \times \frac{\mathbb{C}[T_-]}{(T_-^2 - 2)} \times \frac{\mathbb{C}[\Theta]}{(\Theta^2 + 1)}. \quad (\text{ESH11})$$

To prove the finite splitting, the idempotents in $\mathbb{C}[R]/(R^3 - R)$ are

$$e_0 = 1 - R^2, \quad e_+ = (R^2 + R)/2, \quad e_- = (R^2 - R)/2. \quad (\text{ESH12})$$

They evaluate as the three separate indicator functions at $0, 1, -1$. Their products, sum and squares are consequently the orthogonal idempotent identities: a polynomial remainder of degree at most two vanishing at the three distinct roots is zero. Applying those idempotents to the relation $T^2 = 3R^2 - 1$ gives the first three factors in (ESH11). The infinity root is the disjoint fourth root of the projective root fibre, and (ESH8) gives its fourth factor. Each quadratic has two distinct roots, so each is reduced and of dimension two. The total dimension is eight and every geometric point is reduced.

The eight points are

$$(0, \pm i), \quad (1, \pm\sqrt{2}), \quad (-1, \pm\sqrt{2}), \quad (\infty, \Theta = \pm i). \quad (\text{ESH13})$$

The first entry marks the projective root; the second is its retained sign coordinate. The source chart includes the first six and $(\infty, -i)$. At the latter, FS13 gives the original point $(\alpha, y, z, w) = (0, 0, i/2, 0)$. At (∞, i) , the factor coordinates are exactly

$$(\alpha, b, c, d, e, f) = (0, -i, i, 0, -i, 0), \quad (\text{ESH14})$$

by (ESH8) and the product coefficients. This is the omitted sign in FC44b, not an additional root or an identification of two labels.

3. Both original involutions, with coefficient conjugation separate

The completed sign deck involution is

$$\Sigma : (R, T) \mapsto (R, -T), \quad (v, \Theta) \mapsto (v, -\Theta). \quad (\text{ESH15})$$

Its formulas agree through (ESH5), preserve the equations, and square to the identity. They are the completed version of the simultaneous factor sign in FC43–FC44. Let κ conjugate coefficients and fix the displayed coordinate generators; it is defined on a fibre over real (A, B, C, D) . The conjugate-deck involution is $\#_\Sigma = \Sigma\kappa$, so it negates T, Θ and fixes the real root variable R , while conjugating scalars.

The other original involution is

$$J(\alpha, y, z, w) = (-\alpha, -y, z, -w). \quad (\text{ESH16})$$

Substituting in (ESH1) sends the factor tuple to $(-\alpha, b, c, -d, e, -f)$, so its target map and completion charts are

$$J_{\text{out}}(A, B, C, D) = (-A, -B, C, -D), \quad (\text{ESH17})$$

$$J : (R, T) \mapsto (-R, -T), \quad (v, \Theta) \mapsto (-v, \Theta). \quad (\text{ESH18})$$

The first formula follows from (ESH6), and the second from (ESH5). To verify the equations on their whole domains, one has $F_{J_{\text{out}}\mathbf{U}}(-R) = -F_{\mathbf{U}}(R)$, $F'_{J_{\text{out}}\mathbf{U}}(-R) = F'_{\mathbf{U}}(R)$, and the identical two identities for k at $-v$. Thus (ESH18) is the full map, including completion points. The target $\mathbf{U}_*$ is fixed, so $\#_J = J\kappa$ is an involution of this fibre. It is different from $\#_\Sigma$.

4. The complete trace forms and their signatures

For the quadratic algebra $K_\mu = \mathbb{C}[T]/(T^2 - \mu)$, with real $\mu \neq 0$, multiplication in the unchanged basis $(1, T)$ gives

$$\text{Tr}_{K_\mu}(c + dT) = 2c. \quad (\text{ESH19})$$

Indeed multiplication by T has matrix $\begin{pmatrix} 0 & \mu \\ 1 & 0 \end{pmatrix}$. Hence the coefficient-conjugation and conjugate-deck Hermitian forms have respective Gram matrices

$$[Q_\kappa] = \text{diag}(2, 2\mu), \quad [Q_{\#_\Sigma}] = \text{diag}(2, -2\mu), \quad Q_\iota(x, y) = \text{Tr}(\iota(x)y). \quad (\text{ESH20})$$

These formulas follow also by multiplying $(\bar{c} + \bar{d}T)(e + fT)$ or $(\bar{c} - \bar{d}T)(e + fT)$ and taking twice the scalar coefficient. They verify the full forms in both arguments.

In the factor order of (ESH11), with each basis $(1, T)$ retained, the whole Gram matrices are

$$[Q_\kappa] = \text{diag}(2, -2, 2, 4, 2, 4, 2, -2), \quad (\text{ESH21})$$

$$[Q_{\#_\Sigma}] = \text{diag}(2, 2, 2, -4, 2, -4, 2, 2). \quad (\text{ESH22})$$

Both happen to have inertia $(6, 2, 0)$, but their negative subspaces belong to different factors. Here and below inertia lists positive, negative and radical dimensions. The complex bilinear trace without an antilinear involution has no ordered signature; (ESH21) is its coefficient-conjugation Hermitian version, or its real coefficient restriction.

The native involution has a different complete matrix. In the ordered basis

$$(e_0, e_0T, e_+, e_-, e_+T, e_-T, e_\infty, e_\infty\Theta)$$

one has

$$[Q_{\#_J}] = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \oplus \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & -4 \\ -4 & 0 \end{pmatrix} \oplus \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}. \quad (\text{ESH23})$$

For proof, (ESH18) fixes the root 0 and negates its sign, interchanges e_+, e_- and negates T , and fixes the infinity coordinate Θ . Thus, for example, $\#_J(e_+T) = -e_-T$, whose pairing with e_-T is $-\text{Tr}(2e_-) = -4$. The mixed products between different root factors vanish, and (ESH20) gives the remaining blocks. Each off-diagonal two-by-two block has one eigenvalue of each sign. Therefore

$$\text{inertia}(Q_{\#_J}) = (5, 3, 0). \quad (\text{ESH24})$$

Neither calculation assumes that one of these finite forms is the global Weil form.

5. Omitting precisely one of the eight states

Let $m = (\infty, i)$ be the missing point. Its idempotent, supported in the infinity factor, is

$$e_m = (0, 0, 0, (1 - i\Theta)/2), \quad e_n = (0, 0, 0, (1 + i\Theta)/2), \quad (\text{ESH25})$$

where $n = (\infty, -i)$ is the retained point. Evaluation at $\Theta = \pm i$ verifies that these are the respective indicator functions. Therefore the exact original seven-state quotient is

$$q : \mathcal{A}_* \longrightarrow \mathcal{A}_7 := \mathcal{A}_*/\mathbb{C}e_m, \quad \mathcal{A}_* \cong \mathcal{A}_7 \times \mathbb{C}. \quad (\text{ESH26})$$

The last factor is evaluation at m . Its inverse inserts that value at the omitted coordinate, while the canonical nonunital section of q inserts zero there. These maps are the full coordinate maps, not a count alone.

Coefficient conjugation in (ESH25) gives the exact stability results

$$\#_\Sigma(e_m) = e_m, \quad \Sigma(e_m) = \kappa(e_m) = e_n, \quad J(e_m) = e_m, \quad \#_J(e_m) = e_n. \quad (\text{ESH27})$$

Thus $\#_\Sigma$ descends to $\mathcal{A}_7$, whereas Σ , κ , and $\#_J$ individually do not. Pure native J does descend. For the descended conjugate-deck trace one has, on arbitrary lifts f, g ,

$$Q_{\#_\Sigma, \tau}(qf, qg) = Q_{\#_\Sigma, *}(f, g) - \overline{f(m)}g(m). \quad (\text{ESH28})$$

This follows from the direct-product trace and the fact that m is fixed by the point action of $\#_\Sigma$. The removed summand is positive. Consequently

$$\text{inertia}(Q_{\#_\Sigma, \tau}) = (5, 2, 0). \quad (\text{ESH29})$$

In particular the two explicit retained vectors e_+T, e_-T still have value -4 , and are mutually orthogonal by (ESH22). Removing the omitted infinity point has not removed either negative direction.

For the native Hermitian form there is instead the restriction to the retained corner $(1 - e_m)\mathcal{A}_*$. Its infinity line $\mathbb{C}e_n$ is radical because $\#_J(e_n) = e_m$, whose product with every retained element is zero. The other six-dimensional block has inertia $(4, 2, 0)$ from (ESH23), so this restricted form has

$$\text{inertia}(Q_{\#_J}|_{(1-e_m)\mathcal{A}_*}) = (4, 2, 1). \quad (\text{ESH30})$$

It is not a trace form of a descended native involution. Indeed $Q_{\#_J}(e_m, e_n) = \text{Tr}(e_n) = 1$, so adding a representative from the killed ideal can change the original pairing. This proves directly why that full form cannot be defined on the quotient by choosing arbitrary lifts.

The exact native-stable quotient generated by this omission is also determined. The smallest $\#_J$ -stable ideal containing $\mathbb{C}e_m$ is

$$\mathcal{I}_{\text{stable}} = \mathbb{C}e_m \oplus \mathbb{C}e_n = e_\infty\mathcal{A}_*. \quad (\text{ESH30a})$$

Every stable ideal containing e_m must contain its image e_n ; conversely their sum is an ideal and is stable by (ESH27). This proves minimality and the universal quotient property: any algebra quotient with an induced native involution and killing e_m factors uniquely through $\mathcal{A}_6 = \mathcal{A}_*/\mathcal{I}_{\text{stable}}$, the product of the three finite root factors. Its native trace form has inertia $(4, 2, 0)$ by the first three blocks of (ESH23). The restricted seven-dimensional form (ESH30) factors through the further projection $\mathcal{A}_7 \rightarrow \mathcal{A}_6$, whose kernel is its radical $\mathbb{C}e_n$. This gives the complete projection and radical maps, rather than an involution assigned to a nonstable quotient.

Restoring the eighth coordinate has a different exact effect. In the infinity point basis (e_m, e_n) , the native trace block is

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (\text{ESH30b})$$

The diagonal entries vanish because the two different point idempotents multiply to zero after applying $\#_J$, and the cross entries are the trace-one values just computed. Its eigenvectors $e_m + e_n$ and $e_m - e_n$ have positive and negative values 2 and -2 . Completing the eighth state therefore restores the full nondegenerate hyperbolic pair, whose restriction to the retained one-dimensional infinity corner was radical. Under $\#_\Sigma$, by contrast, these two point idempotents are separately fixed and form two positive scalar summands, as (ESH27)–(ESH29) show.

There is a literal occurrence of the fraction $1/8$ in the normalized trace, with a different meaning. If $\text{tr}_8 = \text{Tr}/8$, then

$$\text{tr}_8(1) = 1, \quad \text{tr}_8(e_m) = 1/8, \quad \text{tr}_8(e_\infty) = 1/4. \quad (\text{ESH31})$$

These follow because the three idempotents have ranks eight, one and two as multiplication projectors. The value $1/8$ is the normalized mass of that one reduced state. It supplies no parameter evolution by itself. The next sections give actual parameter maps and calculate their full forms.

6. Exact boundary maps to the HEB quadratic forms

Write the HEB algebra and involution as

$$E_{a_H} = \mathbb{C}[r_H]/(r_H^2 + a_H), \quad r_H^\# = -r_H, \quad z^\# = \bar{z} \ (z \in \mathbb{C}). \quad (\text{ESH32})$$

Its full regular trace form in basis $(1, r_H)$ is $\text{diag}(2, 2a_H)$, by the same multiplication argument as (ESH19)–(ESH20). The original heat clock is $a_H(h) = -1/4 + 2h$.

For the infinity algebra $I = \mathbb{C}[\Theta]/(\Theta^2 + 1)$, the exact conjugate-deck isomorphism is

$$\phi_\Sigma : E_{1/4} \xrightarrow{\sim} I, \quad r_H \mapsto \Theta/2, \quad \phi_\Sigma^{-1}(\Theta) = 2r_H. \quad (\text{ESH33})$$

The squared relation follows from $\Theta^2 = -1$, and substitution gives both inverse compositions. Since $\#_\Sigma(\Theta) = -\Theta$, the map preserves the involution. An algebra isomorphism conjugates multiplication matrices in the corresponding bases, so preserves regular trace and hence the full reflected pairing. Equivalently the target matrix $\text{diag}(2, 2)$ pulls back under $r_H = \Theta/2$ to $\text{diag}(2, 1/2)$, exactly the HEB matrix at $a_H = 1/4$. That value corresponds to heat time $h = 1/4$, not the nilpotent time $h = 1/8$.

For the native involution on the same infinity algebra there is a different exact comparison:

$$\phi_J : E_{-1/4} \xrightarrow{\sim} I, \quad r_H \mapsto i\Theta/2, \quad \phi_J^{-1}(\Theta) = -2ir_H. \quad (\text{ESH34})$$

Here $(i\Theta/2)^2 = 1/4$, so the source relation is respected. Native $\#_J$ fixes Θ and conjugates i , hence negates $i\Theta/2$. Both inverse compositions and involution compatibility follow by substitution, and the algebra isomorphism again preserves trace. Its form has inertia $(1, 1, 0)$, corresponding to HEB heat time $h = 0$. The scalar i is part of the map: this is a complex algebra and conjugate-linear-involution comparison, not an identification of the coefficient-fixed real presentations. In particular (ESH33) must not be used as a native-involution isometry.

The factor projection $\mathcal{A}_* \rightarrow I$ and the corner inclusion $I \rightarrow \mathcal{A}_*$, with zero in the three other factors, are explicit algebra maps. The projection is unital; the inclusion sends its unit to e_∞ . The inclusion preserves regular trace and the corresponding involution, while the projection retains only its factor contribution to the total trace. No unital trace-preserving map from the entire rank-eight algebra to this rank-two algebra could exist, since the traces of their units are eight and two. The full trace comparison is the sum of the four factor traces in (ESH11), not the projection alone.

The reduced infinity pair is not the collision algebra E_0 , and their precise morphisms show what is lost. Every unital complex algebra map $I \simeq \mathbb{C}^2 \rightarrow E_0$ is one of the two point evaluations followed by the scalar inclusion. Indeed an idempotent in E_0 has form $c + dr_H$; its square equation gives $c^2 = c$ and $(2c - 1)d = 0$, hence it is 0 or 1. The two complementary idempotents of I must therefore map to 0 and 1. Conversely each choice is a map. Every unital map $E_0 \rightarrow I$ kills r_H , since its image would be nilpotent in the reduced algebra I ; hence it is the reduction map followed by the diagonal scalar inclusion. Thus every composite from E_0 through I kills its nonzero nilpotent. The retained dual-number direction is genuinely present in the other heat receiver constructed next.

7. A target path whose marked-root quotient is exactly the original heat algebra

Define the following explicit path in the unchanged quartic coefficient target:

$$\gamma_+(h) = (0, 0, -1 - 16h, 16h), \quad F_h(R) = R^3 - (1 + 16h)R + 16h = (R - 1)(R^2 + R - 16h). \quad (\text{ESH35})$$

It begins at $\mathbf{U}_*$. This is a target path chosen to pull back the HEB clock; it is not asserted to be a native PDE evolution of the whole quartic. The marked root $R = 1$ is retained for every parameter, and

$$F'_h(1) = 2 - 16h = -8a_H(h). \quad (\text{ESH36})$$

The finite signed algebra over this path is

$$\mathcal{B} = \frac{\mathbb{C}[h, R, T]}{(F_h(R), T^2 - (3R^2 - 1 - 16h))}. \quad (\text{ESH37})$$

It is free of rank six over $\mathbb{C}[h]$: first divide by the monic cubic in R , then by the monic quadratic in T , giving basis $1, R, R^2, T, TR, TR^2$. Evaluation at the marked root gives the surjective algebra map

$$\pi_1 : \mathcal{B} \longrightarrow \mathcal{K} = \mathbb{C}[h, T]/(T^2 - 2 + 16h), \quad R \longmapsto 1, \quad T \longmapsto T, \quad \ker \pi_1 = (R - 1). \quad (\text{ESH38})$$

The kernel assertion follows by taking the quotient of the displayed presentation by $R - 1$; the resulting presentation is exactly $\mathcal{K}$. It supplies the entire scheme-theoretic marked-root quotient even where the root is multiple.

There is a full family isomorphism

$$\mathbb{C}[h, r_H]/(r_H^2 - 1/4 + 2h) \xrightarrow{\sim} \mathcal{K}, \quad r_H \longmapsto \frac{T}{2\sqrt{2}}, \quad \text{inverse: } T \longmapsto 2\sqrt{2}r_H. \quad (\text{ESH39})$$

The relation follows because $T^2/8 = 1/4 - 2h$; the inverse substitution recovers both generators. The scale $2\sqrt{2}$ is retained. For real h , $\#_\Sigma$ fixes the marked root and sends T to $-T$, so the quotient and this isomorphism preserve the HEB involution. The isomorphism preserves the regular trace of the rank-two receiver by conjugating its multiplication matrices. The projection selects that receiver; its relation to the whole trace is given separately below. In particular the receiver's pulled-back full trace form is exactly

$$Q_{\text{heat}, h} = 2|c|^2 + 2a_H(h)|d|^2 \quad \text{on } c + dr_H. \quad (\text{ESH40})$$

This is an actual comparison with the HEB threshold $h = 1/8$, arising from a finite marked-root factor, not from the missing infinity sign.

Away from $h = 1/8$, the marked factor is an actual direct summand of the finite fibre. An explicit root idempotent is

$$e_1(R; h) = \frac{R^2 + R - 16h}{2 - 16h}. \quad (\text{ESH41})$$

It has value 1 at $R = 1$ and vanishes on the other quadratic factor. Since the two factors are comaximal when $2 - 16h \neq 0$, these values prove its idempotence in the quotient; alternatively multiply the polynomial identities and reduce by F_h . The inverse to the resulting factor decomposition inserts the rank-two algebra by multiplying its representatives by e_1 . Its regular trace equals the whole-fibre trace on that corner because multiplication is zero on the complementary factors. At $h = 1/8$, the denominator vanishes and that idempotent expression cannot be retained; (ESH38) remains the valid quotient.

The infinity component stays constant along the entire path: $A = 0$, $k'_h(0) = 1$, and $\Theta^2 = -1$ by (ESH8). Thus the completed whole fibre is the finite signed algebra of (ESH37) plus this constant rank-two infinity factor. The omission of $\Theta = i$ is unchanged throughout the path. In the original source its included point is

$$(\alpha, y, z, w) = \left(0, 0, \frac{i(1 + 16h)}{2}, -16h\right), \quad (\text{ESH42})$$

obtained from the original $\alpha = 0$ formula $P = (0, y, 2iz + 7y^2, -4iyz - w - 3y^3)$. Its target is exactly $\gamma_+(h)$. This supplies a concrete check that the old infinity chart and its omitted sign remain different from the moving finite-root collision.

8. The full eight-state trace on the chosen path

For $h \geq 0$, put $s_h = \sqrt{1 + 64h} > 0$. The three finite roots are

$$R_1 = 1, \quad R_+(h) = \frac{-1 + s_h}{2}, \quad R_-(h) = \frac{-1 - s_h}{2}. \quad (\text{ESH43})$$

They are distinct except when $h = 1/8$, where $R_+ = R_1 = 1$ and $R_- = -2$. This follows directly from their formulas; $R_- \leq -1$ never meets the other two. The exact derivative values are

$$\mu_1 = 2 - 16h, \quad \mu_+ = \frac{s_h(s_h - 3)}{2}, \quad \mu_- = \frac{s_h(s_h + 3)}{2} > 0. \quad (\text{ESH44})$$

For proof, differentiate (ESH35). At a root of its quadratic factor the product derivative is $(R - 1)(2R + 1)$, and substitution of (ESH43) gives the last two expressions; the first is (ESH36).

For distinct roots the complete fibre decomposition is

$$\mathcal{A}_h \simeq \mathbb{C}[T_1]/(T_1^2 - \mu_1) \times \mathbb{C}[T_+]/(T_+^2 - \mu_+) \times \mathbb{C}[T_-]/(T_-^2 - \mu_-) \times I. \quad (\text{ESH45})$$

The projections evaluate R at the three displayed roots and use the infinity chart for I . The inverse maps use the three Lagrange idempotents $e_j(R) = \prod_{k \neq j} (R - R_k)/(R_j - R_k)$, multiplying a sign polynomial by its root idempotent. Their denominators are nonzero exactly in the stated distinct-root domain. Evaluations prove their orthogonal-idempotent relations, sum and inverse compositions, as in (ESH12). Thus the entire decomposition and all factor trace comparisons are explicit.

Each real root factor is preserved by $\#_\Sigma$. By (ESH20), a derivative $\mu < 0$ contributes two positive dimensions, while $\mu > 0$ contributes one positive and one negative dimension. The infinity factor always contributes two positive dimensions. For $0 \leq h < 1/8$ the derivative signs in the order (ESH44) are $(+, -, +)$; for $h > 1/8$ they are $(-, +, +)$. Hence

$$\boxed{\text{inertia}(Q_{\#_\Sigma, \mathcal{A}_h}) = (6, 2, 0) \quad (h \geq 0, h \neq 1/8).} \quad (\text{ESH46})$$

The sign gained by the fixed marked-root heat factor is exactly accompanied by the opposite change in the R_+ factor. The R_- factor remains indefinite throughout. An explicit negative vector on that persistent factor is e_-T , with value

$$Q_{\#_\Sigma}(e_-T, e_-T) = -2\mu_- < 0. \quad (\text{ESH47})$$

This follows because its square has scalar coefficient μ_-e_- , the involution negates its sign, and the trace of e_- is 2. Removing the one infinity sign merely removes a positive dimension, so the seven-dimensional quotient has inertia $(5, 2, 0)$ for these noncritical parameters as well.

9. The full collision algebra and its marked-root quotient

At $h_c = 1/8$,

$$F_{h_c}(R) = R^3 - 3R + 2 = (R - 1)^2(R + 2). \quad (\text{ESH48})$$

Near the repeated root write $\epsilon = R - 1$. The other factor $R + 2 = 3 + \epsilon$ is a unit in that local root algebra, so the root relation is exactly $\epsilon^2 = 0$. Differentiating gives $F'_{h_c}(1 + \epsilon) = 6\epsilon + 3\epsilon^2$, hence the signed local algebra is

$$\mathcal{L} = \frac{\mathbb{C}[\epsilon, T]}{(\epsilon^2, T^2 - 6\epsilon)} \xrightarrow{\sim} \mathbb{C}[T]/(T^4), \quad \epsilon \mapsto T^2/6. \quad (\text{ESH49})$$

The inverse sends T to its original sign coordinate. Squaring its relation gives $T^4 = 36\epsilon^2 = 0$, and the two substitutions recover ϵ, T . This proves the isomorphism with its full coefficient 6 retained. Its basis is $1, T, T^2, T^3$, or equivalently $1, \epsilon, T, \epsilon T$, so its dimension is four.

For a fully explicit whole-fibre decomposition at this parameter, the root idempotent for $R = -2$ is

$$e_{-2} = (R - 1)^2/9, \quad e_{\text{dbl}} = 1 - e_{-2}. \quad (\text{ESH50})$$

The first is 1 at -2 and zero in the double-root algebra; therefore it is idempotent modulo (ESH48). These two factors are comaximal because the retained difference is 3, so they give inverse product projections. Since $F'_{h_c}(-2) = 9$, the complete signed fibre is

$$\mathcal{A}_{h_c} \simeq \mathcal{L} \times \mathbb{C}[T_-]/(T_-^2 - 9) \times I. \quad (\text{ESH51})$$

The maps are given by e_{dbl}, e_{-2} , the local substitution (ESH49), and the infinity factor; this retains all eight dimensions.

Under $\#_\Sigma$, ϵ is fixed and T is negated. For any $f \in \mathcal{L}$, multiplication by its nonconstant part is nilpotent: it lies in the nilpotent ideal (T) , and its multiplication operator has fourth power zero. Thus its trace is zero, whereas its constant part acts on four basis vectors. This proves the complete local form

$$\text{Tr}_{\mathcal{L}}(f) = 4f(0), \quad Q_{\#_\Sigma, \mathcal{L}}(f, g) = 4\overline{f(0)}g(0), \quad \text{rad } Q = (T). \quad (\text{ESH52})$$

Its inertia is $(1, 0, 3)$. The simple -2 factor contributes $(1, 1, 0)$, and infinity contributes $(2, 0, 0)$. Therefore

$$\boxed{\text{inertia}(Q_{\#_\Sigma, \mathcal{A}_{h_c}}) = (4, 1, 3)}. \quad (\text{ESH53})$$

In particular the entire collision is still not a positive form: the vector supported on the -2 sign coordinate has value -18 .

The exact HEB receiver at the collision is the further quotient

$$0 \longrightarrow (T^2)/(T^4) \longrightarrow \mathcal{L} \xrightarrow{\bar{\pi}_1} \mathbb{C}[T]/(T^2) \longrightarrow 0, \quad r_H \mapsto T/(2\sqrt{2}). \quad (\text{ESH54})$$

The kernel is a two-dimensional square-zero ideal with basis T^2, T^3 . This follows directly from (ESH49), since the marked-root quotient sets $\epsilon = 0$, equivalently $T^2 = 0$. Both the kernel and quotient are stable under the stated involution. The quotient has trace $2f(0)$, so its trace is related to the full local trace by

$$\text{Tr}_{\mathcal{L}}(f) = 2 \text{Tr}_{E_0}(\bar{\pi}_1 f), \quad Q_{\mathcal{L}}(f, g) = 2Q_{E_0}(\bar{\pi}_1 f, \bar{\pi}_1 g). \quad (\text{ESH55})$$

The factors of two are required by the lengths four and two. The map sends a nonzero nilpotent to the HEB nilpotent and kills the specified further two-dimensional ideal; it does not identify the length-four object with dual numbers.

Deleting only the omitted infinity state from the completed collision gives a seven-dimensional algebra with inertia $(3, 1, 3)$. The original affine source removes more at this parameter: FS15 removes the ramification component $T = 0$, whose open complement is empty on $\mathcal{L}$ because T is nilpotent. Thus its actual three reduced points are the two signs over -2 and the included infinity sign. Their conjugate-deck trace has inertia $(2, 1, 0)$. These assertions follow by deleting the stated direct factors in (ESH51); they distinguish the seven-dimensional completed quotient from the original affine fibre at ramification.

10. The native involution relates two different target paths

The native map in (ESH17) sends the path (ESH35) to

$$\gamma_-(h) = (0, 0, -1 - 16h, -16h), \quad F_h^-(R) = (R + 1)(R^2 - R - 16h). \quad (\text{ESH56})$$

For $h \neq 0$ these are different target points. The exact completion map between their fibres remains

$$(R, T) \mapsto (-R, -T), \quad (v, \Theta) \mapsto (-v, \Theta), \quad (\text{ESH57})$$

as proved in (ESH18). It sends the fixed marked root $+1$ of γ_+ to the fixed marked root -1 of γ_- , whose derivative is also $2 - 16h$. Under the two marked-root isomorphisms (ESH39), it sends the HEB sign generator to its negative; adding coefficient conjugation gives the HEB involution as an antilinear map **between** those two receivers. It is not a self-involution of the whole $\gamma_+(h)$ fibre for nonzero h . At the initial common target it instead exchanges its two different root factors, giving the mixed native form (ESH23). This is why the full-path calculation (ESH46) is specifically for $\#_\Sigma$, without silently substituting the native involution or the global Weil involution.

11. What the two occurrences of one eighth retain

The one-eighth normalized trace value in (ESH31) belongs to one reduced infinity state. The HEB boundary at $h = 1/8$ belongs, under (ESH35)–(ESH39), to a derivative becoming zero at a finite marked root. The explicit maps prove a useful relation to the same eight-state completion, but they locate these events in different factors. The infinity sign pair remains reduced and unramified throughout the path; the finite collision retains the larger length-four algebra (ESH49) and its exact dual-number quotient.

For any fixed support lattice L , the commutative ring quotients and isomorphisms above induce $G_L(f) : (x, \lambda) \mapsto (f(x), \lambda)$. Preservation of the amplitude operations follows from the ring map, and support joins and meets are unchanged, proving these are the actual support-semiring maps. In (ESH54), the retained nonzero amplitude T has supported square e_L in the dual-number quotient, distinct from τ_L . Its nonzero squared amplitude in the length-four source lies in the explicitly killed ideal. The absence of the infinity state is instead the idempotent quotient (ESH26), with the whole kernel $\mathbb{C}e_m$ displayed. The two maps therefore retain different, exactly specified information.

The complete calculation yields no positive whole eight-state trace at the proposed one-eighth boundary: the conjugate-deck inertias are (ESH46) and (ESH53), and the native initial inertia is (ESH24). The positive HEB factor after the boundary is real mathematical data, with the additional indefinite factor and all nilpotent multiplicities retained in the same calculation.

12. The whole trace in a polynomial basis through the collision

The root idempotents in (ESH45) have denominators that vanish at the collision. The following calculation instead uses the free polynomial basis of (ESH37) over the entire parameter line. Introduce the coefficient abbreviations

$$c_h = 1 + 16h, \quad d_h = 16h, \quad K_h = \mathbb{C}[R]/(R^3 - c_h R + d_h), \quad B_h = K_h[T]/(T^2 - 3R^2 + c_h). \quad (\text{ESH58})$$

These are exactly the finite root and signed fibre algebras already used, with the original root and sign coordinates unchanged. The notation also defines free algebras over $\mathbb{C}[h]$ before specializing h . The infinity factor I remains the one in (ESH33), so the whole fibre is $\mathcal{A}_h = B_h \times I$. Use the ordered bases

$$\begin{aligned} \mathcal{E}_R &= (1, R, R^2), & \mathcal{E}_B &= (1, R, R^2, T, TR, TR^2), \\ \mathcal{E}_8 &= ((1, 0), (R, 0), (R^2, 0), (T, 0), (TR, 0), (TR^2, 0), (0, 1), (0, \Theta)). \end{aligned} \quad (\text{ESH59})$$

The entries of $\mathcal{E}_8$ lie in $B_h \times I$; in particular its first entry is the finite-component unit, and its seventh entry is the infinity-component unit. Their sum is the whole unit. This fixed decomposition uses the disjoint finite and infinity root components proved after (ESH41), without splitting the moving finite roots.

Let M_R be multiplication by R in $\mathcal{E}_R$, with each column the image of its corresponding basis vector. The defining monic cubic gives

$$M_R = \begin{pmatrix} 0 & 0 & -d_h \\ 1 & 0 & c_h \\ 0 & 1 & 0 \end{pmatrix}, \quad M_R^3 = c_h M_R - d_h I_3. \quad (\text{ESH60})$$

Thus, writing $p_j = \text{Tr}_{K_h}(R^j) = \text{tr}(M_R^j)$, direct traces of I_3, M_R, M_R^2 and multiplication of (ESH60) by further powers give

$$\begin{aligned} p_0 &= 3, & p_1 &= 0, & p_2 &= 2c_h, & p_j &= c_h p_{j-2} - d_h p_{j-3} \quad (j \geq 3), \\ p_3 &= -3d_h, & p_4 &= 2c_h^2, & p_5 &= -5c_h d_h, & p_6 &= 2c_h^3 + 3d_h^2. \end{aligned} \quad (\text{ESH61})$$

For example the recurrence gives $p_3 = c_h p_1 - d_h p_0 = -3d_h$, then gives each subsequent displayed term by substitution. The calculation is in $\mathbb{C}[h]$, so remains valid on every fibre, including nonreduced ones.

Define the two root-trace matrices in the unchanged basis by

$$\begin{aligned} P_h &= (p_{i+j})_{0 \leq i, j \leq 2} = \begin{pmatrix} 3 & 0 & 2c_h \\ 0 & 2c_h & -3d_h \\ 2c_h & -3d_h & 2c_h^2 \end{pmatrix}, \\ Q_h &= (3p_{i+j+2} - c_h p_{i+j})_{0 \leq i, j \leq 2} = \begin{pmatrix} 3c_h & -9d_h & 4c_h^2 \\ -9d_h & 4c_h^2 & -12c_h d_h \\ 4c_h^2 & -12c_h d_h & 4c_h^3 + 9d_h^2 \end{pmatrix}. \end{aligned} \quad (\text{ESH62})$$

Every entry follows from (ESH61). The second is precisely $\text{Tr}_{K_h}((3R^2 - c_h)R^{i+j})$, retaining the derivative multiplier $3R^2 - c_h = F'_h(R)$.

For clarity, these root traces determine the full signed trace without a reducedness assumption. Multiplication by $a \in K_h$ acts on the two summands $K_h \oplus TK_h$ by the same matrix M_a , so its trace in B_h is $2 \text{Tr}_{K_h}(a)$. Multiplication by Tb , for $b \in K_h$, has block matrix

$$L_{Tb} = \begin{pmatrix} 0 & M_{(3R^2 - c_h)b} \\ M_b & 0 \end{pmatrix} \quad \text{on } K_h \oplus TK_h. \quad (\text{ESH63})$$

Its trace is zero. Even-even products therefore give $2P_h$, mixed products give zero, and odd-odd products give $2Q_h$ for the complex bilinear regular trace. For real h , coefficient conjugation fixes R, T , while $\#_\Sigma$ fixes R and negates T . Consequently their complete Hermitian matrices in $\mathcal{E}_8$ are

$$\begin{aligned} [Q_{\kappa, \mathcal{A}_h}]_{\mathcal{E}_8} &= \text{diag}(2P_h, 2Q_h, 2, -2), \\ [Q_{\#_\Sigma, \mathcal{A}_h}]_{\mathcal{E}_8} &= \text{diag}(2P_h, -2Q_h, 2, 2). \end{aligned} \quad (\text{ESH64})$$

The last two entries are obtained by multiplying in $I = \mathbb{C}[\Theta]/(\Theta^2 + 1)$ with the respective involutions, as in (ESH20). These are polynomial matrix formulas for the whole eight-dimensional family. They do not substitute the native involution $\#_J$, whose distinct target map is still (ESH57).

12.1. The critical ranks and the complete radical

At $h_c = 1/8$, the retained coefficients are $c_{h_c} = 3$ and $d_{h_c} = 2$. Put

$$u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}.$$

Direct substitution in (ESH62) gives

$$\begin{aligned} P_{h_c} &= \begin{pmatrix} 3 & 0 & 6 \\ 0 & 6 & -6 \\ 6 & -6 & 18 \end{pmatrix} = 2uu^\top + vv^\top, \\ Q_{h_c} &= \begin{pmatrix} 9 & -18 & 36 \\ -18 & 36 & -72 \\ 36 & -72 & 144 \end{pmatrix} = 9vv^\top. \end{aligned} \quad (\text{ESH65})$$

For a coefficient vector $z \in \mathbb{C}^3$, $z^* P_{h_c} z = 2|u^\top z|^2 + |v^\top z|^2$. The two rows $u^\top$ and $v^\top$ are independent, since their first entries agree and their second entries differ. Thus P_{h_c} is positive semidefinite of rank two; its kernel is the simultaneous solution of these two linear equations. The second matrix has rank one because $v \neq 0$. The kernels, with all coordinates retained, are

$$\begin{aligned} \ker P_{h_c} &= \mathbb{C}(-2, 1, 1)^\top, \\ \ker Q_{h_c} &= \{(z_0, z_1, z_2)^\top : z_0 - 2z_1 + 4z_2 = 0\} \\ &= \text{span}_{\mathbb{C}}\{(2, 1, 0)^\top, (-2, 1, 1)^\top\}. \end{aligned} \quad (\text{ESH66})$$

The displayed vectors satisfy the equations and are independent; the ranks just proved show that they exhaust the kernels. It follows immediately from the blocks of (ESH64) that the finite conjugate-deck form has inertia $(2, 1, 3)$, and its whole eight-state form has inertia $(4, 1, 3)$. This proves (ESH53) directly in a basis that is defined across the collision.

More explicitly, write any element of $\mathcal{A}_{h_c}$ uniquely as $(p(R) + Tq(R), \eta_0 + \eta_1\Theta)$, with p, q of degree at most two. For two such elements $x = (p + Tq, \eta_0 + \eta_1\Theta)$ and $y = (\tilde{p} + T\tilde{q}, \tilde{\eta}_0 + \tilde{\eta}_1\Theta)$, (ESH64)–(ESH65) give the full pairing

$$Q_{\#_\Sigma}(x, y) = 4\overline{p(1)}\tilde{p}(1) + 2\overline{p(-2)}\tilde{p}(-2) - 18\overline{q(-2)}\tilde{q}(-2) + 2\overline{\eta_0}\tilde{\eta}_0 + 2\overline{\eta_1}\tilde{\eta}_1. \quad (\text{ESH67})$$

Each independent coefficient test in this formula can be realized by a degree-at-most-two polynomial, or by one infinity basis vector. For example the two constant and linear polynomials interpolate arbitrary values at 1, -2 , and a constant polynomial realizes any prescribed $q(-2)$. Hence the radical is exactly the subspace with $p(1) = p(-2) = q(-2) = \eta_0 = \eta_1 = 0$, not the larger cone of vectors whose single diagonal value happens to vanish. Factoring the degree-at-most-two polynomials gives the following complete radical in the original finite coordinates:

$$\mathfrak{N} = \text{rad } Q_{\#_\Sigma, \mathcal{A}_{h_c}} = \text{span}_{\mathbb{C}}\{n_1, n_2, n_3\}, \quad (\text{ESH68})$$

$$n_1 = ((R-1)(R+2), 0), \quad n_2 = (T(R+2), 0), \quad n_3 = (T(R-1)(R+2), 0).$$

Indeed the first polynomial spans the degree-at-most-two polynomials vanishing at both roots, while $R+2$ and $(R-1)(R+2)$ are independent and span the polynomials vanishing at -2 . These vectors are independent in the fixed free basis (ESH59).

12.2. The radical maps to the retained length-four local algebra

The critical product projection (ESH51) sends the three vectors of (ESH68) to zero in the simple -2 and infinity factors. In its length-four factor retain $R = 1 + \epsilon$, $\epsilon^2 = 0$, $T^2 = 6\epsilon$. Their exact images are

$$n_1 \mapsto T^2/2, \quad n_2 \mapsto 3T + T^3/6, \quad n_3 \mapsto T^3/2. \quad (\text{ESH69})$$

For the first image, $(R-1)(R+2) = \epsilon(3+\epsilon) = 3\epsilon = T^2/2$. Multiplication by T , or expansion of $T(3+\epsilon)$, proves the other two. Conversely the local classes T, T^2, T^3 , extended by zero to the other factors, are the images of

$$T \longleftrightarrow n_2/3 - n_3/9, \quad T^2 \longleftrightarrow 2n_1, \quad T^3 \longleftrightarrow 2n_3. \quad (\text{ESH70})$$

Both compositions are checked by substituting (ESH69). Thus the radical is exactly the entire nilpotent ideal (T) of $\mathbb{C}[T]/(T^4)$, extended by zero to the reduced factors. This identifies it as an ideal and as the nilradical of the whole critical algebra, not merely as a vector subspace of zero trace. The connecting maps preserve $\#_\Sigma$: n_1 is fixed, n_2, n_3 are negated, and the respective powers T^2, T, T^3 have those same signs.

The complete multiplication of the three retained generators is

$$n_1^2 = n_1 n_3 = n_2 n_3 = n_3^2 = 0, \quad n_2^2 = 18n_1, \quad n_1 n_2 = 3n_3, \quad (\text{ESH71})$$

$$\mathfrak{N}^2 = \text{span}_{\mathbb{C}}(n_1, n_3), \quad \mathfrak{N}^3 = \mathbb{C}n_3, \quad \mathfrak{N}^4 = 0.$$

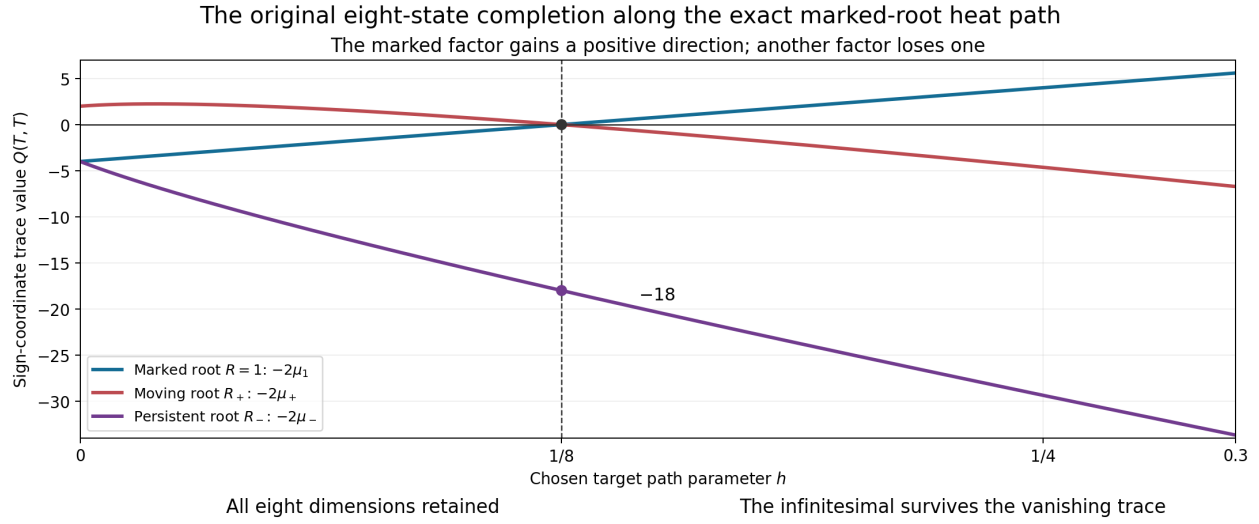
To prove every product, substitute (ESH69), multiply in $\mathbb{C}[T]/(T^4)$, and use the inverse (ESH70). The other components are zero. For instance $(3T + T^3/6)^2 = 9T^2 = 18(T^2/2)$ and $(T^2/2)(3T + T^3/6) = 3T^3/2$. All stated zero products have degree at least four. The displayed nonzero products, together with the independence of the three generators, prove the exact ideal powers and their dimensions. In particular $n_2^3 = 54n_3 \neq 0$ and $n_2^4 = 0$.

Finally, restrict the marked-root map (ESH38) at the collision to this radical. It evaluates $R = 1$ and imposes $T^2 = 0$, so

$$\pi_1(n_1) = 0, \quad \pi_1(n_2) = 3T = 6\sqrt{2}r_H, \quad \pi_1(n_3) = 0, \quad (\text{ESH72})$$

$$0 \longrightarrow \mathfrak{N}^2 \longrightarrow \mathfrak{N} \xrightarrow{\pi_1} \mathbb{C}r_H \longrightarrow 0.$$

Here $\mathbb{C}r_H$ is the nonzero nilpotent ideal in the HEB dual-number receiver, with its original scale $r_H = T/(2\sqrt{2})$. The map is onto since its value on n_2 is nonzero; (ESH68) and these images show that its kernel is precisely $\text{span}(n_1, n_3) = \mathfrak{N}^2$. This is an exact sequence of the corresponding nonunital ideals and of complex vector spaces, with their inherited involutions. It exhibits the full retained infinitesimal, its two-dimensional kernel, and the actual heat quotient in coordinates valid at the collision. No root idempotent with a vanishing denominator enters (ESH58)–(ESH72).



Parameter	Positive	Negative	Radical
$0 \leq h < 1/8$	6	2	0
$h = 1/8$	4	1	3
$h > 1/8$	6	2	0

Involution: $\#_z(T) = -T$, with scalar conjugation.

Infinity stays $\Theta^2 = -1$: two positive states.

The omitted infinity state is unramified.
It is separate from this finite-root collision.

$$F_{1/8}(R) = (R - 1)^2(R + 2)$$

$$\mathcal{L} = \mathbb{C}[T]/(T^4)$$

$$\varepsilon = R - 1 = T^2/6$$

$$\downarrow \text{kernel } (T^2)/(T^4)$$

$$E_0 = \mathbb{C}[T]/(T^2), \quad r_H = T/(2\sqrt{2})$$

$$Q_{\mathcal{L}}(f, g) = 2Q_{E_0}(\tilde{r}_1 f, \tilde{r}_1 g)$$

Figure 7: The full eight-state conjugate-deck trace along the specified target path. ESH35–ESH55 prove the three derivative functions, their sign exchange, the persistent negative factor, and the complete length-four collision with its dual-number quotient.

Sources, exact proof locations, and reading scope

This addition develops the established tau/split-zero base, the original escaping inverse fibre, and the actual quotient-size interpolation. Every new algebraic and analytic calculation appears in full in the accompanying complete proof files. The exact operator input retains its earlier public definitions and proofs; those are linked below. This is not a historical novelty claim or a proof of the Riemann hypothesis.

Earlier programme proofs

- The supported-zero prime and chain are prior programme results. Paper 1 is *An Algebraic Structure Incorporating a $\mathbb{Z}/1\mathbb{Z}$ -Symmetric Element Adjoined to the Integers: Construction, Analysis, and Generalizations*, original `11.tex`, complete public source, labels `thm:prime_ideals_S`, `thm:krull_dim_S`, `eq:maximal_chain`. The source was read completely, 1–1833. The user attributes this result to Gemini Pro 2.5; this is received attribution, not an inferred author field. The present proof of the full-support statement is FSR1–FSR7.
- *Split Support Geometry: Universal Zero Fibres, Arithmetic Curves, and Frobenius-Perfect Quantization*, The Clankers, June 2026, version `split_support_geometry_arithmetic_curve_v11910.tex`, SHA256 8443cc0401b18d373939d992f0a2f5384ee2fa1130f059753d5f3972eec5052d. Root reading covers 1–1910; used definitions and proofs are 1180–1871. This is the received full-lattice foundation. The complete receiving proof, including all needed definitions, is FSR1–FSR36, not an assertion that the entire source was read.
- The actual escaping map and heat constants are in Source mechanism transfer, labels `eq:retained-node-map`, `eq:retained-arithmetic-extension`, and Incompressible fibre heat, labels `eq:ifh-original`, `eq:ifh-chart`, `eq:ifh-target-flow`, `eq:ifh-triple`. Both local source versions were read completely; EFI records their exact hashes. The new derivation EFI1–EFI68 proves every received polynomial identity it uses.
- The original metric and observation are in Observed current plane, OCP1 and OCP5–OCP7. Read coverage is 35–169. The published exact incoming family and support maps are Observed deformation and current, DF1–DF16, and Support spectrum and terminal mass, ST1–ST20, both read completely and byte-matched to their public sources.
- The subsequent complete incoming proofs are Observed collision, OC1–OC17; Observed support propagation, OSP1–OSP51; and Observed cotangent and Frobenius, OCF1–OCF23 and OIF1–OIF36. ISP and EFI give the new exact trace/current and escaping-family comparisons.

Supplied original LaTeX on separated zeta interpolation

The source-reading ledger is keyed to canonical IDs from the existing disk index. Its search results were used for routing; they were not counted as reading.

Source	Exact version and root reading	Result used here
<i>Secondary Note on Split-Zero Globalization</i> , The Clankers; PUBUNIT-B1A260A5E731EA568C353AA4	<code>split_zero_secondary_note.tex</code> , SHA256 8e1aa2f934c49fc21e7a3d946ac6a465bc1a68896ec184afcb3d989cc9f8b364; complete 1–1997	<code>def:hurwitz-specialization</code> , <code>thm:universal-taylor-deformation</code> , <code>thm:arithmetic-kernels-riemann</code> , <code>cor:first-log-jet-explicit</code> ; NL1–NL18, SW1–SW42 give the complete receiving proofs.
<i>Split-Zero Support Repair</i> , source author field “The Clankers / prepared in continuation”; PUBUNIT-C4735C0E0B05FF5D811C3F89	<code>some_considerations.tex</code> , SHA256 3f98c8eab35f87996a594d0afa7d77ba561cbc28b632fa33e16d2d6228416702; 1473–1739	Conservative ideal/zeta side and shifted deformation. The multiplicative ideal-norm hypothesis is retained explicitly in NL.
Split-zero defect preprint, blank source author field; PUBUNIT-6C3FFDFC49FDF75FE8EB43E3	<code>split_zero_defect_preprint(1).tex</code> , SHA256 1d5f15f9999b402356d6ba46286f2473ee6c857f55e091c836c905ca15b4695e; 480–854	Endpoint residues and the invariant/anti-invariant decomposition. SW distinguishes function residues from logarithmic-derivative multiplicities and derives the reflection defect.

Source	Exact version and root reading	Result used here
Primitive defect preprint; PUBUNIT- 2B18D475F69385B03F1E2B30	<code>primitive_defect_preprint.tex</code> , SHA256 2f3289f6a04105d3f15fe5f7c36dde286515e61f465caf1582f9808e19f3d2f7; 1–385; the two supplied copies are identical	Retained augmentation kernel; not used as a positivity theorem.
Primitive support defects; PUBUNIT- E4607F9536B56699EB520E23	<code>primitive_support_defects_preprint.tex</code> , SHA256 f12f2c1c36effbf89740e308733d8434989e69cc088a0fc1c686281a157b42c5; 1–440	Source comparison for signed support defects.
Split-zero moonshine repair; PUBUNIT- 396B23CD772DEC7617B4A2A5	<code>split_zero_moonshine_repair.tex</code> , SHA256 7e672f7f1c21ae8dca327a65d8395d1a8b201bbddebb4d0317a96ab9e2382538; complete 1–826	Character-coefficient positivity and retained negative polar data were read; no moonshine theorem is applied to infer Weil positivity.

These received manuscripts are identified as programme sources. The independently written complete receiving proofs are included here. No private correspondence is reproduced.

Human sources

- Alain Connes and Caterina Consani, *Weil positivity and Trace formula, the archimedean place*, arXiv:2006.13771v1; original author source archive, `weil-compo.tex`, Appendix A “Fourier versus Mellin transforms” (`appenmellinapp`) and Appendix B “Explicit formula” (`appendix2`), 2011–2070. This exact source block was read. Labels `bombieriexplicit`, `bombieriexplicit1`, `burnolexplicit1` fix the signs and conventions. The original archive is retained privately unchanged. WA1–WA30 proves the extension to the actual entire test functions, with convergence.
- Terence Tao, *A digestion of the Jacobian conjecture counterexample*, 21 July 2026, displayed original polynomial map; David Speyer, *The new counterexample to the Jacobian conjecture*, 20 July 2026, marked-factor discussion, as cited by the programme’s source. The first displayed map was checked live. Reading those posts does not claim a fresh audit of every original counterexample argument. EFI1–EFI3 and its exact check script verify the polynomial map and its inverse chart directly.
- NIST DLMF, §25.11, definitions, recurrence, special values and parameter derivative, equations 25.11.1, 25.11.3, 25.11.13, 25.11.14, 25.11.17, were checked in the primary HTML. Equation-TeX downloads returned HTTP 403 and the browser extractor rejected the TeX content type; no source download is claimed, and no PDF was substituted. SW3–SW8 supplies the needed continuation and derivative proofs independently.

The historical cotangent and prism foundations remain Luc Illusie’s regular-immersion and transitivity theorems and Bhatt–Scholze’s prism definitions, at the exact original-source locators retained in the linked OCF/OIF paper. This addition proves the specified polynomial and coefficient maps; it does not assert an unproved equality between complex Weil pairings and prismatic cohomology.

ES and endpoint heat source use

The ES inputs belong to the established Erdős–Straus collection. The source read is its original TeX, including the raw quarter projectors, forced metric and tetrahedral actions, original spatial affine map, simplex linking algebra and free-shell carrier. The new derivations reproduce every finite matrix and every connecting map they use. They do not attribute these older ES constructions to this addition.

The exact 488-page reading edition is pinned at 20709bee872ad10e4cc7d7b976d975e24532a803. The original TeX/build ZIP contains `reader/erdos_straus_project_reader.tex`, SHA256 678b3f70ee7d2a6d00f6c588d7778e390118ab4743a5759612dbba6430984c78. The intact local archive has SHA256 521582fa5d25eae3ab8771fc0668ca52374f21ff0f8871a72e652aa090849dd8. The local PDF was byte-matched to the pinned public Git blob.

Exact reader-source blocks read for this addition are 4981–5121 (raw quarter spectrum and projectors), 20151–20370 (the full simplex 1–2–3 linking map and its proof), and 23426–23960 (ordered shell blades,

simplex support, directed carrier and complete control-length/determinant proof). The corresponding labels are `cor:raw-blade-cells-quarter-three-plus-one`, `eq:raw-negative-cell-projector`, `thm:simplex-123-modular-intertwiner`, `cor:general-shell-simplex-Lorentz-pullback`, and `thm:general-shell-control-determinant-volume`. Figures 10 and 12 are the supplied images' linking and shell constructions. Further source reading in the current ES manuscript covered the exact forced metric, D3 action, regular tetrahedral completion and moving spatial map at the labels recorded in ESQ. This is bounded reading, not a claim to have reread the entire collection.

The endpoint convention is the same primary-source explicit formula already read in Connes–Consani's original author TeX, at the Appendix B locators above. HEB13–HEB17 prove the exact endpoint isomorphism; HEB26–HEB30 prove its rational completion correction. The original polynomial map and heat speed retain the public source-mechanism and incompressible-fibre citations above. No new claim is made for a global arithmetic identification of the ES operator or its infinite fibre trace.

The original eight-state source is the finite signed-root completion, FS1–FS31, read in full. Its source SHA256 is `eaf980a16e7d639e21953e5779e7f76abf18e8bdb80ae923c4160b456c22da6f`. The original polynomial and escaping arcs are FC36–FC47, read at public-source lines 590–769, SHA256 `10906ba3bd21c06645571560e4c7b0c3948c0ccaf65d8b9e525abf7cd732794c`. ESH preserves these original constructions and proves the new heat receivers and whole-fibre trace comparisons.